

Question ID d029366d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: d029366d

Data set A consists of **10** positive integers less than **60**. The list shown gives **9** of the integers from data set A.
43, 45, 44, 43, 38, 39, 40, 46, 40

The mean of these **9** integers is **42**. If the mean of data set A is an integer that is greater than **42**, what is the value of the largest integer from data set A?

ID: d029366d Answer

Correct Answer: 52

Rationale

The correct answer is **52**. The mean of a data set is calculated by dividing the sum of the data values by the number of values. It's given that data set A consists of **10** values, **9** of which are shown. Let ***x*** represent the **10th** data value in data set A, which isn't shown. The mean of data set A can be found using the expression $\frac{43+45+44+43+38+39+40+46+40+x}{10}$, or $\frac{378+x}{10}$. It's given that the mean of the **9** values shown is **42** and that the mean of all **10** numbers is greater than **42**. Consequently, the **10th** data value, ***x***, is larger than **42**. It's also given that the data values in data set A are positive integers less than **60**. Thus, $42 < x < 60$. Finally, it's given that the mean of data set A is an integer. This means that the sum of the **10** data values, $378 + x$, is divisible by **10**. Thus, $378 + x$ must have a ones digit of **0**. It follows that ***x*** must have a ones digit of **2**. Since $42 < x < 60$ and ***x*** has a ones digit of **2**, the only possible value of ***x*** is **52**. Since **52** is larger than any of the integers shown, the largest integer from data set A is **52**.

Question Difficulty: Hard

Question ID 8f4f6c69

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 8f4f6c69

$a, 26, 29, b, 31, 47, c$

For the given data set, the data values are listed in ascending order, where a , b , and c are constants. For this data set, the mean is **36**, the median is **29**, and the range is **72**. What is the value of c ?

- A. 54
- B. 72
- C. 81
- D. 98

ID: 8f4f6c69 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the data set containing $a, 26, 29, b, 31, 47, c$ is listed in ascending order and has a mean of **36**, a median of **29**, and a range of **72**. Since there are **7** values in the given data set, it follows that the median is the value in the fourth position. It's given that b is in the fourth position in the given data set, so it follows that $b = 29$. Since the mean of the data set is **36**, it follows that $\frac{(a+26+29+29+31+47+c)}{7} = 36$. Multiplying both sides of this equation by **7** yields $a + 26 + 29 + 29 + 31 + 47 + c = 252$, which is equivalent to $a + c + 162 = 252$. Subtracting **162** from both sides of this equation yields $a + c = 90$. The range is the difference between the greatest and least values in a data set. Since c is the greatest value and a is the least value in the given data set, it follows that $c - a = 72$, or $-a + c = 72$. Adding the equations $a + c = 90$ and $-a + c = 72$ yields $2c = 162$. Dividing both sides of this equation by **2** yields $c = 81$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 35fb60c1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 35fb60c1

The ratio **140** to ***m*** is equivalent to the ratio **4** to **28**. What is the value of ***m***?

ID: 35fb60c1 Answer

Correct Answer: 980

Rationale

The correct answer is **980**. It's given that the ratio **140** to ***m*** is equivalent to the ratio **4** to **28**. Therefore, the value of ***m*** can be found by solving the equation $\frac{140}{m} = \frac{4}{28}$. Multiplying each side of this equation by ***m*** yields **140** = $\frac{4m}{28}$. Multiplying each side of this equation by **28** yields **3,920** = **4*m***. Dividing each side of this equation by **4** yields **980** = ***m***. Therefore, the value of ***m*** is **980**.

Question Difficulty: Medium

Question ID 1f7a4a78

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 1f7a4a78

A grove has **6** rows of birch trees and **5** rows of maple trees. Each row of birch trees has **8** trees **20** feet or taller and **6** trees shorter than **20** feet. Each row of maple trees has **9** trees **20** feet or taller and **7** trees shorter than **20** feet. A tree from one of these rows will be selected at random. What is the probability of selecting a maple tree, given that the tree is **20** feet or taller?

- A. $\frac{9}{164}$
- B. $\frac{3}{10}$
- C. $\frac{15}{31}$
- D. $\frac{9}{17}$

ID: 1f7a4a78 Answer

Correct Answer: C

Rationale

Choice C is correct. If a tree from one of these rows is selected at random, the probability of selecting a maple tree, given that the tree is **20** feet or taller, is equal to the number of maple trees that are **20** feet or taller divided by the total number of trees that are **20** feet or taller. It's given that there are **6** rows of birch trees, and each row of birch trees has **8** trees that are **20** feet or taller. This means that there are a total of **6(8)**, or **48**, birch trees that are **20** feet or taller. It's given that there are **5** rows of maple trees, and each row of maple trees has **9** trees that are **20** feet or taller. This means that there are a total of **5(9)**, or **45**, maple trees that are **20** feet or taller. It follows that there are a total of **48 + 45**, or **93**, trees that are **20** feet or taller. Therefore, the probability of selecting a maple tree, given that the tree is **20** feet or taller, is $\frac{45}{93}$, or $\frac{15}{31}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID ac2871c5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: ac2871c5

After **20%** of the original number of marbles in a group were removed from the group, **360** marbles remained in the group. How many marbles were removed from the group?

- A. **72**
- B. **90**
- C. **450**
- D. **1,800**

ID: ac2871c5 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that **20%** of the original number of marbles were removed from the group. Let x represent the original number of marbles in the group. It follows that $\frac{20}{100}x$, or $0.20x$, marbles were removed from the group. Therefore, $x - 0.20x$ marbles remained in the group. It's also given that **360** marbles remained in the group. Thus, $x - 0.20x = 360$, or $0.80x = 360$. Dividing both sides of this equation by **0.80** yields $x = 450$. Substituting **450** for x in the expression $0.20x$ yields $0.20(450)$, or **90**. Therefore, **90** marbles were removed from the group.

Choice A is incorrect. This is **20%** of the remaining number of marbles.

Choice C is incorrect. This is the original number of marbles, not the number of marbles that were removed.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID bfb0396d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: bfb0396d

There are **450** tiles in a box. Of these tiles, **6%** are black. How many black tiles are in the box?

ID: bfb0396d Answer

Correct Answer: 27

Rationale

The correct answer is **27**. It's given that **6%** of the **450** tiles in a box are black. Therefore, the number of black tiles in the box can be calculated by multiplying the number of tiles in the box by $\frac{6}{100}$, which is equivalent to **450** $\left(\frac{6}{100}\right)$, or **27**.

Question Difficulty: Medium

Question ID 80d9d387

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 80d9d387

A company designs and makes handbags. To estimate the mean weight of the handbags made by the company on a particular day, a sample of the handbags made by the company on that day was selected at random. Based on the sample, it is estimated that the mean weight of all handbags made by the company on that day is **27.8 ounces (oz)**, with an associated margin of error of **0.02 oz**. Based on this estimate and associated margin of error, which of the following is the most plausible conclusion?

- A. The mean weight of all handbags made by the company on that day is between **27.78 oz** and **27.82 oz**.
- B. The actual weights of all handbags made by the company on that day are between **27.78 oz** and **27.82 oz**.
- C. The actual weights of all handbags from the sample are between **27.78 oz** and **27.82 oz**.
- D. The mean weight of all handbags made by the company on that day is **27.8 oz**.

ID: 80d9d387 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the estimated mean weight of all handbags made by the company on a particular day is **27.8 oz**, with an associated margin of error of **0.02 oz**. It follows that plausible values for the mean weight are between **(27.8 – 0.02) oz** and **(27.8 + 0.02) oz**. Therefore, the most plausible conclusion is that the mean weight of all handbags made by the company on that day is between **27.78 oz** and **27.82 oz**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID ab974dd3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: ab974dd3

Based on a random sample from a population, a researcher estimated that the mean value of a certain variable for the population is **20.5**, with an associated margin of error of **1**. Which of the following is the most appropriate conclusion?

- A. It is plausible that the actual mean value of the variable for the population is between **19.5** and **21.5**.
- B. It is not possible that the mean value of the variable for the population is less than **19.5** or greater than **21.5**.
- C. Every value of the variable in the population is between **19.5** and **21.5**.
- D. The mean value of the variable for the population is **20.5**.

ID: ab974dd3 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that based on a random sample from a population, the estimated mean value for a certain variable for the population is **20.5**, with an associated margin of error of **1**. This means that it is plausible that the actual mean value of the variable for the population is between **20.5 – 1** and **20.5 + 1**. Therefore, the most appropriate conclusion is that it is plausible that the actual mean value of the variable for the population is between **19.5** and **21.5**.

Choice B is incorrect. The estimated mean value and associated margin of error describe only plausible values, not all the possible values, for the actual mean value of the variable, so this is not an appropriate conclusion.

Choice C is incorrect. The estimated mean value and associated margin of error describe only plausible values for the actual mean value of the variable, not all the possible values of the variable, so this is not an appropriate conclusion.

Choice D is incorrect. Since **20.5** is the estimated mean value of the variable based on a random sample, the actual mean value of the variable may not be exactly **20.5**. Therefore, this is not an appropriate conclusion.

Question Difficulty: Medium

Question ID 8054710f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 8054710f

At a conference, there are a total of **275** attendees. Each attendee is assigned to either group A, group B, or group C. If one of these attendees is selected at random, the probability of selecting an attendee who is assigned to group A is **0.44** and the probability of selecting an attendee who is assigned to group B is **0.24**. How many attendees are assigned to group C?

ID: 8054710f Answer

Correct Answer: 88

Rationale

The correct answer is **88**. It's given that there are a total of **275** attendees and each attendee is assigned to either group A, group B, or group C. It's also given that if one of these attendees is selected at random, the probability of selecting an attendee who is assigned to group A is **0.44** and the probability of selecting an attendee who is assigned to group B is **0.24**. It follows that there are **0.44(275)**, or **121**, attendees who are assigned to group A and **0.24(275)**, or **66**, attendees who are assigned to group B. The number of attendees who are assigned to group C is the number of attendees who are not assigned to group A or group B. In other words, the number of attendees who are assigned to group C is the total number of attendees minus the number of attendees who are assigned to group A and group B. Therefore, the number of attendees who are assigned to group C is **275 — 121 — 66**, or **88**.

Question Difficulty: Hard

Question ID 8e6acf26

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 8e6acf26

On average, a certain plant grows **87** millimeters every ***m*** months. At this rate, which expression represents the number of millimeters, on average, the plant grows every ***k*** years?

- A. $\frac{29m}{4k}$
- B. $\frac{29k}{4m}$
- C. $\frac{1,044m}{k}$
- D. $\frac{1,044k}{m}$

ID: 8e6acf26 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that on average, a certain plant grows **87** millimeters every ***m*** months. This average rate can be written as $\frac{87}{m}$ millimeters per month. Since there are **12** months in **1** year, it follows that there are **12*k*** months in ***k*** years. Therefore, the number of millimeters, on average, the plant grows every ***k*** years can be calculated by multiplying the rate $\frac{87}{m}$ millimeters per month by **12*k*** months, which yields $(\frac{87}{m})(12k)$, or $\frac{1,044k}{m}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 25a06893

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 25a06893

The table shows selected values from function f .

x	$f(x)$
-1	16
0	17
1	18
2	19

Which of the following is the best description of function f ?

- A. Decreasing linear
- B. Increasing linear
- C. Decreasing exponential
- D. Increasing exponential

ID: 25a06893 Answer

Correct Answer: B

Rationale

Choice B is correct. The given values show that as x increases, $f(x)$ also increases, which means that f is an increasing function. Furthermore, $f(x)$ increases at a constant rate of 1 for each increase of x by 1. A function with a constant rate of change is linear. Thus, the function f can be described as an increasing linear function.

Choice A is incorrect. For a decreasing linear function, as x increases, $f(x)$ decreases rather than increases.

Choice C is incorrect. For a decreasing exponential function, for each increase of x by 1, $f(x)$ decreases by a fixed percentage rather than increases at a constant rate.

Choice D is incorrect. For an increasing exponential function, for each increase of x by 1, $f(x)$ increases by a fixed percentage rather than at a constant rate.

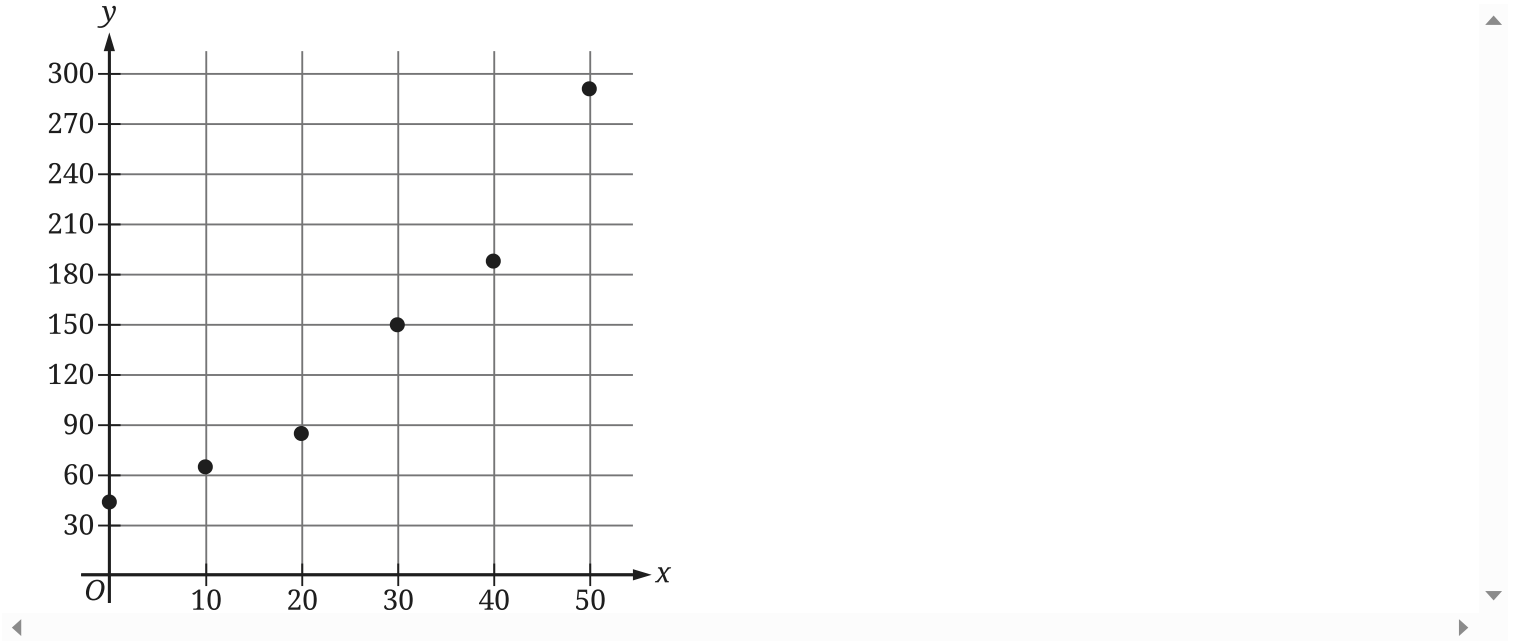
Question Difficulty: Medium

Question ID 8f6ec805

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 8f6ec805

The scatterplot shows the relationship between the number of swans, y , in a certain population and the number of years since 1970, x , where $0 \leq x \leq 50$.



Which of the following equations is the most appropriate exponential model for the data shown in the scatterplot?

- A. $y = 43(1.04)^{-x}$
- B. $y = 43(1.04)^x$
- C. $y = 288(1.04)^{-x}$
- D. $y = 288(1.04)^x$

ID: 8f6ec805 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the scatterplot shows the relationship between the number of swans, y , and the number of years since 1970, x , where $0 \leq x \leq 50$. According to the scatterplot, the number of swans increases as the number of years since 1970 increases, and the data shows an exponential growth pattern rather than a linear pattern. When $a > 0$ and $b > 1$, an exponential model of the form $y = a(b)^x$ represents exponential growth and a model of the form $y = a(b)^{-x}$ represents exponential decay. The models in all four choices show $a > 0$ and $b > 1$. Choices B and D show models of the form $y = a(b)^x$ and thus represent exponential growth. The initial value a should approximate the y -value when $x = 0$, which is a little more than 40, so $a = 43$ is reasonable and consistent with the graph. Therefore, the equation $y = 43(1.04)^x$ is the most appropriate exponential model for the data.

Choice A is incorrect. The model $y = 43(1.04)^{-x}$ represents exponential decay, rather than exponential growth, which doesn't match the pattern in the scatterplot.

Choice C is incorrect. The model $y = 288(1.04)^{-x}$ represents exponential decay rather than exponential growth, which doesn't match the pattern in the scatterplot.

Choice D is incorrect. The model $y = 288(1.04)^x$ has an initial value of **288** when $x = 0$, so the initial value doesn't match the scatterplot.

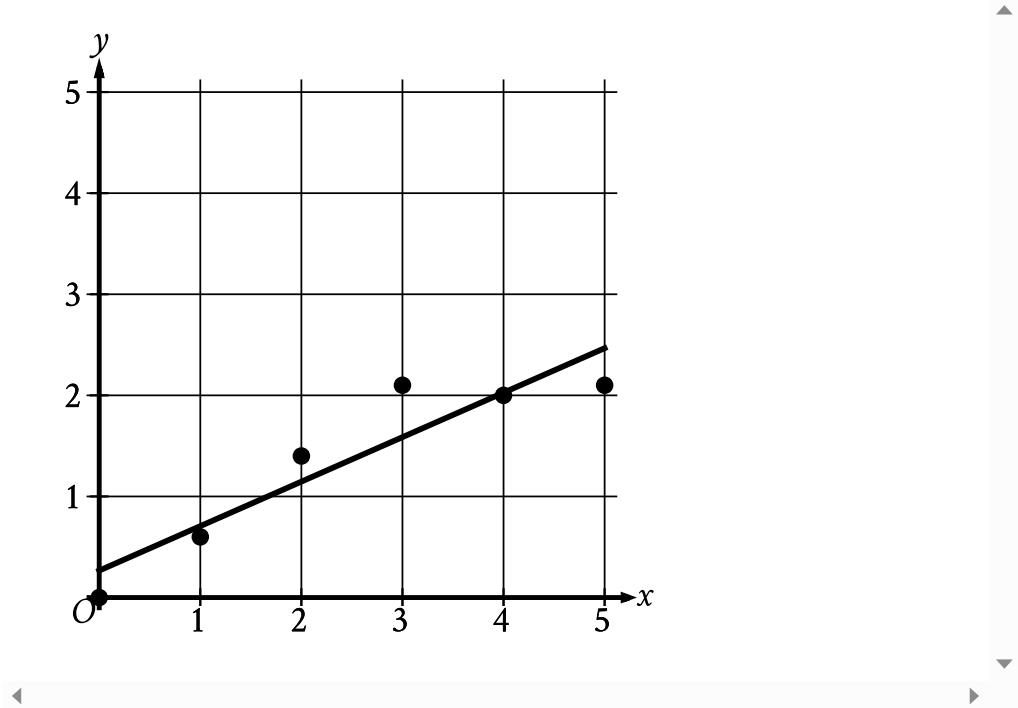
Question Difficulty: Medium

Question ID 65af419a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 65af419a

The scatterplot shows the relationship between x and y . A line of best fit is also shown.



Which of the following is closest to the slope of the line of best fit shown?

- A. -2.27
- B. -0.44
- C. 0.44
- D. 2.27

ID: 65af419a Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the scatterplot shows the relationship between two variables, x and y , and a line of best fit is shown. For the line of best fit shown, for each increase in the value of x by 1, the corresponding value of y increases by a constant rate. It follows that the relationship between the variables x and y has a positive linear trend. A line in the xy -plane that passes through the points (a, b) and (c, d) has a slope of $\frac{d-b}{c-a}$. The line of best fit shown passes approximately through the points $(0, 0.25)$ and $(4, 2)$. It follows that the slope of this line is approximately $\frac{2-0.25}{4-0}$, which is equivalent to 0.4375 . Therefore, of the given choices, 0.44 is closest to the slope of the line of best fit shown.

Choice A is incorrect. This is the slope of a line of best fit for a relationship between x and y that has a negative, rather than a positive, linear trend.

Choice B is incorrect. This is the slope of a line of best fit for a relationship between x and y that has a negative, rather than a positive, linear trend.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 3a8ffcf3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3a8ffcf3

If $\frac{x}{y} = 4$ and $\frac{24x}{ny} = 4$, what is the value of n ?

ID: 3a8ffcf3 Answer

Correct Answer: 24

Rationale

The correct answer is **24**. The equation $\frac{24x}{ny} = 4$ can be rewritten as $\left(\frac{24}{n}\right)\left(\frac{x}{y}\right) = 4$. It's given that $\frac{x}{y} = 4$. Substituting **4** for $\frac{x}{y}$ in the equation $\left(\frac{24}{n}\right)\left(\frac{x}{y}\right) = 4$ yields $\left(\frac{24}{n}\right)(4) = 4$. Multiplying both sides of this equation by n yields $(24)(4) = 4n$. Dividing both sides of this equation by **4** yields **24** = n . Therefore, the value of n is **24**.

Question Difficulty: Hard

Question ID 8f5c6dcd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 8f5c6dcd

A special camera is used for underwater ocean research. When the camera is at a depth of **58** fathoms, what is the camera's depth in feet? (**1 fathom** = **6 feet**)

ID: 8f5c6dcd Answer

Correct Answer: 348

Rationale

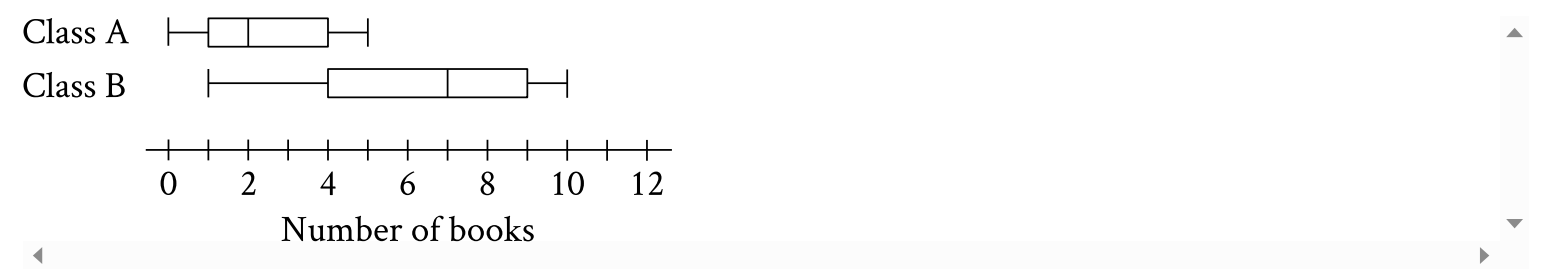
The correct answer is **348**. It's given that **1** fathom is equivalent to **6** feet. Therefore, **58** fathoms is equivalent to **(58 fathoms)**($\frac{6 \text{ feet}}{1 \text{ fathom}}$), or **348** feet. Thus, when the camera is at a depth of **58** fathoms, the camera's depth, in feet, is **348**.

Question Difficulty: Easy

Question ID 31511d69

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 31511d69



The two box plots show the distribution of number of books read over the summer by the students in two different English classes. What is the positive difference between the ranges of number of books read over the summer for the two classes?

ID: 31511d69 Answer

Correct Answer: 4

Rationale

The correct answer is **4**. It's given that the two boxplots show the distribution of number of books read over the summer by the students in two different English classes. In a boxplot, the first vertical line represents the minimum value of the data set and the last vertical line represents the maximum value of the data set. The range of a data set is the difference between its maximum value and its minimum value. In class A, the maximum number of books read is **5** and the minimum number of books read is **0**. The difference between those values is **5 – 0**, or **5**. Therefore, the range of the number of books read in class A is **5**. In class B, the maximum number of books read is **10** and the minimum number of books read is **1**. The difference between those values is **10 – 1**, or **9**. Therefore, the range of the number of books read in class B is **9**. To find the positive difference between the ranges of the number of books read for the two classes, the smaller range must be subtracted from the larger range. Therefore, the positive difference between the ranges of number of books read over the summer for the two classes is **9 – 5**, or **4**.

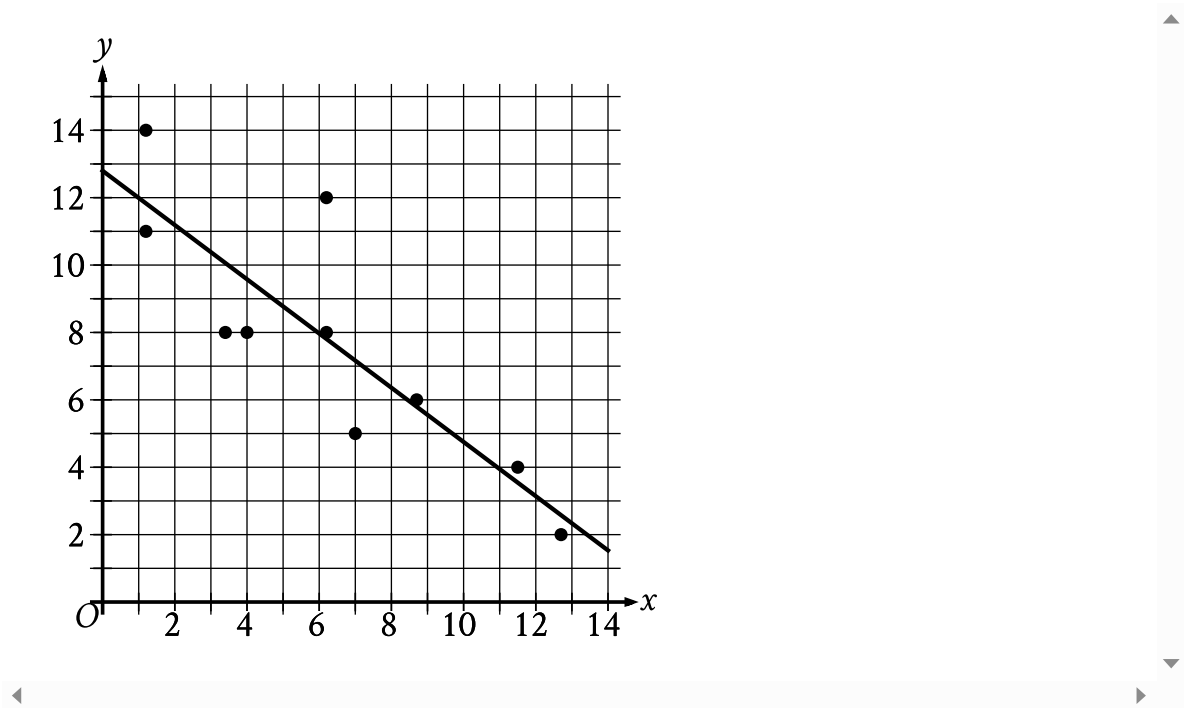
Question Difficulty: Hard

Question ID d50aa895

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: d50aa895

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following is closest to the slope of the line of best fit shown?

- A. -2.4
- B. -0.8
- C. 0.8
- D. 2.4

ID: d50aa895 Answer

Correct Answer: B

Rationale

Choice B is correct. A line of best fit is shown in the scatterplot such that as the value of x increases, the value of y decreases. Thus, the slope of the line of best fit shown is negative. The slope of a line passing through two points, (x_1, y_1) and (x_2, y_2) , can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $(1, 12)$ and $(11, 4)$. Substituting $(1, 12)$ and $(11, 4)$ for (x_1, y_1) and (x_2, y_2) , respectively, in $\frac{y_2 - y_1}{x_2 - x_1}$ gives $\frac{4 - 12}{11 - 1}$, which is equivalent to $-\frac{8}{10}$, or -0.8 . Therefore, of the given choices, -0.8 is closest to the slope of the line of best fit shown.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice D is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Question ID 177bca79

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 177bca79

A sample of oak has a density of **807** kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of **0.90** meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- A. **588**
- B. **726**
- C. **897**
- D. **1,107**

ID: 177bca79 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the sample is in the shape of a cube with edge lengths of **0.9** meters. Therefore, the volume of the sample is **0.90³**, or **0.729**, cubic meters. It’s also given that the sample has a density of **807** kilograms per **1** cubic meter. Therefore, the mass of this sample is **0.729 cubic meters** $\left(\frac{807 \text{ kilograms}}{1 \text{ cubic meter}}\right)$, or **588.303** kilograms. Rounding this mass to the nearest whole number gives **588** kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is **588**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

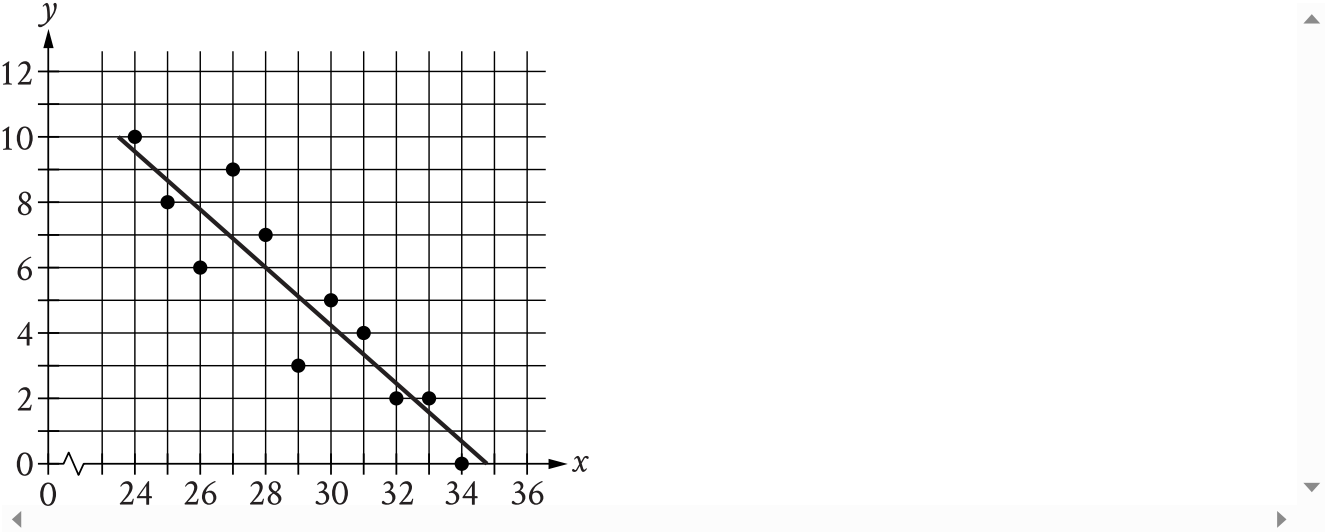
Question Difficulty: Hard

Question ID 2b2e51a9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 2b2e51a9

The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown.



At $x = 32$, which of the following is closest to the y -value predicted by the line of best fit?

- A. 0.4
- B. 1.5
- C. 2.4
- D. 3.3

ID: 2b2e51a9 Answer

Correct Answer: C

Rationale

Choice C is correct. At $x = 32$, the line of best fit has a y -value between 2 and 3. The only choice with a value between 2 and 3 is choice C.

Choice A is incorrect. This is the difference between the y -value predicted by the line of best fit and the actual y -value at $x = 32$ rather than the y -value predicted by the line of best fit at $x = 32$.

Choice B is incorrect. This is the y -value predicted by the line of best fit at $x = 31$ rather than at $x = 32$.

Choice D is incorrect. This is the y -value predicted by the line of best fit at $x = 33$ rather than at $x = 32$.

Question Difficulty: Medium

Question ID 12595509

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 12595509

An area of **46.00** square nautical miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k ? (**1 nautical mile = 1.852 kilometers**)

ID: 12595509 Answer

Correct Answer: 157.8, 789/5

Rationale

The correct answer is **157.8**. It's given that an area of **46.00** square nautical miles is equivalent to k square kilometers. Since **1** nautical mile is equal to **1.852** kilometers, it follows that an area of **1** square nautical mile is equivalent to $(1.852)^2$, or **3.429904**, square kilometers. Multiplying **46.00** by **3.429904** yields **157.775584** square kilometers. Rounding this value to the nearest tenth yields **157.8**. Therefore, the value of k is **157.8**.

Question Difficulty: Hard

Question ID 9764f80c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 9764f80c

An object travels at a constant speed of **12** centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel **108** centimeters?

- A. **9**
- B. **96**
- C. **120**
- D. **972**

ID: 9764f80c Answer

Correct Answer: A

Rationale

Choice A is correct. If the object travels **108** centimeters at a speed of **12** centimeters per second, the time of travel can be determined by dividing the total distance by the speed. This results in $\frac{108 \text{ centimeters}}{12 \text{ centimeters/second}}$, which is **9** seconds.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 3654ed1c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 3654ed1c

The speed of a vehicle is increasing at a rate of **7.3** meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth? (Use **1 mile = 1,609 meters**.)

- A. **0.3**
- B. **16.3**
- C. **195.8**
- D. **220.4**

ID: 3654ed1c Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the speed of a vehicle is increasing at a rate of **7.3** meters per second squared. It's given to use **1 mile = 1,609 meters**. There are **60** seconds in **1** minute; therefore, **60²** or **3,600** seconds squared is equal to **1** minute squared. It follows that the rate of **7.3** meters per second squared is equivalent to $\left(\frac{7.3 \text{ meters}}{1 \text{ second squared}}\right)\left(\frac{1 \text{ mile}}{1,609 \text{ meters}}\right)\left(\frac{3,600 \text{ seconds squared}}{1 \text{ minute squared}}\right)$, or approximately **16.33 miles per minute squared**. The rate, in **miles per minute squared**, rounded to the nearest tenth is **16.3**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 9fb9a483

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 9fb9a483

How many yards are equivalent to **1,116** inches? (**1 yard = 36 inches**)

ID: 9fb9a483 Answer

Correct Answer: 31

Rationale

The correct answer is **31**. It's given that **1** yard is equal to **36** inches. Therefore, **1,116** inches is equivalent to **(1,116 inches)** $\left(\frac{1 \text{ yard}}{36 \text{ inches}}\right)$, or **31** yards.

Question Difficulty: Easy

Question ID c81eba04

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: c81eba04

What is 10% of 370?

- A. 27
- B. 37
- C. 333
- D. 360

ID: c81eba04 Answer

Correct Answer: B

Rationale

Choice B is correct. 10% of a quantity means $\frac{10}{100}$ times the quantity. Therefore, 10% of 370 can be represented as $\frac{10}{100}(370)$, which is equivalent to $0.10(370)$, or 37. Therefore, 10% of 370 is 37.

Choice A is incorrect. This is 10% of 270, not 10% of 370.

Choice C is incorrect. This is 90% of 370, not 10% of 370.

Choice D is incorrect. This is $370 - 10$, not 10% of 370.

Question Difficulty: Easy

Question ID 1a471048

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 1a471048

- The positive number a is 2,241% of the sum of the positive numbers b and c , and b is 83% of c . What percent of b is a ?
- A. 23.24%
 - B. 49.41%
 - C. 2,324%
 - D. 4,941%

ID: 1a471048 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that a is 2,241% of the sum of b and c . This can be represented by the equation $a = \left(\frac{2,241}{100}\right)(b + c)$, or $a = 22.41(b + c)$. It's also given that that b is 83% of c . This can be represented by the equation $b = \left(\frac{83}{100}\right)c$, or $b = 0.83c$. Dividing both sides of this equation by 0.83 yields $\frac{b}{0.83} = c$. Substituting $\frac{b}{0.83}$ for c in the equation $a = 22.41(b + c)$ yields $a = 22.41\left(b + \frac{b}{0.83}\right)$, or $a = 22.41\left(\frac{1.83b}{0.83}\right)$, which is equivalent to $a = \frac{41.0103b}{0.83}$, or $a = 49.41b$. This equation is equivalent to $a = \left(\frac{4,941}{100}\right)b$; therefore, a is 4,941% of b .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 931657ae

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 931657ae

The table summarizes the distribution of color and shape for **100** tiles of equal area.

	Red	Blue	Yellow	Total
Square	10	20	25	55
Pentagon	20	10	15	45
Total	30	30	40	100

If one of these tiles is selected at random, what is the probability of selecting a red tile? (Express your answer as a decimal or fraction, not as a percent.)

ID: 931657ae Answer

Correct Answer: 0.3, 3/10

Rationale

The correct answer is $\frac{3}{10}$. It's given that there are a total of **100** tiles of equal area, which is the total number of possible outcomes. According to the table, there are a total of **30** red tiles. The probability of an event occurring is the ratio of the number of favorable outcomes to the total number of possible outcomes. By definition, the probability of selecting a red tile is given by $\frac{30}{100}$, or $\frac{3}{10}$. Note that 3/10 and .3 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 7d72e3e6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 7d72e3e6

ID: 7d72e3e6 Answer

What is 20% of 440?

Correct Answer: B

Rationale

- A. 44
- Choice B is correct. 20% of 440 can be calculated as $(\frac{20}{100})(440)$, which is equivalent to $\frac{8,800}{100}$, or 88.
- B. 88
- Choice A is incorrect. This is 10%, not 20%, of 440.
- C. 880
- Choice C is incorrect. This is 200%, not 20%, of 440.
- D. 1,760
- Choice D is incorrect. This is 400%, not 20%, of 440.

Question Difficulty: Easy

Question ID 057faaa2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 057faaa2

The positive number a is **230%** of the number b , and a is **60%** of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

- A. 138
- B. 217
- C. 283
- D. 383

ID: 057faaa2 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that a is **230%** of b . It follows that $a = \frac{230}{100}b$. It's also given that a is **60%** of c . It follows that $a = \frac{60}{100}c$. Since $a = \frac{230}{100}b$ and $a = \frac{60}{100}c$, it follows that $\frac{230}{100}b = \frac{60}{100}c$. Multiplying each side of this equation by $\frac{100}{60}$ yields $\frac{23}{6}b = c$. If c is $p\%$ of b , then $c = \frac{p}{100}b$. It follows that $\frac{23}{6} = \frac{p}{100}$. Multiplying each side of this equation by **100** yields $\frac{2,300}{6} = p$. It follows that the value of p is approximately **383.33**. Therefore, of the given choices, **383** is closest to the value of p .

Choice A is incorrect. This is closest to the value of p if b is **230%** of a , rather than if a is **230%** of b , and if b is $p\%$ of c , rather than if c is $p\%$ of b .

Choice B is incorrect. This is closest to the value of p if a is **230%** greater than b , rather than **230%** of b .

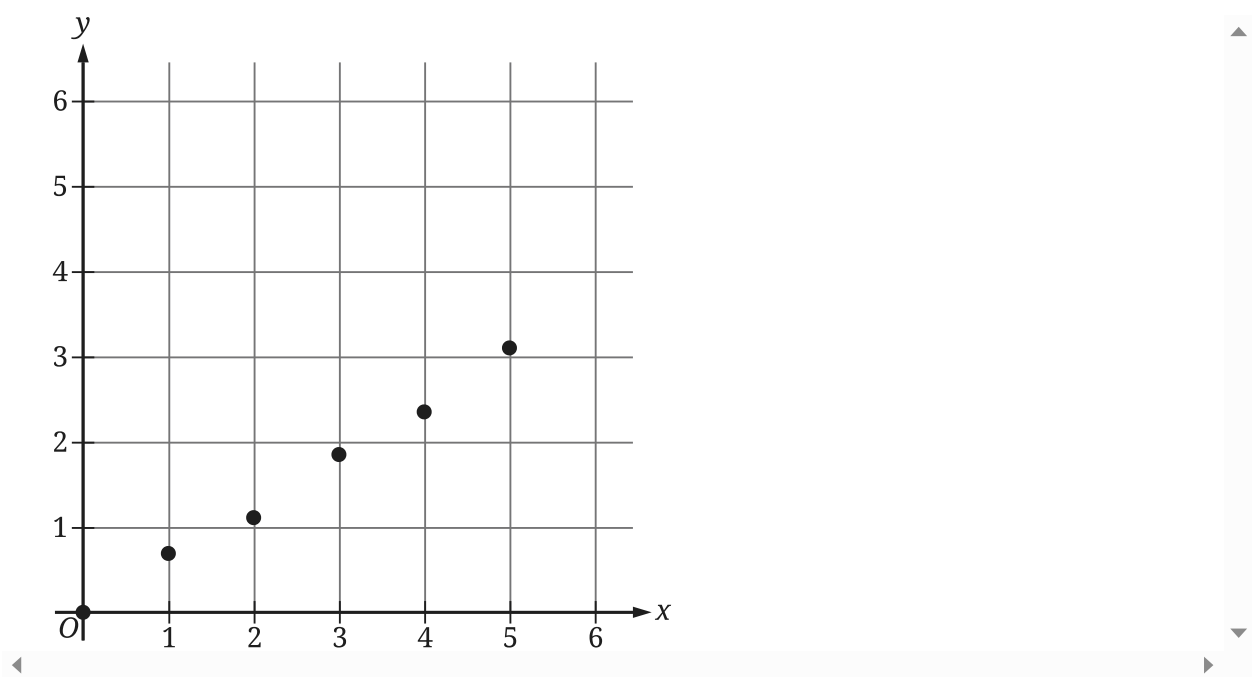
Choice C is incorrect. This is closest to the value of p if c is $p\%$ greater than b , rather than $p\%$ of b .

Question Difficulty: Hard

Question ID 4cc31f82

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 4cc31f82



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

- A. $y = -0.606x$
- B. $y = 0.606 - 1.001x$
- C. $y = 0.606x$
- D. $y = 0.606 + 1.001x$

ID: 4cc31f82 Answer

Correct Answer: C

Rationale

Choice C is correct. A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The graph of an appropriate linear model for the data shown in the scatterplot passes near the points $(0, 0)$ and $(4, 2.4)$ in the xy -plane. Two points on a line, (x_1, y_1) and (x_2, y_2) , can be used to find the slope of the line using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, 0)$ and $(4, 2.4)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{2.4 - 0}{4 - 0}$, or $m = 0.60$. Therefore, the value of m for an appropriate linear model is approximately **0.60**. Substituting **0.60** for m in $y = mx + b$ yields $y = 0.60x + b$. Since an appropriate linear model passes near the point $(0, 0)$, it follows that b is approximately **0** and an appropriate linear model for the data is $y = 0.60x + 0$. Thus, of the given choices, $y = 0.606x$ is the most appropriate linear model for these data.

Alternate approach: A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The scatterplot shows that as the x -values of the data points increase, the y -values of the data points increase, which means the graph of an appropriate linear model has a positive slope. The scatterplot also shows that

an appropriate linear model would pass close to the point $(0, 0)$. Of the given choices, $y = 0.606x$ is the only linear model whose graph has a positive slope and a y-intercept of $(0, 0)$.

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope, and a y-intercept that is not $(0, 0)$.

Choice D is incorrect. The graph of this model has a slope that is too steep to fit the data, and a y-intercept that is not $(0, 0)$.

Question Difficulty: Medium

Question ID f52c74df

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: f52c74df

In a scale model of an office building, a length of **1** inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. If a length in the actual office building measures x feet, which expression represents the corresponding length, in inches, in the model in terms of x ?

- A. $\frac{3x}{(17)(12)}$
- B. $\frac{17x}{(12)(3)}$
- C. $\frac{3x}{17}$
- D. $\frac{17x}{3}$

ID: f52c74df Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that in a scale model, a length of **1** inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. This means that the ratio of a length, in feet, in the actual office building to the corresponding length, in inches, in the model is $\frac{17}{3}$ to **1**. Therefore, the length, in inches, in the model corresponding to a length in the actual office building of x feet can be found by dividing x by $\frac{17}{3}$. Dividing by $\frac{17}{3}$ is equivalent to multiplying by $\frac{3}{17}$, so the corresponding length, in inches, in the model in terms of x can be represented by the expression $\frac{3x}{17}$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This expression represents the corresponding length, in inches, in the model in terms of x for a model where **1** inch in the model is equivalent to a length of $\frac{3}{17}$, not $\frac{17}{3}$, feet in the actual building.

Question Difficulty: Hard

Question ID 89727eba

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 89727eba

The ratio x to y is equivalent to the ratio 9 to 5. If the value of x is 162, what is the value of y ?

ID: 89727eba Answer

Correct Answer: 90

Rationale

The correct answer is 90. It's given that the ratio of x to y is equivalent to the ratio 9 to 5. It follows that $\frac{x}{y} = \frac{9}{5}$. Multiplying each side of this equation by $5y$ yields $\frac{(5y)x}{y} = \frac{9(5y)}{5}$, or $5x = 9y$. Dividing each side of this equation by 9 yields $\frac{5x}{9} = y$. Substituting 162 for x in this equation yields $\frac{5(162)}{9} = y$, which is equivalent to $\frac{810}{9} = y$, or $90 = y$. Therefore, if the value of x is 162, the value of y is 90.

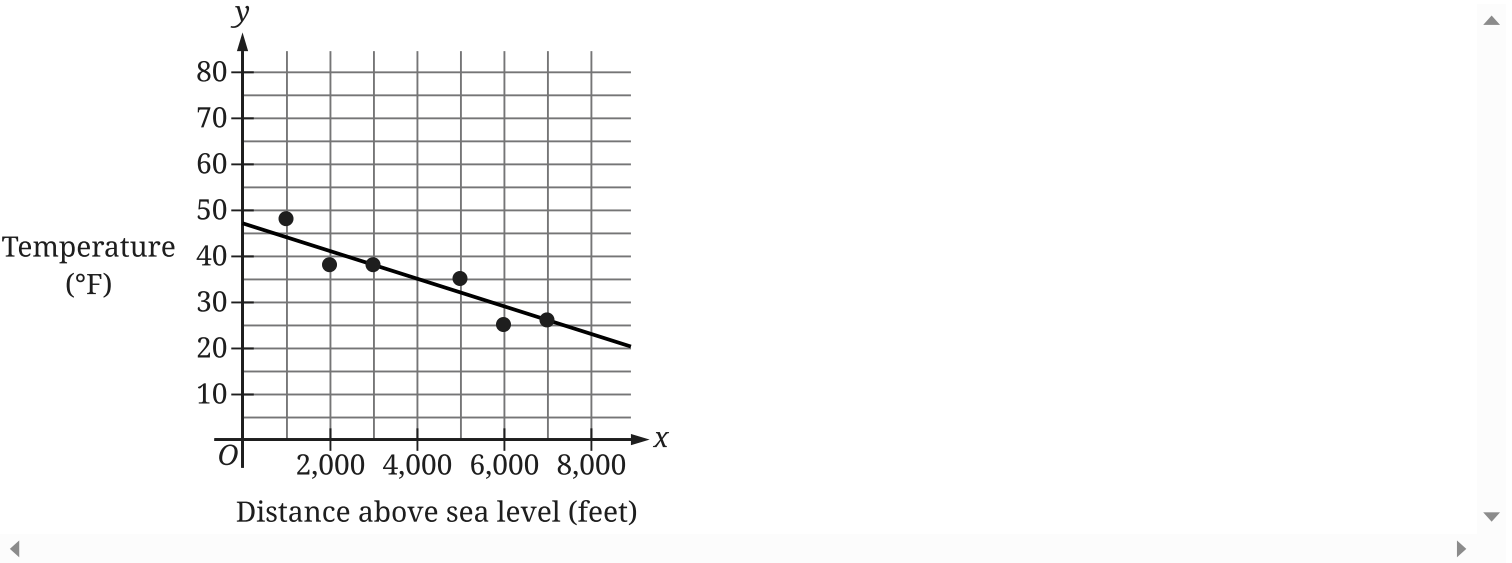
Question Difficulty: Medium

Question ID a6e5c021

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: a6e5c021

The scatterplot shows the temperature, **in degrees Fahrenheit ($^{\circ}\text{F}$)**, and the distance above sea level, in feet, measured at **6** locations on Mount Jefferson. A line of best fit is also shown.



At a distance of **4,000** feet above sea level, what is the temperature, **in $^{\circ}\text{F}$** , predicted by the line of best fit?

- A. **47**
- B. **35**
- C. **25**
- D. **0**

ID: a6e5c021 Answer

Correct Answer: B

Rationale

Choice B is correct. In the given scatterplot, the x-values represent the distance above sea level, in feet, and the y-values represent the temperature, in $^{\circ}\text{F}$. The point on the line of best fit with an x-value of **4,000** has a corresponding y-value of **35**. Therefore, at a distance of **4,000** feet above sea level, the temperature predicted by the line of best fit is **35°F** .

Choice A is incorrect. This is the temperature, in $^{\circ}\text{F}$, predicted by the line of best fit at a distance of **0** feet above sea level.

Choice C is incorrect. This is the measured temperature, in $^{\circ}\text{F}$, at a distance of **6,000** feet above sea level.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 11d0ca85

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 11d0ca85

ID: 11d0ca85 Answer

Correct Answer: A
What is 23% of 100?
Rationale

- A. 23
- Choice A is correct. 23% of 100 can be calculated by multiplying $\frac{23}{100}$ by 100, which yields $(\frac{23}{100})100$, or 23.
- B. 46
- Choice B is incorrect. This is 46%, not 23%, of 100.
- C. 77
- Choice C is incorrect. This is 23% less than 100, not 23% of 100.
- D. 123
- Choice D is incorrect. This is 23% greater than 100, not 23% of 100.

Question Difficulty: Easy

Question ID 9d9d9065

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 9d9d9065

The regular price of a shirt at a store is **\$11.70**. The sale price of the shirt is **80%** less than the regular price, and the sale price is **30%** greater than the store's cost for the shirt. What was the store's cost, in dollars, for the shirt? (Disregard the **\$** sign when entering your answer. For example, if your answer is **\$4.97**, enter **4.97**)

ID: 9d9d9065 Answer

Correct Answer: 1.8, 9/5

Rationale

The correct answer is **1.8**. It's given that the regular price of a shirt at a store is **\$11.70**, and the sale price of the shirt is **80%** less than the regular price. It follows that the sale price of the shirt is **$\$11.70(1 - \frac{80}{100})$** , or **$\$11.70(1 - 0.8)$** , which is equivalent to **\$2.34**. It's also given that the sale price of the shirt is **30%** greater than the store's cost for the shirt. Let **x** represent the store's cost for the shirt. It follows that **$2.34 = (1 + \frac{30}{100})x$** , or **$2.34 = 1.3x$** . Dividing both sides of this equation by **1.3** yields **$x = 1.80$** . Therefore, the store's cost, in dollars, for the shirt is **1.80**. Note that 1.8 and 9/5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 1da384a8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 1da384a8

For **100** buttons, the table summarizes the distribution by group and diameter.

Group	Diameter (millimeters)		
	Less than 20	20 to 30	Greater than 30
Group 1	12	8	3
Group 2	0	17	18
Group 3	10	32	0

One of these buttons will be selected at random. What is the probability of selecting a button with a diameter that is less than or equal to **30** millimeters, given that it is **not** in group 2? (Express your answer as a decimal or fraction, not as a percent.)

ID: 1da384a8 Answer

Correct Answer: .9538, 62/65

Rationale

The correct answer is $\frac{62}{65}$. It's given that one button will be selected at random out of a total of **100** buttons and that the table shows the distribution of buttons by group and diameter. The probability of selecting a button with a diameter less than or equal to **30** millimeters, given that it is not in group 2, is the number of buttons with a diameter less than or equal to **30** millimeters that are not in group 2 divided by the total number of buttons not in group 2. Based on the table, the buttons not in group 2 are those in group 1 and group 3. In group 1, there are **12** buttons with a diameter less than **20** millimeters and **8** buttons with a diameter from **20** to **30** millimeters, which yields $12 + 8$, or **20**, buttons with a diameter less than or equal to **30** millimeters. In group 3, there are **10** buttons with a diameter less than **20** millimeters and **32** buttons with a diameter from **20** to **30** millimeters, which yields $10 + 32$, or **42**, buttons with a diameter less than or equal to **30** millimeters. Therefore, the total number of buttons with a diameter less than or equal to **30** millimeters that are not in group 2 is $20 + 42$, or **62**. The total number of buttons that are not in group 2 is the sum of all buttons in group 1 and group 3. There are $12 + 8 + 3$, or **23**, buttons in group 1 and $10 + 32 + 0$, or **42**, buttons in group 3. Therefore, the total number of buttons in groups 1 and 3 is $23 + 42$, or **65**. It follows that the probability of selecting a button with a diameter less than or equal to **30** millimeters, given that it is not in group 2, is $\frac{62}{65}$. Note that 62/65 or .9538 are examples of ways to enter a correct answer.

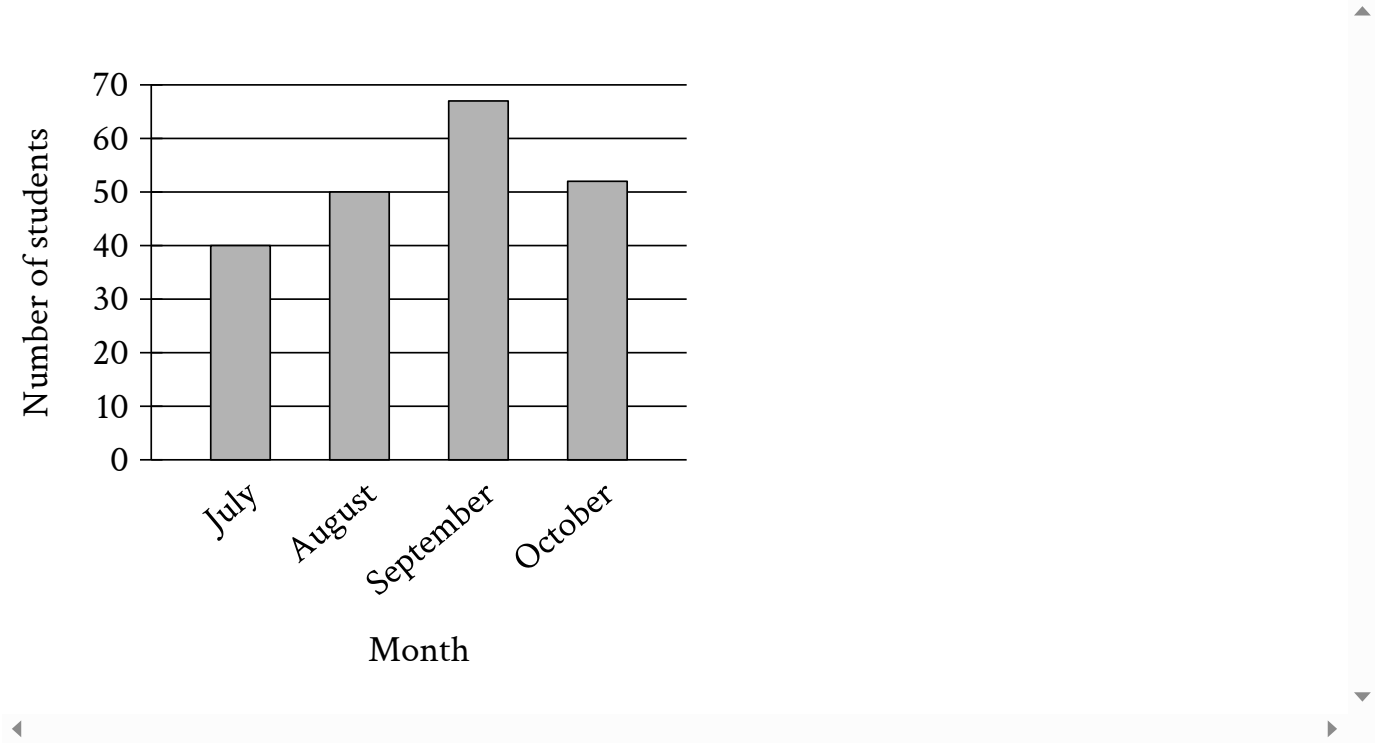
Question Difficulty: Hard

Question ID 88842fe9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 88842fe9

The bar graph shows the distribution of the number of students from one school who were born in one of four months.



How many more students were born in August than were born in July?

- A. 90
- B. 50
- C. 40
- D. 10

ID: 88842fe9 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the bar graph shows the number of students from one school who were born in either July, August, September, or October. The bar representing the number of students born in August has a height of **50**; therefore, **50** students were born in August. The bar representing the number of students born in July has a height of **40**; therefore, **40** students were born in July. Thus, there were **50 — 40**, or **10** more students born in August than in July.

Choice A is incorrect. This is the total number of students born in July and August.

Choice B is incorrect. This is the number of students born in August.

Choice C is incorrect. This is the number of students born in July.

Question ID b20f365f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: b20f365f

The value of a collectible comic book increased by **167%** from the end of **2011** to the end of **2012** and then decreased by **16%** from the end of **2012** to the end of **2013**. What was the net percentage increase in the value of the collectible comic book from the end of **2011** to the end of **2013**?

- A. **124.28%**
- B. **140.28%**
- C. **151.00%**
- D. **209.72%**

ID: b20f365f Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the value of the comic book increased by **167%** from the end of **2011** to the end of **2012**. Therefore, if the value of the comic book at the end of **2011** was x dollars, then the value, in dollars, of the comic book at the end of **2012** was $x + \left(\frac{167}{100}\right)x$, which can be rewritten as $1x + 1.67x$, or $2.67x$. It's also given that the value of the comic book decreased by **16%** from the end of **2012** to the end of **2013**. Therefore, the value, in dollars, of the comic book at the end of **2013** was $2.67x - 2.67x\left(\frac{16}{100}\right)$, which can be rewritten as $2.67x - (2.67x)(0.16)$, or $2.2428x$. Thus, if the value of the comic book at the end of **2011** was x dollars, and the value of the comic book at the end of **2013** was $2.2428x$ dollars, then from the end of **2011** to the end of **2013**, the value of the comic book increased by $2.2428x - 1x$, or $1.2428x$, dollars. Therefore, the increase in the value of the comic book from the end of **2011** to the end of **2013** is equal to **1.2428** times the value of the comic book at the end of **2011**. It follows that from the end of **2011** to the end of **2013**, the net percentage increase in the value of the comic book was $(1.2428)(100)\%$, or **124.28%**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the difference between the net percentage increase in the value of the comic book from the end of **2011** to the end of **2012** and the net percentage decrease in the value of the comic book from the end of **2012** to the end of **2013**, not the net percentage increase in the value of the comic book from the end of **2011** to the end of **2013**.

Choice D is incorrect. This is the net percentage increase in the value of the comic book from the end of **2011** to the end of **2013**, if the value of the comic book increased by **167%** from the end of **2011** to the end of **2012** and then increased, not decreased, by **16%** from the end of **2012** to the end of **2013**.

Question Difficulty: Hard

Question ID 0a4927b1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 0a4927b1

A printer produces posters at a constant rate of **42** posters per minute. At what rate, in posters per hour, does the printer produce the posters?

ID: 0a4927b1 Answer

Correct Answer: 2520

Rationale

The correct answer is **2,520**. There are **60** minutes in one hour. At a rate of **42** posters per minute, the number of posters produced in one hour can be determined by $\left(\frac{42 \text{ posters}}{1 \text{ minute}}\right)\left(\frac{60 \text{ minutes}}{1 \text{ hour}}\right)$, which is **2,520** posters per hour.

Question Difficulty: Easy

Question ID f0ca5e13

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: f0ca5e13

A distance of **61** furlongs is equivalent to how many feet? (**1 furlong = 220 yards and 1 yard = 3 feet**)

ID: f0ca5e13 Answer

Correct Answer: 40260

Rationale

The correct answer is **40,260**. It's given that **1 furlong = 220 yards** and **1 yard = 3 feet**. It follows that a distance of **61** furlongs is equivalent to **(61 furlongs)** $\left(\frac{220 \text{ yards}}{1 \text{ furlong}}\right)$ $\left(\frac{3 \text{ feet}}{1 \text{ yard}}\right)$, or **40,260** feet.

Question Difficulty: Medium

Question ID 275bc8e8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 275bc8e8

The table gives the distribution of votes for a new school mascot and grade level for 80 students.

Mascot	Grade level			
	Sixth	Seventh	Eighth	Total
Badger	4	9	9	22
Lion	9	2	9	20
Longhorn	4	6	4	14
Tiger	6	9	9	24
Total	23	26	31	80

If one of these students is selected at random, what is the probability of selecting a student whose vote for new mascot was for a lion?

- A. $\frac{1}{9}$
- B. $\frac{1}{5}$
- C. $\frac{1}{4}$
- D. $\frac{2}{3}$

ID: 275bc8e8 Answer

Correct Answer: C

Rationale

Choice C is correct. If one of these students is selected at random, the probability of selecting a student whose vote for the new mascot was for a lion is given by the number of votes for a lion divided by the total number of votes. The given table indicates that the number of votes for a lion is **20** votes, and the total number of votes is **80** votes. The table gives the distribution of votes for **80** students, and the table shows a total of 80 votes were counted. It follows that each of the **80** students voted exactly once. Thus, the probability of selecting a student whose vote for the new mascot was for a lion is $\frac{20}{80}$, or $\frac{1}{4}$.

Choice A is incorrect and may result from conceptual or computational errors.

Choice B is incorrect and may result from conceptual or computational errors.

Choice D is incorrect and may result from conceptual or computational errors.

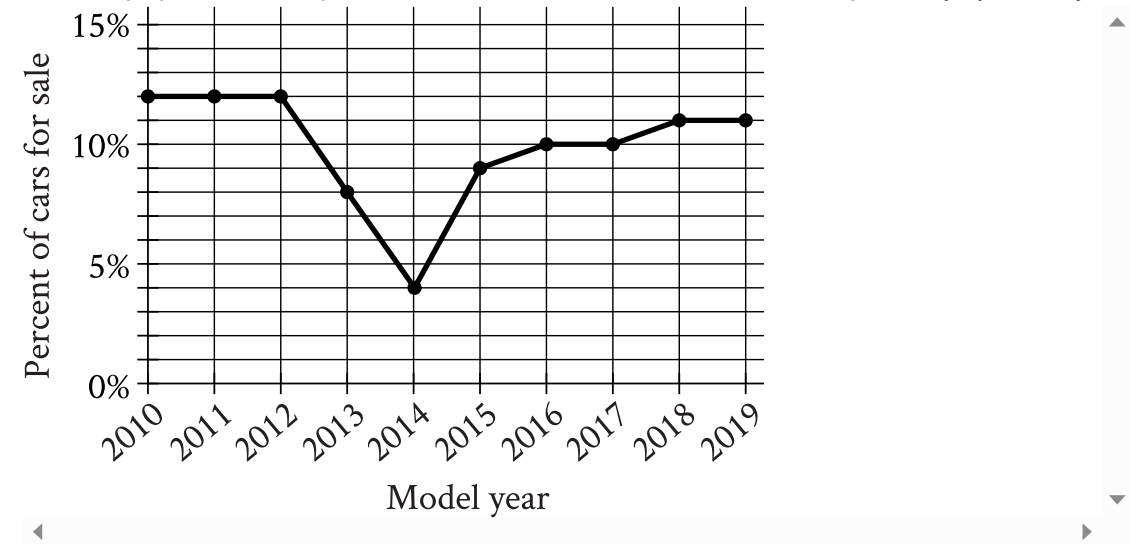
Question Difficulty: Easy

Question ID c1832bfe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: c1832bfe

The line graph shows the percent of cars for sale at a used car lot on a given day by model year.



For what model year is the percent of cars for sale the smallest?

- A. 2012
- B. 2013
- C. 2014
- D. 2015

ID: c1832bfe Answer

Correct Answer: C

Rationale

Choice C is correct. For the given line graph, the percent of cars for sale at a used car lot on a given day is represented on the vertical axis. The percent of cars for sale is the smallest when the height of the line graph is the lowest. The lowest height of the line graph occurs for cars with a model year of **2014**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

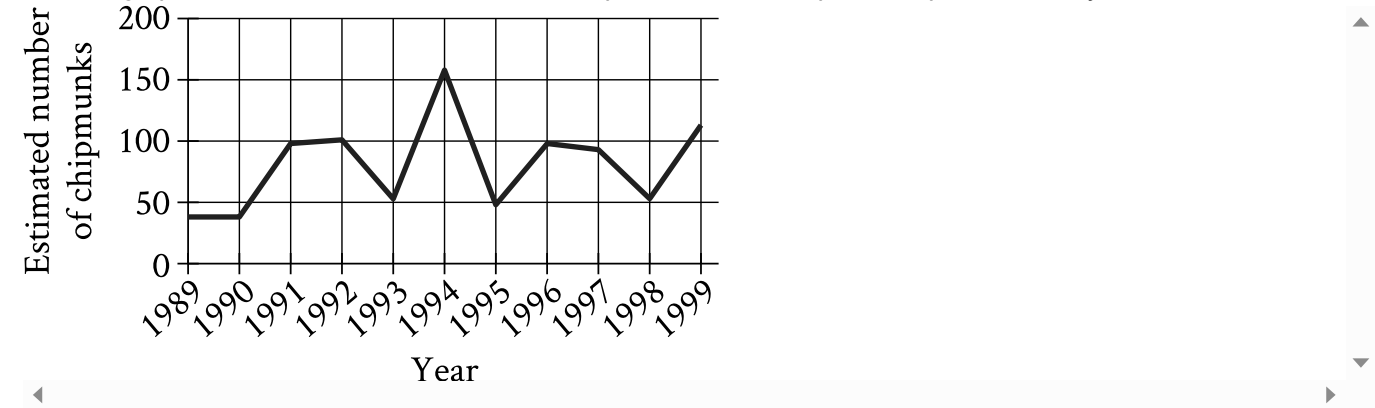
Question Difficulty: Easy

Question ID 3317d95c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 3317d95c

The line graph shows the estimated number of chipmunks in a state park on April 1 of each year from 1989 to 1999.



Based on the line graph, in which year was the estimated number of chipmunks in the state park the greatest?

- A. 1989
- B. 1994
- C. 1995
- D. 1998

ID: 3317d95c Answer

Correct Answer: B

Rationale

Choice B is correct. For the given line graph, the estimated number of chipmunks is represented on the vertical axis. The greatest estimated number of chipmunks in the state park is indicated by the greatest height in the line graph. This height is achieved when the year is **1994**.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 70a4d05d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 70a4d05d

Of **300,000** paper clips, **234,000** are size large. What percentage of the paper clips are size large?

- A. **22%**
- B. **33%**
- C. **66%**
- D. **78%**

ID: 70a4d05d Answer

Correct Answer: D

Rationale

Choice D is correct. The proportion of the paper clips that are size large can be written as $\frac{234,000}{300,000}$, or **0.78**. Therefore, the percentage of the paper clips that are size large is **0.78(100)**, or **78%**.

Choice A is incorrect. This is the percentage of the paper clips that are not size large.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 3781fc55

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 3781fc55

Which of the following lists represents a data set with the smallest standard deviation?

- A. 51, 52, 54, 56, 57
- B. 52, 52, 54, 56, 56
- C. 52, 53, 54, 55, 56
- D. 53, 54, 54, 54, 55

ID: 3781fc55 Answer

Correct Answer: D

Rationale

Choice D is correct. Standard deviation is a measure of the spread of the values in a data set from the mean of the data set, and the smaller the spread, the smaller the standard deviation. Each of the four choices represents a data set that is symmetric with respect to the value **54**. Therefore, **54** is the mean of each of these data sets. All the values in the data set in choice D differ from **54** by either **0** or **1**. In each of the other three data sets, the values in the data set, other than the value of **54**, differ from **54** by at least **1**. Therefore, the data set in choice D has the smallest spread and thus the smallest standard deviation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

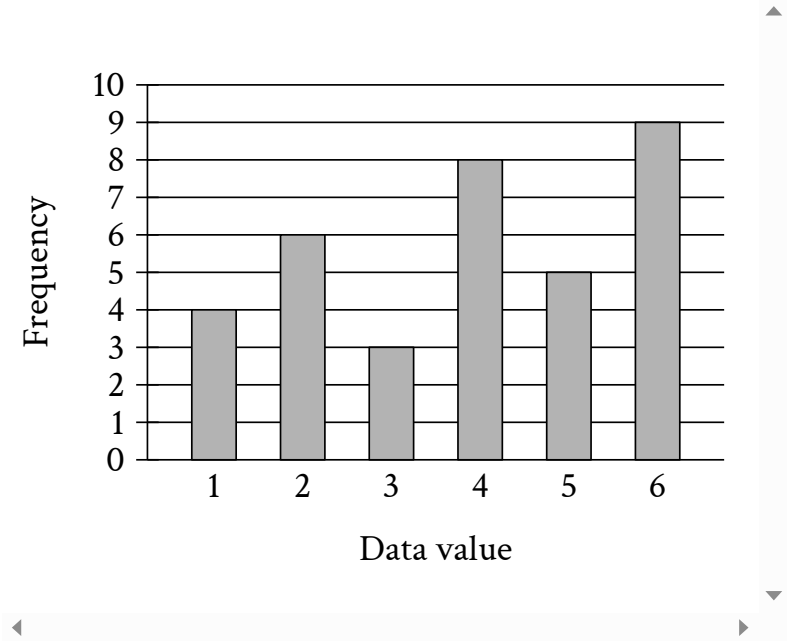
Question Difficulty: Medium

Question ID 76e664ae

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 76e664ae

The bar graph shows the frequency of each data value in a data set.



What is the frequency of data value **2** in this data set?

ID: 76e664ae Answer

Correct Answer: 6

Rationale

The correct answer is **6**. It's given that the bar graph shows the frequency of each data value in a data set. The bar corresponding to the data value **2** has a height of **6**. Therefore, the frequency of the data value **2** in this data set is **6**.

Question Difficulty: Easy

Question ID 60526ddc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 60526ddc

The population of Greenville increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k ?

- A. 0.07
- B. 0.7
- C. 1.07
- D. 1.7

ID: 60526ddc Answer

Correct Answer: C

Rationale

Choice C is correct. Let x be the 2015 population of Greenville. It's given that the population increased by 7% from 2015 to 2016. The increase in population can be written as $(0.07)x$. The 2016 population of Greenville is given as the sum of the 2015 population of Greenville and the increase in population from 2015 to 2016. This can be rewritten as $x + (0.07)x$, or $1.07x$. Therefore, the value of k is 1.07.

Choice A is incorrect. This is the percent, represented as a decimal, that the population increased from 2015 to 2016, not the value of k .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of k if the population increased by 70%, not 7%, from 2015 to 2016.

Question Difficulty: Hard

Question ID 124cbdca

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 124cbdca

The table shows the frequency of values in a data set.

Value	Frequency
19	7
21	1
23	7
25	4

What is the minimum value of the data set?

ID: 124cbdca Answer

Correct Answer: 19

Rationale

The correct answer is **19**. The minimum value of a data set is the least value in the data set. The frequency refers to the number of times a value occurs. The given table shows that for this data set, the value **19** occurs **7** times, the value **21** occurs **1** time, the value **23** occurs **7** times, and the value **25** occurs **4** times. Therefore, of the values **19, 21, 23,** and **25** given in the data set, the minimum value of the data set is **19**.

Question Difficulty: Hard

Question ID 2d68e3c3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 2d68e3c3

A customer spent **\$27** to purchase oranges at **\$3** per pound. How many pounds of oranges did the customer purchase?

ID: 2d68e3c3 Answer

Correct Answer: 9

Rationale

The correct answer is **9**. It’s given that the customer spent **\$27** to purchase oranges at **\$3** per pound. Therefore, the number of pounds of oranges the customer purchased is **$\$27\left(\frac{1\text{ pound}}{\$3}\right)$** , or **9** pounds.

Question Difficulty: Easy

Question ID 9b7d2104

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 9b7d2104

There are **55** students in Spanish club. A sample of the Spanish club students was selected at random and asked whether they intend to enroll in a new study program. Of those surveyed, **20%** responded that they intend to enroll in the study program. Based on this survey, which of the following is the best estimate of the total number of Spanish club students who intend to enroll in the study program?

- A. **11**
- B. **20**
- C. **44**
- D. **55**

ID: 9b7d2104 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that **20%** of the students surveyed responded that they intend to enroll in the study program. Therefore, the proportion of students in Spanish club who intend to enroll in the study program, based on the survey, is **0.20**. Since there are **55** total students in Spanish club, the best estimate for the total number of these students who intend to enroll in the study program is **55(0.20)**, or **11**.

Choice B is incorrect. This is the best estimate for the percentage, rather than the total number, of students in Spanish club who intend to enroll in the study program.

Choice C is incorrect. This is the best estimate for the total number of Spanish club students who do not intend to enroll in the study program.

Choice D is incorrect. This is the total number of students in Spanish club.

Question Difficulty: Easy

Question ID ab74cba9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: ab74cba9

Value	Data set A frequency	Data set B frequency
30	2	9
34	4	7
38	5	5
42	7	4
46	9	2

Data set A and data set B each consist of **27** values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

- A. The mean of data set A is greater than the mean of data set B.
- B. The mean of data set A is less than the mean of data set B.
- C. The mean of data set A is equal to the mean of data set B.
- D. There is not enough information to compare the means of the data sets.

ID: ab74cba9 Answer

Correct Answer: A

Rationale

Choice A is correct. The mean value of a data set is the sum of the values of the data set divided by the number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set A, the sum of products of each value and its frequency is $30(2) + 34(4) + 38(5) + 42(7) + 46(9)$, or **1,094**. It's given that there are **27** values in data set A. Therefore, the mean of data set A is $\frac{1,094}{27}$, or approximately **40.52**. Similarly, the mean of data B is $\frac{958}{27}$, or approximately **35.48**. Therefore, the mean of data set A is greater than the mean of data set B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

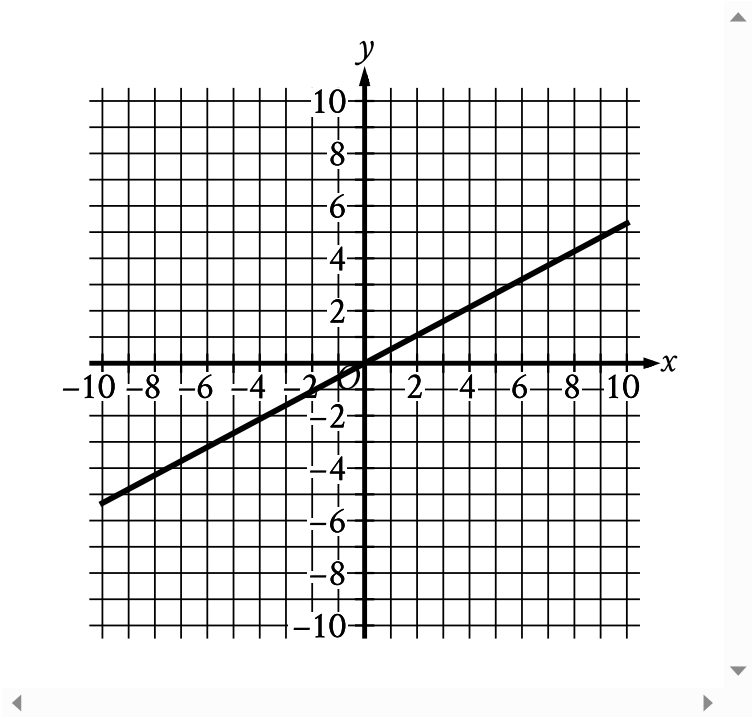
Question Difficulty: Hard

Question ID e392a4aa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: e392a4aa

The graph of function f is shown, where $y = f(x)$.



Which of the following describes function f ?

- A. Increasing linear
- B. Decreasing linear
- C. Increasing exponential
- D. Decreasing exponential

ID: e392a4aa Answer

Correct Answer: A

Rationale

Choice A is correct. The graph of function f shows that as x increases, $f(x)$ also increases, which means $f(x)$ is an increasing function. The graph of f is a line, which indicates a constant rate of change. A function that has a constant rate of change is a linear function. Therefore, function f can be described as increasing linear.

Choice B is incorrect. For a decreasing function, as x increases, $f(x)$ decreases, rather than increases.

Choice C is incorrect. The graph of an exponential function isn't a line.

Choice D is incorrect. For a decreasing function, as x increases, $f(x)$ decreases, rather than increases, and the graph of an exponential function isn't a line.

Question Difficulty: Easy

Question ID a34277bd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: a34277bd

4, 10, 18, 4, 4, 5, 6, 5

What is the median of the data set shown?

- A. 4
- B. 5
- C. 7
- D. 14

ID: a34277bd Answer

Correct Answer: B

Rationale

Choice B is correct. If a data set contains an even number of data values, when the data values are listed in ascending or descending order, the median is between the two middle values. The given data set contains 8 values. When listed in ascending order, the data set is 4, 4, 4, 5, 5, 6, 10, 18 and the two middle values are 5 and 5. Since the two middle values are the same, the median must be 5.

Choice A is incorrect. This value is between the two middle values in the list shown, not the two middle values when the data values are listed in ascending or descending order.

Choice C is incorrect. This is the mean, not the median, of the data set.

Choice D is incorrect. This is the range, not the median, of the data set.

Question Difficulty: Medium

Question ID 673fe587

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 673fe587

The number of audiobooks in a library increased by **131%** from last year to this year. The number of audiobooks in the library last year was n , and the number of audiobooks in the library this year is bn , where b and n are constants. What is the value of b ?

ID: 673fe587 Answer

Correct Answer: 2.31

Rationale

The correct answer is **2.31**. It's given that the number of audiobooks in the library increased by **131%** from last year to this year. Converting **131%** to a decimal yields **1.31**. It's also given that the number of audiobooks in the library last year was n , where n is a constant. It follows that the number of audiobooks in the library this year can be represented by n plus **131%** of n , or $n + 1.31n$. Combining like terms in this expression yields **2.31** n . Since the number of audiobooks in the library this year is represented as bn , where b is a constant, it follows that the value of b is **2.31**.

Question Difficulty: Hard

Question ID b24ca9e9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: b24ca9e9

A scientist measured the lengths of **240** gray seals from Muskeget Island and **120** gray seals from Sable Island. The scientist determined that the mean length of the **240** gray seals from Muskeget Island was **88** inches and the mean length of the **120** gray seals from Sable Island was **94** inches. What was the mean length of all **360** gray seals the scientist measured for this study?

- A. **89**
- B. **90**
- C. **91**
- D. **92**

ID: b24ca9e9 Answer

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is calculated by dividing the sum of the values in the data set by the number of values in the data set. It's given that the mean length of the **240** gray seals from Muskeget Island was **88** inches. This can be represented by the equation $\frac{x}{240} = 88$, where x represents the sum of the lengths, in inches, of the **240** gray seals from Muskeget Island. Multiplying both sides of this equation by **240** yields $x = (240)(88)$, or $x = 21,120$ inches. It's also given that the mean length of the **120** gray seals from Sable Island was **94** inches. This can be represented by the equation $\frac{y}{120} = 94$, where y represents the sum of the lengths, in inches, of the **120** gray seals from Sable Island. Multiplying both sides of this equation by **120** yields $y = (120)(94)$, or $y = 11,280$ inches. Therefore, the sum of the lengths, in inches, of all **360** gray seals is $21,120 + 11,280$, or $32,400$. Dividing this sum by **360** yields $\frac{32,400}{360}$, or **90** inches. Therefore, the mean length of all **360** gray seals the scientist measured for this study is **90** inches.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the mean length of **88** inches and **94** inches, not the mean length of all **360** gray seals the scientist measured for this study.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 64b9b4bf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 64b9b4bf

What number is **40%** greater than **115**?

ID: 64b9b4bf Answer

Correct Answer: 161

Rationale

The correct answer is **161**. For a number to be **40%** greater than **115**, it follows that the number is **(100% of 115) + (40% of 115)**, which can be written as $\frac{100}{100}(115) + \frac{40}{100}(115)$. This expression is equivalent to $1(115) + 0.4(115)$, or $1.4(115)$, which is equal to **161**. Therefore, **161** is **40%** greater than **115**.

Question Difficulty: Medium

Question ID cf16b539

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: cf16b539

For an electric field passing through a flat surface perpendicular to it, the electric flux of the electric field through the surface is the product of the electric field’s strength and the area of the surface. A certain flat surface consists of two adjacent squares, where the side length, in meters, of the larger square is **3** times the side length, in meters, of the smaller square. An electric field with strength **29.00** volts per meter passes uniformly through this surface, which is perpendicular to the electric field. If the total electric flux of the electric field through this surface is **4,640 volts · meters**, what is the electric flux, in **volts · meters**, of the electric field through the larger square?

ID: cf16b539 Answer

Correct Answer: 4176

Rationale

The correct answer is **4,176**. It’s given that the side length of the larger square is **3** times the side length of the smaller square. This means that the area of the larger square is **3²**, or **9**, times the area of the smaller square. If the area of the smaller square is represented by ***x***, then the area of the larger square can be represented by **9*x***. Therefore, the flat surface of the two adjacent squares has a total area of ***x* + 9*x***, or **10*x***. It’s given that an electric field with strength **29.00** volts per meter passes uniformly through this surface and the total electric flux of the electric field through this surface is **4,640 volts · meters**. Since it’s given that the electric flux is the product of the electric field’s strength and the area of the surface, the equation **29.00(10*x*) = 4,640**, or **290*x* = 4,640**, can be used to represent this situation. Dividing each side of this equation by **290** yields ***x* = 16**. Substituting **16** for ***x*** in the expression for the area of the larger square, **9*x***, yields **9(16)**, or **144**, square meters. Since the area of the larger square is **144** square meters, the electric flux, in **volts · meters**, of the electric field through the larger square can be determined by multiplying the area of the larger square by the strength of the electric field. Thus, the electric flux is **(144 square meters)($\frac{29.00 \text{ volts}}{\text{meter}}$)**, or **4,176 volts · meters**.

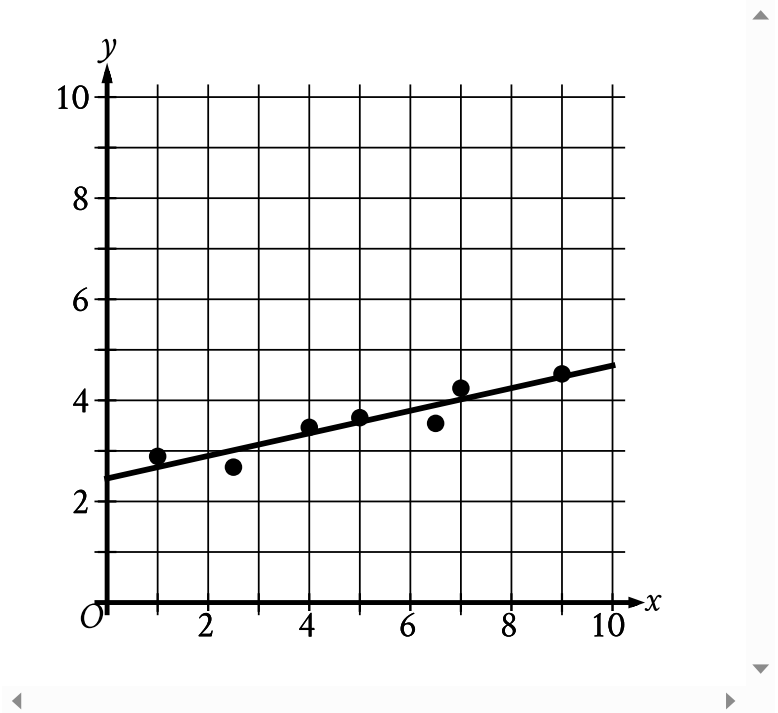
Question Difficulty: Hard

Question ID 1b483347

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 1b483347

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit?

- A. -2.45
- B. -0.22
- C. 0.22
- D. 2.45

ID: 1b483347 Answer

Correct Answer: C

Rationale

Choice C is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 2.5)$ and $(10, 4.7)$. It follows that the slope of the line of best fit is approximately $\frac{4.7 - 2.5}{10 - 0}$, which is equivalent to $\frac{2.2}{10}$, or 0.22 . Therefore, of the given choices, 0.22 is closest to the slope of the line of best fit.

Choice A is incorrect. This is closest to the negative y -coordinate of the y -intercept, not to the slope, of the line of best fit.

Choice B is incorrect. This is closest to the negative slope, not to the slope, of the line of best fit.

Choice D is incorrect. This is closest to the y -coordinate of the y -intercept, not to the slope, of the line of best fit.

Question ID b218d8dc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: b218d8dc

Type of store	Average number of employees
Warehouse store	365
Department store	213
Supermarket	130

For a certain region, the table shows the average number of store employees in **2016** by type of store. Based on the table, how much greater was the average number of store employees in warehouse stores than in supermarkets?

- A. 83
- B. 152
- C. 235
- D. 495

ID: b218d8dc Answer

Correct Answer: C

Rationale

Choice C is correct. The table shows that for a certain region in **2016**, the average number of store employees in warehouse stores was **365** and the average number of store employees in supermarkets was **130**. Subtracting **130** from **365** yields **365 — 130**, or **235**. Therefore, the average number of store employees was **235** greater in warehouse stores than in supermarkets.

Choice A is incorrect. For this region in **2016**, this is how much greater the average number of store employees was in department stores than in supermarkets.

Choice B is incorrect. For this region in **2016**, this is how much greater the average number of store employees was in warehouse stores than in department stores.

Choice D is incorrect. For this region in **2016**, this is the sum of the average number of store employees in warehouse stores and in supermarkets.

Question Difficulty: Easy

Question ID cd194614

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: cd194614

At a movie theater, there are a total of **350** customers. Each customer is located in either theater A, theater B, or theater C. If one of these customers is selected at random, the probability of selecting a customer who is located in theater A is **0.48**, and the probability of selecting a customer who is located in theater B is **0.24**. How many customers are located in theater C?

- A. 28
- B. 40
- C. 84
- D. 98

ID: cd194614 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that at a movie theater, there are a total of **350** customers and that each customer is located in either theater A, theater B, or theater C. If the probability of selecting a customer in theater A is **0.48**, then **(0.48)(350)**, or **168**, customers are located in theater A. If the probability of selecting a customer in theater B is **0.24**, then **(0.24)(350)**, or **84**, customers are located in theater B. It follows that there are **168 + 84**, or **252**, customers in theater A and theater B. Therefore, there are **350 – 252**, or **98**, customers in theater C.

Choice A is incorrect. This is the percent, not the number, of the customers that are located in theater C.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of customers that are located in theater B, not theater C.

Question Difficulty: Medium

Question ID e20db239

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: e20db239

A distance of **354** furlongs is equivalent to how many feet? (**1 furlong = 220 yards and 1 yard = 3 feet**)

- A. **306**
- B. **402**
- C. **25,960**
- D. **233,640**

ID: e20db239 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that **1 furlong = 220 yards** and **1 yard = 3 feet**. It follows that a distance of **354** furlongs is equivalent to **(354 furlongs)** $\left(\frac{220 \text{ yards}}{1 \text{ furlong}}\right)$ $\left(\frac{3 \text{ feet}}{1 \text{ yard}}\right)$, or **233,640** feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 41eb05a0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 41eb05a0

From a population of **50,000** people, **1,000** were chosen at random and surveyed about a proposed piece of legislation. Based on the survey, it is estimated that **35%** of people in the population support the legislation, with an associated margin of error of **3%**. Based on these results, which of the following is a plausible value for the total number of people in the population who support the proposed legislation?

- A. **350**
- B. **650**
- C. **16,750**
- D. **31,750**

ID: 41eb05a0 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that an estimated **35%** of people in the population support the legislation, with an associated margin of error of **3%**. Subtracting and adding the margin of error from the estimate gives an interval of plausible values for the true percentage of people in the population who support the legislation. Therefore, it's plausible that between **32%** and **38%** of people in this population support the legislation. The corresponding numbers of people represented by these percentages in the population can be calculated by multiplying the total population, **50,000**, by **0.32** and by **0.38**, which gives **50,000(0.32) = 16,000** and **50,000(0.38) = 19,000**, respectively. It follows that any value in the interval **16,000** to **19,000** is a plausible value for the total number of people in the population who support the proposed legislation. Of the choices given, only **16,750** is in this interval.

Choice A is incorrect. This is the number of people in the sample, rather than in the population, who support the legislation.

Choice B is incorrect. This is the number of people in the sample who do not support the legislation.

Choice D is incorrect. This is a plausible value for the total number of people in the population who do not support the proposed legislation.

Question Difficulty: Hard

Question ID 5bcefcbe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 5bcefcbe

How many teaspoons are equivalent to ~~44~~ tablespoons? (**3 teaspoons = 1 tablespoon**)

- A. ~~47~~
- B. ~~88~~
- C. **132**
- D. ~~176~~

ID: 5bcefcbe Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that **3** teaspoons is equivalent to **1** tablespoon. Therefore, ~~44~~ tablespoons is equivalent to **(~~44~~ tablespoons) $\left(\frac{3 \text{ teaspoons}}{1 \text{ tablespoon}}\right)$, or **132** teaspoons.**

Choice A is incorrect. This is equivalent to approximately **15.66** tablespoons, not ~~44~~ tablespoons.

Choice B is incorrect. This is equivalent to approximately **29.33** tablespoons, not ~~44~~ tablespoons.

Choice D is incorrect. This is equivalent to approximately **58.66** tablespoons, not ~~44~~ tablespoons.

Question Difficulty: Easy

Question ID 46e38539

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 46e38539

Out of **300** seeds that were planted, **80%** sprouted. How many of these seeds sprouted?

ID: 46e38539 Answer

Correct Answer: 240

Rationale

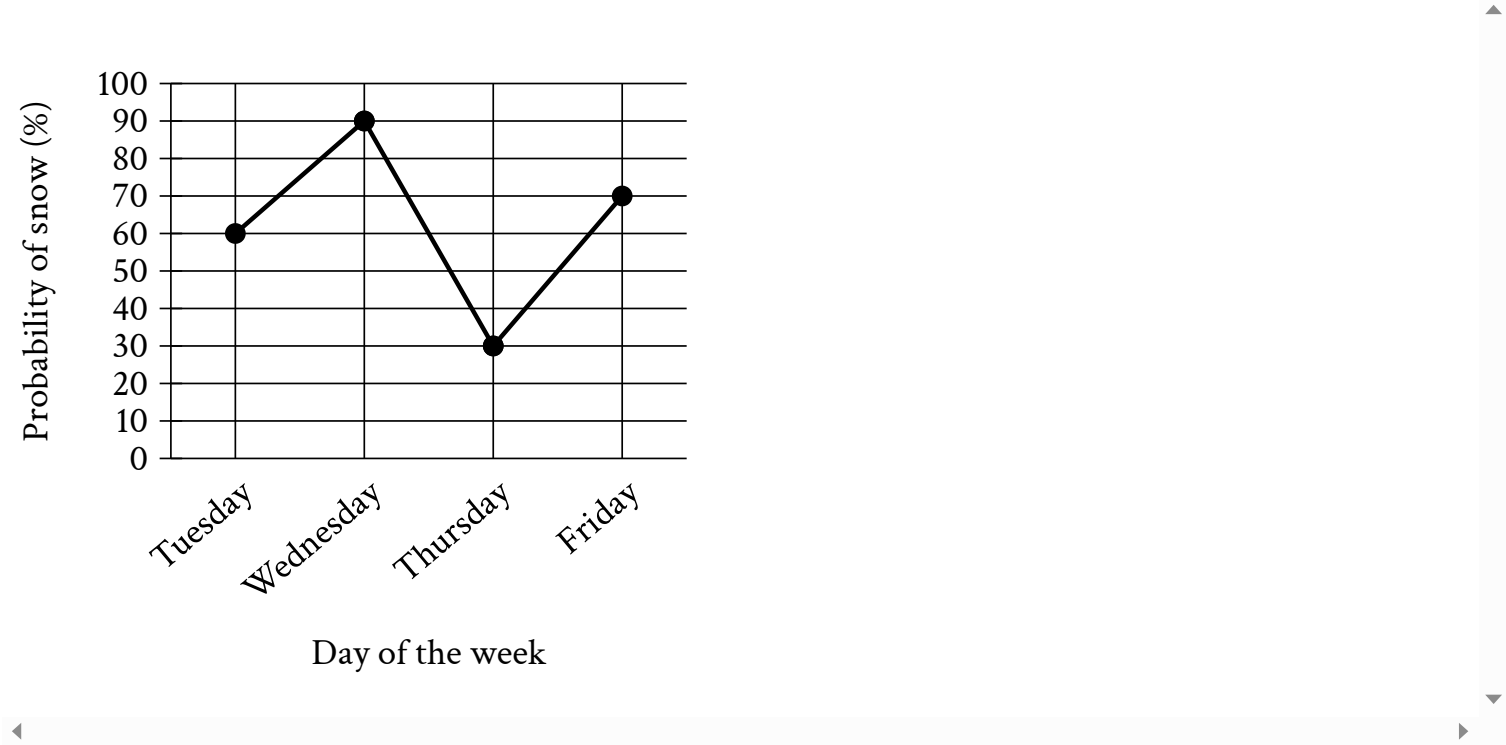
The correct answer is **240**. It's given that **80%** of the **300** seeds sprouted. Therefore, the number of seeds that sprouted can be calculated by multiplying the number of seeds that were planted by $\frac{80}{100}$, which gives **300**($\frac{80}{100}$), or **240**.

Question Difficulty: Easy

Question ID 3c0867f9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 3c0867f9



The line graph shows the probability of snow, as a percent, at a certain location for each day during a four-day period. According to the line graph, for which day during this four-day period is the probability of snow **30%**?

- A. Tuesday
- B. Wednesday
- C. Thursday
- D. Friday

ID: 3c0867f9 Answer

Correct Answer: C

Rationale

Choice C is correct. For the line graph shown, the probability of snow, as a percent, is represented on the vertical axis. According to the line graph, during this four-day period, the probability of snow is **30%** for Thursday.

Choice A is incorrect. The probability of snow on Tuesday is **60%**.

Choice B is incorrect. The probability of snow on Wednesday is **90%**.

Choice D is incorrect. The probability of snow on Friday is **70%**.

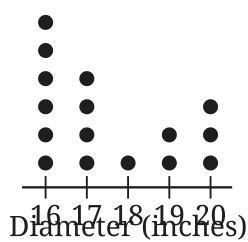
Question Difficulty: Easy

Question ID f2d2782c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: f2d2782c

To study the characteristics of sea stars in a group of tide pools, researchers measured the diameter of the sea stars within the tide pools. The dot plot gives the diameter, to the nearest inch, of each of the sea stars in these tide pools.



Based on the dot plot, how many sea stars had a diameter, to the nearest inch, of **16** inches?

- A. **16**
- B. **6**
- C. **4**
- D. **1**

ID: f2d2782c Answer

Correct Answer: B

Rationale

Choice B is correct. The dot plot gives the diameter, to the nearest inch, of each sea star in a group of tide pools. The number of dots above a given diameter represents the number of sea stars with that diameter, to the nearest inch. There are **6** dots above **16** inches. Therefore, **6** sea stars had a diameter, to the nearest inch, of **16** inches.

Choice A is incorrect and may result from calculation or conceptual errors.

Choice C is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of **17** inches, not **16** inches.

Choice D is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of **18** inches, not **16** inches.

Question Difficulty: Easy

Question ID 6a29e4b6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 6a29e4b6

What percentage of **300** is **75**?

- A. **25%**
- B. **50%**
- C. **75%**
- D. **225%**

ID: 6a29e4b6 Answer

Correct Answer: A

Rationale

Choice A is correct. Let x represent the percentage of **300** that is **75**. This can be written as $\frac{x}{100}(300) = 75$, or $3x = 75$. Dividing both sides of this equation by **3** yields $x = 25$. Therefore, **25%** of **300** is **75**.

Choice B is incorrect. **50%** of **300** is **150**, not **75**.

Choice C is incorrect. **75%** of **300** is **225**, not **75**.

Choice D is incorrect. **225%** of **300** is **675**, not **75**.

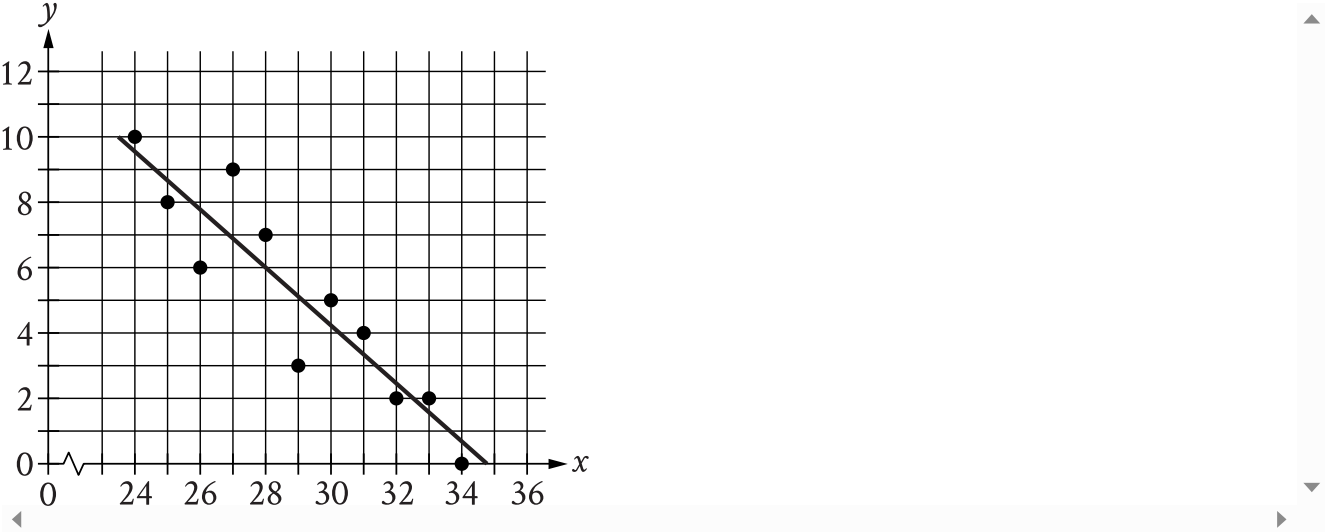
Question Difficulty: Easy

Question ID 304e91b7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 304e91b7

The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown.



At $x = 25.5$, which of the following is closest to the y -value predicted by the line of best fit?

- A. 6.2
- B. 7.3
- C. 8.2
- D. 9.1

ID: 304e91b7 Answer

Correct Answer: C

Rationale

Choice C is correct. On the line of best fit, an x -value of **25.5** corresponds to a y -value between **8** and **8.5**. Therefore, at $x = 25.5$, **8.2** is closest to the y -value predicted by the line of best fit.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 182fa115

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 182fa115

2, 9, 14, 23, 32

What is the mean of the data shown?

- A. 14
- B. 16
- C. 17
- D. 32

ID: 182fa115 Answer

Correct Answer: B

Rationale

Choice B is correct. The mean of a set of data values is the sum of all the data values divided by the number of data values in the set. The sum of the data values shown is $2 + 9 + 14 + 23 + 32$, or **80**. Since there are **5** data values in the set, the mean of the data shown is $\frac{80}{5}$, or **16**.

Choice A is incorrect. This is the median, not the mean, of the data shown.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the maximum, not the mean, of the data shown.

Question Difficulty: Medium

Question ID bd38029d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: bd38029d

The number a is **70%** less than the positive number b . The number c is **80%** greater than a . The number c is how many times b ?

ID: bd38029d Answer

Correct Answer: .54, 27/50

Rationale

The correct answer is **.54**. It's given that the number a is **70%** less than the positive number b . Therefore, $a = (1 - \frac{70}{100})b$, which is equivalent to $a = (1 - 0.70)b$, or $a = 0.30b$. It's also given that the number c is **80%** greater than a . Therefore, $c = (1 + \frac{80}{100})a$, which is equivalent to $c = (1 + 0.80)a$, or $c = 1.80a$. Since $a = 0.30b$, substituting $0.30b$ for a in the equation $c = 1.80a$ yields $c = 1.80(0.30b)$, or $c = 0.54b$. Thus, c is **0.54** times b . Note that .54 and 27/50 are examples of ways to enter a correct answer.

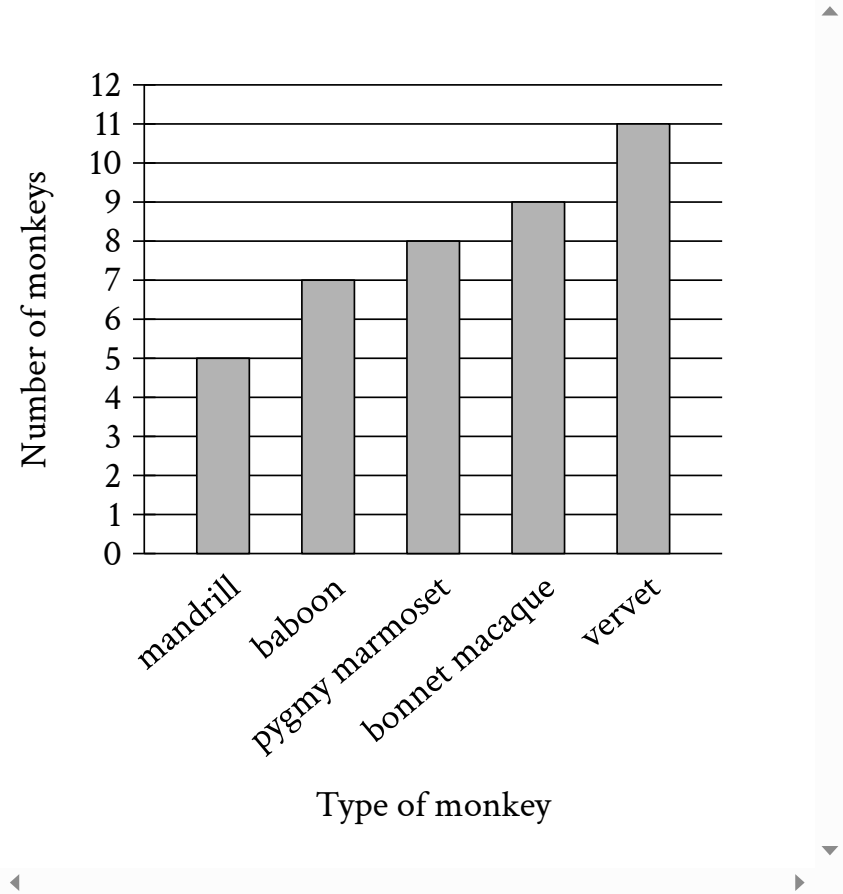
Question Difficulty: Hard

Question ID 5dea66f2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 5dea66f2

The bar graph shows the number of each type of monkey in a sanctuary.



How many more vervets are in this sanctuary than mandrills?

- A. 11
- B. 6
- C. 5
- D. 3

ID: 5dea66f2 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the bar graph shows the number of each type of monkey in a sanctuary. The bar representing the number of mandrills has a height of **5**; therefore, there are **5** mandrills in the sanctuary. The bar representing vervets has a height of **11**; therefore, there are **11** vervets in the sanctuary. Therefore, there are **11 – 5**, or **6**, more vervets in this sanctuary than mandrills.

Choice A is incorrect. This is the number of vervets in the sanctuary.

Choice C is incorrect. This is the number of mandrills in the sanctuary.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 36e94dbb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 36e94dbb

An airplane descends from an altitude of **9,500** feet to **5,000** feet at a constant rate of **400** feet per minute. What type of function best models the relationship between the descending airplane's altitude and time?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

ID: 36e94dbb Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the airplane descends at a constant rate of **400 feet per minute**. Since the altitude decreases by a constant amount during each fixed time period, the relationship between the airplane's altitude and time is linear. Since the airplane descends from an altitude of **9,500 feet** to **5,000 feet**, the airplane's altitude is decreasing with time. Thus, the relationship is best modeled by a decreasing linear function.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

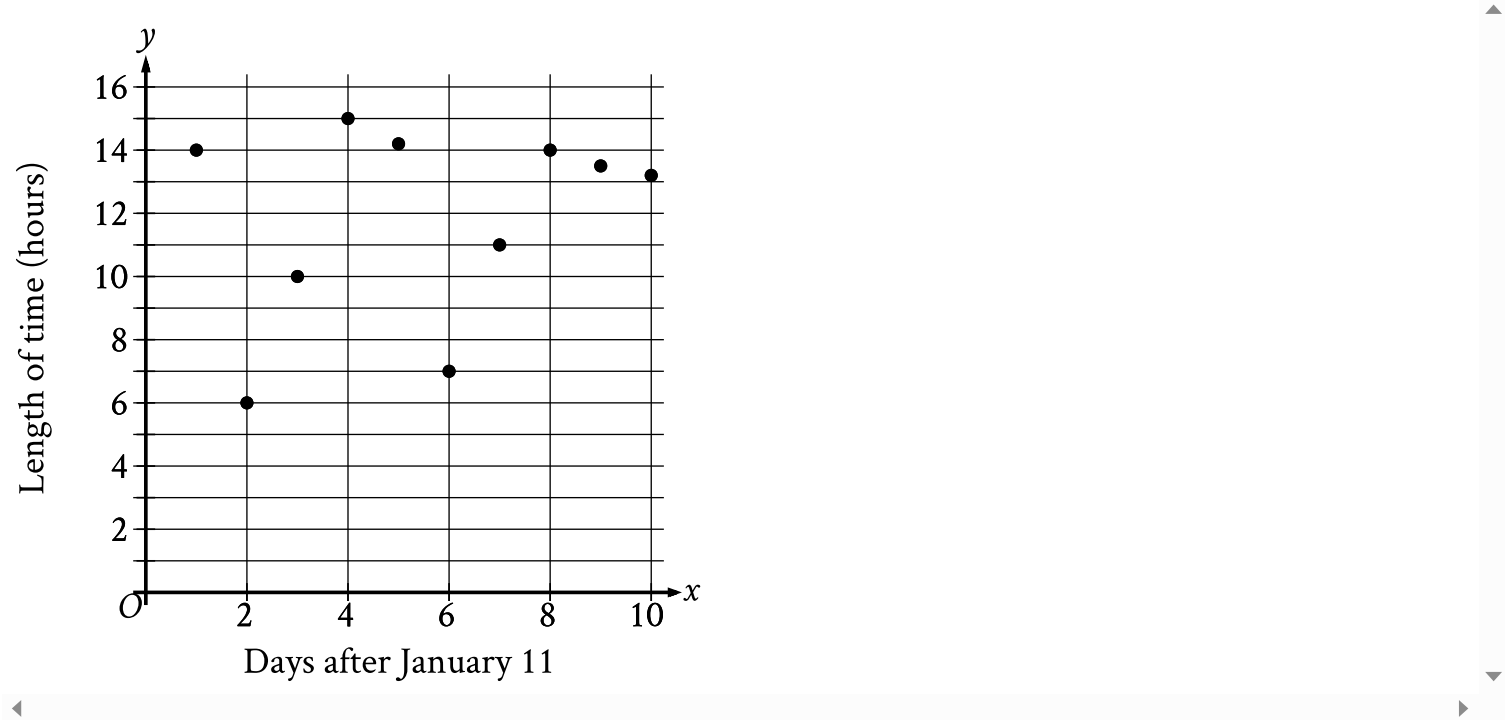
Question Difficulty: Medium

Question ID f96ee12b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: f96ee12b

The scatterplot shows the relationship between the length of time y , in hours, a certain bird spent in flight and the number of days after January 11, x .



What is the average rate of change, in hours per day, of the length of time the bird spent in flight on January 13 to the length of time the bird spent in flight on January 15?

ID: f96ee12b Answer

Correct Answer: 4.5, 9/2

Rationale

The correct answer is $\frac{9}{2}$. It's given that the scatterplot shows the relationship between the length of time y , in hours, a certain bird spent in flight and the number of days after January 11, x . Since January 13 is 2 days after January 11, it follows that January 13 corresponds to an x -value of 2 in the scatterplot. In the scatterplot, when $x = 2$, the corresponding value of y is 6. In other words, on January 13, the bird spent 6 hours in flight. Since January 15 is 4 days after January 11, it follows that January 15 corresponds to an x -value of 4 in the scatterplot. In the scatterplot, when $x = 4$, the corresponding value of y is 15. In other words, on January 15, the bird spent 15 hours in flight. Therefore, the average rate of change, in hours per day, of the length of time the bird spent in flight on January 13 to the length of time the bird spent in flight on January 15 is the difference in the length of time, in hours, the bird spent in flight divided by the difference in the number of days after January 11, or $\frac{15-6}{4-2}$, which is equivalent to $\frac{9}{2}$. Note that 9/2 and 4.5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 47be3162

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 47be3162

A giant armadillo has a mass of **39** kilograms. What is the giant armadillo's mass in **grams**? (**1 kilogram = 1,000 grams**)

ID: 47be3162 Answer

Correct Answer: 39000

Rationale

The correct answer is **39,000**. It's given that the giant armadillo has a mass of **39** kilograms. Since **1** kilogram is equal to **1,000** grams, **39** kilograms is equal to **39 kilograms** $\left(\frac{1,000 \text{ grams}}{1 \text{ kilogram}}\right)$, or **39,000** grams. Therefore, the giant armadillo's mass, in grams, is **39,000**.

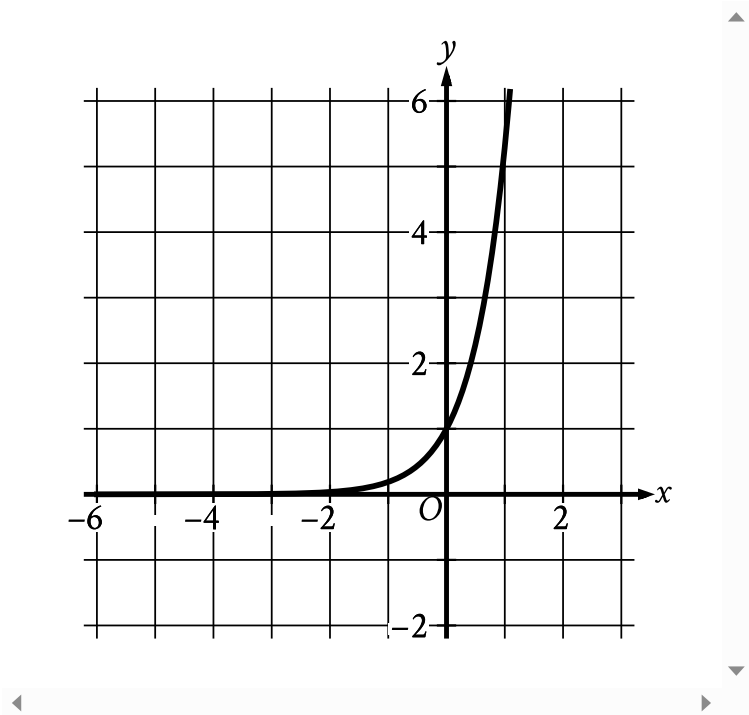
Question Difficulty: Easy

Question ID 68547ecc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 68547ecc

The graph of the function f is shown, where $y = f(x)$.



Which of the following best describes the function f ?

- A. Decreasing exponential
- B. Increasing exponential
- C. Decreasing linear
- D. Increasing linear

ID: 68547ecc Answer

Correct Answer: B

Rationale

Choice B is correct. The graph of function f shows that as x increases, $f(x)$ also increases, which means that f is an increasing function. The graph of function f is curved upward and increases at an increasing rate, not a constant rate. This behavior is characteristic of exponential functions, not linear functions. Therefore, of the given choices, function f is best described as increasing exponential.

Choice A is incorrect. For a decreasing function, as x increases, $f(x)$ decreases rather than increases.

Choice C is incorrect. For a decreasing function, as x increases, $f(x)$ decreases rather than increases, and the graph of a linear function isn't curved.

Choice D is incorrect. The graph of a linear function isn't curved.

Question ID 70bf7e21

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 70bf7e21

In 2008, Zinah earned 14% more than in 2007, and in 2009 Zinah earned 4% more than in 2008. If Zinah earned y times as much in 2009 as in 2007, what is the value of y ?

- A. 0.5600
- B. 1.0056
- C. 1.1800
- D. 1.1856

ID: 70bf7e21 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that in 2008 Zinah earned 14% more than in 2007. Let h represent the amount Zinah earned in 2007 and let j represent the amount that Zinah earned in 2008. This situation can be represented by the equation $j = (1 + \frac{14}{100})h$, or $j = 1.14h$. It's also given that in 2009 Zinah earned 4% more than in 2008. Let k represent the amount Zinah earned in 2009. This situation can be represented by the equation $k = (1 + \frac{4}{100})j$, or $k = 1.04j$. Substituting $1.14h$ for j in the equation $k = 1.04j$ yields $k = (1.04)(1.14h)$, or $k = 1.1856h$. If Zinah earned y times as much in 2009 as in 2007, then the value of y is 1.1856.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

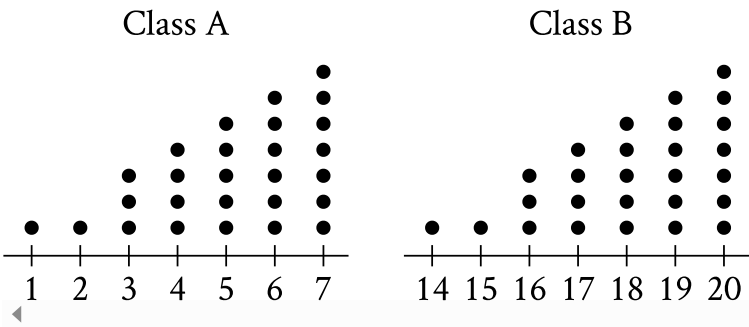
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e6f6cf0d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: e6f6cf0d



Each of the dot plots shown represents the number of glue sticks brought in by each student for two classes, class A and class B. Which statement best compares the standard deviations of the numbers of glue sticks brought in by each student for these two classes?

- A. The standard deviation of the number of glue sticks brought in by each student for class A is less than the standard deviation of the number of glue sticks brought in by each student for class B.
- B. The standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.
- C. The standard deviation of the number of glue sticks brought in by each student for class A is greater than the standard deviation of the number of glue sticks brought in by each student for class B.
- D. There is not enough information to compare these standard deviations.

ID: e6f6cf0d Answer

Correct Answer: B

Rationale

Choice B is correct. Standard deviation is a measure of the spread of a data set from its mean. The dot plot for class A and the dot plot for class B have the same shape. Thus, the frequency distributions for both class A and class B are the same. Since both class A and class B have the same frequency distribution of glue sticks brought in by each student, it follows that both class A and class B have the same spread of the number of glue sticks brought in by each student from their respective means. Therefore, the standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 3ce69ff2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 3ce69ff2

On a street with **7** houses, **2** houses are blue. If a house from this street is selected at random, what is the probability of selecting a house that is blue?

- A. $\frac{1}{7}$
- B. $\frac{2}{7}$
- C. $\frac{5}{7}$
- D. $\frac{7}{7}$

ID: 3ce69ff2 Answer

Correct Answer: B

Rationale

Choice B is correct. If a house from the street is selected at random, the probability of selecting a house that is blue is equal to the number of houses on the street that are blue divided by the total number of houses on the street. Since there are **2** blue houses on a street with **7** total houses, the probability of selecting a house that is blue from this street is $\frac{2}{7}$.

Choice A is incorrect. This is the probability of selecting a house that is blue from a street on which **1** of the **7** houses is blue.

Choice C is incorrect. This is the probability of selecting a house that is not blue from this street.

Choice D is incorrect. This is the probability of selecting a house that is blue from a street on which all the houses are blue.

Question Difficulty: Easy

Question ID 84ddc307

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 84ddc307

A reseller buys certain books for a purchase price of **5.00** dollars each and then marks them each for sale at a consumer price that is **270%** of the purchase price. After **4** months, any remaining books not yet sold are marked at a discounted price that is **70%** off the consumer price. What is the discounted price of each of the remaining books, in dollars?

ID: 84ddc307 Answer

Correct Answer: 4.05, 81/20

Rationale

The correct answer is **4.05**. It's given that the purchase price for certain books is **5.00** dollars each. It's also given that each book is marked for sale at a consumer price that is **270%** of the purchase price. Since the consumer price is **270%** of the purchase price of **5.00** dollars, it follows that the consumer price is **(2.7)(5.00)**, or **13.50**, dollars. It's given that after **4** months, any remaining books are discounted at **70%** off the consumer price. Thus, the discount amount is **(0.7)(13.50)**, or **9.45**, dollars. Subtracting the discount amount from the consumer price gives the discounted price of each of the remaining books: **13.50 — 9.45 = 4.05**. Note that 4.05 and 81/20 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID e8aa0765

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: e8aa0765

Five *Eretmochelys imbricata*, a type of sea turtle, each have a nest. The table shows an original data set of the number of eggs that each turtle laid in its nest.

Nest	Number of eggs
A	149
B	144
C	148
D	136
E	139

A sixth nest with **121** eggs is added to create a new data set. Which of the following correctly compares the means of the two data sets?

- A. The mean of the original data set is greater than the mean of the new data set.
- B. The mean of the original data set is less than the mean of the new data set.
- C. The means of both data sets are equal.
- D. There is not enough information to compare the means.

ID: e8aa0765 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the table shows an original data set of **5** values. It's also given that a sixth value is added to create a new data set. The new data set consists of the **5** values in the original data set and one additional value, **121**. Since the additional value, **121**, is less than any value in the original data set, the mean of the original data set is greater than the mean of the new data set.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID c8a2954a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: c8a2954a

The number w is 110% greater than the number z . The number z is 55% less than 50. What is the value of w ?

ID: c8a2954a Answer

Correct Answer: 189/4, 47.25

Rationale

The correct answer is **47.25**. It's given that the number w is 110% greater than the number z . It follows that $w = (1 + \frac{110}{100})z$, or $w = 2.1z$. It's also given that the number z is 55% less than 50. It follows that $z = (1 - \frac{55}{100})(50)$, or $z = 0.45(50)$, which yields $z = 22.5$. Substituting **22.5** for z in the equation $w = 2.1z$ yields $w = 2.1(22.5)$, which is equivalent to $w = 47.25$. Therefore, the value of w is **47.25**. Note that 47.25 and 189/4 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID b1e2590f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: b1e2590f

432 is 96% of what number?

ID: b1e2590f Answer

Correct Answer: 450

Rationale

The correct answer is **450**. Let x represent the number that **432** is **96%** of. This can be written as $(\frac{96}{100})x = 432$, or $0.96x = 432$. Dividing both sides of this equation by **0.96** yields $x = 450$. Therefore, **432** is **96%** of **450**.

Question Difficulty: Medium

Question ID a5c1f801

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: a5c1f801

Researchers studied a sample of a species of fish from a certain location. The researchers found that the mean length of fish in the sample was **16.59** millimeters. Based on the sample mean and margin of error, the researchers estimated that the mean length of all fish of this species from this location was between **15.86** and **17.32** millimeters. What is the margin of error, in millimeters?

- A. **0.73**
- B. **2.19**
- C. **15.86**
- D. **16.59**

ID: a5c1f801 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the mean length of fish in the sample was **16.59** millimeters. It's also given that based on the sample mean and margin of error, the researchers estimated that the mean length of all fish of this species from this location was between **15.86** and **17.32** millimeters. These values are calculated by subtracting the margin of error from **16.59** millimeters to get the lower bound, **15.86** millimeters, and by adding the margin of error to **16.59** millimeters to get the upper bound, **17.32** millimeters. It follows that the margin of error, in millimeters, can be calculated as the difference between **16.59** and **15.86** or between **17.32** and **16.59**. The difference between **16.59** and **15.86** is **16.59 – 15.86**, or **0.73**. Therefore, the margin of error, in millimeters, is **0.73**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the lower bound of the researchers' estimate of the mean length of all fish of this species from this location, not the margin of error.

Choice D is incorrect. This is the mean length of fish in the sample, not the margin of error.

Question Difficulty: Medium

Question ID 7af63c22

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 7af63c22

An object has a mass of **168** grams and a volume of **24** cubic centimeters. What is the density, in grams per cubic centimeter, of the object?

- A. **7**
- B. **144**
- C. **192**
- D. **4,032**

ID: 7af63c22 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the object has a mass of **168** grams and a volume of **24** cubic centimeters. Dividing the mass, in grams, of the object by the volume, in cubic centimeters, of the object gives the density, in grams per cubic centimeter, of the object. It follows that the density of the object is $\frac{168 \text{ grams}}{24 \text{ cubic centimeters}}$, which is equivalent to $\frac{168}{24}$ grams per cubic centimeter, or **7** grams per cubic centimeter.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 95522e4d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 95522e4d

The table summarizes the number of objects in each group.

Group	Number of objects
A	375
B	54
C	690
D	81
Total	1,200

The number of objects in group C is $p\%$ of the number of objects in group A. What is the value of p ?

ID: 95522e4d Answer

Correct Answer: 184

Rationale

The correct answer is **184**. It's given in the table that there are **375** objects in group A and **690** objects in group C. It's also given that the number of objects in group C is $p\%$ of the number of objects in group A. Therefore, **690** is $p\%$ of **375**, which can be represented by $690 = \left(\frac{p}{100}\right)(375)$, or $690 = 3.75p$. Dividing both sides of this equation by **3.75** yields $184 = p$. Therefore, the value of p is **184**.

Question Difficulty: Hard

Question ID cd21510c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: cd21510c

The weight of sample A is **468%** of the weight of sample B, and the weight of sample A is **0.780%** of the weight of sample C. If the weight of sample C is **$p\%$** of the weight of sample B, what is the value of **p** ?

ID: cd21510c Answer

Correct Answer: 60000

Rationale

The correct answer is **60,000**. It's given that the weight of sample A is **468%** of the weight of sample B and **0.780%** of the weight of sample C. Let the weight of sample B be **b** and the weight of sample C be **c** . Then, the weight of sample A is equivalent to $\frac{468b}{100}$, or **4.68 **b**** , and also equivalent to $\frac{0.780c}{100}$, or **0.00780 **c**** . Setting these expressions equal to each other yields **4.68 **b** = 0.00780 **c**** . Dividing both sides of this equation by **0.00780** yields $\frac{4.68b}{0.00780} = c$, or **$c = 600b$** . If the weight of sample C is **$p\%$** of the weight of sample B, then **$c = \frac{p}{100}b$** . Since **$c = 600b$** and **$c = \frac{p}{100}b$** , it follows that **600 **b** = $\frac{p}{100}b$** , or **600 = $\frac{p}{100}$** . Multiplying both sides of this equation by **100** yields **60,000 = p** . Therefore, the value of **p** is **60,000**.

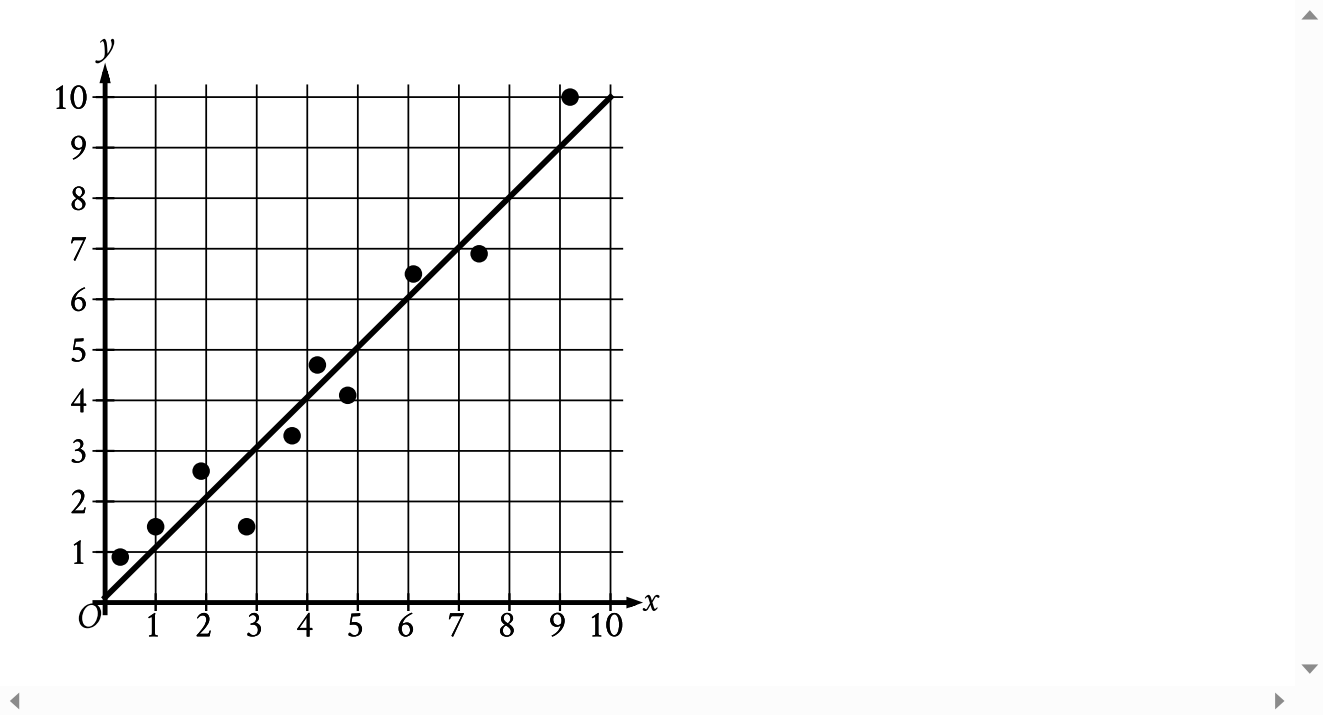
Question Difficulty: Hard

Question ID ea939862

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: ea939862

The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown.



For how many of the **10** data points is the actual y -value greater than the y -value predicted by the line of best fit?

- A. **3**
- B. **4**
- C. **6**
- D. **7**

ID: ea939862 Answer

Correct Answer: C

Rationale

Choice C is correct. Any data point that's located above the line of best fit has a y -value that's greater than the y -value predicted by the line of best fit. For the scatterplot shown, **6** of the data points are above the line of best fit. Therefore, **6** of the data points have an actual y -value that's greater than the y -value predicted by the line of best fit.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of data points that have an actual y -value that's less than the y -value predicted by the line of best fit.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID aa798de4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: aa798de4

	Live east of the river	Live west of the river	Total
Less than 40 years old	17	11	28
At least 40 years old	18	89	107
Total	35	100	135

The table summarizes members of a local organization by age and whether they live east or west of the river. If a member of the organization is selected at random, what is the probability that the selected member is at least 40 years old?

- A. $\frac{28}{135}$
- B. $\frac{35}{135}$
- C. $\frac{100}{135}$
- D. $\frac{107}{135}$

ID: aa798de4 Answer

Correct Answer: D

Rationale

Choice D is correct. If a member of the organization is selected at random, the probability that the selected member is at least 40 years old is equal to the number of members who are at least 40 years old divided by the total number of members. According to the table, there are a total of 135 members of the organization, and 107 of these members are at least 40 years old. Therefore, the probability that the selected member is at least 40 years old is $\frac{107}{135}$.

Choice A is incorrect. This is the probability that the selected member is less than 40 years old.

Choice B is incorrect. This is the probability that the selected member lives east of the river.

Choice C is incorrect. This is the probability that the selected member lives west of the river.

Question Difficulty: Easy

Question ID 95c6c2a2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 95c6c2a2

In a group, **40%** of the items are red. Of all the red items in the group, **30%** also have stripes. What percentage of the items in the group are red with stripes?

- A. **10%**
- B. **12%**
- C. **70%**
- D. **75%**

ID: 95c6c2a2 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that in a group, **40%** of the items are red. It follows that the number of red items in the group can be represented by **0.4x**, where **x** represents the total number of items in the group. It's also given that of all the red items in the group, **30%** also have stripes. It follows that the number of items in the group that are red and have stripes can be represented by **0.3(0.4x)**, or **0.12x**. The expression **0.12x** represents **12%** of **x**. Since **x** represents the total number of items in the group, it follows that **12%** of the items in the group are red and have stripes.

Choice A is incorrect and may result from subtracting **30%** from **40%** rather than calculating **30%** of **40%**.

Choice C is incorrect and may result from adding **30%** and **40%** rather than calculating **30%** of **40%**.

Choice D is incorrect and may result from calculating the percentage that **30%** is of **40%** rather than calculating **30%** of **40%**.

Question Difficulty: Medium

Question ID 579ff7ac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Percentages	

ID: 579ff7ac

At a recreation center, visitors used the exercise room for a total of t hours in July. At the same recreation center, visitors used the exercise room for $3.49t$ hours in August. What is the percent increase in the number of hours visitors used the exercise room from July to August?

- A. ~~2.49~~%
- B. ~~3.49~~%
- C. ~~249~~%
- D. ~~349~~%

ID: 579ff7ac Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the total number of hours visitors used the exercise room increased from t hours in July to $3.49t$ hours in August. It follows that the number of hours increased by $3.49t - t$, or $2.49t$. Let $p\%$ represent the percent increase in the exercise room use from July to August, or the percentage that $2.49t$ is of t . The value of p can be found using the proportion $\frac{2.49t}{t} = \frac{p}{100}$, or $2.49 = \frac{p}{100}$. Multiplying both sides of this equation by 100 yields $p = 249$. Therefore, the percent increase in the number of hours visitors used the exercise room from July to August is **249%**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

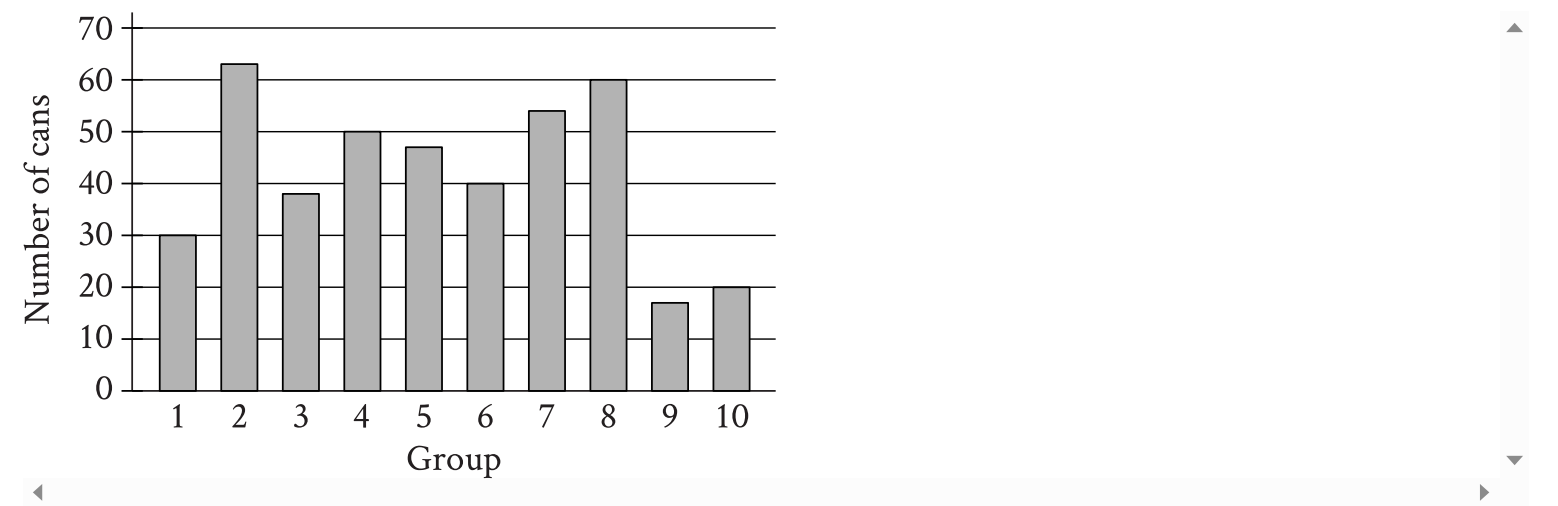
Choice D is incorrect. The exercise room use in August is **349%** of the exercise room use in July, but this doesn’t represent the percent increase from July to August.

Question Difficulty: Hard

Question ID de34ffaa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: de34ffaa



The bar graph shows the distribution of **419** cans collected by **10** different groups for a food drive. How many cans were collected by group **6**?

ID: de34ffaa Answer

Correct Answer: 40

Rationale

The correct answer is **40**. The height of each bar in the bar graph shown represents the number of cans collected by the group specified at the bottom of the bar. The bar for group **6** reaches a height of **40**. Therefore, group **6** collected **40** cans.

Question Difficulty: Easy

Question ID c963a231

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c963a231

A distance of **112** furlongs is equivalent to how many feet? (**1 furlong = 220 yards and 1 yard = 3 feet**)

ID: c963a231 Answer

Correct Answer: 73920

Rationale

The correct answer is **73,920**. It's given that **1 furlong = 220 yards** and **1 yard = 3 feet**. It follows that a distance of **112** furlongs is equivalent to **(112 furlongs) $\left(\frac{220 \text{ yards}}{1 \text{ furlong}}\right) \left(\frac{3 \text{ feet}}{1 \text{ yard}}\right)$** , or **73,920** feet.

Question Difficulty: Medium

Question ID 9dcc971f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 9dcc971f

A cluster fly is traveling at a velocity of **76** centimeters per second. What is this velocity, in **millimeters** per second? (**1 centimeter = 10 millimeters**)

- A. 7,600
- B. 836
- C. 760
- D. 86

ID: 9dcc971f Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that **1 centimeter = 10 millimeters**. It follows that a velocity of **76** centimeters per second is equivalent to $\left(\frac{76 \text{ centimeters}}{1 \text{ second}}\right)\left(\frac{10 \text{ millimeters}}{1 \text{ centimeter}}\right)$, which is equivalent to **(76)(10)**, or **760**, millimeters per second.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 6d36fc79

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 6d36fc79

A mechanical device in a workshop produces items at a constant rate of **60** items per hour. At this rate, how many items will the mechanical device produce in **3** hours?

ID: 6d36fc79 Answer

Correct Answer: 180

Rationale

The correct answer is **180**. It's given that a mechanical device produces items at a constant rate of **60** items per hour. This rate can be written as $\frac{60 \text{ items}}{1 \text{ hour}}$. Let x represent the number of items the mechanical device will produce in **3** hours at the given rate. It follows that $\frac{60 \text{ items}}{1 \text{ hour}} = \frac{x \text{ items}}{3 \text{ hours}}$, which can be written as $\frac{60}{1} = \frac{x}{3}$, or $60 = \frac{x}{3}$. Multiplying each side of this equation by **3** yields $180 = x$. Therefore, at the given rate, the mechanical device will produce **180** items in **3** hours.

Alternate approach: It's given that a mechanical device produces items at a constant rate of **60** items per hour. At this rate, the mechanical device will produce $(\frac{60 \text{ items}}{1 \text{ hour}})(3 \text{ hours})$, or **180** items in **3** hours.

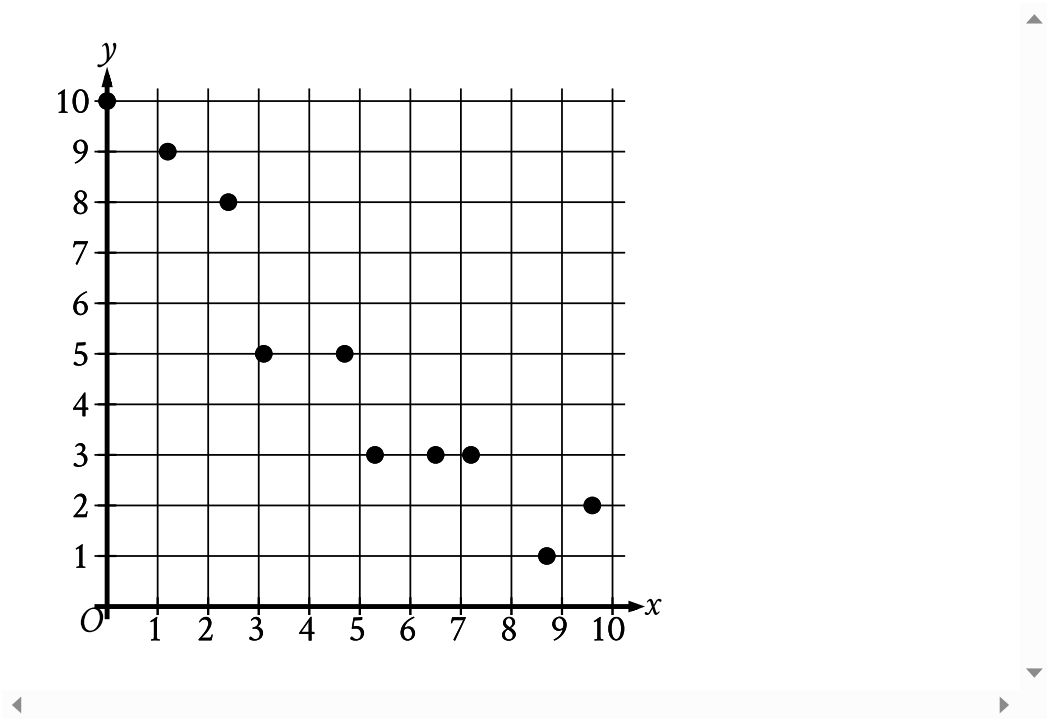
Question Difficulty: Easy

Question ID f10a805e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: f10a805e

The scatterplot shows the relationship between two variables, x and y .



Which of the following equations is the most appropriate linear model for the data shown?

- A. $y = 0.9 + 9.4x$
- B. $y = 0.9 - 9.4x$
- C. $y = 9.4 + 0.9x$
- D. $y = 9.4 - 0.9x$

ID: f10a805e Answer

Correct Answer: D

Rationale

Choice D is correct. The data points suggest that as the variable x increases, the variable y decreases, which implies that an appropriate linear model for the data has a negative slope. The data points also show that when x is close to 0 , y is greater than 9 . Therefore, the y-intercept of the graph of an appropriate linear model has a y-coordinate greater than 9 . The graph of an equation of the form $y = a + bx$, where a and b are constants, has a y-intercept with a y-coordinate of a and has a slope of b . Of the given choices, only choice D represents a graph that has a negative slope, -0.9 , and a y-intercept with a y-coordinate greater than 9 , 9.4 .

Choice A is incorrect. The graph of this equation has a positive slope, not a negative slope, and a y-intercept with a y-coordinate less than 1 , not greater than 9 .

Choice B is incorrect. The graph of this equation has a y-intercept with a y-coordinate less than 1 , not greater than 9 .

Choice C is incorrect. The graph of this equation has a positive slope, not a negative slope.

Question Difficulty: Medium

Question ID 90a442f6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 90a442f6

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

- A. 9
- B. $9\sqrt{2}$
- C. 18
- D. $18\sqrt{2}$

ID: 90a442f6 Answer

Correct Answer: C

Rationale

Choice C is correct. The perimeter of a triangle is the sum of the lengths of its sides. Since the given triangle is an isosceles right triangle, the length of each leg is the same and the length of the hypotenuse is equal to $\sqrt{2}$ times the length of a leg. Let x represent the length, in inches, of a leg of this isosceles right triangle. Therefore, the perimeter, in inches, of the triangle is $x + x + x\sqrt{2}$, or $2x + x\sqrt{2}$, which is equivalent to $x(2 + \sqrt{2})$. It's given that the perimeter of this triangle is $18 + 18\sqrt{2}$ inches. Thus, $x(2 + \sqrt{2}) = 18 + 18\sqrt{2}$. Dividing both sides of this equation by $2 + \sqrt{2}$ yields $x = \frac{18+18\sqrt{2}}{2+\sqrt{2}}$. Multiplying the right-hand side of this equation by $\frac{2-\sqrt{2}}{2-\sqrt{2}}$ yields $x = \frac{36+36\sqrt{2}-18\sqrt{2}-36}{2}$, or $x = 9\sqrt{2}$. It follows that the length, in inches, of a leg of this isosceles right triangle is $9\sqrt{2}$. Therefore, the length, in inches, of the hypotenuse of this isosceles right triangle is $(9\sqrt{2})(\sqrt{2})$, or 18.

Choice A is incorrect. If this were the length of the hypotenuse, the perimeter would be $9 + 9\sqrt{2}$ inches.

Choice B is incorrect. This is the length, in inches, of a leg of this triangle, not the hypotenuse.

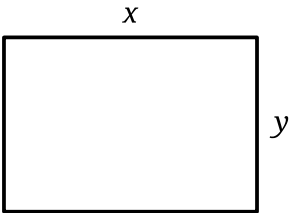
Choice D is incorrect. If this were the length of the hypotenuse, the perimeter would be $36 + 18\sqrt{2}$ inches.

Question Difficulty: Hard

Question ID 36fa8c3a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 36fa8c3a



Note: Figure not drawn to scale.

In the rectangle shown, $x = 7$ centimeters and $y = 5$ centimeters. What is the area, in square centimeters, of the rectangle?

- A. 7
- B. 12
- C. 35
- D. 40

ID: 36fa8c3a Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $x = 7$ centimeters and $y = 5$ centimeters for the rectangle. The area, A , of a rectangle can be found using the formula $A = \ell w$, where ℓ represents the length and w represents the width. In this rectangle, x represents the length and y represents the width. Substituting 7 for ℓ and 5 for w in the equation yields $A = (7)(5)$, or $A = 35$. Therefore, the area of the rectangle is 35 square centimeters.

Choice A is incorrect. This is the length, not the area, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a889e145

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a889e145

What is the area, in square inches, of a rectangle with a length of **7** inches and a width of **6** inches?

- A. **13**
- B. **20**
- C. **42**
- D. **84**

ID: a889e145 Answer

Correct Answer: C

Rationale

Choice C is correct. The area, ***A***, of a rectangle is given by the formula **$A = \ell w$** , where **ℓ** represents the length of the rectangle and **w** represents its width. It's given that the rectangle has a length of **7** inches and a width of **6** inches. Substituting **7** for **ℓ** and **6** for **w** in the formula **$A = \ell w$** yields **$A = (7)(6)$** , or **$A = 42$** . Thus, the area, in square inches, of the rectangle is **42**.

Choice A is incorrect. This is the sum, not the product, of the length and width of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is twice the area, in square inches, of the rectangle.

Question Difficulty: Easy

Question ID f707da8d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f707da8d

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

- A. $-14x + 1,540$
- B. $14x - 320$
- C. $-28x + 1,720$
- D. 360

ID: f707da8d Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. Since there are two parallel lines intersected by a transversal, all four acute angles have the same measure and all four obtuse angles have the same measure. Additionally, each acute angle is supplementary to each obtuse angle. It's given that the measure of one of these eight angles is $(7x - 250)^\circ$. It follows that a supplementary angle has measure $(180 - (7x - 250))^\circ$, or $(-7x + 430)^\circ$. It's also given that the sum of the measures of four of the eight angles is k° . It follows that the possible values of k are $4(7x - 250)$; $(7x - 250) + 3(-7x + 430)$; $2(7x - 250) + 2(-7x + 430)$; $3(7x - 250) + (-7x + 430)$; and $4(-7x + 430)$. These values are equivalent to $28x - 1,000$; $-14x + 1,040$; 360 ; $14x - 320$; and $-28x + 1,720$, respectively. It follows that of the given choices, only $-14x + 1,540$ could NOT be equivalent to k , for all values of x .

Choice B is incorrect. This is the sum of three angles with measure $(7x - 250)^\circ$ and one angle with measure $(-7x + 430)^\circ$.

Choice C is incorrect. This is the sum of four angles with measure $(-7x + 430)^\circ$.

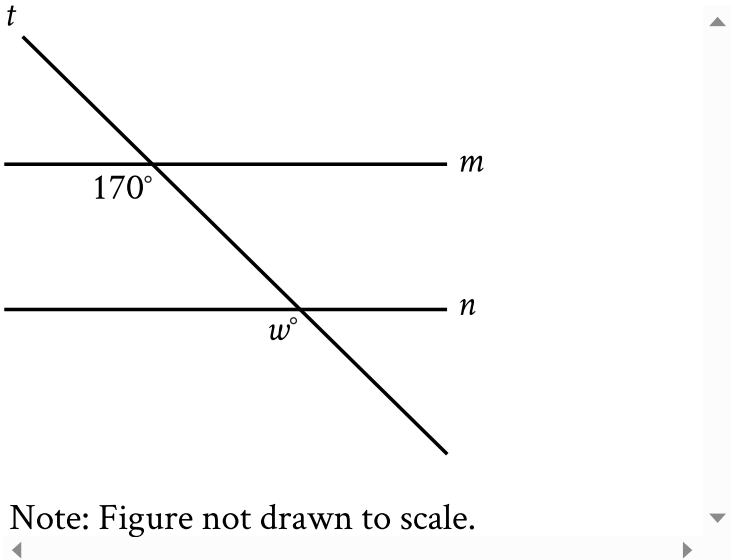
Choice D is incorrect. This is the sum of two angles with measure $(7x - 250)^\circ$ and two angles with measure $(-7x + 430)^\circ$.

Question Difficulty: Hard

Question ID 9a9f8e6a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 9a9f8e6a



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 17
- B. 30
- C. 70
- D. 170

ID: 9a9f8e6a Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n , it's a transversal. The angles in the figure marked as 170° and w° are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w^\circ = 170^\circ$. Therefore, the value of w is 170.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID ec5df4b7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: ec5df4b7

In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively.
- D. The measures of angles M and T are 70° and 50° , respectively.

ID: ec5df4b7 Answer

Correct Answer: D

Rationale

Choice D is correct. Two triangles are similar if they have three pairs of congruent corresponding angles. It's given that angles L and R each measure 60° , and so these corresponding angles are congruent. If angle M is 70° , then angle N must be 50° so that the sum of the angles in triangle LMN is 180° . If angle T is 50° , then angle S must be 70° so that the sum of the angles in triangle RST is 180° . Therefore, if the measures of angles M and T are 70° and 50° , respectively, then corresponding angles M and S are both 70° , and corresponding angles N and T are both 50° . It follows that triangles LMN and RST have three pairs of congruent corresponding angles, and so the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle LMN is similar to triangle RST is that the measures of angles M and T are 70° and 50° , respectively.

Choice A is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the two sides given are not proportional and the angle given is not included by the given sides.

Choice B is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the angle given is not included between the proportional sides.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b1f4ff0c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: b1f4ff0c

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

- A. 20°
- B. 45°
- C. 135°
- D. 160°

ID: b1f4ff0c Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that triangles EFG and JKL are congruent such that angle E corresponds to angle J . Corresponding angles of congruent triangles are congruent, so angle E and angle J must be congruent. Therefore, if the measure of angle E is 45° , then the measure of angle J is also 45° .

Choice A is incorrect. This is the measure of angle K , not angle J .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 959b5c59

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 959b5c59

A right rectangular prism has a length of **11** meters, a width of **8** meters, and a height of **10** meters. What is the volume, in cubic meters, of the prism?

ID: 959b5c59 Answer

Correct Answer: 880

Rationale

The correct answer is **880**. The volume, V , of a right rectangular prism is given by the formula $V = \ell wh$, where ℓ is the length, w is the width, and h is the height of the prism. It's given that a right rectangular prism has a length of **11** meters, a width of **8** meters, and a height of **10** meters. Substituting **11** for ℓ , **8** for w , and **10** for h in the formula $V = \ell wh$ yields $V = (11)(8)(10)$, or $V = 880$. Therefore, the volume, in cubic meters, of the prism is **880**.

Question Difficulty: Medium

Question ID 12001820

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 12001820

The perimeter of an equilateral triangle is **852** centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?

ID: 12001820 Answer

Correct Answer: 284/3, 94.66, 94.67

Rationale

The correct answer is $\frac{284}{3}$. Since the perimeter of a triangle is the sum of the lengths of its sides, and the given triangle is equilateral, the length of each side is $\frac{852}{3}$, or **284**, centimeters (cm). Right triangle **AMO** can be formed, where **M** is the midpoint of one of the triangle's sides, **A** is one of this side's endpoints, and **O** is the center of the circle. It follows that **AM** is $\frac{284}{2}$, or **142**, cm. Additionally, triangle **AMO** has angles measuring **30°**, **60°**, and **90°**, where the measure of angle **OMA** is **90°** and the measure of angle **OAM** is **30°**. It follows that the length of side **MO** is half the length of hypotenuse **AO**, and the length of side **AM** is $\sqrt{3}$ times the length of side **MO**. It's given that **AO** = $w\sqrt{3}$ cm. Therefore, **MO** = $\frac{w\sqrt{3}}{2}$ cm and **AM** = $\frac{w\sqrt{3}\sqrt{3}}{2}$ cm, which is equivalent to **AM** = $\frac{3w}{2}$ cm. Since **AM** = **142** cm, it follows that $\frac{3w}{2} = 142$. Multiplying both sides of this equation by **2** yields **3w** = **284**. Dividing both sides of this equation by **3** yields $w = \frac{284}{3}$. Note that 284/3, 94.66, and 94.67 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID b7e6ffc2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: b7e6ffc2

The floor of a ballroom has an area of **600** square meters. An architect creates a scale model of the floor of the ballroom, where the length of each side of the model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of the ballroom. What is the area, in square meters, of the scale model?

- A. **6**
- B. **10**
- C. **60**
- D. **150**

ID: b7e6ffc2 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the length of each side of a scale model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of a ballroom. Therefore, the area of the scale model is $\left(\frac{1}{10}\right)^2$, or $\frac{1}{100}$, times the area of the actual floor of the ballroom. It’s given that the area of the floor of the ballroom is **600** square meters. Therefore, the area, in square meters, of the scale model is $\left(\frac{1}{100}\right)(600)$, or **6**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 1aa2b038

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 1aa2b038

A right circular cylinder has a volume of **432** cubic centimeters. The area of the base of the cylinder is **24** square centimeters. What is the height, in centimeters, of the cylinder?

- A. 18
- B. 24
- C. 216
- D. 10,368

ID: 1aa2b038 Answer

Correct Answer: A

Rationale

Choice A is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where πr^2 is the area of the base of the cylinder and h is the height. It's given that a right circular cylinder has a volume of **432** cubic centimeters and the area of the base is **24** square centimeters. Substituting **432** for V and **24** for πr^2 in the formula $V = \pi r^2 h$ yields **432 = 24h**. Dividing both sides of this equation by **24** yields **18 = h**. Therefore, the height of the cylinder, in centimeters, is **18**.

Choice B is incorrect. This is the area of the base, in square centimeters, not the height, in centimeters, of the cylinder.

Choice C is incorrect. This is the height, in centimeters, of a cylinder if its volume is **432** cubic centimeters and the area of its base is **2**, not **24**, cubic centimeters.

Choice D is incorrect. This is the height, in centimeters, of a cylinder if its volume is **432** cubic centimeters and the area of its base is $\frac{1}{24}$, not **24**, cubic centimeters.

Question Difficulty: Medium

Question ID 35efc692

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 35efc692

In triangle ABC , angle B is a right angle. The length of side AB is $10\sqrt{37}$ and the length of side BC is $24\sqrt{37}$. What is the length of side AC ?

- A. $14\sqrt{37}$
- B. $26\sqrt{37}$
- C. $34\sqrt{37}$
- D. $\sqrt{34 \cdot 37}$

ID: 35efc692 Answer

Correct Answer: B

Rationale

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that in triangle ABC , angle B is a right angle. Therefore, triangle ABC is a right triangle, where the hypotenuse is side AC and the legs are sides AB and BC . It's given that the lengths of sides AB and BC are $10\sqrt{37}$ and $24\sqrt{37}$, respectively. Substituting these values for a and b in the formula $c^2 = a^2 + b^2$ yields $c^2 = (10\sqrt{37})^2 + (24\sqrt{37})^2$, which is equivalent to $c^2 = 100(37) + 576(37)$, or $c^2 = 676(37)$. Taking the square root of both sides of this equation yields $c = \pm 26\sqrt{37}$. Since c represents the length of the hypotenuse, side AC , c must be positive. Therefore, the length of side AC is $26\sqrt{37}$.

Choice A is incorrect. This is the result of solving the equation $c = 24\sqrt{37} - 10\sqrt{37}$, not $c^2 = (10\sqrt{37})^2 + (24\sqrt{37})^2$.

Choice C is incorrect. This is the result of solving the equation $c = 10\sqrt{37} + 24\sqrt{37}$, not $c^2 = (10\sqrt{37})^2 + (24\sqrt{37})^2$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 162d2e92

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 162d2e92

Rectangle P has an area of **72** square inches. If a rectangle with an area of **20** square inches is removed from rectangle P, what is the area, in square inches, of the resulting figure?

- A. **92**
- B. **84**
- C. **80**
- D. **52**

ID: 162d2e92 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that rectangle P has an area of **72** square inches. If a rectangle with an area of **20** square inches is removed from rectangle P, the area, in square inches, of the resulting figure is **72 — 20**, or **52**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

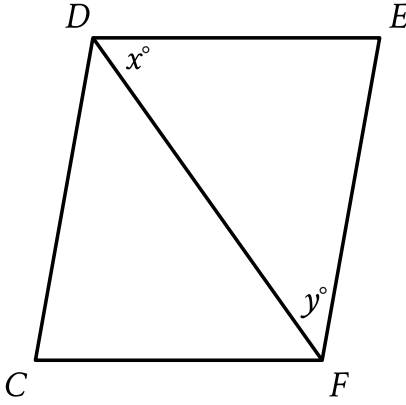
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a2c8a744

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a2c8a744



Note: Figure not drawn to scale.

In the figure shown, $CD = EF$ and $FC = DE$. If $x = 65$ and $y = 30$, what is the measure of $\angle CDE$?

- A. 30°
- B. 65°
- C. 95°
- D. 130°

ID: a2c8a744 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $CD = EF$ and $FC = DE$, and the figure shows that $\overline{DF}$ is a side of both $\triangle CDF$ and $\triangle EFD$. Since $\triangle CDF$ and $\triangle EFD$ have three pairs of congruent sides, the triangles are congruent, where $\angle CFD$ corresponds to $\angle EDF$ and $\angle CDF$ corresponds to $\angle EFD$. It's given that the measure of $\angle EDF$ is 65° and the measure of $\angle EFD$ is 30° . Since corresponding angles of congruent triangles are congruent, $\angle CFD$ is congruent to $\angle EDF$ and $\angle CDF$ is congruent to $\angle EFD$. Therefore, the measure of $\angle CFD$ is 65° and the measure of $\angle CDF$ is 30° . Since $\angle CDE$ is made up of $\angle CDF$ and $\angle EDF$, it follows that the measure of $\angle CDE$ is equal to the sum of the measures of $\angle CDF$ and $\angle EDF$. Therefore, the measure of $\angle CDE$ is equal to $(30 + 65)^\circ$, or 95° .

Choice A is incorrect. This is the measure of $\angle CDF$, not the measure of $\angle CDE$.

Choice B is incorrect. This is the measure of $\angle CFD$, not the measure of $\angle CDE$.

Choice D is incorrect. This is the sum of the measures of $\angle CFD$ and $\angle EDF$, not $\angle CDF$ and $\angle EDF$.

Question Difficulty: Medium

Question ID 57616839

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 57616839

The length of a rectangle’s diagonal is $5\sqrt{17}$, and the length of the rectangle’s shorter side is 5. What is the length of the rectangle’s longer side?

- A. $\sqrt{17}$
- B. 20
- C. $15\sqrt{2}$
- D. 400

ID: 57616839 Answer

Correct Answer: B

Rationale

Choice B is correct. A rectangle’s diagonal divides a rectangle into two congruent right triangles, where the diagonal is the hypotenuse of both triangles. It’s given that the length of the diagonal is $5\sqrt{17}$ and the length of the rectangle’s shorter side is 5. Therefore, each of the two right triangles formed by the rectangle’s diagonal has a hypotenuse with length $5\sqrt{17}$, and a shorter leg with length 5. To calculate the length of the longer leg of each right triangle, the Pythagorean theorem, $a^2 + b^2 = c^2$, can be used, where a and b are the lengths of the legs and c is the length of the hypotenuse of the triangle. Substituting 5 for a and $5\sqrt{17}$ for c in the equation $a^2 + b^2 = c^2$ yields $5^2 + b^2 = (5\sqrt{17})^2$, which is equivalent to $25 + b^2 = 25(17)$, or $25 + b^2 = 425$. Subtracting 25 from each side of this equation yields $b^2 = 400$. Taking the positive square root of each side of this equation yields $b = 20$. Therefore, the length of the longer leg of each right triangle formed by the diagonal of the rectangle is 20. It follows that the length of the rectangle’s longer side is 20.

Choice A is incorrect and may result from dividing the length of the rectangle’s diagonal by the length of the rectangle’s shorter side, rather than substituting these values into the Pythagorean theorem.

Choice C is incorrect and may result from using the length of the rectangle’s diagonal as the length of a leg of the right triangle, rather than the length of the hypotenuse.

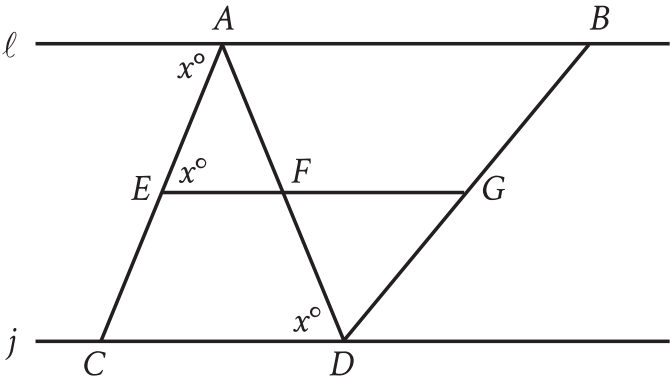
Choice D is incorrect. This is the square of the length of the rectangle’s longer side.

Question Difficulty: Hard

Question ID bd9f0496

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: bd9f0496



Note: Figure not drawn to scale.

In the figure, $\overline{EG}$ and $\overline{AD}$ intersect at point F , and line ℓ is parallel to line j . If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?

- I. $AB = 2.25EF$
- II. $FG = 0.5AB$
- III. $\frac{EF}{FG} = \frac{AF}{FD}$

- A. I only
- B. II only
- C. I and II
- D. II and III

ID: bd9f0496 Answer

Correct Answer: B

Rationale

Choice B is correct. In the figure, $\overline{AE}$ intersects line ℓ and $\overline{EG}$, and alternate interior angles are both marked x° . It follows that $\overline{EG}$ is parallel to line ℓ . It's given that line ℓ is parallel to line j , so $\overline{EG}$ is also parallel to line j . Thus, the following angles also have measure x° : $\angle EFA$, $\angle ACD$, $\angle BAD$, and $\angle GFD$. Triangles AEF and ACD each have two angles with measure x° , so triangles AEF and ACD are similar. It's also given that $EF = 0.5CD$, so $AF = 0.5AD$. It follows that $AF = FD$. Triangles DFG and DAB each have one angle with measure x° and share $\angle FDG$, so triangles DFG and DAB are similar. Since $AF = FD$, it follows that $AD = 2FD$, so $AB = 2FG$. Substituting $2FG$ for AB in statement I yields $2FG = 2.25EF$. It's given that $FG = 2.25CD$. Substituting $2.25CD$ for FG and $0.5CD$ for EF in the equation $2FG = 2.25EF$ yields $2(2.25CD) = 2.25(0.5CD)$, or $4.5CD = 1.125CD$. Thus, statement I is not true. Since $AB = 2FG$, it follows that $0.5AB = FG$, or $FG = 0.5AB$. Therefore, statement II must be true. Substituting $0.5CD$ for EF , $2.25CD$ for FG , and FD for AF in statement III yields $\frac{0.5CD}{2.25CD} = \frac{FD}{FD}$, or $\frac{2}{9} = 1$. Thus, statement III is not true. Therefore, statement II only must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

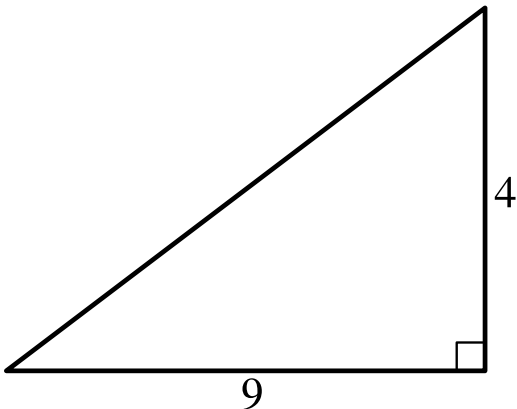
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID c8131bca

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: c8131bca



Note: Figure not drawn to scale.

The figure shows the lengths, in units, of two sides of a triangle. What is the area, in square units, of the triangle?

- A. 18
- B. 13
- C. 9
- D. 4

ID: c8131bca Answer

Correct Answer: A

Rationale

Choice A is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. In the right triangle shown, the length of the base of the triangle is 9 units, and its height is 4 units. Substituting 9 for b and 4 for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(9)(4)$, or $A = 18$. Therefore, the area of the triangle, in square units, is 18.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of the base of the triangle, in units, not the area of the triangle, in square units.

Choice D is incorrect. This is the height of the triangle, in units, not the area of the triangle, in square units.

Question Difficulty: Medium

Question ID f1d15e84

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f1d15e84

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 140)^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

- A. $(-18x + 280)^\circ$
- B. $(-18x + 640)^\circ$
- C. $(18x - 280)^\circ$
- D. 180°

ID: f1d15e84 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that two lines intersect at exactly one point, forming two acute angles and two obtuse angles. When two lines intersect, opposite angles are congruent and adjacent angles are supplementary. It's given that the measure of one of the angles is $(9x - 140)^\circ$, so the measure of the opposite angle is $(9x - 140)^\circ$, and the measure of each adjacent angle is $(180 - (9x - 140))^\circ$, or $(-9x + 320)^\circ$. The possible sums of the measures of any two of the angles are $(9x - 140)^\circ + (9x - 140)^\circ$, or $(18x - 280)^\circ$, $(-9x + 320)^\circ + (-9x + 320)^\circ$, or $(-18x + 640)^\circ$, and $(9x - 140)^\circ + (-9x + 320)^\circ$, or 180° . Of the given choices, only $(-18x + 280)^\circ$ is not equivalent to any of these expressions and could not be the sum of the measures of any two of the angles.

Choice B is incorrect. This is the sum of the measures of two angles that are adjacent to the given angle.

Choice C is incorrect. This is the sum of the measure of the given angle and the measure of its opposite angle.

Choice D is incorrect. This is the sum of the measure of the given angle and the measure of an angle adjacent to the given angle.

Question Difficulty: Hard

Question ID 3b22a4a0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 3b22a4a0

The length of a side of square X is **9** centimeters. The area of rectangle Y is **32** square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

- A. **145**
- B. **113**
- C. **82**
- D. **81**

ID: 3b22a4a0 Answer

Correct Answer: B

Rationale

Choice B is correct. The area of a square is s^2 , where s is the side length of the square. It's given that the length of a side of square X is **9** centimeters. It follows that the area of square X is **9²**, or **81** square centimeters. It's also given that the area of rectangle Y is **32** square centimeters. Therefore, the total area, in square centimeters, of square X and rectangle Y is **81 + 32**, or **113**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area of square X, not the total area of square X and rectangle Y, in square centimeters.

Question Difficulty: Medium

Question ID 995a6673

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 995a6673

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **6 cm**. The volume of this cone is $n\pi \text{ cm}^3$. What is the value of n ?

ID: 995a6673 Answer

Correct Answer: 66

Rationale

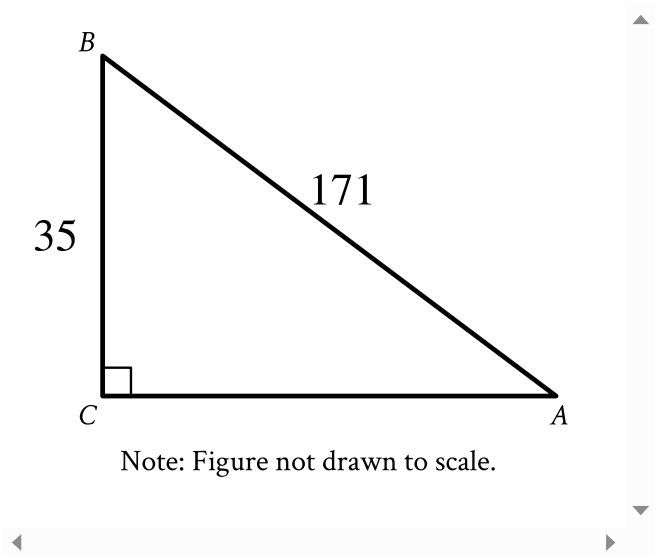
The correct answer is **66**. It's given that the right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **6 cm**. Since the diameter of the base of the cone is **6 cm**, the radius of the base is **3 cm**. The volume V , **in cm³**, of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where h is the height, **in cm**, and r is the radius, **in cm**, of the base of the cone. Substituting **22** for h and **3** for r in this formula yields $V = \frac{1}{3}\pi(3)^2(22)$, or $V = 66\pi$. Therefore, the volume of the cone is $66\pi \text{ cm}^3$. It's given that the volume of the cone is $n\pi \text{ cm}^3$. Therefore, the value of n is **66**.

Question Difficulty: Hard

Question ID 2935ddfa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 2935ddfa



In the right triangle shown, what is the value of $\sin A$?

- A. $\frac{1}{171}$
- B. $\frac{35}{171}$
- C. $\frac{171}{35}$
- D. 171

ID: 2935ddfa Answer

Correct Answer: B

Rationale

Choice B is correct. The sine of an acute angle in a right triangle is the ratio of the length of the side opposite that angle to the length of the hypotenuse. The hypotenuse of a right triangle is the side opposite the right angle. In right triangle ABC , side BC is the side opposite angle A and side AB is the hypotenuse. It's given that the length of side BC is 35 units and the length of side AB is 171 units. Therefore, the value of $\sin A$ is $\frac{35}{171}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the ratio of the length of the hypotenuse to the length of the side opposite angle A rather than the ratio of the length of the side opposite angle A to the length of the hypotenuse.

Choice D is incorrect. This is the length of the hypotenuse rather than $\sin A$.

Question Difficulty: Medium

Question ID 52e433bb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 52e433bb

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 27° . What is the measure, in degrees, of angle F ?

ID: 52e433bb Answer

Correct Answer: 63

Rationale

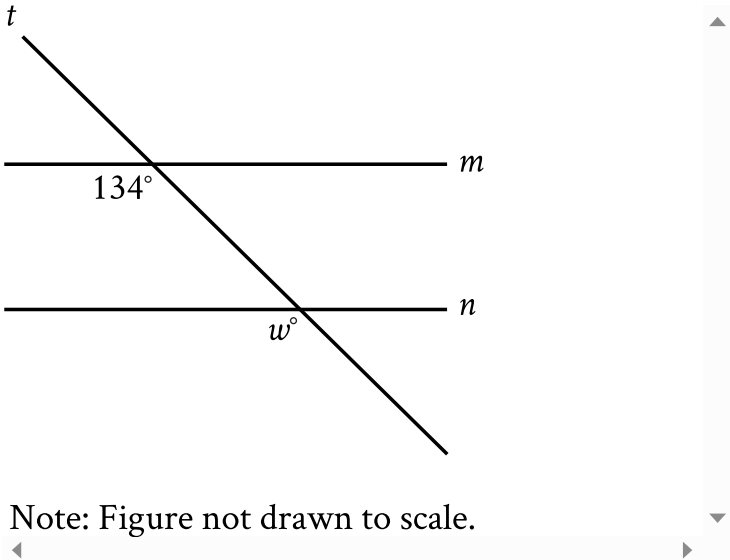
The correct answer is **63**. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is 27° . It follows that the measure of angle D is also 27° . It's given that angle E is a right angle. Therefore, the measure of angle E is 90° . Let x represent the measure, in degrees, of angle F . Since the measures of the angles in a triangle sum to 180° , it follows that $27 + 90 + x = 180$, or $117 + x = 180$. Subtracting **117** from both sides of this equation yields $x = \mathbf{63}$. Therefore, the measure, in degrees, of angle F is **63**.

Question Difficulty: Medium

Question ID ca3863dc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: ca3863dc



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 13
- B. 34
- C. 66
- D. 134

ID: ca3863dc Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n , it's a transversal. The angles in the figure marked as 134° and w° are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w^\circ = 134^\circ$. Therefore, the value of w is **134**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

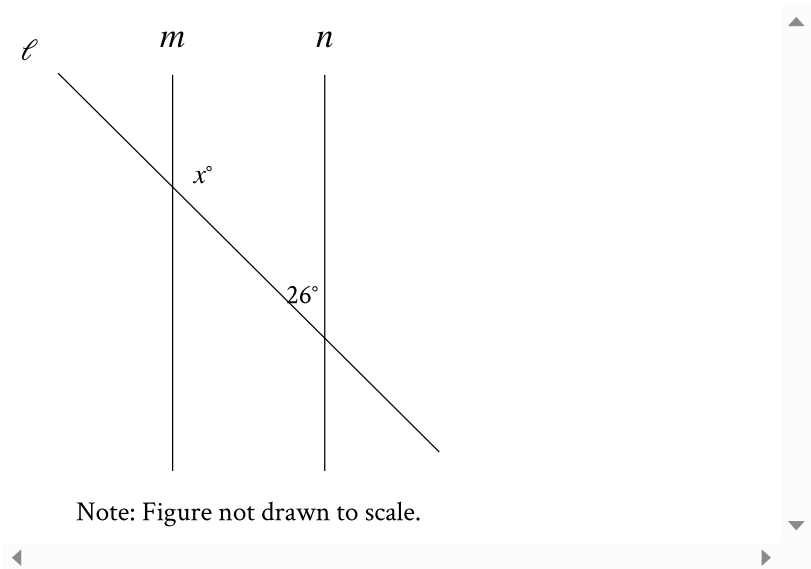
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 49dceeb4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 49dceeb4



In the figure shown, line *m* is parallel to line *n*. What is the value of *x*?

- A. 13
- B. 26
- C. 52
- D. 154

ID: 49dceeb4 Answer

Correct Answer: D

Rationale

Choice D is correct. The sum of consecutive interior angles between two parallel lines and on the same side of the transversal is **180** degrees. Since it's given that line *m* is parallel to line *n*, it follows that $x + 26 = 180$. Subtracting **26** from both sides of this equation yields **154**. Therefore, the value of *x* is **154**.

Choice A is incorrect. This is half of the given angle measure.

Choice B is incorrect. This is the value of the given angle measure.

Choice C is incorrect. This is twice the value of the given angle measure.

Question Difficulty: Easy

Question ID 6e1ab4a5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 6e1ab4a5

A cube has a volume of **474,552** cubic units. What is the surface area, in square units, of the cube?

ID: 6e1ab4a5 Answer

Correct Answer: 36504

Rationale

The correct answer is **36,504**. The volume of a cube can be found using the formula $V = s^3$, where s represents the edge length of a cube. It's given that this cube has a volume of **474,552** cubic units. Substituting **474,552** for V in $V = s^3$ yields **474,552** = s^3 . Taking the cube root of both sides of this equation yields **78** = s . Thus, the edge length of the cube is **78** units. Since each face of a cube is a square, it follows that each face has an edge length of **78** units. The area of a square can be found using the formula $A = s^2$. Substituting **78** for s in this formula yields $A = 78^2$, or $A = 6,084$. Therefore, the area of one face of this cube is **6,084** square units. Since a cube has **6** faces, the surface area, in square units, of this cube is **6(6,084)**, or **36,504**.

Question Difficulty: Hard

Question ID 44ab70ec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 44ab70ec

The length of the base of a triangle is **8** inches, and the height of the triangle is **12** inches. What is the area, in square inches, of the triangle?

- A. **20**
- B. **48**
- C. **60**
- D. **96**

ID: 44ab70ec Answer

Correct Answer: B

Rationale

Choice B is correct. The formula for the area, ***A***, of a triangle is $A = \frac{1}{2}bh$, where ***b*** is the length of the base of the triangle and ***h*** is the height of the triangle. It's given that the length of the base of a triangle is **8** inches and the height of the triangle is **12** inches. Substituting **8** for ***b*** and **12** for ***h*** in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(8)(12)$, or $A = 48$. Therefore, the area, in square inches, of the triangle is **48**.

Choice A is incorrect. This is the sum of the length of the base of the triangle and the height of the triangle, not the area of the triangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the product of the length of the base of the triangle and the height of the triangle, not the area of the triangle.

Question Difficulty: Medium

Question ID 8d6e840e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 8d6e840e

A right triangle has legs with lengths of **11** centimeters and **9** centimeters. What is the length of this triangle's hypotenuse, in centimeters?

- A. $\sqrt{40}$
- B. $\sqrt{202}$
- C. 20
- D. 202

ID: 8d6e840e Answer

Correct Answer: B

Rationale

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that a right triangle has legs with lengths of **11** centimeters and **9** centimeters. Substituting **11** for a and **9** for b in the formula $c^2 = a^2 + b^2$ yields $c^2 = 11^2 + 9^2$, which is equivalent to $c^2 = 121 + 81$, or $c^2 = 202$. Taking the square root of each side of this equation yields $c = \pm\sqrt{202}$. Since c represents a length, c must be positive. Therefore, the length of the triangle's hypotenuse, in centimeters, is $\sqrt{202}$.

Choice A is incorrect. This is the result of solving the equation $c^2 = 11(2) + 9(2)$, not $c^2 = 11^2 + 9^2$.

Choice C is incorrect. This is the result of solving the equation $c(2) = 11(2) + 9(2)$, not $c^2 = 11^2 + 9^2$.

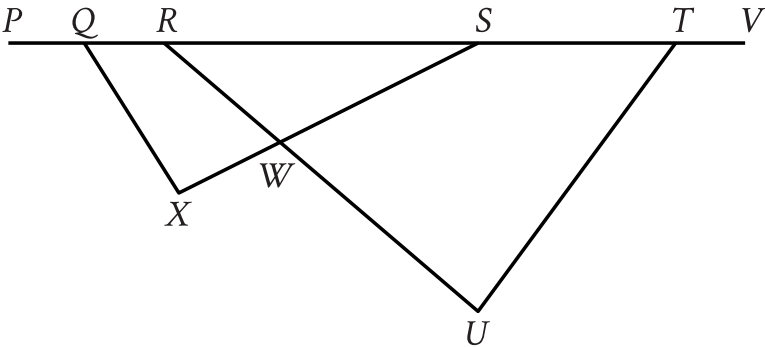
Choice D is incorrect. This is the result of solving the equation $c = 11^2 + 9^2$, not $c^2 = 11^2 + 9^2$.

Question Difficulty: Medium

Question ID f4bbb880

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f4bbb880



Note: Figure not drawn to scale.

In the figure shown, points Q , R , S , and T lie on line segment PV , and line segment RU intersects line segment SX at point W . The measure of $\angle SQX$ is 48° , the measure of $\angle SXQ$ is 86° , the measure of $\angle SWU$ is 85° , and the measure of $\angle VTU$ is 162° . What is the measure, in degrees, of $\angle TUR$?

ID: f4bbb880 Answer

Correct Answer: 123

Rationale

The correct answer is **123**. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is **180** degrees. It's given that the measure of $\angle SQX$ is 48° and the measure of $\angle SXQ$ is 86° . Since points S , Q , and X form a triangle, it follows from the triangle angle sum theorem that the measure, in degrees, of $\angle QSX$ is $180 - 48 - 86$, or **46**. It's also given that the measure of $\angle SWU$ is 85° . Since $\angle SWU$ and $\angle SWR$ are supplementary angles, the sum of their measures is **180** degrees. It follows that the measure, in degrees, of $\angle SWR$ is $180 - 85$, or **95**. Since points R , S , and W form a triangle, and $\angle RSW$ is the same angle as $\angle QSX$, it follows from the triangle angle sum theorem that the measure, in degrees, of $\angle WRS$ is $180 - 46 - 95$, or **39**. It's given that the measure of $\angle VTU$ is 162° . Since $\angle VTU$ and $\angle STU$ are supplementary angles, the sum of their measures is **180** degrees. It follows that the measure, in degrees, of $\angle STU$ is $180 - 162$, or **18**. Since points R , T , and U form a triangle, and $\angle URT$ is the same angle as $\angle WRS$, it follows from the triangle angle sum theorem that the measure, in degrees, of $\angle TUR$ is $180 - 39 - 18$, or **123**.

Question Difficulty: Hard

Question ID 34a7512e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 34a7512e

Which of the following expressions is equivalent to $(\sin 24^\circ)(\cos 66^\circ) + (\cos 24^\circ)(\sin 66^\circ)$?

- A. $2(\cos 66^\circ)(\sin 24^\circ)$
- B. $2(\cos 66^\circ) + 2(\cos 24^\circ)$
- C. $(\cos 66^\circ)^2 + (\cos 24^\circ)^2$
- D. $(\cos 66^\circ)^2 + (\sin 24^\circ)^2$

ID: 34a7512e Answer

Correct Answer: C

Rationale

Choice C is correct. The sine of an angle is equal to the cosine of its complementary angle. Since angles with measures 24° and 66° are complementary to each other, $\sin 24^\circ$ is equal to $\cos 66^\circ$ and $\sin 66^\circ$ is equal to $\cos 24^\circ$. Substituting $\cos 66^\circ$ for $\sin 24^\circ$ and $\cos 24^\circ$ for $\sin 66^\circ$ in the given expression yields $(\cos 66^\circ)(\cos 66^\circ) + (\cos 24^\circ)(\cos 24^\circ)$, or $(\cos 66^\circ)^2 + (\cos 24^\circ)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID fb70c83e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: fb70c83e

One leg of a right triangle has a length of **43.2** millimeters. The hypotenuse of the triangle has a length of **196.8** millimeters. What is the length of the other leg of the triangle, in millimeters?

- A. **43.2**
- B. **120**
- C. **192**
- D. **201.5**

ID: fb70c83e Answer

Correct Answer: C

Rationale

Choice C is correct. The Pythagorean theorem states that for a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. It's given that one leg of a right triangle has a length of **43.2** millimeters. It's also given that the hypotenuse of the triangle has a length of **196.8** millimeters. Let **b** represent the length of the other leg of the triangle, in millimeters. Therefore, by the Pythagorean theorem, $43.2^2 + b^2 = 196.8^2$, or $1,866.24 + b^2 = 38,730.24$. Subtracting **1,866.24** from both sides of this equation yields $b^2 = 36,864$. Taking the positive square root of both sides of this equation yields **b = 192**. Therefore, the length of the other leg of the triangle, in millimeters, is **192**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID f0284dd1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f0284dd1

Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

- A. 204
- B. 182
- C. 60
- D. 12

ID: f0284dd1 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. It follows that side AB corresponds to side XY and side BC corresponds to side YZ . Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{XY}{AB} = \frac{YZ}{BC}$. Substituting 170 for AB , 60 for YZ , and 850 for BC in this equation yields $\frac{XY}{170} = \frac{60}{850}$. Multiplying each side of this equation by 170 yields $XY = 12$. Therefore, the length of $\overline{XY}$ is 12.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

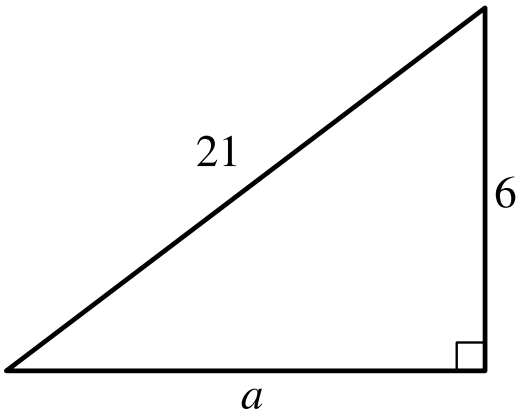
Choice C is incorrect. This is the length of $\overline{YZ}$, not $\overline{XY}$.

Question Difficulty: Medium

Question ID 87bce857

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 87bce857



Note: Figure not drawn to scale.

For the triangle shown, which expression represents the value of a ?

- A. $\sqrt{21^2 - 6^2}$
- B. $21^2 - 6^2$
- C. $\sqrt{21 - 6}$
- D. $21 - 6$

ID: 87bce857 Answer

Correct Answer: A

Rationale

Choice A is correct. For the right triangle shown, the lengths of the legs are a units and 6 units, and the length of the hypotenuse is 21 units. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Therefore, $a^2 + 6^2 = 21^2$. Subtracting 6^2 from both sides of this equation yields $a^2 = 21^2 - 6^2$. Taking the square root of both sides of this equation yields $a = \pm\sqrt{21^2 - 6^2}$. Since a is a length, a must be positive. Therefore, $a = \sqrt{21^2 - 6^2}$. Thus, for the triangle shown, $\sqrt{21^2 - 6^2}$ represents the value of a .

Choice B is incorrect. For the triangle shown, this expression represents the value of a^2 , not a .

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 45f4c00c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 45f4c00c

What is the area, in square centimeters, of a rectangle with a length of **36** centimeters and a width of **34** centimeters?

- A. **70**
- B. **140**
- C. **1,156**
- D. **1,224**

ID: 45f4c00c Answer

Correct Answer: D

Rationale

Choice D is correct. The area **A** , in square centimeters, of a rectangle can be found using the formula **$A = \ell w$** , where ℓ is the length, in centimeters, of the rectangle and w is its width, in centimeters. It's given that the rectangle has a length of **36** centimeters and a width of **34** centimeters. Substituting **36** for ℓ and **34** for w in the formula **$A = \ell w$** yields **$A = 36(34)$** , or **$A = 1,224$** . Therefore, the area, in square centimeters, of this rectangle is **1,224**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, in centimeters, not the area, in square centimeters, of the rectangle.

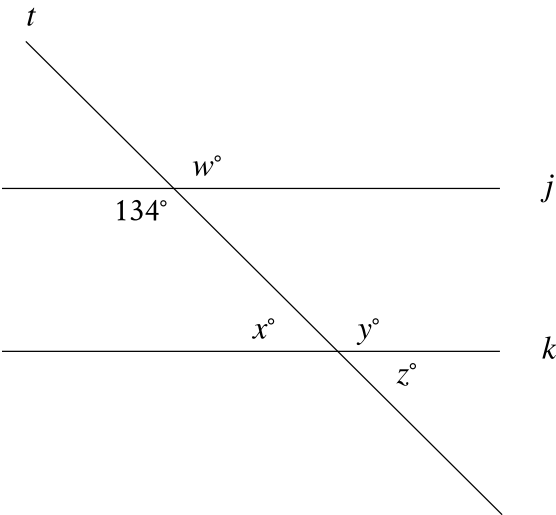
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e7f986ab

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: e7f986ab



Note: . igure not drawn to scale.

◀ ▶

In the figure, line t intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- A. $x = 46$
- B. $y = 46$
- C. $w = 134$
- D. $z = 134$

ID: e7f986ab Answer

Correct Answer: A

Rationale

Choice A is correct. In the figure shown, line t , which is a transversal, intersects lines j and k . If two lines are intersected by a transversal and the measures of a pair of same-side interior angles sum to 180° , then the lines are parallel. In the figure, the angle measuring 134° and the angle measuring x° are a pair of same-side interior angles. If $x = 46$, $(134 + x)^\circ = (134 + 46)^\circ$, or $(134 + x)^\circ = 180^\circ$. Therefore, $x = 46$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring y° and the angle measuring 134° are alternate interior angles. If $y = 46$ rather than $y = 134$, then lines j and k would not be parallel.

Choice C is incorrect. The angle measuring w° and the angle measuring 134° are vertical angles. Thus, $w = 134$ whether lines j and k are parallel or not.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 17332082

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 17332082

In triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

- A. $\frac{15}{38}$
- B. $\frac{15}{23}$
- C. $\frac{23}{15}$
- D. $\frac{38}{15}$

ID: 17332082 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that in triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. Therefore, the measure of angle BDC is 90° . It follows that angle B is congruent to angle D and angle C is congruent to angle C . By the angle-angle similarity postulate, triangle ABC is similar to triangle BDC . Since triangles ABC and BDC are similar, it follows that $\frac{AC}{AB} = \frac{BC}{BD}$. It's also given that the length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. Therefore, the length of $\overline{AC}$ is $15 + 23$, or 38 . Substituting 15 for AB and 38 for AC in the equation $\frac{AC}{AB} = \frac{BC}{BD}$ yields $\frac{38}{15} = \frac{BC}{BD}$. Therefore, the value of $\frac{BC}{BD}$ is $\frac{38}{15}$.

Choice A is incorrect. This is the value of $\frac{BD}{BC}$.

Choice B is incorrect and may result from conceptual or calculation errors.

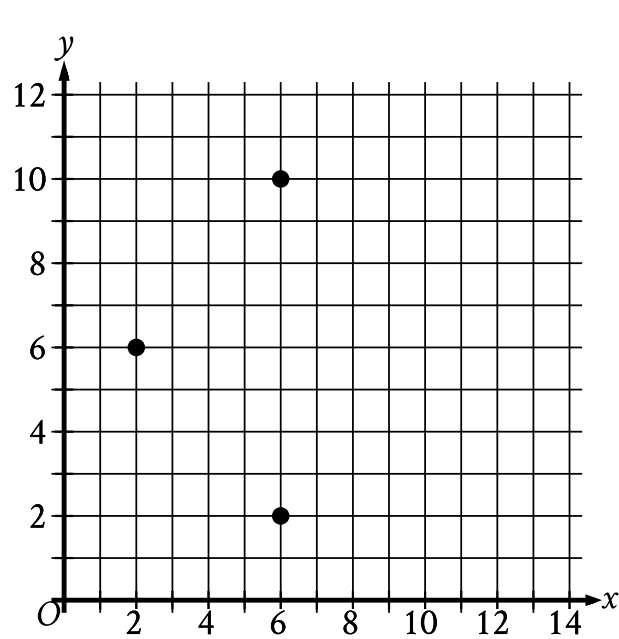
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 55c9a02c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 55c9a02c



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

ID: 55c9a02c Answer

Correct Answer: 8

Rationale

The correct answer is **8**. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point **(6, 6)** is halfway between the points **(6, 2)** and **(6, 10)**, and is a distance of **4** units from each of those two points. The point **(6, 6)** is also a distance of **4** units from **(2, 6)**. Because the point **(6, 6)** is the same distance from all three points shown, it must be the center of the circle. Since that distance is **4**, it follows that the radius of the circle is **4**. The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the given circle is $2\pi(4)$, or 8π . It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is **8**.

Question Difficulty: Hard

Question ID e85d3221

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: e85d3221

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P, Q, R , and S correspond to P', Q', R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

- A. 10°
- B. 30°
- C. 40°
- D. 90°

ID: e85d3221 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P, Q, R , and S correspond to P', Q', R' , and S' , respectively. Since corresponding angles of similar quadrilaterals are congruent, it follows that the measure of angle P is equal to the measure of angle P' . It's given that the measure of angle P is 30° . Therefore, the measure of angle P' is 30° .

Choice A is incorrect. This is $\frac{1}{3}$ the measure of angle P' .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is 3 times the measure of angle P' .

Question Difficulty: Hard

Question ID 3d8c0156

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 3d8c0156

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

What is the value of n ?

- A. 2,100
- B. 1,680
- C. 840
- D. 420

ID: 3d8c0156 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that rectangle A and rectangle B are similar. Therefore, there is a positive constant k such that the length of each side of rectangle B is k times the length of the corresponding side of rectangle A and the area of rectangle B is k^2 times the area of rectangle A. It’s also given that the area of rectangle A is 630 square inches and that the area of rectangle B is 2,520 square inches. It follows that $2,520 = 630k^2$. Dividing both sides of this equation by 630 yields $4 = k^2$, or $k^2 = 4$. Taking the positive square root of each side of this equation yields $k = 2$. Therefore, the length of each side of rectangle B is 2 times the length of the corresponding side of rectangle A. It follows that the perimeter of rectangle B is 2 times the perimeter of rectangle A. Since the perimeter of rectangle A is 210 inches, the perimeter of rectangle B is $2(210)$, or 420, inches. Therefore, $n = 420$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from using the area scale factor instead of the side length scale factor.

Question Difficulty: Hard

Question ID 5460398e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5460398e

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F ?

- A. 18°
- B. 72°
- C. 90°
- D. 162°

ID: 5460398e Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that triangle ABC is congruent to triangle DEF . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that angle A corresponds to angle D , and that the measure of angle A is 18° . It's also given that the measures of angles B and E are 90° . Since these angles have equal measure, they are corresponding angles. It follows that angle C corresponds to angle F . Let x° represent the measure of angle C . Since the sum of the measures of the interior angles of a triangle is 180° , it follows that $18^\circ + 90^\circ + x^\circ = 180^\circ$, or $108^\circ + x^\circ = 180^\circ$. Subtracting 108° from both sides of this equation yields $x^\circ = 72^\circ$. Therefore, the measure of angle C is 72° . Since angle C corresponds to angle F , it follows that the measure of angle F is also 72° .

Choice A is incorrect. This is the measure of angle D , not the measure of angle F .

Choice C is incorrect. This is the measure of angle E , not the measure of angle F .

Choice D is incorrect. This is the sum of the measures of angles E and F , not the measure of angle F .

Question Difficulty: Medium

Question ID ca118f8a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: ca118f8a

Circle K has a radius of **4 millimeters (mm)**. Circle L has an area of **$100\pi \text{ mm}^2$** . What is the total area, **in mm^2** , of circles K and L ?

- A. **14π**
- B. **28π**
- C. **56π**
- D. **116π**

ID: ca118f8a Answer

Correct Answer: D

Rationale

Choice D is correct. The area, A , of a circle is given by the formula $A = \pi r^2$, where r represents the radius of the circle. It's given that circle K has a radius of **4 millimeters (mm)**. Substituting **4** for r in the formula $A = \pi r^2$ yields $A = \pi(4)^2$, or $A = 16\pi$. Therefore, the area of circle K is **$16\pi \text{ mm}^2$** . It's given that circle L has an area of **$100\pi \text{ mm}^2$** . Therefore, the total area, **in mm^2** , of circles K and L is **$16\pi + 100\pi$** , or **116π** .

Choice A is incorrect. This is the sum of the radii, **in mm**, of circles K and L multiplied by π , not the total area, **in mm^2** , of the circles.

Choice B is incorrect. This is the sum of the diameters, **in mm**, of circles K and L multiplied by π , not the total area, **in mm^2** , of the circles.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8a0648de

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 8a0648de

In $\triangle JKL$, the measures of both $\angle J$ and $\angle K$ are equal and the measure of $\angle L$ is 118° . What is the measure of $\angle J$?

- A. 31°
- B. 59°
- C. 62°
- D. 90°

ID: 8a0648de Answer

Correct Answer: A

Rationale

Choice A is correct. The sum of the measures of the three angles in a triangle is 180° . It's given that in $\triangle JKL$, the measure of $\angle L$ is 118° . Let x represent the measure of $\angle J$. It's also given that the measures of $\angle J$ and $\angle K$ are equal, so x also represents the measure of $\angle K$. It follows that $x + x + 118 = 180$, or $2x + 118 = 180$. Subtracting 118 from both sides of this equation yields $2x = 62$. Dividing both sides of this equation by 2 yields $x = 31$. Therefore, the measure of $\angle J$ is 31° .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures of $\angle J$ and $\angle K$, not the measure of $\angle J$.

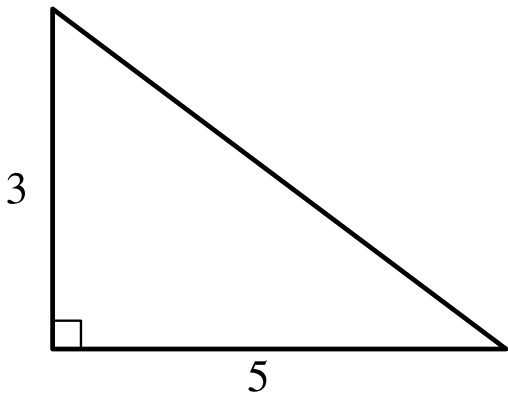
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID b458eea8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: b458eea8



Note: Figure not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

ID: b458eea8 Answer

Correct Answer: 7.5, 15/2

Rationale

The correct answer is $\frac{15}{2}$. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. In the right triangle shown, the length of the base of the triangle is **5** inches, and the height is **3** inches. It follows that $b = 5$ and $h = 3$. Substituting **5** for b and **3** for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(5)(3)$, which is equivalent to $A = \frac{1}{2}(15)$, or $A = \frac{15}{2}$. Therefore, the area of the triangle, in square inches, is $\frac{15}{2}$. Note that 15/2 and 7.5 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 8c12e4da

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 8c12e4da

In triangle ABC , the measure of angle B is 52° and the measure of angle C is 17° . What is the measure of angle A ?

- A. 21°
- B. 35°
- C. 69°
- D. 111°

ID: 8c12e4da Answer

Correct Answer: D

Rationale

Choice D is correct. The sum of the angle measures of a triangle is 180° . Adding the measures of angles B and C gives $52 + 17 = 69^\circ$. Therefore, the measure of angle A is $180 - 69 = 111^\circ$.

Choice A is incorrect and may result from subtracting the sum of the measures of angles B and C from 90° , instead of from 180° .

Choice B is incorrect and may result from subtracting the measure of angle C from the measure of angle B .

Choice C is incorrect and may result from adding the measures of angles B and C but not subtracting the result from 180° .

Question Difficulty: Medium

Question ID c1ed0c96

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: c1ed0c96

A right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. What is the slant height, in centimeters, of this cone?

- A. 12
- B. 36
- C. 77
- D. 85

ID: c1ed0c96 Answer

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a right circular cone is given by the formula $V = \frac{1}{3}\pi r^2 h$, where πr^2 is the area of the circular base of the cone and h is the height. It's given that this right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. Substituting $71,148\pi$ for V and $5,929\pi$ for πr^2 in the formula $V = \frac{1}{3}\pi r^2 h$ yields $71,148\pi = (\frac{1}{3})(5,929\pi)(h)$. Dividing each side of this equation by $5,929\pi$ yields $12 = \frac{h}{3}$. Multiplying each side of this equation by 3 yields $36 = h$. Let s represent the slant height, in centimeters, of this cone. A right triangle is formed by the radius, r , height, h , and slant height, s , of this cone, where r and h are the legs of the triangle and s is the hypotenuse. Using the Pythagorean theorem, the equation $r^2 + h^2 = s^2$ represents this relationship. Because $5,929\pi$ is the area of the base and the area of the base is πr^2 , it follows that $5,929\pi = \pi r^2$. Dividing both sides of this equation by π yields $5,929 = r^2$. Substituting 5,929 for r^2 and 36 for h in the equation $r^2 + h^2 = s^2$ yields $5,929 + 36^2 = s^2$, which is equivalent to $5,929 + 1,296 = s^2$, or $7,225 = s^2$. Taking the positive square root of both sides of this equation yields $85 = s$. Therefore, the slant height of the cone is 85 centimeters.

Choice A is incorrect. This is one-third of the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice B is incorrect. This is the height, in centimeters, not the slant height, in centimeters, of this cone.

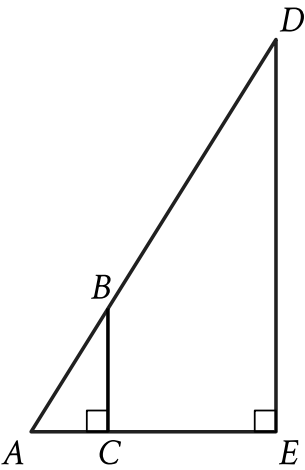
Choice C is incorrect. This is the radius, in centimeters, of the base, not the slant height, in centimeters, of this cone.

Question Difficulty: Hard

Question ID 88475d5e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 88475d5e



Note: Figure not drawn to scale.

In the figure shown, $AB = \sqrt{34}$ units, $AC = 3$ units, and $CE = 21$ units. What is the area, in square units, of triangle ADE ?

ID: 88475d5e Answer

Correct Answer: 480

Rationale

The correct answer is **480**. It's given in the figure that angle ACB and angle AED are right angles. It follows that angle ACB is congruent to angle AED . It's also given that angle BAC and angle DAE are the same angle. It follows that angle BAC is congruent to angle DAE . Since triangles ABC and ADE have two pairs of congruent angles, the triangles are similar. Sides AB and AC in triangle ABC correspond to sides AD and AE , respectively, in triangle ADE . Corresponding sides in similar triangles are proportional. Therefore, $\frac{AD}{AB} = \frac{AE}{AC}$. It's given that $AC = 3$ units and $CE = 21$ units. Therefore, $AE = 24$ units. It's also given that $AB = \sqrt{34}$ units. Substituting 3 for AC , 24 for AE , and $\sqrt{34}$ for AB in the equation $\frac{AD}{AB} = \frac{AE}{AC}$ yields $\frac{AD}{\sqrt{34}} = \frac{24}{3}$, or $\frac{AD}{\sqrt{34}} = 8$. Multiplying each side of this equation by $\sqrt{34}$ yields $AD = 8\sqrt{34}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Since triangle ADE is a right triangle, it follows that AD represents the length of the hypotenuse, c , and DE and AE represent the lengths of the legs, a and b . Substituting 24 for b and $8\sqrt{34}$ for c in the equation $a^2 + b^2 = c^2$ yields $a^2 + (24)^2 = (8\sqrt{34})^2$, which is equivalent to $a^2 + 576 = 64(34)$, or $a^2 + 576 = 2,176$. Subtracting 576 from both sides of this equation yields $a^2 = 1,600$. Taking the square root of both sides of this equation yields $a = \pm 40$. Since a represents a length, which must be positive, the value of a is 40 . Therefore, $DE = 40$. Since DE and AE represent the lengths of the legs of triangle ADE , it follows that DE and AE can be used to calculate the area, in square units, of the triangle as $\frac{1}{2}(40)(24)$, or **480**. Therefore, the area, in square units, of triangle ADE is **480**.

Question Difficulty: Hard

Question ID 0f0355a9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 0f0355a9

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D , and angles C and F are right angles. If $\cos B = \frac{1}{22}$, what is the value of $\cos E$?

- A. $\frac{1}{22}$
- B. $\frac{1}{23}$
- C. $\frac{21}{22}$
- D. $\frac{22}{23}$

ID: 0f0355a9 Answer

Correct Answer: A

Rationale

Choice A is correct. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to that angle to the length of the hypotenuse. It's given that angle C is a right angle in triangle ABC and that angle F is a right angle in triangle DEF . Therefore, $\cos B$ is equal to the ratio of the length of side BC to the length of side AB , and $\cos E$ is equal to the ratio of the length of side EF to the length of side DE . It's also given that triangle ABC is similar to triangle DEF , where angle A corresponds to angle D . Since similar triangles have proportional side lengths, $\frac{BC}{AB} = \frac{EF}{DE}$. Therefore, the value of $\cos B$ is equal to the value of $\cos E$. Since $\cos B = \frac{1}{22}$, the value of $\cos E$ is $\frac{1}{22}$.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

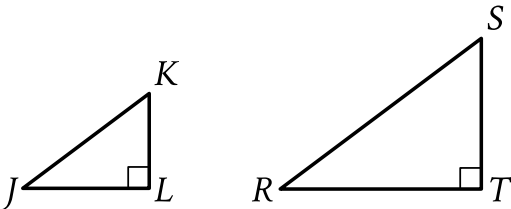
Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID dfeef4a1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: dfeef4a1



Note: Figure not drawn to scale.

In the figure shown, triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . The length of $\overline{JK}$ is 15, and the perimeter of triangle JKL is 36. The length of $\overline{RS}$ is 135. What is the perimeter of triangle RST ?

- A. 324
- B. 540
- C. 2,916
- D. 8,100

ID: dfeef4a1 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . It follows that $\overline{JK}$ corresponds to $\overline{RS}$. If two triangles are similar, then the scale factor between their perimeters is equal to the scale factor between the lengths of their corresponding sides. It's given that the length of $\overline{JK}$ is 15 and the length of $\overline{RS}$ is 135. Therefore, the scale factor from the length of $\overline{JK}$ to the length of $\overline{RS}$ is $\frac{135}{15}$, or 9. It's given that the perimeter of triangle JKL is 36. Let p represent the perimeter of triangle RST . It follows that $\frac{p}{36} = 9$. Multiplying each side of this equation by 36 yields $p = 324$. Therefore, the perimeter of triangle RST is 324.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 4803dc51

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 4803dc51

In triangles ABC and DEF , angles B and E each have measure 27° and angles C and F each have measure 41° . Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF ?

- A. The measure of angle A
- B. The length of side AB
- C. The lengths of sides BC and EF
- D. No additional information is necessary.

ID: 4803dc51 Answer

Correct Answer: C

Rationale

Choice C is correct. Since angles B and E each have the same measure and angles C and F each have the same measure, triangles ABC and DEF are similar, where side BC corresponds to side EF . To determine whether two similar triangles are congruent, it is sufficient to determine whether one pair of corresponding sides are congruent. Therefore, to determine whether triangles ABC and DEF are congruent, it is sufficient to determine whether sides BC and EF have equal length. Thus, the lengths of BC and EF are sufficient to determine whether triangle ABC is congruent to triangle DEF .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. The given information is sufficient to determine that triangles ABC and DEF are similar, but not whether they are congruent.

Question Difficulty: Hard

Question ID d7260964

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: d7260964

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?

ID: d7260964 Answer

Correct Answer: 31

Rationale

The correct answer is **31**. The circumference of a circle is equal to $2\pi r$ centimeters, where r represents the radius, in centimeters, of the circle, and the diameter of the circle is equal to $2r$ centimeters. It's given that a circle has a circumference of 31π centimeters. Therefore, $31\pi = 2\pi r$. Dividing both sides of this equation by π yields $31 = 2r$. Since the diameter of the circle is equal to $2r$ centimeters, it follows that the diameter, in centimeters, of the circle is **31**.

Question Difficulty: Hard

Question ID 42fab20a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 42fab20a

The length of each edge of a box is **29** inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

ID: 42fab20a Answer

Correct Answer: 4205

Rationale

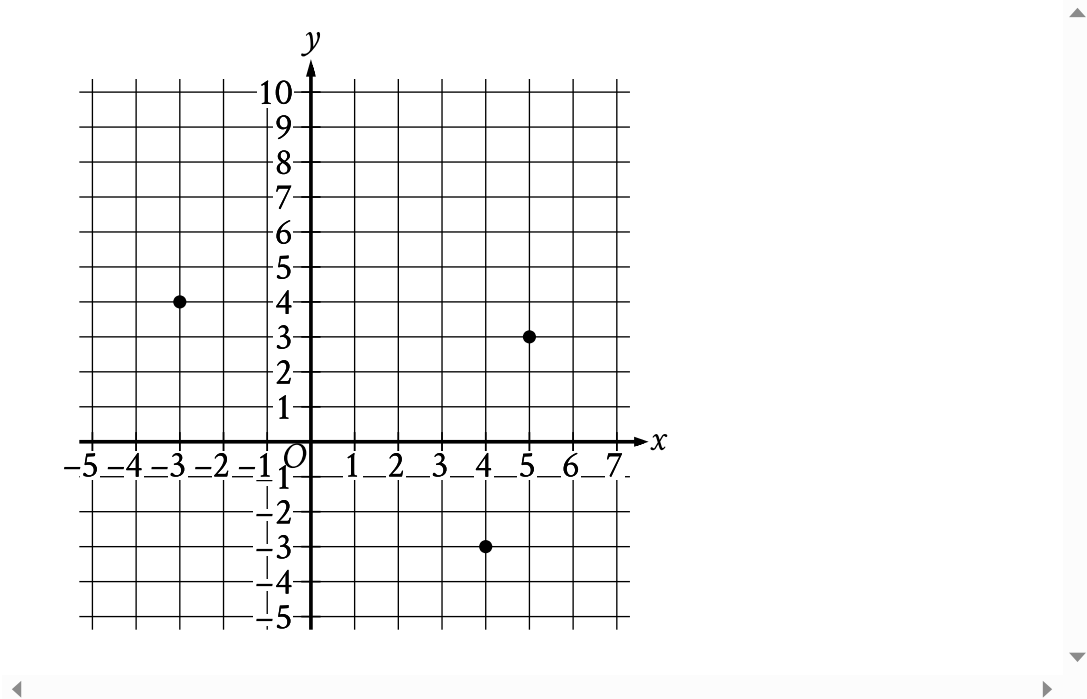
The correct answer is **4,205**. The exterior surface area of a figure is the sum of the areas of all its faces. It's given that the box does not have a lid and that each side of the box is in the shape of a square. Therefore, the box consists of **5** congruent square faces. It's also given that the length of each edge is **29** inches. Let *s* represent the length of an edge of a square. It follows that the area of a square is equal to *s*². Therefore, the area of each of the **5** square faces is equal to **29²**, or **841**, square inches. Since the box consists of **5** congruent square faces, it follows that the sum of the areas of all its faces, or the exterior surface area of this box without a lid, is **5(841)**, or **4,205**, square inches.

Question Difficulty: Hard

Question ID b57ba22f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: b57ba22f



What is the area, in square units, of the triangle formed by connecting the three points shown?

ID: b57ba22f Answer

Correct Answer: 24.5, 49/2

Rationale

The correct answer is **24.5**. It's given that a triangle is formed by connecting the three points shown, which are $(-3, 4)$, $(5, 3)$, and $(4, -3)$. Let this triangle be triangle A. The area of triangle A can be found by calculating the area of the rectangle that circumscribes it and subtracting the areas of the three triangles that are inside the rectangle but outside triangle A. The rectangle formed by the points $(-3, 4)$, $(5, 4)$, $(5, -3)$, and $(-3, -3)$ circumscribes triangle A. The width, in units, of this rectangle can be found by calculating the distance between the points $(5, 4)$ and $(5, -3)$. This distance is $4 - (-3)$, or 7. The length, in units, of this rectangle can be found by calculating the distance between the points $(5, 4)$ and $(-3, 4)$. This distance is $5 - (-3)$, or 8. It follows that the area, in square units, of the rectangle is $(7)(8)$, or 56. One of the triangles that lies inside the rectangle but outside triangle A is formed by the points $(-3, 4)$, $(5, 4)$, and $(5, 3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(5, 4)$ and $(5, 3)$. This distance is $4 - 3$, or 1. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(5, 4)$ and $(-3, 4)$. This distance is $5 - (-3)$, or 8. It follows that the area, in square units, of this triangle is $\frac{1}{2}(8)(1)$, or 4. A second triangle that lies inside the rectangle but outside triangle A is formed by the points $(4, -3)$, $(5, 3)$, and $(5, -3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(5, 3)$ and $(5, -3)$. This distance is $3 - (-3)$, or 6. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(5, -3)$ and $(4, -3)$. This distance is $5 - 4$, or 1. It follows that the area, in square units, of this triangle is $\frac{1}{2}(1)(6)$, or 3. The third triangle that lies inside the rectangle but outside triangle A is formed by the points $(-3, 4)$, $(-3, -3)$, and $(4, -3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(4, -3)$ and $(-3, -3)$. This distance is $4 - (-3)$, or 7. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(-3, 4)$ and $(-3, -3)$. This distance is $4 - (-3)$, or 7. It follows that the area, in square units, of this triangle is $\frac{1}{2}(7)(7)$, or 24.5. Thus, the area, in square units, of

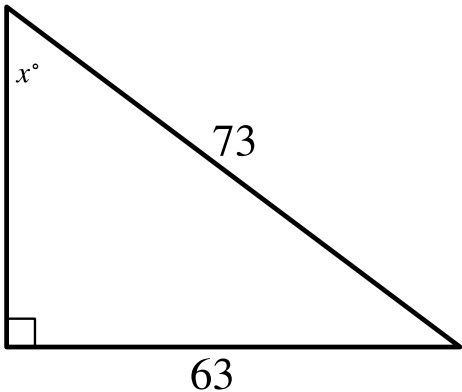
the triangle formed by connecting the three points shown is $56 - 4 - 3 = 24.5$, or 24.5 . Note that 24.5 and $49/2$ are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID b43dc59a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: b43dc59a



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of $\sin x^\circ$?

- A. $\frac{1}{73}$
- B. $\frac{10}{73}$
- C. $\frac{63}{73}$
- D. $\frac{136}{73}$

ID: b43dc59a Answer

Correct Answer: C

Rationale

Choice C is correct. The sine of an acute angle in a right triangle is the ratio of the length of the side opposite that angle to the length of the hypotenuse. In the right triangle shown, it's given that the length of the side opposite the angle with measure x° is **63** units and the length of the hypotenuse is **73** units. Therefore, the value of $\sin x^\circ$ is $\frac{63}{73}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID c3565327

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: c3565327

Which expression represents the area, in square inches, of a square with a side length of **35** inches?

- A. $\sqrt{35}$
- B. $35 + 35$
- C. $(4)(35)$
- D. 35^2

ID: c3565327 Answer

Correct Answer: D

Rationale

Choice D is correct. The area of a square is given by s^2 , where s is the length of each side of the square. It's given that the length of each side of a square is **35** inches. Substituting **35** for s in the expression s^2 yields 35^2 . Therefore, the expression that represents the area, in square inches, of the square is 35^2 .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

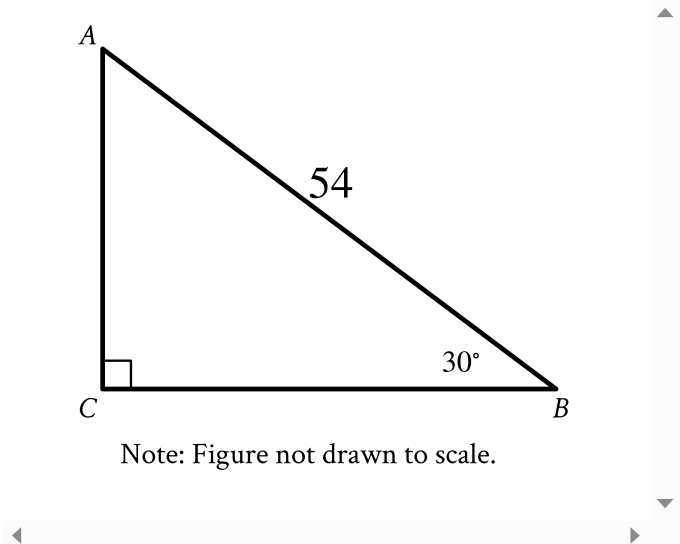
Choice C is incorrect. This expression represents the perimeter, in inches, of the square rather than the area, in square inches.

Question Difficulty: Medium

Question ID 7e881644

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 7e881644



Right triangle ABC is shown. What is the value of $\tan A$?

- A. $\frac{\sqrt{3}}{54}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. $27\sqrt{3}$

ID: 7e881644 Answer

Correct Answer: C

Rationale

Choice C is correct. In the triangle shown, the measure of angle B is 30° and angle C is a right angle, which means that it has a measure of 90° . Since the sum of the angles in a triangle is equal to 180° , the measure of angle A is equal to $180^\circ - (30^\circ + 90^\circ)^\circ$, or 60° . In a right triangle whose acute angles have measures 30° and 60° , the lengths of the legs can be represented by the expressions x , $x\sqrt{3}$, and $2x$, where x is the length of the leg opposite the angle with measure 30° , $x\sqrt{3}$ is the length of the leg opposite the angle with measure 60° , and $2x$ is the length of the hypotenuse. In the triangle shown, the hypotenuse has a length of 54 . It follows that $2x = 54$, or $x = 27$. Therefore, the length of the leg opposite angle B is 27 and the length of the leg opposite angle A is $27\sqrt{3}$. The tangent of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. The length of the leg opposite angle A is $27\sqrt{3}$ and the length of the leg adjacent to angle A is 27 . Therefore, the value of $\tan A$ is $\frac{27\sqrt{3}}{27}$, or $\sqrt{3}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $\frac{1}{\tan A}$, not the value of $\tan A$.

Choice D is incorrect. This is the length of the leg opposite angle A , not the value of $\tan A$.

Question Difficulty: Hard

Question ID bc0df5a6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: bc0df5a6

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k ?

- A. $\frac{29}{28}$
- B. 1
- C. $\frac{28}{29}$
- D. 0

ID: bc0df5a6 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. Since the length of each side of triangle T is greater than the length of each side of triangle S, the scale factor of k must be greater than 1. Of the given choices, only $\frac{29}{28}$ is greater than 1.

Choice B is incorrect. If each side of equilateral triangle S is multiplied by a scale factor of 1, the length of each side of triangle T would be equal to the length of each side of triangle S.

Choice C is incorrect. If each side of equilateral triangle S is multiplied by a scale factor of $\frac{28}{29}$, the length of each side of triangle T would be less than the length of each side of triangle S.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 5226980b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5226980b

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . What is the measure of $\angle Z$?

- A. 58°
- B. 74°
- C. 122°
- D. 212°

ID: 5226980b Answer

Correct Answer: A

Rationale

Choice A is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180° . It's given that in $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . It follows that the measure of $\angle Z$ is $(180 - 24 - 98)^\circ$, or 58° .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

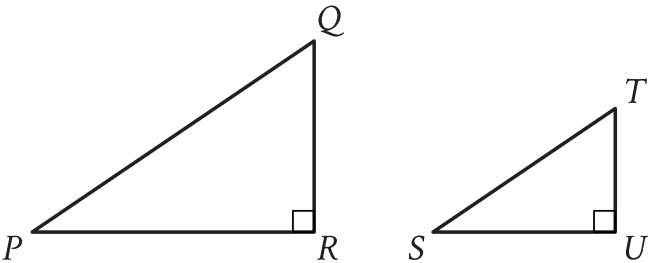
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 7a12d469

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 7a12d469



Note: Figures not drawn to scale.

Right triangles PQR and STU are similar, where P corresponds to S . If the measure of angle Q is 18° , what is the measure of angle S ?

- A. 18°
- B. 72°
- C. 82°
- D. 162°

ID: 7a12d469 Answer

Correct Answer: B

Rationale

Choice B is correct. In similar triangles, corresponding angles are congruent. It's given that right triangles PQR and STU are similar, where angle P corresponds to angle S . It follows that angle P is congruent to angle S . In the triangles shown, angle R and angle U are both marked as right angles, so angle R and angle U are corresponding angles. It follows that angle Q and angle T are corresponding angles, and thus, angle Q is congruent to angle T . It's given that the measure of angle Q is 18° , so the measure of angle T is also 18° . Angle U is a right angle, so the measure of angle U is 90° . The sum of the measures of the interior angles of a triangle is 180° . Thus, the sum of the measures of the interior angles of triangle STU is 180 degrees. Let s represent the measure, in degrees, of angle S . It follows that $s + 18 + 90 = 180$, or $s + 108 = 180$. Subtracting 108 from both sides of this equation yields $s = 72$. Therefore, if the measure of angle Q is 18 degrees, then the measure of angle S is 72 degrees.

Choice A is incorrect. This is the measure of angle T .

Choice C is incorrect and may result from conceptual or calculation errors.

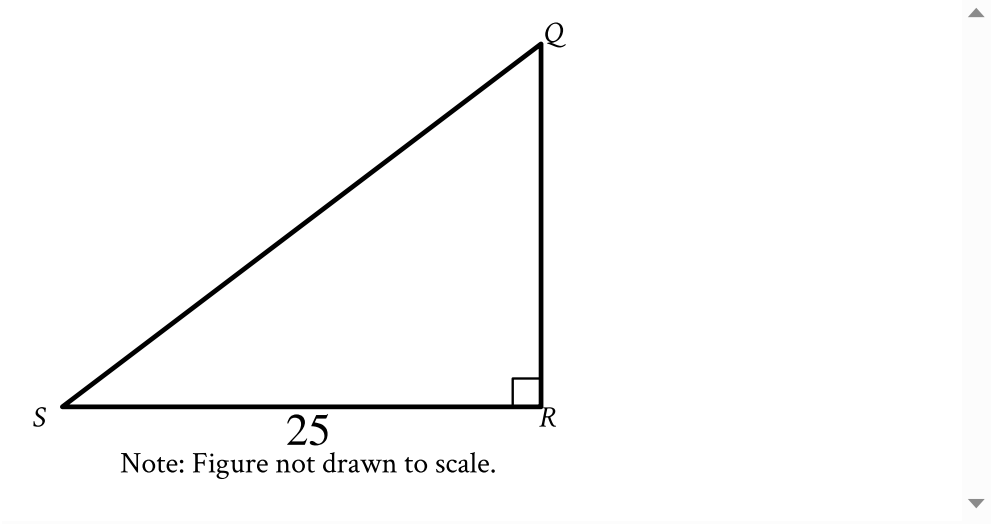
Choice D is incorrect. This is the sum of the measures of angle S and angle U .

Question Difficulty: Medium

Question ID b75ae100

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: b75ae100



In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- A. $25 \cos Q$
- B. $25 \sin Q$
- C. $\frac{25}{\cos Q}$
- D. $\frac{25}{\sin Q}$

ID: b75ae100 Answer

Correct Answer: D

Rationale

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 25 . Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS , an expression representing QS can be found by substituting 25 for RS in this equation, which yields $\sin Q = \frac{25}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 25$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{25}{\sin Q}$. Therefore, the expression $\frac{25}{\sin Q}$ represents the length of $\overline{QS}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a298a9b6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a298a9b6

Square X has a side length of **12** centimeters. The perimeter of square Y is **2** times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

- A. **6**
- B. **10**
- C. **14**
- D. **24**

ID: a298a9b6 Answer

Correct Answer: D

Rationale

Choice D is correct. The perimeter, P , of a square can be found using the formula $P = 4s$, where s is the length of each side of the square. It's given that square X has a side length of **12** centimeters. Substituting **12** for s in the formula for the perimeter of a square yields $P = 4(12)$, or $P = 48$. Therefore, the perimeter of square X is **48** centimeters. It's also given that the perimeter of square Y is **2** times the perimeter of square X. Therefore, the perimeter of square Y is **2(48)**, or **96**, centimeters. Substituting **96** for P in the formula $P = 4s$ gives $96 = 4s$. Dividing both sides of this equation by **4** gives $24 = s$. Therefore, the length of one side of square Y is **24** centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 3ee76077

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 3ee76077

A hemisphere is half of a sphere. If a hemisphere has a radius of **27** inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- A. **1,500**
- B. **6,100**
- C. **30,900**
- D. **41,200**

ID: 3ee76077 Answer

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a sphere is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Since a hemisphere is half of a sphere, it follows that the volume, V , of a hemisphere is given by $V = \left(\frac{1}{2}\right)\left(\frac{4}{3}\right)\pi r^3$, or $V = \frac{2}{3}\pi r^3$. Substituting **27** for r in this formula yields $V = \frac{2}{3}\pi(27)^3$, which gives $V = 13,122\pi$, or V is approximately equal to **41,223.98**. Therefore, the choice that is closest to the volume, in cubic inches, of this hemisphere is **41,200**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 93f7842a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 93f7842a

The perimeter of an equilateral triangle is **624** centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?

ID: 93f7842a Answer

Correct Answer: 104

Rationale

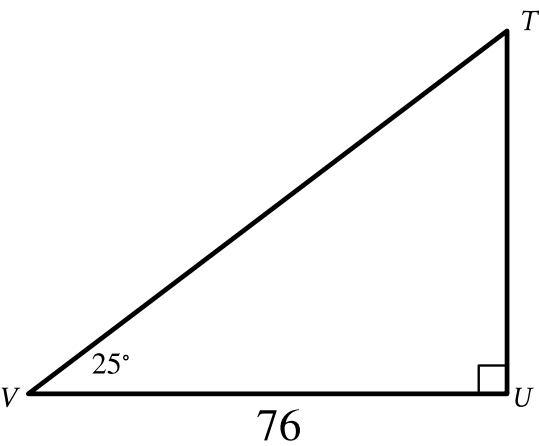
The correct answer is **104**. An equilateral triangle is a triangle in which all three sides have the same length and all three angles have a measure of **60°**. The height of the triangle, $k\sqrt{3}$, is the length of the altitude from one vertex. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the **60°** angle in each 30-60-90 right triangle. Since the altitude has a length of $k\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each **30°** angle has a length of k and each hypotenuse has a length of $2k$. In this case, the hypotenuse of each 30-60-90 right triangle is a side of the equilateral triangle; therefore, each side length of the equilateral triangle is $2k$. The perimeter of a triangle is the sum of the lengths of each side. It's given that the perimeter of the equilateral triangle is **624**; therefore, $2k + 2k + 2k = 624$, or $6k = 624$. Dividing both sides of this equation by **6** yields $k = 104$.

Question Difficulty: Hard

Question ID f49534f6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f49534f6



Note: . igure not drawn to scale.

Triangle TUV is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$ (not shown), where T corresponds to T' and U corresponds to U' . What is the measure, in degrees, of angle V' ?

- A. 25
- B. 38
- C. 50
- D. 76

ID: f49534f6 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that triangle TUV is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$, where T corresponds to T' and U corresponds to U' . A dilation is a transformation that multiplies all side lengths by the same scale factor while preserving angle measures. Therefore, corresponding angles in triangles TUV and $T'U'V'$ are equal. Since V corresponds to V' , the measure of angle V' is equal to the measure of angle V . The figure shows that the measure of angle V is 25° , so it follows that the measure of angle V' is also 25° .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length of $\overline{VU}$, not the measure, in degrees, of angle V' .

Question Difficulty: Hard

Question ID 15709a24

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 15709a24

A triangle has a base length of **40** centimeters and a height of **90** centimeters. What is the area, in square centimeters, of the triangle?

ID: 15709a24 Answer

Correct Answer: 1800

Rationale

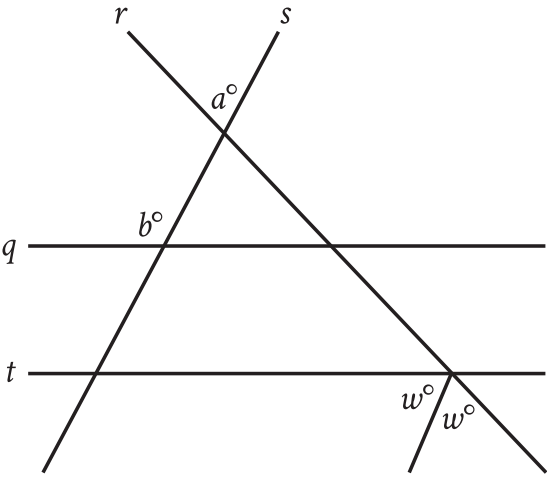
The correct answer is **1,800**. The area, ***A***, of a triangle can be found using the formula **$A = \frac{1}{2}bh$** , where ***b*** is the base length of the triangle and ***h*** is the height of the triangle. It's given that the triangle has a base length of **40** centimeters and a height of **90** centimeters. Substituting **40** for ***b*** and **90** for ***h*** in the formula **$A = \frac{1}{2}bh$** yields **$A = \frac{1}{2}(40)(90)$** , or **$A = 1,800$** . Therefore, the area, in square centimeters, of the triangle is **1,800**.

Question Difficulty: Medium

Question ID 5789afdb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5789afdb



Note: Figure not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43$ and $b = 122$, what is the value of w ?

ID: 5789afdb Answer

Correct Answer: 101/2, 50.5

Rationale

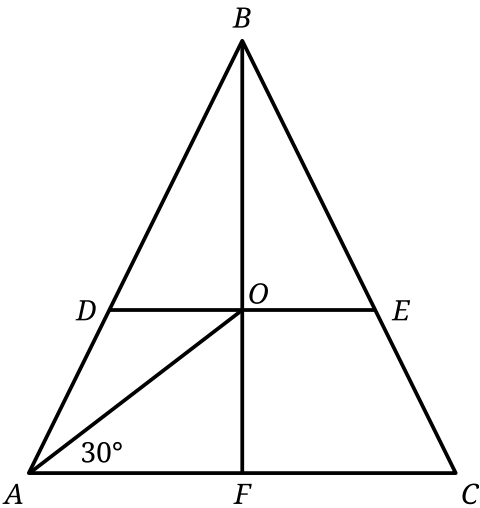
The correct answer is $\frac{101}{2}$. In the figure, lines q , r , and s form a triangle. One interior angle of this triangle is vertical to the angle marked a° ; therefore, the interior angle also has measure a° . It's given that $a = 43$. Therefore, the interior angle of the triangle has measure 43° . A second interior angle of the triangle forms a straight line, q , with the angle marked b° . Therefore, the sum of the measures of these two angles is 180° . It's given that $b = 122$. Therefore, the angle marked b° has measure 122° and the second interior angle of the triangle has measure $(180 - 122)^\circ$, or 58° . The sum of the interior angles of a triangle is 180° . Therefore, the measure of the third interior angle of the triangle is $(180 - 43 - 58)^\circ$, or 79° . It's given that parallel lines q and t are intersected by line r . It follows that the triangle's interior angle with measure 79° is congruent to the same side interior angle between lines q and t formed by lines t and r . Since this angle is supplementary to the two angles marked w° , the sum of 79° , w° , and w° is 180° . It follows that $79 + w + w = 180$, or $79 + 2w = 180$. Subtracting 79 from both sides of this equation yields $2w = 101$. Dividing both sides of this equation by 2 yields $w = \frac{101}{2}$. Note that 101/2 and 50.5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID c7efd1cd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c7efd1cd



Note: Figure not drawn to scale.

In triangle ABC , $AB = BC$. Points D and E lie on line segments AB and BC , respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE , and the measure of $\angle OAC$ is 30° . Point F lies on line segment AC , and line segment BF passes through point O . If $AB = 81$ and $AO = BO$, what is the length of line segment BE ?

ID: c7efd1cd Answer

Correct Answer: 54

Rationale

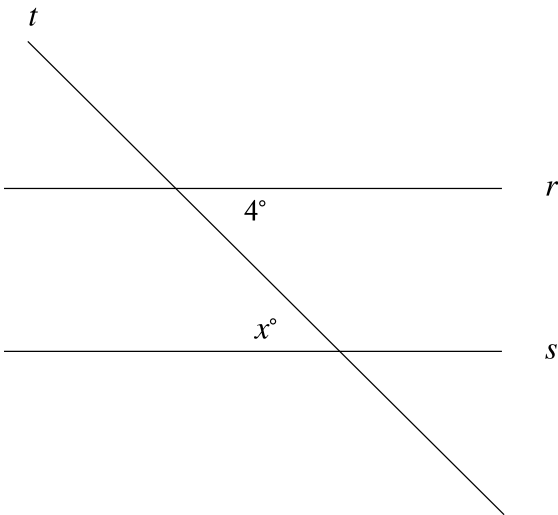
The correct answer is **54**. It's given that $AB = BC$, so triangle ABC is isosceles. It's also given that line segments DE and AC are parallel. Since line segments DE and AC are parallel, triangle BDE is similar to triangle BAC , which means triangle BDE is also isosceles. It's given that point O is the midpoint of line segment DE . Since triangle BDE is isosceles, line segment BO splits triangle BDE into two congruent triangles, where $\angle BOD$ is a right angle. Therefore, $\angle BFA$ is also a right angle. The figure shows that the measure of $\angle OAF$ is 30° . It follows that triangle AOF is a 30-60-90 triangle, where the measure of $\angle AOF$ is 60° . In a 30-60-90 triangle, the side opposite the 30° angle is half the length of the side opposite the 90° angle. Let x represent the length of line segment OF and let $2x$ represent the length of line segment AO . It's given that $AO = BO$. Since $AO = BO$, the length of line segment BO is also $2x$. It follows that the length of line segment BF is $BO + OF$, which is $2x + x$, or $3x$. Since triangle BDE is similar to triangle BAC , the ratio of corresponding side lengths is equal to the ratio of the heights of the triangles. This means that $\frac{BE}{BC} = \frac{BO}{BF}$. Substituting $2x$ for BO and $3x$ for BF in this equation yields $\frac{BE}{BC} = \frac{2x}{3x}$, or $\frac{BE}{BC} = \frac{2}{3}$. It's given that $AB = 81$ and $AB = BC$, so $BC = 81$. Substituting 81 for BC in the equation $\frac{BE}{BC} = \frac{2}{3}$ yields $\frac{BE}{81} = \frac{2}{3}$. Multiplying both sides of this equation by 81 yields $BE = \frac{2}{3}(81)$, or $BE = 54$.

Question Difficulty: Hard

Question ID cc069d95

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: cc069d95



Note: . igure not drawn to scale.

In the figure, line t intersects lines r and s . Which additional piece of information is sufficient to prove that lines r and s are parallel?

- A. $x < 24$
- B. $x = 24$
- C. $x > 24$
- D. $x - 24 = 360$

ID: cc069d95 Answer

Correct Answer: B

Rationale

Choice B is correct. In the figure shown, transversal t intersects lines r and s . Two lines intersected by a transversal are parallel if and only if a pair of alternate interior angles have equal measure. The angle marked 24° and the angle marked x° form a pair of alternate interior angles. Therefore, $x = 24$ is sufficient to prove that lines r and s are parallel.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID fd1a9b2e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: fd1a9b2e

A right circular cylinder has a base diameter of **22** centimeters and a height of **6** centimeters. What is the volume, in cubic centimeters, of the cylinder?

- A. **132** π
- B. **264** π
- C. **726** π
- D. **2,904** π

ID: fd1a9b2e Answer

Correct Answer: C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of **6** centimeters. Therefore, $h = 6$. It's also given that the cylinder has a base diameter of **22** centimeters. The radius of a circle is half the diameter of the circle. Since the base of a right circular cylinder is a circle, it follows that the radius of the base of the right circular cylinder is $\frac{22}{2}$, or **11**, centimeters. Therefore, $r = 11$. Substituting **11** for r and **6** for h in the formula $V = \pi r^2 h$ yields $V = \pi(11)^2(6)$, which is equivalent to $V = \pi(121)(6)$, or $V = 726\pi$. Therefore, the volume, in cubic centimeters, of the cylinder is **726** π .

Choice A is incorrect. This is the volume of a right circular cylinder that has a base diameter of $2\sqrt{22}$, not **22**, centimeters and a height of **6** centimeters.

Choice B is incorrect. This is the volume of a right circular cylinder that has a base diameter of $4\sqrt{11}$, not **22**, centimeters and a height of **6** centimeters.

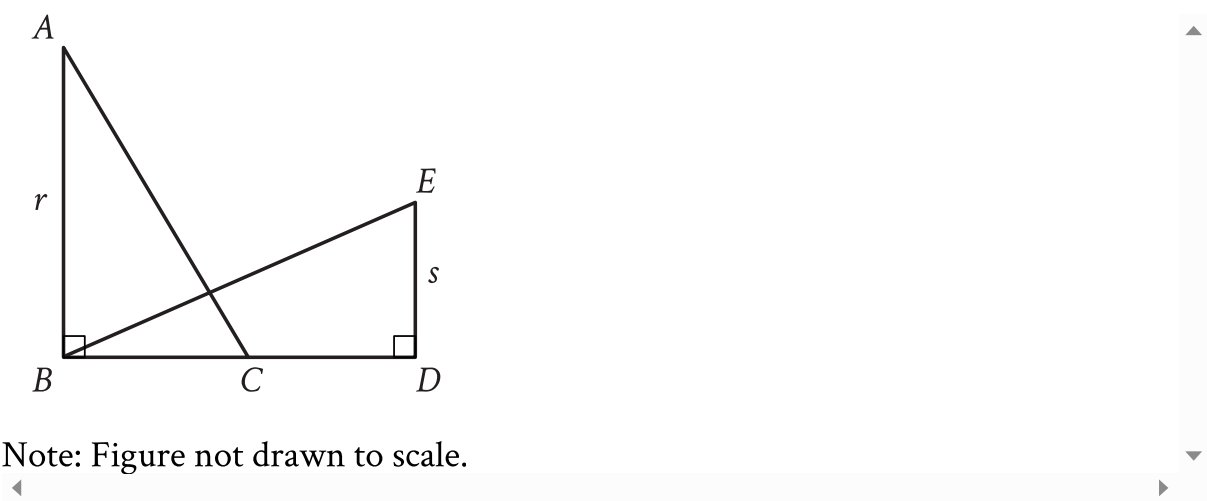
Choice D is incorrect. This is the volume of a right circular cylinder that has a base diameter of **44**, not **22**, centimeters and a height of **6** centimeters.

Question Difficulty: Medium

Question ID 64ffeb74

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 64ffeb74



Note: Figure not drawn to scale.

In the figure shown, $\triangle ABC$ is congruent to $\triangle BDE$. If $r = 54$ and $s = 26$, what is the length of $\overline{CD}$?

- A. 26
- B. 27
- C. 28
- D. 54

ID: 64ffeb74 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $\triangle ABC$ is congruent to $\triangle BDE$. Corresponding sides of congruent triangles are equal in length. It's given that r , which represents the length of $\overline{AB}$, is equal to 54 and s , which represents the length of $\overline{DE}$, is equal to 26 . Since $\overline{AB}$ and $\overline{DE}$ aren't equal in length, it follows that $\overline{AB}$ corresponds to $\overline{BD}$ and $\overline{BC}$ corresponds to $\overline{DE}$. Therefore, the length of $\overline{BD}$ is 54 and the length of $\overline{BC}$ is 26 . The points B , C , and D lie on a straight line, so the length of $\overline{BD}$ can be expressed as the sum of the lengths of $\overline{BC}$ and $\overline{CD}$. This relationship can be written as $BD = BC + CD$. Substituting 54 for BD and 26 for BC in this equation yields $54 = 26 + CD$. Subtracting 26 from both sides of this equation yields $CD = 28$. Therefore, the length of $\overline{CD}$ is 28 .

Choice A is incorrect. This is the length of $\overline{BC}$, not $\overline{CD}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length of $\overline{BD}$, not $\overline{CD}$.

Question Difficulty: Hard

Question ID 1f041079

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 1f041079

A right rectangular prism has a base area of $24t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the rectangular prism is 15 cm. Which expression represents the surface area, in cm^2 , of the right rectangular prism?

- A. $48t + 160$
- B. $318t + 80$
- C. $1,968t + 80$
- D. $360t$

ID: 1f041079 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the base area of the right rectangular prism is $24t \text{ cm}^2$ and the length of the base is $\frac{8}{3}$ cm. Dividing the base area by the length yields the width: $\frac{24t}{\frac{8}{3}}$, or $(24t)(\frac{3}{8})$, or $9t$ cm. It's also given that the height of the prism is 15 cm. A right rectangular prism has two rectangular bases and four rectangular lateral faces. The total area of the two bases is $2(24t)$, or $48t \text{ cm}^2$. The four lateral faces include two with dimensions $\frac{8}{3}$ cm by 15 cm and two with dimensions $9t$ cm by 15 cm. The total area of these four lateral faces is $2(\frac{8}{3})(15) + 2(9t)(15) \text{ cm}^2$, which is equivalent to $80 + 270t \text{ cm}^2$. Adding this to the total area of the two bases yields $48t + 270t + 80$, or $318t + 80 \text{ cm}^2$. Thus, the expression, which represents the surface area, in cm^2 , of the right rectangular prism is $318t + 80$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the volume, in cm^3 , not the surface area, in cm^2 .

Question Difficulty: Hard

Question ID a954e998

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: a954e998

The length of a rectangle’s diagonal is $3\sqrt{17}$, and the length of the rectangle’s shorter side is 3 . What is the length of the rectangle’s longer side?

ID: a954e998 Answer

Correct Answer: 12

Rationale

The correct answer is 12 . The diagonal of a rectangle forms a right triangle, where the shorter side and the longer side of the rectangle are the legs of the triangle and the diagonal of the rectangle is the hypotenuse of the triangle. It's given that the length of the rectangle's diagonal is $3\sqrt{17}$ and the length of the rectangle's shorter side is 3 . Thus, the length of the hypotenuse of the right triangle formed by the diagonal is $3\sqrt{17}$ and the length of one of the legs is 3 . By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting $3\sqrt{17}$ for c and 3 for b in this equation yields $a^2 + (3)^2 = (3\sqrt{17})^2$, or $a^2 + 9 = 153$. Subtracting 9 from both sides of this equation yields $a^2 = 144$. Taking the square root of both sides of this equation yields $a = \pm\sqrt{144}$, or $a = \pm 12$. Since a represents a length, which must be positive, the value of a is 12 . Thus, the length of the rectangle's longer side is 12 .

Question Difficulty: Hard

Question ID 02402514

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 02402514

A right triangle has legs with lengths of **28** centimeters and **20** centimeters. What is the length of this triangle's hypotenuse, in centimeters?

- A. $8\sqrt{6}$
- B. $4\sqrt{74}$
- C. 48
- D. 1,184

ID: 02402514 Answer

Correct Answer: B

Rationale

Choice B is correct. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. It's given that the right triangle has legs with lengths of **28** centimeters and **20** centimeters. Let c represent the length of this triangle's hypotenuse, in centimeters. Therefore, by the Pythagorean theorem, $28^2 + 20^2 = c^2$, or $1,184 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{1,184} = c$, or $4\sqrt{74} = c$. Therefore, the length of this triangle's hypotenuse, in centimeters, is $4\sqrt{74}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the square of the length of the triangle's hypotenuse.

Question Difficulty: Medium

Question ID 470626ec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 470626ec

A cube has an edge length of **68** inches. A solid sphere with a radius of **34** inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

- A. **149,796**
- B. **164,500**
- C. **190,955**
- D. **310,800**

ID: 470626ec Answer

Correct Answer: A

Rationale

Choice A is correct. The volume of a cube can be found by using the formula $V = s^3$, where V is the volume and s is the edge length of the cube. Therefore, the volume of the given cube is $V = 68^3$, or **314,432** cubic inches. The volume of a sphere can be found by using the formula $V = \frac{4}{3}\pi r^3$, where V is the volume and r is the radius of the sphere. Therefore, the volume of the given sphere is $V = \frac{4}{3}\pi(34)^3$, or approximately **164,636** cubic inches. The volume of the space in the cube not taken up by the sphere is the difference between the volume of the cube and volume of the sphere. Subtracting the approximate volume of the sphere from the volume of the cube gives **314,432 – 164,636 = 149,796** cubic inches.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

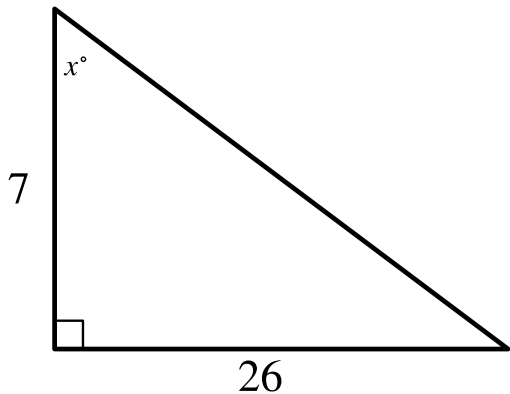
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 7af120f8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 7af120f8



Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\tan x^\circ$?

- A. $\frac{1}{26}$
- B. $\frac{19}{26}$
- C. $\frac{26}{7}$
- D. $\frac{33}{7}$

ID: 7af120f8 Answer

Correct Answer: C

Rationale

Choice C is correct. The tangent of an acute angle in a right triangle is defined as the ratio of the length of the side opposite the angle to the length of the shorter side adjacent to the angle. In the triangle shown, the length of the side opposite the angle with measure x° is **26** units and the length of the side adjacent to the angle with measure x° is **7** units. Therefore, the value of $\tan x^\circ$ is $\frac{26}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID b30fc018

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: b30fc018

In triangle XYZ , angle Z is a right angle and the length of $\overline{YZ}$ is **24** units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ ?

- A. 188
- B. 168
- C. 84
- D. 71

ID: b30fc018 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that angle Z in triangle XYZ is a right angle. Thus, side YZ is the leg opposite angle X and side XZ is the leg adjacent to angle X . The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. It follows that $\tan X = \frac{YZ}{XZ}$. It's given that $\tan X = \frac{12}{35}$ and the length of side YZ is **24** units. Substituting $\frac{12}{35}$ for $\tan X$ and **24** for YZ in the equation $\tan X = \frac{YZ}{XZ}$ yields $\frac{12}{35} = \frac{24}{XZ}$. Multiplying both sides of this equation by **35**(XZ) yields **12**(XZ) = **24**(**35**), or **12**(XZ) = **840**. Dividing both sides of this equation by **12** yields $XZ = 70$. The length XY can be calculated using the Pythagorean theorem, which states that if a right triangle has legs with lengths of a and b and a hypotenuse with length c , then $a^2 + b^2 = c^2$. Substituting **70** for a and **24** for b in this equation yields **70**² + **24**² = c^2 , or **5,476** = c^2 . Taking the square root of both sides of this equation yields $\pm 74 = c$. Since the length of the hypotenuse must be positive, **74** = c . Therefore, the length of XY is **74** units. The perimeter of a triangle is the sum of the lengths of all sides. Thus, (**74** + **70** + **24**) units, or **168** units, is the perimeter of triangle XYZ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the perimeter, in units, for a right triangle where the length of side YZ is **12** units, not **24** units.

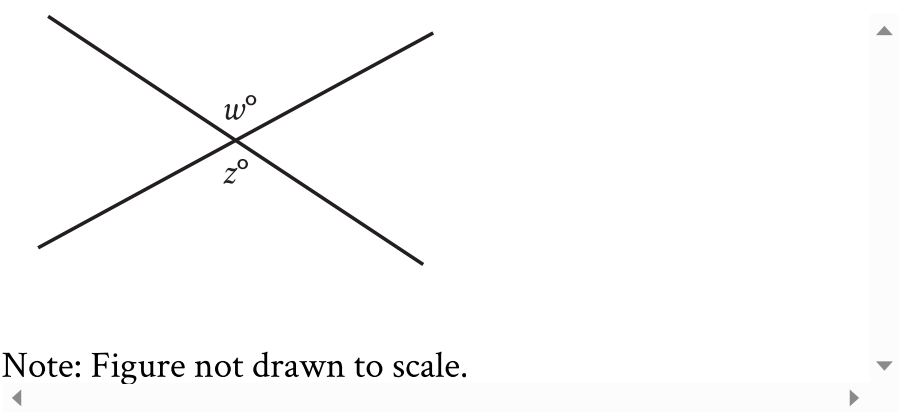
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 1abbf457

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 1abbf457



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

- A. 36
- B. 44
- C. 68
- D. 136

ID: 1abbf457 Answer

Correct Answer: D

Rationale

Choice D is correct. In the figure shown, the angles with measures w° and z° are vertical angles. Since vertical angles are congruent, $w = z$. Therefore, if $w = 136$, the value of z is **136**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure, in degrees, of an angle that's supplementary, not congruent, to the angle with measure w° .

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 02bed84d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 02bed84d

A rectangle has a length of **64** inches and a width of **32** inches. What is the area, in square inches, of the rectangle?

ID: 02bed84d Answer

Correct Answer: 2048

Rationale

The correct answer is **2,048**. The area **A** , in square inches, of a rectangle is equal to the product of its length **ℓ** , in inches, and its width **w** , in inches, or **$A = \ell w$** . It's given that the rectangle has a length of **64** inches and a width of **32** inches. Substituting **64** for **ℓ** and **32** for **w** in the equation **$A = \ell w$** yields **$A = (64)(32)$** , or **$A = 2,048$** . Therefore, the area, in square inches, of the rectangle is **2,048**.

Question Difficulty: Easy

Question ID 30c6df13

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 30c6df13

The length of each side of a square is ~~94~~ centimeters (cm). Which expression gives the area, in **cm²**, of the square?

- A. ~~2~~ · ~~94~~
- B. ~~2~~ · ~~94~~ · ~~94~~
- C. ~~4~~ · ~~94~~
- D. ~~94~~ · ~~94~~

ID: 30c6df13 Answer

Correct Answer: D

Rationale

Choice D is correct. The area of a square is given by **s^2** , where **s** is the length of each side of the square. It's given that the length of each side of a square is ~~94 cm~~. It follows that the area, in **cm²**, of the square is **$(94)^2$** , or **$94 \cdot 94$** . Therefore, the expression that gives the area, in **cm²**, of the square is **$94 \cdot 94$** .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This expression gives the perimeter, in **cm**, of the square.

Question Difficulty: Medium

Question ID 437a6f56

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 437a6f56

In right triangle ABC , angle C is the right angle and $BC = 162$. Point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC and $CE = 2AE$. What is the length of line segment DE ?

ID: 437a6f56 Answer

Correct Answer: 54

Rationale

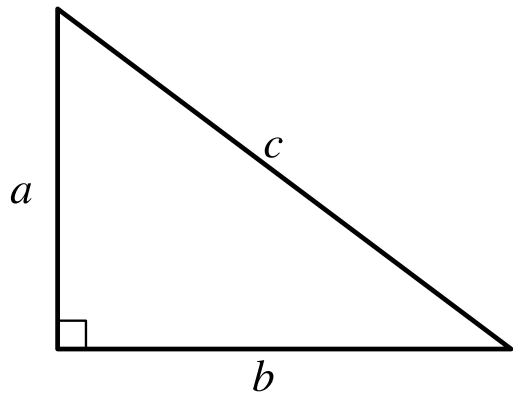
The correct answer is **54**. It's given that in triangle ABC , point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC . It follows that parallel segments DE and BC are intersected by sides AB and AC . If two parallel segments are intersected by a third segment, corresponding angles are congruent. Thus, corresponding angles C and AED are congruent and corresponding angles B and ADE are congruent. Since triangle ADE has two angles that are each congruent to an angle in triangle ABC , triangle ADE is similar to triangle ABC by the angle-angle similarity postulate, where side DE corresponds to side BC , and side AE corresponds to side AC . Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{DE}{BC} = \frac{AE}{AC}$. Since point E lies on side AC , $AE + CE = AC$. It's given that $CE = 2AE$. Substituting $2AE$ for CE in the equation $AE + CE = AC$ yields $AE + 2AE = AC$, or $3AE = AC$. It's given that $BC = 162$. Substituting 162 for BC and $3AE$ for AC in the equation $\frac{DE}{BC} = \frac{AE}{AC}$ yields $\frac{DE}{162} = \frac{AE}{3AE}$, or $\frac{DE}{162} = \frac{1}{3}$. Multiplying both sides of this equation by 162 yields $DE = 54$. Thus, the length of line segment DE is **54**.

Question Difficulty: Hard

Question ID 75f5f9f2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 75f5f9f2



Note: Figure not drawn to scale.

For the right triangle shown, $a = 4$ and $b = 5$. Which expression represents the value of c ?

- A. $4 + 5$
- B. $\sqrt{(4)(5)}$
- C. $\sqrt{4 + 5}$
- D. $\sqrt{4^2 + 5^2}$

ID: 75f5f9f2 Answer

Correct Answer: D

Rationale

Choice D is correct. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $c^2 = a^2 + b^2$. In the right triangle shown, the hypotenuse has length c and the legs have lengths a and b . It's given that $a = 4$ and $b = 5$. Substituting 4 for a and 5 for b in the Pythagorean theorem yields $c^2 = 4^2 + 5^2$. Taking the square root of both sides of this equation yields $c = \pm\sqrt{4^2 + 5^2}$. Since the length of a side of a triangle must be positive, the value of c is $\sqrt{4^2 + 5^2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID f925fa23

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: f925fa23

Circle N has a radius of **7** millimeters (**mm**). Circle M has an area of **64π mm²**. What is the total area, in **mm²**, of circles N and M?

- A. **113π**
- B. **92π**
- C. **78π**
- D. **15π**

ID: f925fa23 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that circle N has a radius of **7 mm**. The area of a circle is given by the expression **πr^2** , where **r** is the radius of the circle. Substituting **7** for **r** in the expression yields **$\pi(7)^2$** , or **49π** , for the area, in **mm²**, of circle N. It's also given that the area of circle M is **64π mm²**. Adding the two areas yields **$49\pi + 64\pi$** , or **113π mm²**. Therefore, the total area, in **mm²**, of circles N and M is **113π** .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID cd856d61

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: cd856d61

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

- A. $\frac{5\pi}{17}$
- B. $\frac{68\pi}{15}$
- C. $\frac{32\pi}{5}$
- D. $\frac{19,652\pi}{375}$

ID: cd856d61 Answer

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a sphere can be found using the formula $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. It's given that the sphere has a radius of $\frac{17}{5}$ feet. Substituting $\frac{17}{5}$ for r in the formula $V = \frac{4}{3}\pi r^3$ yields $V = \frac{4}{3}\pi\left(\frac{17}{5}\right)^3$, which is equivalent to $V = \frac{4}{3}\pi\left(\frac{4,913}{125}\right)$, or $V = \frac{19,652\pi}{375}$. Therefore, the volume, in cubic feet, of the sphere is $\frac{19,652\pi}{375}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the volume, in cubic feet, of a sphere with a radius of $\sqrt[3]{\frac{17}{5}}$ feet.

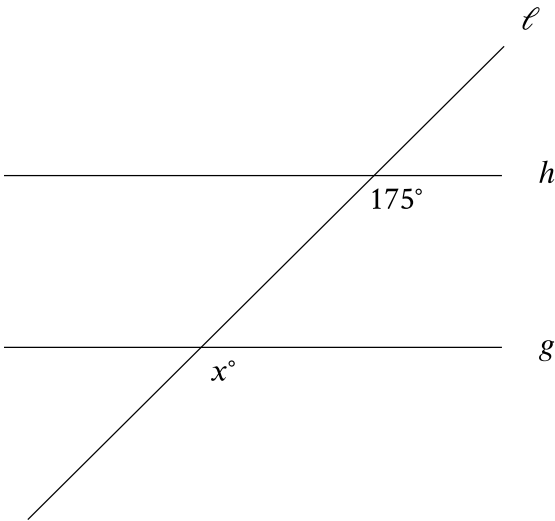
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a46e712c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a46e712c



Note: Figure not drawn to scale.

In the figure shown, line g is parallel to line h . Line ℓ intersects both lines. What is the value of x ?

ID: a46e712c Answer

Correct Answer: 175

Rationale

The correct answer is **175**. It's given that line g is parallel to line h . Since line ℓ intersects both lines, it's a transversal. The angles in the figure marked as **175°** and x° are on the same side of the transversal, where one is an interior angle with line h as a side, and the other is an exterior angle with line g as a side, respectively. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles have equal measure. Therefore, the value of x is **175**.

Question Difficulty: Easy

Question ID 32b09ce9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 32b09ce9

The lengths of two sides of a triangle are **4** centimeters and **6** centimeters. If the perimeter of the triangle is **18** centimeters, what is the length, in centimeters, of the third side of this triangle?

- A. **2**
- B. **8**
- C. **10**
- D. **24**

ID: 32b09ce9 Answer

Correct Answer: B

Rationale

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three of its sides. It's given that the lengths of two sides of a triangle are **4** centimeters and **6** centimeters. Let x represent the length, in centimeters, of the third side of this triangle. The sum of the lengths, in centimeters, of all three sides of the triangle can be represented by the expression $4 + 6 + x$. Since it's given that the perimeter of the triangle is **18** centimeters, it follows that $4 + 6 + x = 18$, or $10 + x = 18$. Subtracting **10** from both sides of this equation yields $x = 8$. Therefore, the length, in centimeters, of the third side of this triangle is **8**.

Choice A is incorrect. If the length of the third side of this triangle were **2** centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 2$, or **12**, not **18**.

Choice C is incorrect. If the length of the third side of this triangle were **10** centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 10$, or **20**, not **18**.

Choice D is incorrect. If the length of the third side of this triangle were **24** centimeters, the perimeter, in centimeters, of the triangle would be $4 + 6 + 24$, or **34**, not **18**.

Question Difficulty: Easy

Question ID 036097b4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 036097b4

An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

- A. 47
- B. $47\sqrt{2}$
- C. 94
- D. $94\sqrt{2}$

ID: 036097b4 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the right triangle is isosceles. In an isosceles right triangle, the two legs have equal lengths, and the length of the hypotenuse is $\sqrt{2}$ times the length of one of the legs. Let x represent the length, in inches, of each leg of the isosceles right triangle. It follows that the length of the hypotenuse is $x\sqrt{2}$ inches. The perimeter of a figure is the sum of the lengths of the sides of the figure. Therefore, the perimeter of the isosceles right triangle is $x + x + x\sqrt{2}$ inches. It's given that the perimeter of the triangle is $94 + 94\sqrt{2}$ inches. It follows that $x + x + x\sqrt{2} = 94 + 94\sqrt{2}$. Factoring the left-hand side of this equation yields $(1 + 1 + \sqrt{2})x = 94 + 94\sqrt{2}$, or $(2 + \sqrt{2})x = 94 + 94\sqrt{2}$. Dividing both sides of this equation by $2 + \sqrt{2}$ yields $x = \frac{94+94\sqrt{2}}{2+\sqrt{2}}$. Rationalizing the denominator of the right-hand side of this equation by multiplying the right-hand side of the equation by $\frac{2-\sqrt{2}}{2-\sqrt{2}}$ yields $x = \frac{(94+94\sqrt{2})(2-\sqrt{2})}{(2+\sqrt{2})(2-\sqrt{2})}$. Applying the distributive property to the numerator and to the denominator of the right-hand side of this equation yields $x = \frac{188-94\sqrt{2}+188\sqrt{2}-94\sqrt{4}}{4-2\sqrt{2}+2\sqrt{2}-\sqrt{4}}$. This is equivalent to $x = \frac{94\sqrt{2}}{2}$, or $x = 47\sqrt{2}$. Therefore, the length, in inches, of one leg of the isosceles right triangle is $47\sqrt{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length, in inches, of the hypotenuse.

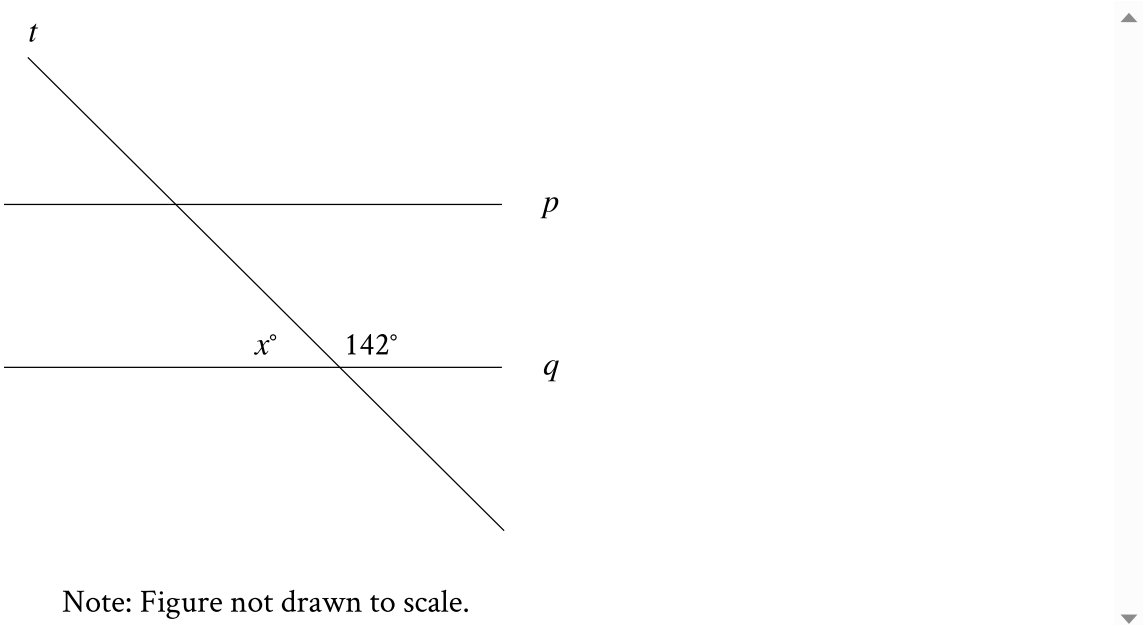
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 8f0aff11

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 8f0aff11



In the figure, line p is parallel to line q , and line t intersects both lines. What is the value of $x + 142$?

- A. 52
- B. 90
- C. 142
- D. 180

ID: 8f0aff11 Answer

Correct Answer: D

Rationale

Choice D is correct. In the figure shown, the angle marked x° and the angle marked 142° form a linear pair of angles. If two angles form a linear pair of angles, the sum of the measures of the angles is 180° . Therefore, the value of $x + 142$ is **180**.

Choice A is incorrect. This is **90** less than **142**, not the sum of x and **142**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the measure, in degrees, of one of the angles shown.

Question Difficulty: Medium

Question ID 8001a32f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 8001a32f

A right rectangular prism has a length of **28 centimeters (cm)**, a width of **15 cm**, and a height of **16 cm**. What is the surface area, **in cm²**, of the right rectangular prism?

ID: 8001a32f Answer

Correct Answer: 2216

Rationale

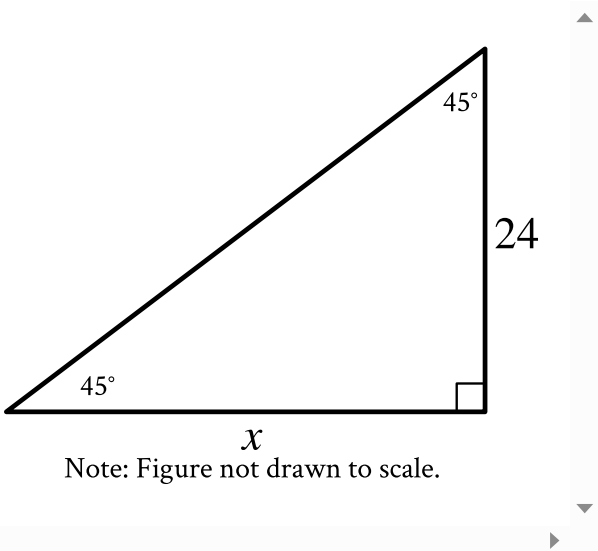
The correct answer is **2,216**. The surface area of a prism is the sum of the areas of all its faces. A right rectangular prism consists of six rectangular faces, where opposite faces are congruent. It's given that this prism has a length of **28 cm**, a width of **15 cm**, and a height of **16 cm**. Thus, for this prism, there are two faces with area **(28)(15) cm²**, two faces with area **(28)(16) cm²**, and two faces with area **(15)(16) cm²**. Therefore, the surface area, **in cm²**, of the right rectangular prism is **2(28)(15) + 2(28)(16) + 2(15)(16)**, or **2,216**.

Question Difficulty: Hard

Question ID 6b0992d8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 6b0992d8



In the triangle shown, what is the value of x ?

- A. 24
- B. 45
- C. 48
- D. 69

ID: 6b0992d8 Answer

Correct Answer: A

Rationale

Choice A is correct. Since the two acute angles have the same measure and the third angle is a right angle, the triangle shown is an isosceles right triangle. In an isosceles right triangle, the two legs have the same length. The figure shows that the length of one leg of the triangle is **24** and the length of the other leg of the triangle is x . It follows that the value of x is **24**.

Choice B is incorrect. This is the measure, in degrees, of one of the angles shown.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID d42965a0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: d42965a0

For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of x ?

- A. 5
- B. 19
- C. 23
- D. 29

ID: d42965a0 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that for two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. For two acute angles, if the sine of one angle is equal to the cosine of the other angle, the angles are complementary. It follows that $\angle Q$ and $\angle R$ are complementary. That is, the sum of the measures of the angles is **90** degrees. It's given that the measure of $\angle Q$ is $x + 61$ degrees and the measure of $\angle R$ is $4x + 4$ degrees. It follows that $(x + 61) + (4x + 4) = 90$. By combining like terms, this equation can be rewritten as $5x + 65 = 90$. Subtracting **65** from each side of this equation yields $5x = 25$. Dividing each side of this equation by **5** yields $x = 5$.

Choice B is incorrect. This would be the value of x if $\cos(Q) = \cos(R)$ rather than $\cos(Q) = \sin(R)$.

Choice C is incorrect. This would be the value of x if $\cos(Q) = -\cos(R)$ rather than $\cos(Q) = \sin(R)$ and if $\angle R$ were obtuse rather than acute.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID c021ed5c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: c021ed5c

A right square prism has a height of **14** units. The volume of the prism is **2,016** cubic units. What is the length, in units, of an edge of the base?

ID: c021ed5c Answer

Correct Answer: 12

Rationale

The correct answer is **12**. The volume, V , of a right square prism can be calculated using the formula $V = s^2h$, where s represents the length of an edge of the base and h represents the height of the prism. It's given that the volume of the prism is **2,016** cubic units and the height is **14** units. Substituting **2,016** for V and **14** for h in the formula $V = s^2h$ yields $2,016 = (s^2)(14)$. Dividing both sides of this equation by **14** yields $144 = s^2$. Taking the square root of both sides of this equation yields two solutions: $-12 = s$ and $12 = s$. The length can't be negative, so $12 = s$. Therefore, the length, in units, of an edge of the base is **12**.

Question Difficulty: Hard

Question ID cd81e894

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: cd81e894

Each side of a square has a length of **45**. What is the perimeter of this square?

ID: cd81e894 Answer

Correct Answer: 180

Rationale

The correct answer is **180**. The perimeter of a polygon is equal to the sum of the lengths of the sides of the polygon. It's given that each side of the square has a length of **45**. Since a square is a polygon with **4** sides, the perimeter of this square is **$45 + 45 + 45 + 45$** , or **180**.

Question Difficulty: Easy

Question ID a4c8c496

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a4c8c496

What is the area of a rectangle with a length of **17 centimeters (cm)** and a width of **7 cm**?

- A. **24 cm²**
- B. **48 cm²**
- C. **119 cm²**
- D. **576 cm²**

ID: a4c8c496 Answer

Correct Answer: C

Rationale

Choice C is correct. The area of a rectangle with length *l* and width *w* can be found using the formula $A = lw$. It's given that the rectangle has a length of **17 cm** and a width of **7 cm**. Therefore, the area of this rectangle is $A = 17(7)$, or **119 cm²**.

Choice A is incorrect. This is the sum of the length and width of the rectangle, not the area.

Choice B is incorrect. This is the perimeter of the rectangle, not the area.

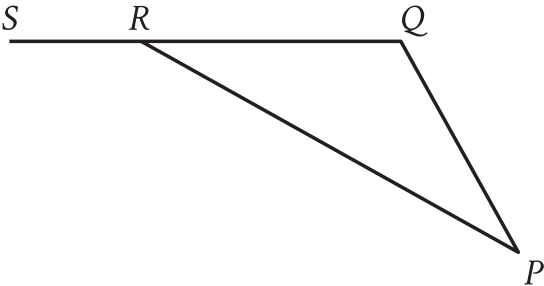
Choice D is incorrect. This is the sum of the length and width of the rectangle squared, not the area.

Question Difficulty: Easy

Question ID 761b22d4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 761b22d4



Note: Figure not drawn to scale.

In triangle PQR , $\overline{QR}$ is extended to point S . The measure of $\angle PQR$ is 132° , and the measure of $\angle PRS$ is 163° . What is the measure of $\angle QPR$?

- A. 48°
- B. 31°
- C. 24°
- D. 17°

ID: 761b22d4 Answer

Correct Answer: B

Rationale

Choice B is correct. In the figure shown, since $\overline{QS}$ is a line segment, the sum of the measures of $\angle PRS$ and $\angle PRQ$ is 180° . It's given that the measure of $\angle PRS$ is 163° . Thus, the measure of $\angle PRQ$ is $(180 - 163)^\circ$, or 17° . The sum of the measures of the interior angles of a triangle is 180° . It's given that the measure of $\angle PQR$ is 132° . Therefore, the measure of $\angle QPR$ is $(180 - 17 - 132)^\circ$, or 31° .

Choice A is incorrect. This is the measure of the supplement of $\angle PQR$, not the measure of $\angle QPR$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure of $\angle PRQ$, not the measure of $\angle QPR$.

Question Difficulty: Medium

Question ID 173cd9a9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 173cd9a9

A rectangular poster has an area of **360** square inches. A copy of the poster is made in which the length and width of the original poster are each increased by **20%**. What is the area of the copy, in square inches?

ID: 173cd9a9 Answer

Correct Answer: 2592/5, 518.4

Rationale

The correct answer is **518.4**. It's given that the area of the original poster is **360** square inches. Let l represent the length, in inches, of the original poster, and let w represent the width, in inches, of the original poster. Since the area of a rectangle is equal to its length times its width, it follows that $360 = lw$. It's also given that a copy of the poster is made in which the length and width of the original poster are each increased by **20%**. It follows that the length of the copy is the length of the original poster plus **20%** of the length of the original poster, which is equivalent to $l + \frac{20}{100}l$ inches. This length can be rewritten as $l + 0.2l$ inches, or $1.2l$ inches. Similarly, the width of the copy is the width of the original poster plus **20%** of the width of the original poster, which is equivalent to $w + \frac{20}{100}w$ inches. This width can be rewritten as $w + 0.2w$ inches, or $1.2w$ inches. Since the area of a rectangle is equal to its length times its width, it follows that the area, in square inches, of the copy is equal to $(1.2l)(1.2w)$, which can be rewritten as $(1.2)(1.2)(lw)$. Since $360 = lw$, the area, in square inches, of the copy can be found by substituting **360** for lw in the expression $(1.2)(1.2)(lw)$, which yields $(1.2)(1.2)(360)$, or **518.4**. Therefore, the area of the copy, in square inches, is **518.4**.

Question Difficulty: Hard

Question ID 9e84779f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 9e84779f

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k ?

- A. 36
- B. 45
- C. 72
- D. 108

ID: 9e84779f Answer

Correct Answer: C

Rationale

Choice C is correct. The sum of the interior angles of a triangle is 180° . It's given that the interior angles of triangle ABC are 54° , 90° , and $\left(\frac{k}{2}\right)^\circ$. It follows that $54 + 90 + \frac{k}{2} = 180$, or $144 + \frac{k}{2} = 180$. Subtracting 144 from each side of this equation yields $\frac{k}{2} = 36$. Multiplying each side of this equation by 2 yields $k = 72$. Therefore, the value of k is 72.

Choice A is incorrect. This is the value of $\frac{k}{2}$, not k .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 327679a4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 327679a4

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n ?

- A. $\frac{9}{7,744}$
- B. $\frac{27}{3,872}$
- C. $\frac{3}{88}$
- D. $\frac{9}{44}$

ID: 327679a4 Answer

Correct Answer: A

Rationale

Choice A is correct. Since all cubes are similar to each other, cube X and cube Y are similar figures. In similar figures, if lengths differ by a scale factor of k , then areas differ by a scale factor of k^2 . It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. It follows that the surface area, in square inches, of cube Y is $\left(\frac{3}{88}\right)^2$, or $\frac{9}{7,744}$, times the surface area, in square inches, of cube X. Therefore, the value of n is $\frac{9}{7,744}$.

Alternate approach: The surface area of a cube is $6s^2$, where s is the edge length. Let x represent the edge length, in inches, of cube X. Substituting x for s in $6s^2$ yields $6x^2$. Therefore, the surface area, in square inches, of cube X can be expressed as $6x^2$. It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. Therefore, the edge length, in inches, of cube Y can be expressed as $\left(\frac{3}{88}\right)x$, or $\frac{3x}{88}$. Substituting $\frac{3x}{88}$ for s in $6s^2$ yields $6\left(\frac{3x}{88}\right)^2$, which is equivalent to $6\left(\frac{9x^2}{7,744}\right)$, or $\frac{27x^2}{3,872}$. Therefore, the surface area, in square inches, of cube Y can be expressed as $\frac{27x^2}{3,872}$. It's given that the surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. It follows that $\frac{27x^2}{3,872} = n\left(6x^2\right)$. Dividing both sides of this equation by $6x^2$ yields $\frac{27x^2}{3,872(6x^2)} = n$, which is equivalent to $\frac{27x^2}{23,232x^2} = n$, or $\frac{9}{7,744} = n$. Therefore, the value of n is $\frac{9}{7,744}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

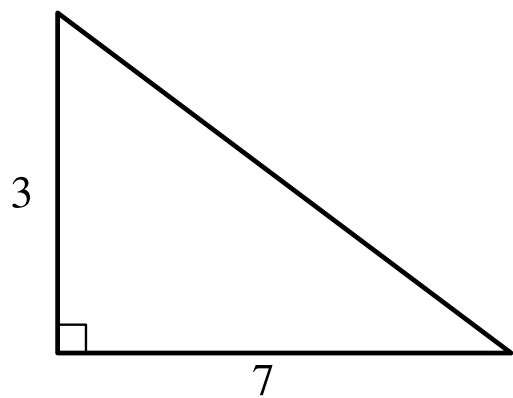
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b1328025

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: b1328025



Note: Figure not drawn to scale.

The lengths of the legs of a right triangle are shown. Which of the following is closest to the length of the triangle's hypotenuse?

- A. 3.2
- B. 5
- C. 7.6
- D. 20

ID: b1328025 Answer

Correct Answer: C

Rationale

Choice C is correct. The Pythagorean theorem states that for a right triangle, $a^2 + b^2 = c^2$, where a and b represent the lengths of the legs of the triangle and c represents the length of its hypotenuse. In the triangle shown, the legs have lengths of **3** and **7**. Substituting **3** for a and **7** for b in the equation $a^2 + b^2 = c^2$ yields $3^2 + 7^2 = c^2$, which is equivalent to $9 + 49 = c^2$, or $58 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{58} = c$. Thus, the value of c is approximately **7.6**. Therefore, of the given choices, **7.6** is the closest to the length of the triangle's hypotenuse.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

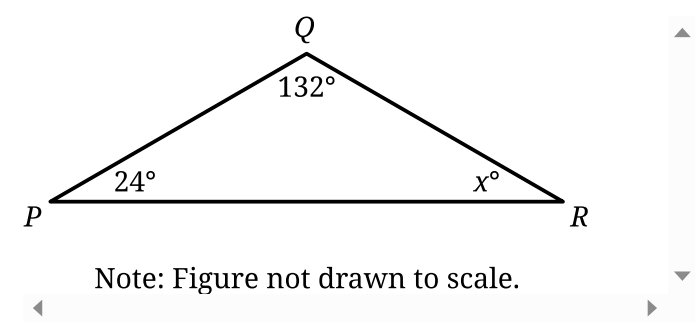
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 0770ad58

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 0770ad58



In the triangle shown, $PQ = QR$. What is the value of x ?

- A. 156
- B. 66
- C. 48
- D. 24

ID: 0770ad58 Answer

Correct Answer: D

Rationale

Choice D is correct. The sum of the angle measures of a triangle is 180° . It's given that the measure of angle Q is 132° , the measure of angle P is 24° , and the measure of angle R is x° . It follows that $x + 132 + 24 = 180$, or $x + 152 = 180$. Subtracting 152 from both sides of this equation yields $x = 24$.

Alternate approach: It's given that in the triangle shown, $PQ = QR$. Thus, triangle PQR is an isosceles triangle. Since base angles of an isosceles triangle are congruent, the measure of angle P is equal to the measure of angle R . Therefore, the measure of angle R is 24° , and the value of x is 24 .

Choice A is incorrect. This is the sum of the measures, in degrees, of angles Q and P , not the measure, in degrees, of angle R .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures, in degrees, of angles P and R , not the measure, in degrees, of angle R .

Question Difficulty: Easy

Question ID 40780ef3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 40780ef3

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1,089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

ID: 40780ef3 Answer

Correct Answer: 66

Rationale

The correct answer is **66**. It's given that each vertex of the rectangle lies on the circumference of the circle. Therefore, the length of the diameter of the circle is equal to the length of the diagonal of the rectangle. The diagonal of a rectangle forms a right triangle with the shortest and longest sides of the rectangle, where the shortest side and the longest side of the rectangle are the legs of the triangle and the diagonal of the rectangle is the hypotenuse of the triangle. Let s represent the length, in units, of the shortest side of the rectangle. Since it's given that the diagonal is twice the length of the shortest side, $2s$ represents the length, in units, of the diagonal of the rectangle. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting s for a and $2s$ for c in this equation yields $s^2 + b^2 = (2s)^2$, or $s^2 + b^2 = 4s^2$. Subtracting s^2 from both sides of this equation yields $b^2 = 3s^2$. Taking the positive square root of both sides of this equation yields $b = s\sqrt{3}$. Therefore, the length, in units, of the rectangle's longest side is $s\sqrt{3}$. The area of a rectangle is the product of the length of the shortest side and the length of the longest side. The lengths, in units, of the shortest and longest sides of the rectangle are represented by s and $s\sqrt{3}$, and it's given that the area of the rectangle is $1,089\sqrt{3}$ square units. It follows that $1,089\sqrt{3} = s(s\sqrt{3})$, or $1,089\sqrt{3} = s^2\sqrt{3}$. Dividing both sides of this equation by $\sqrt{3}$ yields $1,089 = s^2$. Taking the positive square root of both sides of this equation yields $33 = s$. Since the length, in units, of the diagonal is represented by $2s$, it follows that the length, in units, of the diagonal is $2(33)$, or **66**. Since the length of the diameter of the circle is equal to the length of the diagonal of the rectangle, the length, in units, of the diameter of the circle is **66**.

Question Difficulty: Hard

Question ID edaa921d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: edaa921d

A triangle has a base length of **10** centimeters and a corresponding height of **70** centimeters. What is the area, in square centimeters, of the triangle?

- A. **700**
- B. **350**
- C. **175**
- D. **80**

ID: edaa921d Answer

Correct Answer: B

Rationale

Choice B is correct. The area, ***A***, of a triangle is given by **$A = \left(\frac{1}{2}\right)bh$** , where ***b*** is the length of a base of the triangle and ***h*** is the corresponding height of the triangle. It's given that a triangle has a base length of **10** centimeters and a corresponding height of **70** centimeters. Substituting **10** for ***b*** and **70** for ***h*** in the formula **$A = \left(\frac{1}{2}\right)bh$** yields **$A = \left(\frac{1}{2}\right)(10)(70)$** , or **$A = 350$** . Therefore, the area, in square centimeters, of the triangle is **350**.

Choice A is incorrect. This is the product of the given base and height of the triangle, not its area.

Choice C is incorrect and may result from conceptual or calculation errors.

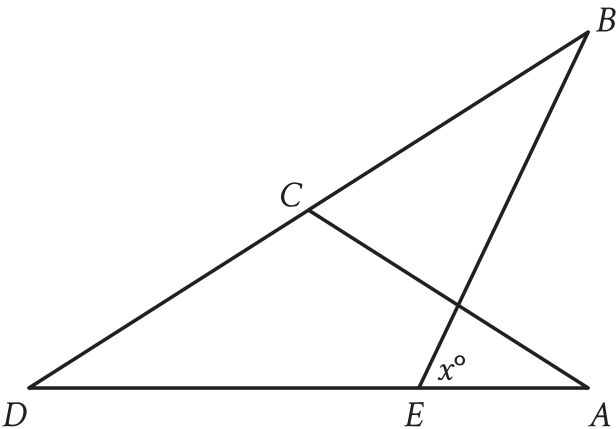
Choice D is incorrect. This is the sum of the given base and height of the triangle, not its area.

Question Difficulty: Medium

Question ID fd643f08

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: fd643f08



Note: Figure not drawn to scale.

In the figure, $AC = CD$. The measure of angle EBC is 45° , and the measure of angle ACD is 104° . What is the value of x ?

ID: fd643f08 Answer

Correct Answer: 83

Rationale

The correct answer is **83**. It's given that in the figure, $AC = CD$. Thus, triangle ACD is an isosceles triangle and the measure of angle CDA is equal to the measure of angle CAD . The sum of the measures of the interior angles of a triangle is 180° . Thus, the sum of the measures of the interior angles of triangle ACD is 180° . It's given that the measure of angle ACD is 104° . It follows that the sum of the measures of angles CDA and CAD is $(180 - 104)^\circ$, or 76° . Since the measure of angle CDA is equal to the measure of angle CAD , the measure of angle CDA is half of 76° , or 38° . The sum of the measures of the interior angles of triangle BDE is 180° . It's given that the measure of angle EBC is 45° . Since the measure of angle BDE , which is the same angle as angle CDA , is 38° , it follows that the measure of angle DEB is $(180 - 45 - 38)^\circ$, or 97° . Since angle DEB and angle AEB form a straight line, the sum of the measures of these angles is 180° . It's given in the figure that the measure of angle AEB is x° . It follows that $97 + x = 180$. Subtracting **97** from both sides of this equation yields $x = 83$.

Question Difficulty: Hard

Question ID a315fc1a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a315fc1a

A right circular cylinder has a height of **16** inches and a volume of **$3,136\pi$** cubic inches. What is the radius, in inches, of this cylinder?

ID: a315fc1a Answer

Correct Answer: 14

Rationale

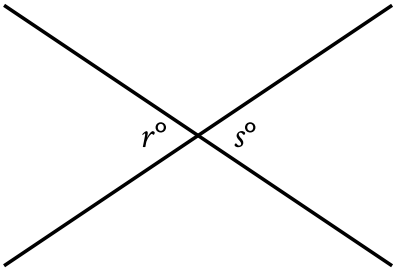
The correct answer is **14**. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that the right circular cylinder has a height of **16** inches and a volume of **$3,136\pi$** cubic inches. Substituting **16** for h and **$3,136\pi$** for V in the formula $V = \pi r^2 h$ yields **$3,136\pi = \pi r^2 (16)$** , or **$3,136\pi = 16\pi r^2$** . Dividing both sides of this equation by **16π** yields **$196 = r^2$** . Since r represents the radius of the cylinder, the value of r is positive. Taking the positive square root of both sides of this equation yields **$r = 14$** . Therefore, the radius, in inches, of the cylinder is **14**.

Question Difficulty: Hard

Question ID af7c7a66

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: af7c7a66



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $r = 53$, what is the value of s ?

- A. 37
- B. 53
- C. 143
- D. 233

ID: af7c7a66 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that two lines intersect at a point. The angles with measures of r° and s° are vertical angles, so it follows that they are equal in measure. Therefore, $s = r$. If $r = 53$, it follows that $s = 53$.

Choice A is incorrect. This is the measure, in degrees, of the complement of an angle with a measure of 53° , not the value of s .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c21cfdfe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c21cfdfe

In triangle ABC , the measure of angle A is 52° and $AC = 30$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

- A. $AB = 50$ and $PQ = 50$.
- B. $AB = 50$ and $QR = 200$.
- C. The measures of angle B and angle R are 32° and 96° , respectively.
- D. The measures of angle B and angle Q are 52° and 32° , respectively.

ID: c21cfdfe Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the measure of angle A is 52° and the measure of angle P is 52° . Therefore, angle A is congruent to angle P . Choice C states that the measures of angle B and angle R are 32° and 96° , respectively. The triangle sum theorem states that the measures of the angles in a triangle sum to 180° . In triangle ABC , if the measure of angle B is 32° , it follows that the measure of angle C is $(180 - (52 + 32))^\circ$, or 96° . In triangle PQR , if the measure of angle R is 96° , it follows that the measure of angle Q is $(180 - (52 + 96))^\circ$, or 32° . Therefore, in triangles ABC and PQR , angle B is congruent to angle Q , and angle C is congruent to angle R . If two triangles have three pairs of congruent angles, then the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle ABC is similar to triangle PQR is that the measures of angle B and angle R are 32° and 96° , respectively.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. If the measure of angle B is 52° , it follows that the measure of angle C is $(180 - (52 + 52))^\circ$, or 76° . If the measure of angle Q is 32° , which isn't equal to any angle measure in triangle ABC , the triangles can't be similar.

Question Difficulty: Hard

Question ID 42a07493

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 42a07493

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is 2 times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is 5 square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

- A. 7
- B. 10
- C. 20
- D. 25

ID: 42a07493 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that parallelogram $ABCD$ is similar to parallelogram $PQRS$. When two parallelograms are similar, if the scale factor between their corresponding side lengths is k , the scale factor between their areas is k^2 . It’s given that the length of each side of parallelogram $PQRS$ is 2 times the length of its corresponding side of parallelogram $ABCD$, so the scale factor between their corresponding side lengths is 2 . It follows that the scale factor between their areas is 2^2 , or 4 . It’s given that the area, in square centimeters, of parallelogram $ABCD$ is 5 . It follows that the area, in square centimeters, of parallelogram $PQRS$ is $5(4)$, or 20 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID ad5516fd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: ad5516fd

In $\triangle RST$, the measure of $\angle R$ is 63° . Which of the following could be the measure, in degrees, of $\angle S$?

- A. 116
- B. 118
- C. 126
- D. 180

ID: ad5516fd Answer

Correct Answer: A

Rationale

Choice A is correct. The sum of the measures of the angles of a triangle is 180° . Therefore, the sum of the measures of $\angle R$, $\angle S$, and $\angle T$ is 180° . It's given that the measure of $\angle R$ is 63° . It follows that the sum of the measures of $\angle S$ and $\angle T$ is $(180 - 63)^\circ$, or 117° . Therefore, the measure of $\angle S$, in degrees, must be less than 117. Of the given choices, only 116 is less than 117. Thus, the measure, in degrees, of $\angle S$ could be 116.

Choice B is incorrect. If the measure of $\angle S$ is 118° , then the sum of the measures of the angles of the triangle is greater than, not equal to, 180° .

Choice C is incorrect. If the measure of $\angle S$ is 126° , then the sum of the measures of the angles of the triangle is greater than, not equal to, 180° .

Choice D is incorrect. This is the sum of the measures of the angles of a triangle, in degrees.

Question Difficulty: Medium

Question ID d15d0186

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: d15d0186



- $\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\overline{ST}$, $\overline{QP}$, $\overline{PR}$, and $\overline{QR}$ in the figure are 14, 15, 20, and 25, respectively. What is the length of $\overline{SR}$?
- A. $\frac{350}{15}$
 - B. $\frac{350}{20}$
 - C. $\frac{210}{20}$
 - D. $\frac{210}{25}$

ID: d15d0186 Answer

Correct Answer: A

Rationale

Choice A is correct. The figure shows that angle P in $\triangle QPR$ and angle T in $\triangle STR$ are right angles. It follows that angle P is congruent to angle T . The figure also shows that the measures of angle QRP and angle SRT are both x° . Therefore, angle QRP is congruent to angle SRT . It's given that $\triangle QPR$ is similar to $\triangle STR$. Since angle P is congruent to angle T , and angle QRP is congruent to angle SRT , it follows that $\overline{QR}$ corresponds to $\overline{SR}$, and $\overline{QP}$ corresponds to $\overline{ST}$. Since corresponding sides of similar triangles are proportional, it follows that $\frac{SR}{QR} = \frac{ST}{QP}$. It's also given that the lengths of $\overline{ST}$, $\overline{QP}$, and $\overline{QR}$ are 14, 15, and 25, respectively. Substituting 14 for ST , 15 for QP , and 25 for QR in the equation $\frac{SR}{QR} = \frac{ST}{QP}$ yields $\frac{SR}{25} = \frac{14}{15}$. Multiplying each side of this equation by 25 yields $SR = \left(\frac{14}{15}\right)(25)$, or $SR = \frac{350}{15}$. Thus, the length of $\overline{SR}$ is $\frac{350}{15}$.

Choice B is incorrect. This is the result of solving the equation $\frac{SR}{25} = \frac{14}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice C is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice D is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{25}$, not $\frac{SR}{25} = \frac{14}{15}$.

Question Difficulty: Hard

Question ID e03c8917

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: e03c8917

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is **6** times the length of the corresponding side of $ABCD$. The area of $ABCD$ is **54** square units. What is the area, in square units, of $EFGH$?

- A. **9**
- B. **36**
- C. **324**
- D. **1,944**

ID: e03c8917 Answer

Correct Answer: D

Rationale

Choice D is correct. The area of a rectangle is given by bh , where b is the length of the base of the rectangle and h is its height. Let x represent the length, in units, of the base of rectangle $ABCD$, and let y represent its height, in units. Substituting x for b and y for h in the formula bh yields xy . Therefore, the area, in square units, of $ABCD$ can be represented by the expression xy . It's given that the length of each side of $EFGH$ is **6** times the length of the corresponding side of $ABCD$. Therefore, the length, in units, of the base of $EFGH$ can be represented by the expression $6x$, and its height, in units, can be represented by the expression $6y$. Substituting $6x$ for b and $6y$ for h in the formula bh yields $(6x)(6y)$, which is equivalent to $36xy$. Therefore, the area, in square units, of $EFGH$ can be represented by the expression $36xy$. It's given that the area of $ABCD$ is **54** square units. Since xy represents the area, in square units, of $ABCD$, substituting **54** for xy in the expression $36xy$ yields **36(54)**, or **1,944**. Therefore, the area, in square units, of $EFGH$ is **1,944**.

Choice A is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{1}{6}}$, not **6**, times the length of the corresponding side of $ABCD$.

Choice B is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{2}{3}}$, not **6**, times the length of the corresponding side of $ABCD$.

Choice C is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{6}$, not **6**, times the length of the corresponding side of $ABCD$.

Question Difficulty: Hard

Question ID f3cfbc3e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: f3cfbc3e

A cylinder has a diameter of **8** inches and a height of **12** inches. What is the volume, in cubic inches, of the cylinder?

- A. **16π**
- B. **96π**
- C. **192π**
- D. **768π**

ID: f3cfbc3e Answer

Correct Answer: C

Rationale

Choice C is correct. The base of a cylinder is a circle with a diameter equal to the diameter of the cylinder. The volume, V , of a cylinder can be found by multiplying the area of the circular base, A , by the height of the cylinder, h , or $V = Ah$. The area of a circle can be found using the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the diameter of the cylinder is **8** inches. Thus, the radius of this circle is **4** inches. Therefore, the area of the circular base of the cylinder is $A = \pi(4)^2$, or **16π** square inches. It's given that the height h of the cylinder is **12** inches. Substituting **16π** for A and **12** for h in the formula $V = Ah$ gives $V = 16\pi(12)$, or **192π** cubic inches.

Choice A is incorrect. This is the area of the circular base of the cylinder.

Choice B is incorrect and may result from using **8**, instead of **16**, as the value of r^2 in the formula for the area of a circle.

Choice D is incorrect and may result from using **8**, instead of **4**, for the radius of the circular base.

Question Difficulty: Medium

Question ID 34bc22bf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 34bc22bf

Triangle XYZ is similar to triangle RST such that X , Y , and Z correspond to R , S , and T , respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

- A. 2°
- B. 10°
- C. 20°
- D. 40°

ID: 34bc22bf Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that triangle XYZ is similar to triangle RST , such that X , Y , and Z correspond to R , S , and T , respectively. Since corresponding angles of similar triangles are congruent, it follows that the measure of $\angle Z$ is congruent to the measure of $\angle T$. It's given that the measure of $\angle Z$ is 20° . Therefore, the measure of $\angle T$ is 20° .

Choice A is incorrect and may result from a conceptual error.

Choice B is incorrect. This is half the measure of $\angle Z$.

Choice D is incorrect. This is twice the measure of $\angle Z$.

Question Difficulty: Hard

Question ID 7ad836d0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 7ad836d0

Triangles PQR and LMN are graphed in the xy -plane. Triangle PQR has vertices P , Q , and R at $(4, 5)$, $(4, 7)$, and $(6, 5)$, respectively. Triangle LMN has vertices L , M , and N at $(4, 5)$, $(4, 7 + k)$, and $(6 + k, 5)$, respectively, where k is a positive constant. If the measure of $\angle Q$ is t° , what is the measure of $\angle N$?

- A. $(90 - (t - k))^\circ$
- B. $(90 - (t + k))^\circ$
- C. $(90 - t)^\circ$
- D. $(90 + k)^\circ$

ID: 7ad836d0 Answer

Correct Answer: C

Rationale

Choice C is correct. Since $P = (4, 5)$ and $Q = (4, 7)$, side PQ is parallel to the y -axis and has a length of 2 . Since $P = (4, 5)$ and $R = (6, 5)$, side PR is parallel to the x -axis and has a length of 2 . Therefore, triangle PQR is a right isosceles triangle, where $\angle P$ has measure 90° and $\angle Q$ and $\angle R$ each have measure 45° . It follows that if the measure of $\angle Q$ is t° , then $t = 45$. Since $L = (4, 5)$ and $M = (4, 7 + k)$, side LM is parallel to the y -axis and has a length of $k + 2$. Since $L = (4, 5)$ and $N = (6 + k, 5)$, side LN is parallel to the x -axis and has a length of $k + 2$. Therefore, triangle LMN is a right isosceles triangle, where $\angle L$ has measure 90° and $\angle M$ and $\angle N$ each have measure 45° . Of the given choices, only $(90 - t)^\circ$ is equal to 45° , so the measure of $\angle N$ is $(90 - t)^\circ$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a19b04bc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a19b04bc

The width of a rectangle is **7** centimeters. The length of the rectangle is **40** centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

- A. **7**
- B. **14**
- C. **54**
- D. **329**

ID: a19b04bc Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the width of this rectangle is **7** centimeters and that the length of this rectangle is **40** centimeters longer than the width. Therefore, the length of this rectangle is **7 + 40**, or **47**, centimeters. The area of a rectangle can be found by multiplying its length and its width. Therefore the area, in square centimeters, of this rectangle is **(7)(47)**, or **329**.

Choice A is incorrect. This is the width, in centimeters, not the area, in square centimeters, of this rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

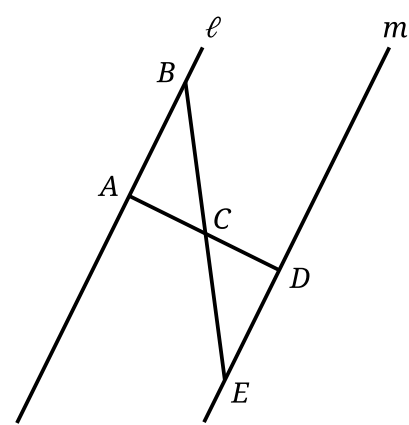
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e5c37ef9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: e5c37ef9



Note: Figure not drawn to scale.

In the figure, segment AD and segment BE intersect at point C . Segment AD is perpendicular to line ℓ at point A . Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC ?

- I. Segment AD is perpendicular to line m at point D .
- II. $AC = 13$ and $DC = 13$

- A. I is sufficient by itself, but II is not.
- B. II is sufficient by itself, but I is not.
- C. I is sufficient by itself, and II is sufficient by itself.
- D. Neither I nor II is sufficient by itself.

ID: e5c37ef9 Answer

Correct Answer: D

Rationale

Choice D is correct. In the figure, it's given that segment AD is perpendicular to line ℓ at point A . Statement I is that segment AD is perpendicular to line m at point D . It follows that angles BAC and EDC are right angles. Therefore, triangles ABC and DEC are right triangles. At least one pair of congruent corresponding sides is needed to determine triangle congruence by the side-angle-side (SAS) condition or the angle-side-angle (ASA) condition, so statement I is not sufficient to determine that triangle ABC is congruent to triangle DEC . Statement II is that $AC = 13$ and $DC = 13$. It follows that triangles ABC and DEC share one pair of congruent corresponding sides. Two additional pairs of congruent corresponding parts are needed to determine triangle congruence by SAS condition, ASA condition, or side-side-side (SSS) condition, so statement II is not sufficient to determine that triangle ABC is congruent to triangle DEC . Therefore, neither I nor II is sufficient by itself to determine that triangle ABC is congruent to triangle DEC .

Choice A is incorrect and may result from conceptual errors.

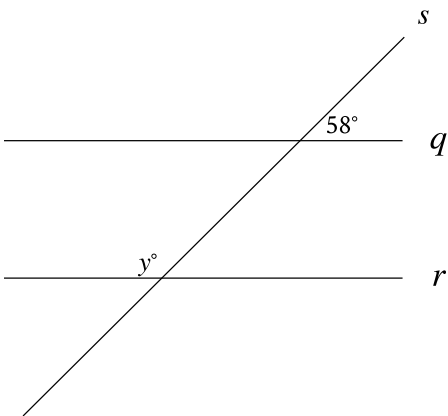
Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID b4029e85

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: b4029e85



Note: Figure not drawn to scale.

In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 8$, what is the value of x ?

ID: b4029e85 Answer

Correct Answer: 57

Rationale

The correct answer is **57**. Based on the figure, the angle with measure y° and the angle vertical to the angle with measure 58° are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure 58° also has measure 58° . It's given that lines q and r are parallel. Therefore, same side interior angles between lines q and r are supplementary. It follows that $y + 58 = 180$. If $y = 2x + 8$, then the value of x can be found by substituting $2x + 8$ for y in the equation $y + 58 = 180$, which yields $(2x + 8) + 58 = 180$, or $2x + 66 = 180$. Subtracting **66** from both sides of this equation yields $2x = 114$. Dividing both sides of this equation by **2** yields $x = 57$. Thus, if $y = 2x + 8$, the value of x is **57**.

Question Difficulty: Hard

Question ID a85da65c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a85da65c

The length of the edge of the base of a right square prism is **6** units. The volume of the prism is **2,880** cubic units. What is the height, in units, of the prism?

- A. $4\sqrt{30}$
- B. **36**
- C. $24\sqrt{5}$
- D. **80**

ID: a85da65c Answer

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a right square prism is given by the formula $V = s^2h$, where s represents the length of the edge of the base and h represents the height of the prism. It's given that the volume of a right square prism is **2,880** cubic units and the length of the edge of the base is **6** units. Substituting **2,880** for V and **6** for s in the formula $V = s^2h$ yields **2,880** = $(6^2)h$, or **2,880** = $36h$. Dividing both sides of this equation by **36** yields **80** = h . Therefore, the height, in units, of the prism is **80**.

Choice A is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is **6** units and the volume of the prism is **144** $\sqrt{30}$, not **2,880**, units.

Choice B is incorrect. This is the area, in square units, of the base, not the height, in units, of the prism.

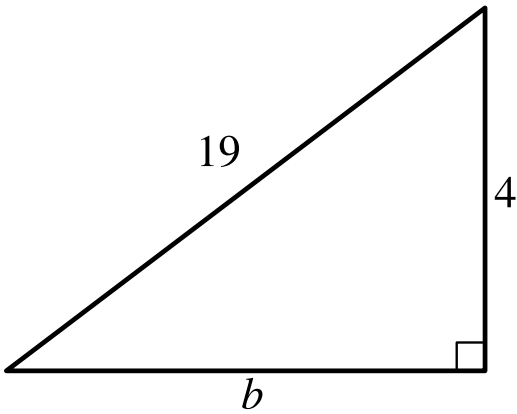
Choice C is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is **6** units and the volume of the prism is **864** $\sqrt{5}$, not **2,880**, units.

Question Difficulty: Hard

Question ID 7a589426

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 7a589426



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

- A. $4b = 19$
- B. $4 + b = 19$
- C. $4^2 + b^2 = 19^2$
- D. $4^2 - b^2 = 19^2$

ID: 7a589426 Answer

Correct Answer: C

Rationale

Choice C is correct. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Therefore, $a^2 + b^2 = c^2$, where a and b are the lengths of the legs and c is the length of the hypotenuse. For the given right triangle, the lengths of the legs are 4 and b , and the length of the hypotenuse is 19 . Substituting 4 for a and 19 for c in the equation $a^2 + b^2 = c^2$ yields $4^2 + b^2 = 19^2$. Thus, the relationship between the side lengths of the given triangle is $4^2 + b^2 = 19^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 3b3a2841

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 3b3a2841

In triangle XYZ , angle Y is a right angle, point P lies on $\overline{XZ}$, and point Q lies on $\overline{YZ}$ such that $\overline{PQ}$ is parallel to $\overline{XY}$. If the measure of angle XZY is 63° , what is the measure, in degrees, of angle XPQ ?

ID: 3b3a2841 Answer

Correct Answer: 153

Rationale

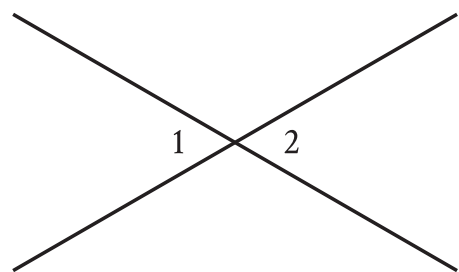
The correct answer is **153**. Since it's given that $\overline{PQ}$ is parallel to $\overline{XY}$ and angle Y is a right angle, angle ZQP is also a right angle. Angle ZPQ is complementary to angle XZY , which means its measure, in degrees, is $90 - 63$, or **27**. Since angle XPQ is supplementary to angle ZPQ , its measure, in degrees, is $180 - 27$, or **153**.

Question Difficulty: Hard

Question ID 4e57ee40

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 4e57ee40



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. Angle **1** and angle **2** are vertical angles. The measure of angle **1** is **72°**. What is the measure of angle **2**?

- A. 72°
- B. 108°
- C. 144°
- D. 288°

ID: 4e57ee40 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that angle **1** and angle **2** are vertical angles, and the measure of angle **1** is **72°**. Vertical angles have equal measures. Therefore, the measure of angle **2** is **72°**.

Choice B is incorrect. This is the measure of an angle that is supplementary, not congruent, to angle **1**.

Choice C is incorrect. This is the sum of the measures of angle **1** and angle **2**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a685dfbb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: a685dfbb

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

ID: a685dfbb Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the equilateral triangle has a height of $39\sqrt{3}$. In equilateral triangles, all three angles have a measure of 60° and the height is equal to the length of an altitude from one vertex to the midpoint of the side across from the vertex. Thus, an altitude in this equilateral triangle has a length of $39\sqrt{3}$. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the 60° angle in each 30-60-90 right triangle. Since the altitude has a length of $39\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each 30° angle has a length of 39 and each hypotenuse has a length of $2(39)$, or 78. Therefore, each side of the equilateral triangle has a length of 78. In equilateral triangles, all three sides have equal length. Therefore, the perimeter of this equilateral triangle is $78 + 78 + 78$, or 234.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 2f004f77

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 2f004f77

A right circular cylinder has a volume of **377** cubic centimeters. The area of the base of the cylinder is **13** square centimeters. What is the height, in centimeters, of the cylinder?

ID: 2f004f77 Answer

Correct Answer: 29

Rationale

The correct answer is **29**. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. Since the base of the cylinder is a circle with radius r , the area of the base of the cylinder is πr^2 . It's given that a right circular cylinder has a volume of **377** cubic centimeters; therefore, $V = 377$. It's also given that the area of the base of the cylinder is **13** square centimeters; therefore, $\pi r^2 = 13$. Substituting **377** for V and **13** for πr^2 in the formula $V = \pi r^2 h$ yields $377 = 13h$. Dividing both sides of this equation by **13** yields $29 = h$. Therefore, the height of the cylinder, in centimeters, is **29**.

Question Difficulty: Medium

Question ID fcf5cbea

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: fcf5cbea

In triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . What is the measure of $\angle J$, in degrees? (Disregard the degree symbol when entering your answer.)

ID: fcf5cbea Answer

Correct Answer: 84

Rationale

The correct answer is **84**. The sum of the measures of the interior angles of a triangle is 180° . It's given that in triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . Adding the measures, in degrees, of $\angle K$ and $\angle L$ gives $48 + 48$, or **96**. Therefore, the measure of $\angle J$, in degrees, is $180 - 96$, or **84**.

Question Difficulty: Medium

Question ID 51a28608

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 51a28608

Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

- A. 37°
- B. 53°
- C. 127°
- D. 143°

ID: 51a28608 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that triangle LMN is similar to triangle PQR . Corresponding angles of similar triangles are congruent. Since angle M and angle Q correspond to each other, they must be congruent. Therefore, if the measure of angle M is 53° , then the measure of angle Q is also 53° .

Choice A is incorrect and may result from concluding that angle M and angle Q are complementary rather than congruent.

Choice C is incorrect and may result from concluding that angle M and angle Q are supplementary rather than congruent.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID d8114972

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: d8114972

The perimeter of triangle ABC is 17 inches, the length of side AB is 4 inches, and the length of side AC is 7 inches. What is the length, in inches, of side BC ?

- A. 4
- B. 6
- C. 7
- D. 11

ID: d8114972 Answer

Correct Answer: B

Rationale

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three sides of the triangle. It's given that the lengths of side AB and side AC are 4 inches and 7 inches, respectively. Let x represent the length, in inches, of side BC . The sum of the lengths, in inches, of all three sides of triangle ABC can be represented by the expression $4 + 7 + x$. Since it's given that the perimeter of triangle ABC is 17 inches, it follows that $17 = 4 + 7 + x$, or $17 = 11 + x$. Subtracting 11 from both sides of this equation yields $6 = x$. Therefore, the length, in inches, of side BC is 6 .

Choice A is incorrect. This is the length, in inches, of side AB .

Choice C is incorrect. This is the length, in inches, of side AC .

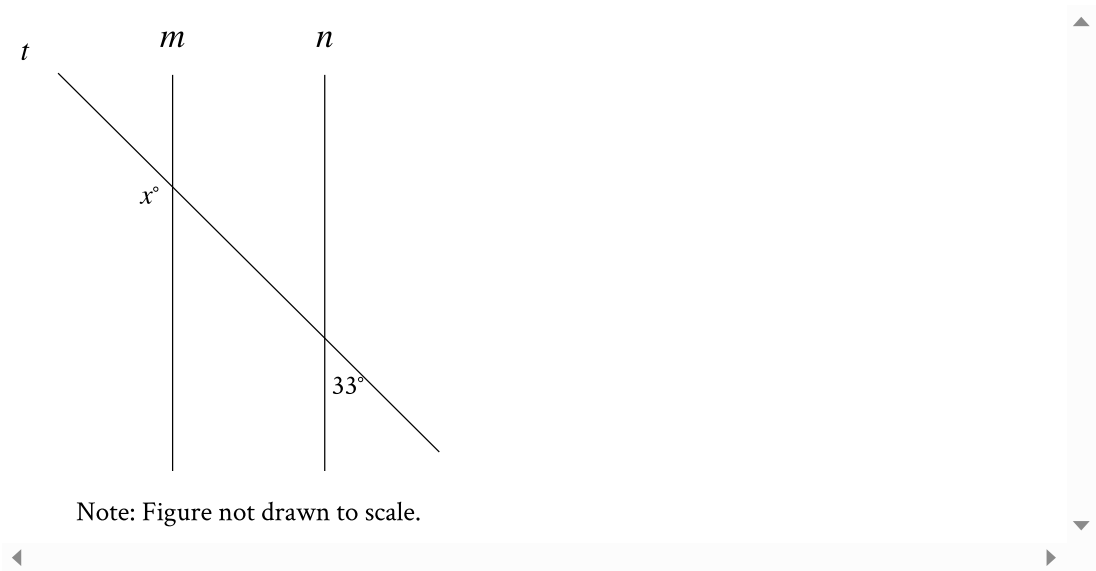
Choice D is incorrect. This is the sum of the lengths, in inches, of sides AB and AC .

Question Difficulty: Easy

Question ID 3423c8cb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 3423c8cb



In the figure, line *m* is parallel to line *n*, and line *t* intersects both lines. What is the value of *x*?

- A. 33
- B. 57
- C. 123
- D. 147

ID: 3423c8cb Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that line *m* is parallel to line *n*, and line *t* intersects both lines. It follows that line *t* is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal are supplementary. Thus, $x + 33 = 180$. Subtracting 33 from both sides of this equation yields $x = 147$. Therefore, the value of *x* is 147.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 716f5a56

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 716f5a56

In triangle XYZ , angle Y is a right angle, the measure of angle Z is 33° , and the length of $\overline{YZ}$ is 26 units. If the area, in square units, of triangle XYZ can be represented by the expression $k \tan 33^\circ$, where k is a constant, what is the value of k ?

ID: 716f5a56 Answer

Correct Answer: 338

Rationale

The correct answer is **338**. The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. In triangle XYZ , it's given that angle Y is a right angle. Thus, $\overline{XY}$ is the leg opposite of angle Z and $\overline{YZ}$ is the leg adjacent to angle Z . It follows that $\tan Z = \frac{XY}{YZ}$. It's also given that the measure of angle Z is 33° and the length of $\overline{YZ}$ is 26 units. Substituting 33° for Z and 26 for YZ in the equation $\tan Z = \frac{XY}{YZ}$ yields $\tan 33^\circ = \frac{XY}{26}$. Multiplying each side of this equation by 26 yields $26 \tan 33^\circ = XY$. Therefore, the length of $\overline{XY}$ is $26 \tan 33^\circ$. The area of a triangle is half the product of the lengths of its legs. Since the length of $\overline{YZ}$ is 26 and the length of $\overline{XY}$ is $26 \tan 33^\circ$, it follows that the area of triangle XYZ is $\frac{1}{2}(26)(26 \tan 33^\circ)$ square units, or $338 \tan 33^\circ$ square units. It's given that the area, in square units, of triangle XYZ can be represented by the expression $k \tan 33^\circ$, where k is a constant. Therefore, **338** is the value of k .

Question Difficulty: Hard

Question ID 9df10dea

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 9df10dea

A circle has a radius of **2.1** inches. The area of the circle is **$b\pi$** square inches, where **b** is a constant. What is the value of **b** ?

ID: 9df10dea Answer

Correct Answer: 4.41, 441/100

Rationale

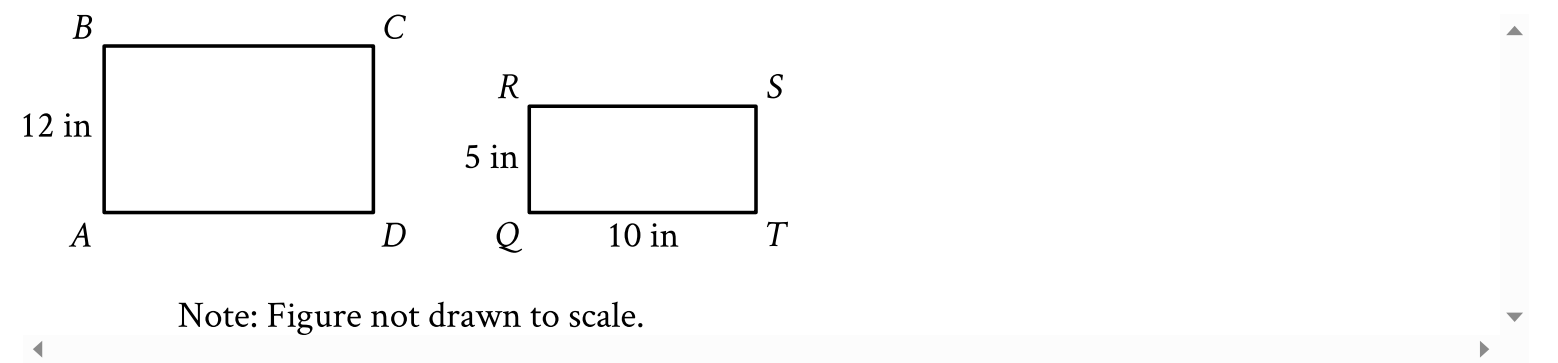
The correct answer is **4.41**. The area, **A** , of a circle is given by the formula **$A = \pi r^2$** , where **r** is the radius of the circle. It's given that the area of the circle is **$b\pi$** square inches, where **b** is a constant, and the radius of the circle is **2.1** inches. Substituting **$b\pi$** for **A** and **2.1** for **r** in the formula **$A = \pi r^2$** yields **$b\pi = \pi(2.1^2)$** . Dividing both sides of this equation by **π** yields **$b = 4.41$** . Therefore, the value of **b** is **4.41**.

Question Difficulty: Hard

Question ID 58a83077

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 58a83077



Rectangles $ABCD$ and $QRST$ shown are similar, where A, B, C , and D correspond to Q, R, S , and T , respectively. What is the length, in inches (**in**), of $\overline{AD}$?

- A. 60
- B. 24
- C. 17
- D. 10

ID: 58a83077 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that rectangles $ABCD$ and $QRST$ are similar, where A, B, C , and D correspond to Q, R, S , and T , respectively. It follows that $\overline{AB}$ corresponds to $\overline{QR}$ and $\overline{AD}$ corresponds to $\overline{QT}$. If two rectangles are similar, then the lengths of their corresponding sides are proportional. It's given in the figure that the length of $\overline{AB}$ is **12** inches, the length of $\overline{QR}$ is **5** inches, and the length of $\overline{QT}$ is **10** inches. If x is the length, in inches, of $\overline{AD}$, then $\frac{12}{5}$ is equivalent to $\frac{x}{10}$. Therefore, the value of x can be found using the equation $\frac{12}{5} = \frac{x}{10}$. Multiplying each side of this equation by **10** yields $\frac{120}{5} = x$, or **24** = x . Therefore, the length, in inches, of $\overline{AD}$ is **24**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length, in inches, of $\overline{QT}$, not $\overline{AD}$.

Question Difficulty: Medium

Question ID 814903d3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 814903d3

The base of a right rectangular pyramid has a length of **27** centimeters and a width of **2** centimeters. The height of this pyramid is **7** centimeters. What is the volume, in cubic centimeters, of this pyramid?

ID: 814903d3 Answer

Correct Answer: 126

Rationale

The correct answer is **126**. It’s given that the base of the right rectangular pyramid has a length of **27** centimeters and a width of **2** centimeters and that the height of the pyramid is **7** centimeters. The volume, V , of a right rectangular pyramid is given by the formula $V = \frac{1}{3} \ell wh$, where ℓ and w are the length and width, respectively, of the pyramid’s base and h is the height of the pyramid. Substituting **27** for ℓ , **2** for w , and **7** for h in this formula yields $V = \frac{1}{3} (27) (2) (7)$, or $V = 126$. Therefore, the volume, in cubic centimeters, of the pyramid is **126**.

Question Difficulty: Hard

Question ID b9c52ff0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: b9c52ff0

A right triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. What is the area of the triangle, in square units?

- A. $8\sqrt{2} + \sqrt{80}$
- B. 12
- C. $24\sqrt{80}$
- D. 24

ID: b9c52ff0 Answer

Correct Answer: B

Rationale

Choice B is correct. The area, A , of a triangle can be found using the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. It's given that the triangle is a right triangle. Therefore, its base and height can be represented by the two legs. It's also given that the triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. Since $\sqrt{80}$ units is the greatest of these lengths, it's the length of the hypotenuse. Therefore, the two legs have lengths $2\sqrt{2}$ and $6\sqrt{2}$ units. Substituting these values for b and h in the formula $A = \frac{1}{2}bh$ gives $A = \frac{1}{2}(2\sqrt{2})(6\sqrt{2})$, which is equivalent to $A = 6\sqrt{4}$ square units, or $A = 12$ square units.

Choice A is incorrect. This expression represents the perimeter, rather than the area, of the triangle.

Choice C is incorrect and may result from conceptual or calculation errors.

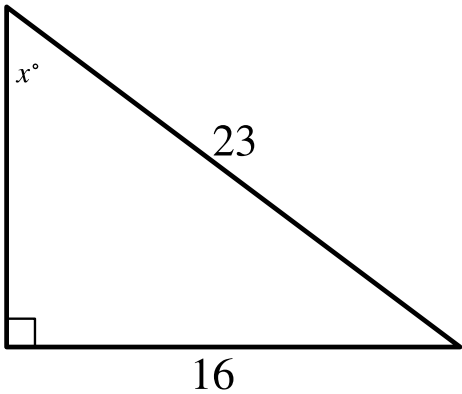
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f0333710

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: f0333710



Note: Figure not drawn to scale.



In the triangle shown, what is the value of $\sin x^\circ$?

ID: f0333710 Answer

Correct Answer: .6956, .6957, 16/23

Rationale

The correct answer is $\frac{16}{23}$. In a right triangle, the sine of an acute angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the length of the side opposite the angle with measure x° is **16** units and the length of the hypotenuse is **23** units. Therefore, the value of $\sin x^\circ$ is $\frac{16}{23}$. Note that 16/23, .6956, .6957, 0.695, and 0.696 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID a375d8c4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a375d8c4

A right circular cylinder has a height of **8 meters (m)** and a base with a radius of **12 m**. What is the volume, **in m³**, of the cylinder?

- A. **8π**
- B. **20π**
- C. **768π**
- D. **$1,152\pi$**

ID: a375d8c4 Answer

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a right circular cylinder is given by $V = \pi r^2 h$, where r is the radius of the circular base and h is the height of the cylinder. It's given that the cylinder has a height of **8 meters** and a base with a radius of **12 meters**. Substituting **12** for r and **8** for h in $V = \pi r^2 h$ yields $V = \pi(12)^2(8)$, or $V = 1,152\pi$. Therefore, the volume, in **m³**, of the cylinder is **1,152π**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the volume, in **m³**, of a cylinder with a radius of **8 meters** and a height of **12 meters**.

Question Difficulty: Medium

Question ID 865680fb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 865680fb

Two identical rectangular prisms each have a height of **90 centimeters (cm)**. The base of each prism is a square, and the surface area of each prism is $K \text{ cm}^2$. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K \text{ cm}^2$. What is the side length, in **cm**, of each square base?

- A. 4
- B. 8
- C. 9
- D. 16

ID: 865680fb Answer

Correct Answer: B

Rationale

Choice B is correct. Let x represent the side length, in **cm**, of each square base. If the two prisms are glued together along a square base, the resulting prism has a surface area equal to twice the surface area of one of the prisms, minus the area of the two square bases that are being glued together, which yields $2K - 2x^2 \text{ cm}^2$. It's given that this resulting surface area is equal to $\frac{92}{47}K \text{ cm}^2$, so $2K - 2x^2 = \frac{92}{47}K$. Subtracting $\frac{92}{47}K$ from both sides of this equation yields $2K - \frac{92}{47}K - 2x^2 = 0$. This equation can be rewritten by multiplying $2K$ on the left-hand side by $\frac{47}{47}$, which yields $\frac{94}{47}K - \frac{92}{47}K - 2x^2 = 0$, or $\frac{2}{47}K - 2x^2 = 0$. Adding $2x^2$ to both sides of this equation yields $\frac{2}{47}K = 2x^2$. Multiplying both sides of this equation by $\frac{47}{2}$ yields $K = 47x^2$. The surface area K , in cm^2 , of each rectangular prism is equivalent to the sum of the areas of the two square bases and the areas of the four lateral faces. Since the height of each rectangular prism is **90 cm** and the side length of each square base is $x \text{ cm}$, it follows that the area of each square base is $x^2 \text{ cm}^2$ and the area of each lateral face is $90x \text{ cm}^2$. Therefore, the surface area of each rectangular prism can be represented by the expression $2x^2 + 4(90x)$, or $2x^2 + 360x$. Substituting this expression for K in the equation $K = 47x^2$ yields $2x^2 + 360x = 47x^2$. Subtracting $2x^2$ and $360x$ from both sides of this equation yields $0 = 45x^2 - 360x$. Factoring x from the right-hand side of this equation yields $0 = x(45x - 360)$. Applying the zero product property, it follows that $x = 0$ and $45x - 360 = 0$. Adding **360** to both sides of the equation $45x - 360 = 0$ yields $45x = 360$. Dividing both sides of this equation by **45** yields $x = 8$. Since a side length of a rectangular prism can't be **0**, the length of each square base is **8 cm**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 26bf11b7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 26bf11b7

In a right triangle, the measure of one of the acute angles is 51° . What is the measure, in degrees, of the other acute angle?

- A. 6
- B. 39
- C. 49
- D. 51

ID: 26bf11b7 Answer

Correct Answer: B

Rationale

Choice B is correct. The sum of the measures of the interior angles of a triangle is **180** degrees. Since the triangle is a right triangle, it has one angle that measures **90** degrees. Therefore, the sum of the measures, in degrees, of the remaining two angles is **180 — 90**, or **90**. It’s given that the measure of one of the acute angles in the triangle is **51** degrees. Therefore, the measure, in degrees, of the other acute angle is **90 — 51**, or **39**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

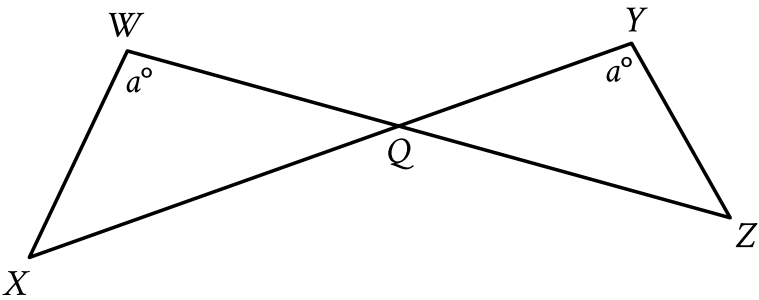
Choice D is incorrect. This is the measure, in degrees, of the acute angle whose measure is given.

Question Difficulty: Medium

Question ID 948e6696

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 948e6696



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

- A. 18
- B. 36
- C. 120
- D. 200

ID: 948e6696 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . It follows that $\angle WQX$ and $\angle YQZ$ are vertical angles, which means they are congruent to each other. The figure shows that $\angle W$ and $\angle Y$ both have a measure of a° , so they are congruent to each other. Therefore, $\triangle QWX$ and $\triangle QYZ$ are similar triangles, where $\overline{WQ}$ corresponds to $\overline{YQ}$ and $\overline{WX}$ corresponds to $\overline{YZ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{YQ} = \frac{WX}{WQ}$. It's given that $YQ = 21$, $WQ = 70$, and $WX = 60$. Substituting 21 for YQ , 70 for WQ , and 60 for WX in the equation $\frac{YZ}{YQ} = \frac{WX}{WQ}$ yields $\frac{YZ}{21} = \frac{60}{70}$, or $\frac{YZ}{21} = \frac{6}{7}$. Multiplying each side of this equation by 21 yields $YZ = 18$. Therefore, the length of $\overline{YZ}$ is 18.

Choice B is incorrect. This is the length of $\overline{QZ}$, not $\overline{YZ}$.

Choice C is incorrect. This is the length of $\overline{XQ}$, not $\overline{YZ}$.

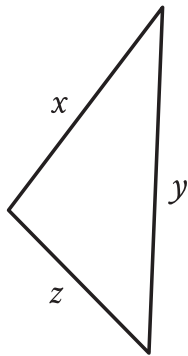
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 5d1eb251

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 5d1eb251



Note: Figure not drawn to scale.

The triangle shown has a perimeter of **22** units. If $x = 9$ units and $y = 7$ units, what is the value of z , in units?

- A. **6**
- B. **7**
- C. **9**
- D. **16**

ID: 5d1eb251 Answer

Correct Answer: A

Rationale

Choice A is correct. The perimeter of a triangle is the sum of the lengths of its three sides. The triangle shown has side lengths x , y , and z . It's given that the triangle has a perimeter of **22** units. Therefore, $x + y + z = 22$. If $x = 9$ units and $y = 7$ units, the value of z , in units, can be found by substituting **9** for x and **7** for y in the equation $x + y + z = 22$, which yields $9 + 7 + z = 22$, or $16 + z = 22$. Subtracting **16** from both sides of this equation yields $z = 6$. Therefore, if $x = 9$ units and $y = 7$ units, the value of z , in units, is **6**.

Choice B is incorrect. This is the value of y , in units, not the value of z , in units.

Choice C is incorrect. This is the value of x , in units, not the value of z , in units.

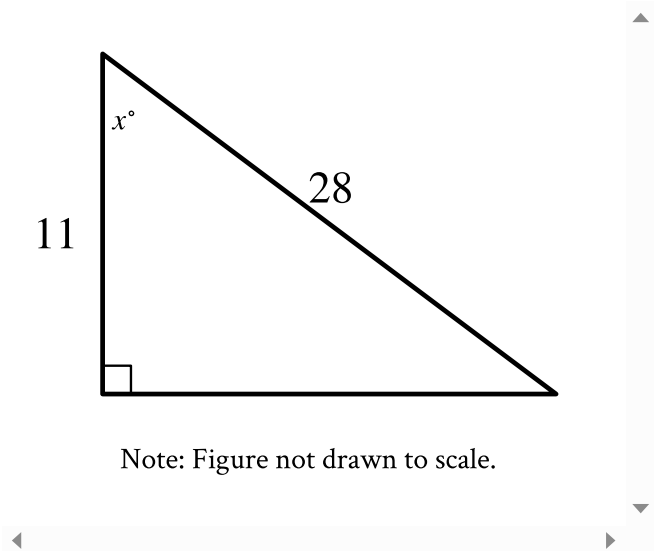
Choice D is incorrect. This is the value of $x + y$, in units, not the value of z , in units.

Question Difficulty: Easy

Question ID 3f6e0eb2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 3f6e0eb2



In the triangle shown, what is the value of $\cos x^\circ$?

ID: 3f6e0eb2 Answer

Correct Answer: .3928, .3929, 11/28

Rationale

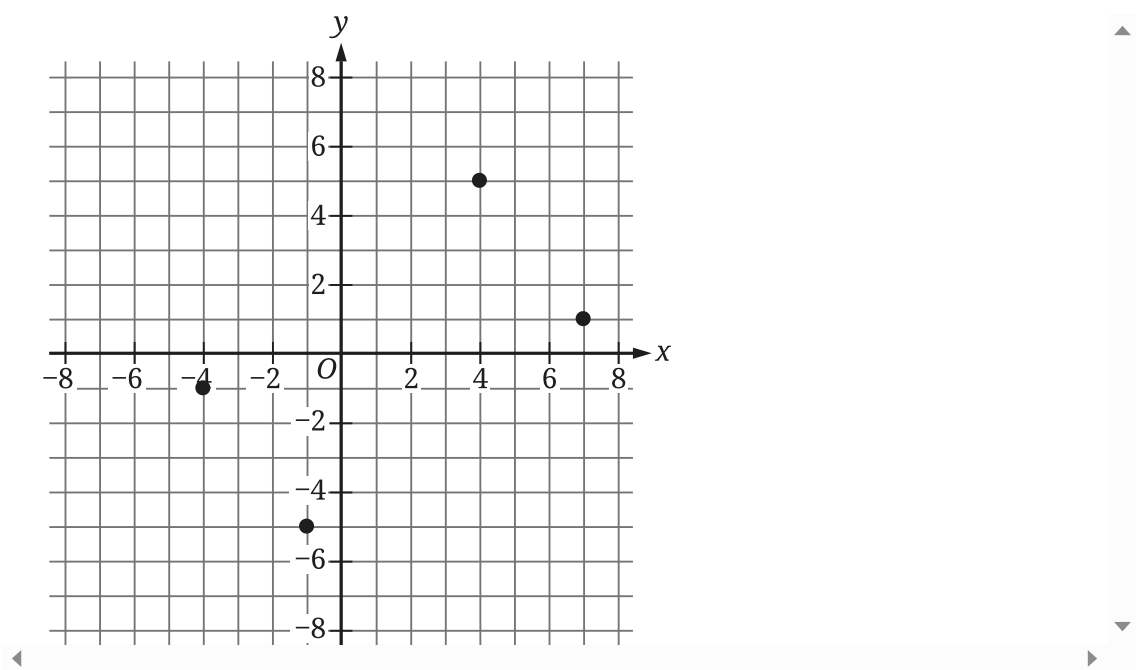
The correct answer is $\frac{11}{28}$. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to the angle with measure x° is **11** units and the length of the hypotenuse is **28** units. Therefore, the value of $\cos x^\circ$ is $\frac{11}{28}$. Note that 11/28, .3928, .3929, 0.392, and 0.393 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID e101f1a1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: e101f1a1



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

ID: e101f1a1 Answer

Correct Answer: 50

Rationale

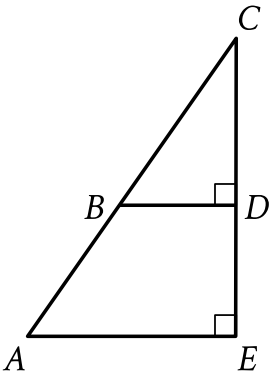
The correct answer is **50**. The area of a rectangle is lw , where l and w are the lengths of two adjacent sides of the rectangle. It's given that the points $(-4, -1)$, $(4, 5)$, $(7, 1)$, and $(-1, -5)$ form a rectangle. The points $(-4, -1)$ and $(4, 5)$ form a side of the rectangle that is adjacent to the side of the rectangle formed by the points $(4, 5)$ and $(7, 1)$. In the xy -plane, the distance between two points, (x_1, y_1) and (x_2, y_2) , can be found as $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Thus, the length, in units, of the side formed by the points $(-4, -1)$ and $(4, 5)$ can be found as $\sqrt{(4 - (-4))^2 + (5 - (-1))^2}$, which can be simplified to $\sqrt{100}$, or **10**. The length, in units, of the side formed by the points $(4, 5)$ and $(7, 1)$ can be found as $\sqrt{(7 - 4)^2 + (1 - 5)^2}$, which can be simplified to $\sqrt{25}$, or **5**. Therefore, the area, in square units, of the rectangle formed by the four points shown is **(10)(5)**, or **50**.

Question Difficulty: Hard

Question ID a18f7911

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a18f7911



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 26(BD)$. What is the measure of angle CAE ?

- A. $(26 \cdot 57)^\circ$
- B. $(26 + 57)^\circ$
- C. 57°
- D. 26°

ID: a18f7911 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that triangle CAE is similar to triangle CBD . Corresponding angles in similar triangles have equal measure. Angle BCD and angle ACE represent the same angle. It follows that angle BCD and angle ACE have equal measure and are corresponding angles. It's given in the figure that angle BDC and angle AEC are right angles and therefore have equal measure. It follows that angle BDC and angle AEC are corresponding angles. Therefore, angle CBD and angle CAE are corresponding angles and have equal measure. It's given that the measure of angle CBD is 57° , so the measure of angle CAE is 57° .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

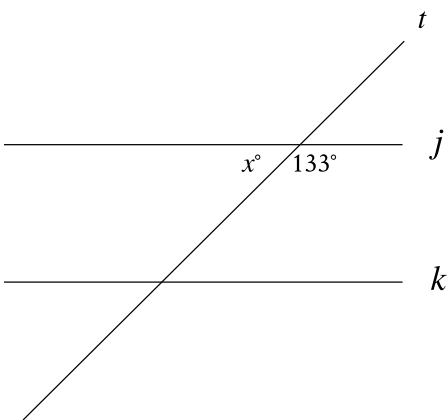
Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID f9ae3b99

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f9ae3b99



Note: Figure not drawn to scale.

In the figure, line *j* is parallel to line *k*. What is the value of *x*?

ID: f9ae3b99 Answer

Correct Answer: 47

Rationale

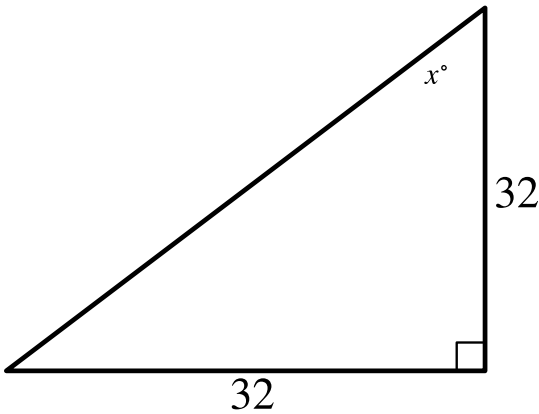
The correct answer is **47**. Based on the figure, the angle with measure ***x*°** and the angle with measure **133°** together form a straight line. Therefore, these two angles are supplementary, so the sum of their measures is **180°**. It follows that ***x* + 133 = 180**. Subtracting **133** from both sides of this equation yields ***x* = 47**.

Question Difficulty: Easy

Question ID 22c99ca6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 22c99ca6



Note: Figure not drawn to scale.



In the triangle shown, what is the value of x ?

ID: 22c99ca6 Answer

Correct Answer: 45

Rationale

The correct answer is **45**. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of **32**. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are **45°** . It follows that the value of x is **45**.

Question Difficulty: Medium

Question ID dbb36b07

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: dbb36b07

What is the perimeter, in inches, of a rectangle with a length of **4** inches and a width of **9** inches?

- A. **13**
- B. **17**
- C. **22**
- D. **26**

ID: dbb36b07 Answer

Correct Answer: D

Rationale

Choice D is correct. The perimeter of a figure is equal to the sum of the measurements of the sides of the figure. It’s given that the rectangle has a length of **4** inches and a width of **9** inches. Since a rectangle has **4** sides, of which opposite sides are parallel and equal, it follows that the rectangle has two sides with a length of **4** inches and two sides with a width of **9** inches. Therefore, the perimeter of this rectangle is **4 + 4 + 9 + 9**, or **26** inches.

Choice A is incorrect. This is the sum, in inches, of the length and the width of the rectangle.

Choice B is incorrect. This is the sum, in inches, of the two lengths and the width of the rectangle.

Choice C is incorrect. This is the sum, in inches, of the length and the two widths of the rectangle.

Question Difficulty: Easy

Question ID e02054c8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: e02054c8

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **18 cm**. The volume of this cone is $n\pi \text{ cm}^3$. What is the value of n ?

ID: e02054c8 Answer

Correct Answer: 594

Rationale

The correct answer is **594**. It's given that the right circular cone has a height of **22 cm** and a base with a diameter of **18 cm**. Since the diameter of the base of the cone is **18 cm**, the radius of the base is **9 cm**. The volume V , in cm^3 , of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where h is the height, in **cm**, of the cone and r is the radius, in **cm**, of the base of the cone. Substituting **22** for h and **9** for r in this formula yields $V = \frac{1}{3}\pi(9)^2(22)$, or $V = 594\pi$. Therefore, the volume of the cone is **$594\pi \text{ cm}^3$** . It's given that the volume of the cone is $n\pi \text{ cm}^3$. Therefore, the value of n is **594**.

Question Difficulty: Hard

Question ID 00181f03

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 00181f03

An isosceles right triangle has a hypotenuse of length 58 inches. What is the perimeter, in inches, of this triangle?

- A. $29\sqrt{2}$
- B. $58\sqrt{2}$
- C. $58 + 58\sqrt{2}$
- D. $58 + 116\sqrt{2}$

ID: 00181f03 Answer

Correct Answer: C

Rationale

Choice C is correct. Since the triangle is an isosceles right triangle, the two sides that form the right angle must be the same length. Let x be the length, in inches, of each of those sides. The Pythagorean theorem states that in a right triangle, $a^2 + b^2 = c^2$, where c is the length of the hypotenuse and a and b are the lengths of the other two sides. Substituting x for a , x for b , and 58 for c in this equation yields $x^2 + x^2 = 58^2$, or $2x^2 = 58^2$. Dividing each side of this equation by 2 yields $x^2 = \frac{58^2}{2}$, or $x^2 = \frac{2 \cdot 58^2}{4}$. Taking the square root of each side of this equation yields two solutions: $x = \frac{58\sqrt{2}}{2}$ and $x = -\frac{58\sqrt{2}}{2}$. The value of x must be positive because it represents a side length. Therefore, $x = \frac{58\sqrt{2}}{2}$, or $x = 29\sqrt{2}$. The perimeter, in inches, of the triangle is $58 + x + x$, or $58 + 2x$. Substituting $29\sqrt{2}$ for x in this expression gives a perimeter, in inches, of $58 + 2(29\sqrt{2})$, or $58 + 58\sqrt{2}$.

Choice A is incorrect. This is the length, in inches, of each of the congruent sides of the triangle, not the perimeter, in inches, of the triangle.

Choice B is incorrect. This is the sum of the lengths, in inches, of the congruent sides of the triangle, not the perimeter, in inches, of the triangle.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 3e9cd2c6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 3e9cd2c6

A triangle has a base length of **56** centimeters and a height of **112** centimeters. What is the area, in square centimeters, of the triangle?

- A. **168**
- B. **1,568**
- C. **3,136**
- D. **6,272**

ID: 3e9cd2c6 Answer

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the base length and h is the height of the triangle. It's given that a triangle has a base length of **56** centimeters and a height of **112** centimeters. Substituting **56** for b and **112** for h in the formula $A = \frac{1}{2}bh$ yields $A = (\frac{1}{2})(56)(112)$, or $A = \mathbf{3,136}$. Therefore, the area, in square centimeters, of the triangle is **3,136**.

Choice A is incorrect. This is the value of $56 + 112$, not $(\frac{1}{2})(56)(112)$.

Choice B is incorrect. This is the value of $(\frac{1}{4})(56)(112)$, not $(\frac{1}{2})(56)(112)$.

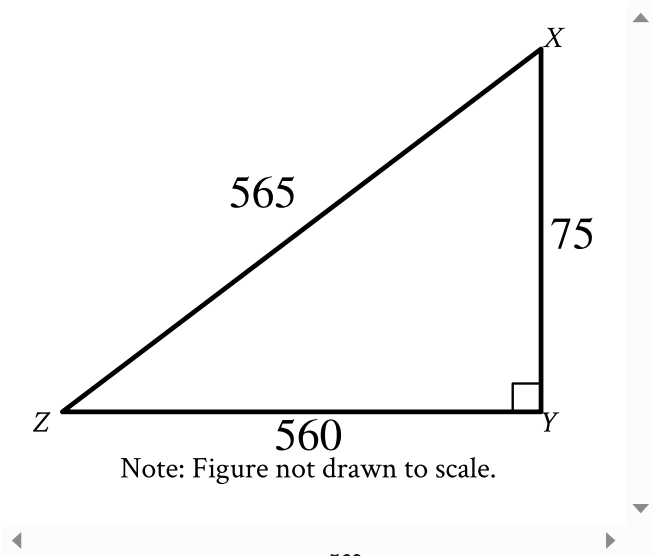
Choice D is incorrect. This is the value of $(56)(112)$, not $(\frac{1}{2})(56)(112)$.

Question Difficulty: Medium

Question ID 61255a55

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 61255a55



In triangle XYZ , $\cos Z = \frac{560}{c}$. What is the value of c ?

ID: 61255a55 Answer

Correct Answer: 565

Rationale

The correct answer is **565**. In triangle XYZ , angle Y is a right angle. Therefore, triangle XYZ is a right triangle. The cosine of an acute angle in a right triangle is the ratio of the length of the side adjacent to that angle to the length of the hypotenuse. In right triangle XYZ , side YZ is the side adjacent to angle Z and side XZ is the hypotenuse. It's given that the length of side YZ is **560** and the length of side XZ is **565**. Therefore, $\cos Z = \frac{560}{565}$. It's also given that $\cos Z = \frac{560}{c}$. It follows that $\frac{560}{c} = \frac{560}{565}$. Therefore, $c = \mathbf{565}$.

Question Difficulty: Medium

Question ID 4f536598

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 4f536598

What is the area of a rectangle with a length of **4 centimeters (cm)** and a width of **2 cm**?

- A. **6 cm²**
- B. **8 cm²**
- C. **12 cm²**
- D. **36 cm²**

ID: 4f536598 Answer

Correct Answer: B

Rationale

Choice B is correct. The area of a rectangle with length l and width w can be found using the formula $A = lw$. It's given that the rectangle has a length of **4 cm** and a width of **2 cm**. Therefore, the area of this rectangle is **(4 cm)(2 cm)**, or **8 cm²**.

Choice A is incorrect. This is the sum, **in cm**, of the length and width of the rectangle, not the area, **in cm²**.

Choice C is incorrect. This is the perimeter, **in cm**, of the rectangle, not the area, **in cm²**.

Choice D is incorrect. This is the sum of the length and width of the rectangle squared, not the area.

Question Difficulty: Easy

Question ID 7f94a1a7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 7f94a1a7

A cube has an edge length of **41** inches. What is the volume, in cubic inches, of the cube?

- A. **164**
- B. **1,681**
- C. **10,086**
- D. **68,921**

ID: 7f94a1a7 Answer

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a cube can be found using the formula $V = s^3$, where s is the edge length of the cube. It's given that a cube has an edge length of **41** inches. Substituting **41** inches for s in this equation yields $V = 41^3$ cubic inches, or $V = \mathbf{68,921}$ cubic inches. Therefore, the volume of the cube is **68,921** cubic inches.

Choice A is incorrect. This is the perimeter, in inches, of the cube.

Choice B is incorrect. This is the area, in square inches, of a face of the cube.

Choice C is incorrect. This is the surface area, in square inches, of the cube.

Question Difficulty: Medium

Question ID c721811f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c721811f

For triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle FGH ?

- I. The measure of angle A is equal to the measure of angle F .
- II. The measure of angle C is equal to the measure of angle H .
- III. The length of $\overline{FH}$ is 3 times the length of $\overline{AC}$.
- A. I is sufficient, II is sufficient, and III is sufficient.
- B. I is sufficient and II is sufficient, but III is not.
- C. II is sufficient and III is sufficient, but I is not.
- D. III is sufficient, but I is not and II is not.

ID: c721811f Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that in triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . For statement I, if the measure of angle A is equal to the measure of angle F , then triangles ABC and FGH have two pairs of congruent angles, angles A and F and angles B and G . When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement I is sufficient. For statement II, if the measure of angle C is equal to the measure of angle H , then triangles ABC and FGH have 2 pairs of congruent angles, angles C and H and angles B and G . When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement II is sufficient. For statement III, if the length of $\overline{FH}$ is 3 times the length of $\overline{AC}$, then the ratio of the length of $\overline{FH}$ to the length of $\overline{AC}$ is 3 . It's given that the length of $\overline{FG}$ is 18 and the length of $\overline{AB}$ is 6 . Therefore, the ratio of the length of $\overline{FG}$ to the length of $\overline{AB}$ is also 3 , so there are two pairs of sides in proportion. However, angle B isn't between sides $\overline{AB}$ and $\overline{AC}$, and angle G isn't between sides $\overline{FG}$ and $\overline{FH}$. Since the given angle measures aren't between the proportional sides, this information isn't sufficient to prove that the triangles are similar. Therefore, statements I and II are each sufficient, but statement III isn't sufficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

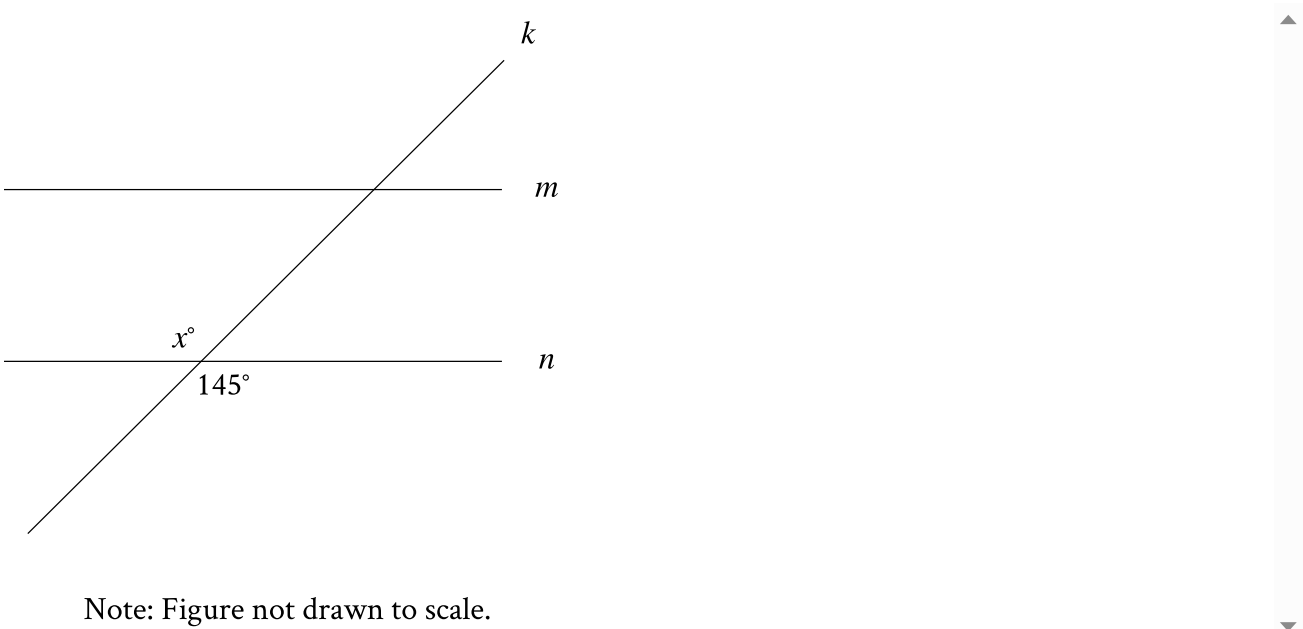
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 9aa57971

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 9aa57971



In the figure, line *m* is parallel to line *n*, and line *k* intersects both lines. Which of the following statements is true?

- A. The value of *x* is less than 145.
- B. The value of *x* is greater than 145.
- C. The value of *x* is equal to 145.
- D. The value of *x* cannot be determined.

ID: 9aa57971 Answer

Correct Answer: C

Rationale

Choice C is correct. Vertical angles, or angles that are opposite each other when two lines intersect, are congruent. It’s given that line *k* intersects line *n*. Based on the figure, the angle with measure *x*° and the angle with measure 145° are vertical angles. Therefore, the value of *x* is equal to 145.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID f8584dd3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: f8584dd3

In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{BC}$ is 136 millimeters. If $\cos A = \frac{3}{5}$, what is the length, in millimeters, of $\overline{AB}$?

- A. 34
- B. 102
- C. 136
- D. 170

ID: f8584dd3 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that in $\triangle ABC$, $\angle B$ is a right angle. Therefore, $\triangle ABC$ is a right triangle, and $\overline{AC}$ is the hypotenuse of the triangle. It's also given that $\cos A = \frac{3}{5}$. Since the cosine of an acute angle in a right triangle is defined as the ratio of the length of the side adjacent to the angle to the length of the hypotenuse, the ratio of the length of $\overline{AB}$ to the length of $\overline{AC}$ is 3 to 5. It follows that the length of $\overline{AB}$ can be represented as $3a$ and the length of $\overline{AC}$ can be represented as $5a$, where a is a constant. The Pythagorean theorem states that the sum of the squares of the length of the legs of a right triangle is equal to the square of the length of its hypotenuse, so it follows that $AB^2 + BC^2 = AC^2$. Substituting $3a$ for AB and $5a$ for AC in this equation yields $(3a)^2 + BC^2 = (5a)^2$, or $9a^2 + BC^2 = 25a^2$. Subtracting $9a^2$ from both sides of this equation yields $BC^2 = 16a^2$, or $BC = 4a$. It follows that the ratio of the length of $\overline{AB}$ to the length of $\overline{BC}$ is 3 to 4. Let x represent the length, in millimeters, of $\overline{AB}$. It's given that the length of $\overline{BC}$ is 136 millimeters. Since the ratio of the length of $\overline{AB}$ to the length of $\overline{BC}$ is 3 to 4, $\frac{x}{136} = \frac{3}{4}$. Multiplying both sides of this equation by 136 yields $x = \frac{3(136)}{4}$, or $x = 102$. Therefore, the length of $\overline{AB}$ is 102 millimeters.

Choice A is incorrect. This is the scale factor by which the 3 to 4 to 5 ratio is multiplied that results in the side lengths of $\triangle ABC$.

Choice C is incorrect. This is the length of $\overline{BC}$, not the length of $\overline{AB}$.

Choice D is incorrect. This is the length of $\overline{AC}$, not the length of $\overline{AB}$.

Question Difficulty: Hard

Question ID 74f32a28

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 74f32a28

The sum of the measures of two angles in a triangle is **64°**. What is the measure of the third angle?

- A. **26°**
- B. **52°**
- C. **116°**
- D. **128°**

ID: 74f32a28 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the sum of the measures of two angles in a triangle is **64°**. The sum of the measures of all three angles in a triangle is **180°**. It follows that the measure of the third angle is the difference between **180°** and **64°**, or **180 – 64**, which is **116°**. Therefore, the measure of the third angle is **116°**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

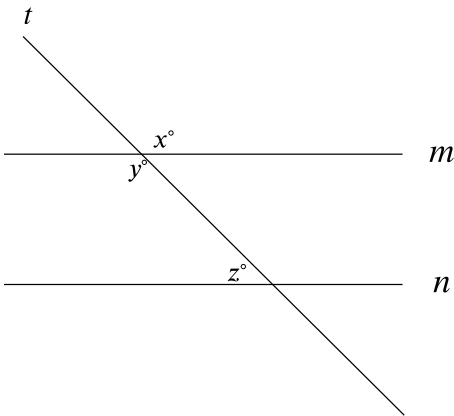
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID e280a2b3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: e280a2b3



Note: Figure not drawn to scale.

In the figure, lines *m* and *n* are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of z ?

- A. 3
- B. 21
- C. 41
- D. 139

ID: e280a2b3 Answer

Correct Answer: C

Rationale

Choice C is correct. Vertical angles, which are angles that are opposite each other when two lines intersect, are congruent. The figure shows that lines *t* and *m* intersect. It follows that the angle with measure x° and the angle with measure y° are vertical angles, so $x = y$. It's given that $x = 6k + 13$ and $y = 8k - 29$. Substituting $6k + 13$ for x and $8k - 29$ for y in the equation $x = y$ yields $6k + 13 = 8k - 29$. Subtracting $6k$ from both sides of this equation yields $13 = 2k - 29$. Adding 29 to both sides of this equation yields $42 = 2k$, or $2k = 42$. Dividing both sides of this equation by 2 yields $k = 21$. It's given that lines *m* and *n* are parallel, and the figure shows that lines *m* and *n* are intersected by a transversal, line *t*. If two parallel lines are intersected by a transversal, then the same-side interior angles are supplementary. It follows that the same-side interior angles with measures y° and z° are supplementary, so $y + z = 180$. Substituting $8k - 29$ for y in this equation yields $8k - 29 + z = 180$. Substituting 21 for k in this equation yields $8(21) - 29 + z = 180$, or $139 + z = 180$. Subtracting 139 from both sides of this equation yields $z = 41$. Therefore, the value of z is 41 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of k , not z .

Choice D is incorrect. This is the value of x or y , not z .

Question Difficulty: Medium

Question ID 279d49a6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 279d49a6

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 6.

	Perimeter
Triangle TUV	50
Triangle XYZ	150

What is the length of $\overline{XY}$?

- A. 2
- B. 6
- C. 18
- D. 56

ID: 279d49a6 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that triangle TUV is similar to triangle XYZ , and $\overline{TU}$ corresponds to $\overline{XY}$. If two triangles are similar, then the ratio of their perimeters is equal to the ratio of their corresponding sides. It's given that the perimeter of triangle TUV is 50, the perimeter of triangle XYZ is 150, and the length of $\overline{TU}$ is 6. Let n represent the length of $\overline{XY}$. It follows that $\frac{50}{150} = \frac{6}{n}$, or $\frac{1}{3} = \frac{6}{n}$. Multiplying each side of this equation by n yields $\frac{n}{3} = 6$. Multiplying each side of this equation by 3 yields $n = 18$. Therefore, the length of $\overline{XY}$ is 18.

Choice A is incorrect. This is the solution to $\frac{3}{1} = \frac{6}{n}$, not $\frac{1}{3} = \frac{6}{n}$.

Choice B is incorrect. This is the length of $\overline{TU}$, not $\overline{XY}$.

Choice D is incorrect. This is the sum of the length of $\overline{TU}$ and the perimeter of triangle TUV , not the length of $\overline{XY}$.

Question Difficulty: Medium

Question ID 92ed0dbe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 92ed0dbe

A right prism has a square base with side lengths of **10** inches. The height of the prism is **58** inches. What is the volume, in cubic inches, of the prism?

ID: 92ed0dbe Answer

Correct Answer: 5800

Rationale

The correct answer is **5,800**. The volume, V , of a right prism with a square base is given by the formula $V = s^2h$, where s is the side length of the square base and h is the height of the prism. It's given that a right prism has a square base with side lengths of **10** inches. It's also given that the height of the prism is **58** inches. Substituting **10** for s and **58** for h in the formula $V = s^2h$ yields $V = (10)^2(58)$, or $V = 5,800$. Therefore, the volume, in cubic inches, of the prism is **5,800**.

Question Difficulty: Hard

Question ID a1eee276

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a1eee276

The area of a triangle is **36** square units. If the height of this triangle is **2** units, what is the length, in units, of the base of this triangle?

ID: a1eee276 Answer

Correct Answer: 36

Rationale

The correct answer is **36**. The area, ***A***, of a triangle is given by the formula **$A = \frac{1}{2}bh$** , where ***b*** represents the length of the base of the triangle and ***h*** represents its height. It's given that the triangle has an area of **36** square units and a height of **2** units. Substituting **36** for ***A*** and **2** for ***h*** in the formula **$A = \frac{1}{2}bh$** yields **$36 = \frac{1}{2}b(2)$** , or **$36 = b$** . Thus, the length, in units, of the base of this triangle is **36**.

Question Difficulty: Medium

Question ID a52db6d0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a52db6d0

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

ID: a52db6d0 Answer

Correct Answer: C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 meters. Therefore, $h = 6$. It's also given that the right circular cylinder has a base with a circumference of 20π meters. The circumference, C , of a circle is given by $C = 2\pi r$, where r is the radius of the circle. Substituting 20π for C in the formula $C = 2\pi r$ yields $20\pi = 2\pi r$. Dividing each side of this equation by 2π yields $10 = r$. Substituting 10 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi(10)^2(6)$, or $V = 600\pi$. Therefore, the volume, in cubic meters, of the cylinder is 600π .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the lateral surface area, not the volume, of the cylinder.

Choice D is incorrect. This is the result of using the diameter, not the radius, for the value of r in the formula $V = \pi r^2 h$.

Question Difficulty: Hard

Question ID ac0a4d27

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: ac0a4d27

A rectangle has a length of **13** and a width of **6**. What is the perimeter of the rectangle?

- A. **12**
- B. **26**
- C. **38**
- D. **52**

ID: ac0a4d27 Answer

Correct Answer: C

Rationale

Choice C is correct. The perimeter of a quadrilateral is the sum of the lengths of its four sides. It's given that the rectangle has a length of **13** and a width of **6**. It follows that the rectangle has two sides with length **13** and two sides with length **6**. Therefore, the perimeter of the rectangle is **13 + 13 + 6 + 6**, or **38**.

Choice A is incorrect. This is the sum of the lengths of the two sides with length **6**, not the sum of the lengths of all four sides of the rectangle.

Choice B is incorrect. This is the sum of the lengths of the two sides with length **13**, not the sum of the lengths of all four sides of the rectangle.

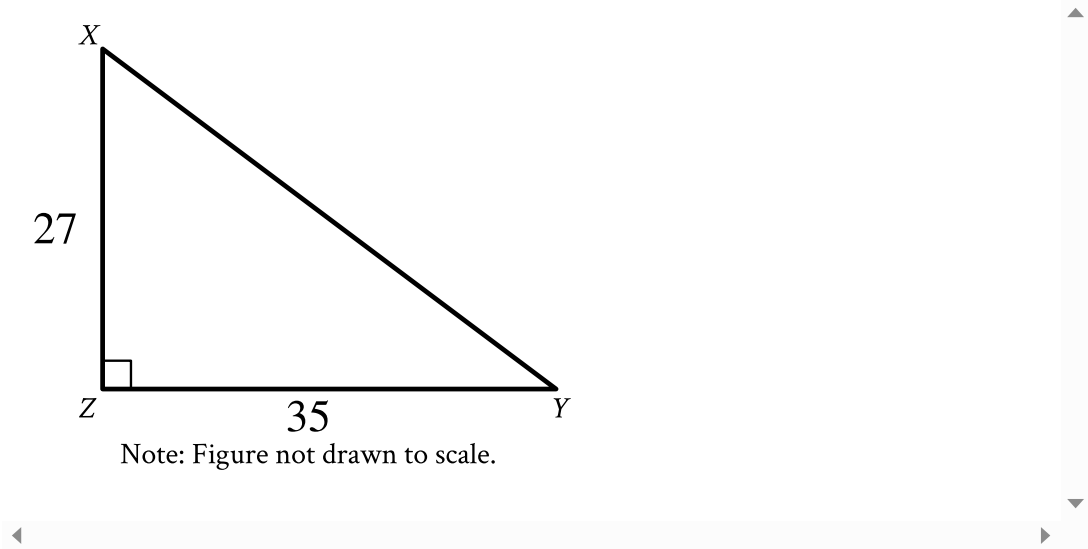
Choice D is incorrect. This is the perimeter of a rectangle that has four sides with length **13**, not two sides with length **13** and two sides with length **6**.

Question Difficulty: Easy

Question ID c20d4271

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: c20d4271



Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

- A. $\tan X$
- B. $\tan Y$
- C. $\cos X$
- D. $\cos Y$

ID: c20d4271 Answer

Correct Answer: D

Rationale

Choice D is correct. The sine of an angle is equal to the cosine of its complementary angle. In the triangle shown, angle Z is a right angle; thus, angles X and Y are complementary angles. Therefore, $\cos Y$ has the same value as $\sin X$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 2095ae72

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 2095ae72

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

- A. 20
- B. $\frac{20\sqrt{2}}{2}$
- C. $20\sqrt{2}$
- D. 40

ID: 2095ae72 Answer

Correct Answer: A

Rationale

Choice A is correct. When a square is inscribed in a circle, a diagonal of the square is a diameter of the circle. It's given that a square is inscribed in a circle and the length of a radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. Therefore, the length of a diameter of the circle is $2\left(\frac{20\sqrt{2}}{2}\right)$ inches, or $20\sqrt{2}$ inches. It follows that the length of a diagonal of the square is $20\sqrt{2}$ inches. A diagonal of a square separates the square into two right triangles in which the legs are the sides of the square and the hypotenuse is a diagonal. Since a square has 4 congruent sides, each of these two right triangles has congruent legs and a hypotenuse of length $20\sqrt{2}$ inches. Since each of these two right triangles has congruent legs, they are both 45-45-90 triangles. In a 45-45-90 triangle, the length of the hypotenuse is $\sqrt{2}$ times the length of a leg. Let s represent the length of a leg of one of these 45-45-90 triangles. It follows that $20\sqrt{2} = \sqrt{2}(s)$. Dividing both sides of this equation by $\sqrt{2}$ yields $20 = s$. Therefore, the length of a leg of one of these 45-45-90 triangles is 20 inches. Since the legs of these two 45-45-90 triangles are the sides of the square, it follows that the side length of the square is 20 inches.

Choice B is incorrect. This is the length of a radius, in inches, of the circle.

Choice C is incorrect. This is the length of a diameter, in inches, of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6b397531

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 6b397531

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 69° . What is the measure, in degrees, of angle F ?

ID: 6b397531 Answer

Correct Answer: 21

Rationale

The correct answer is **21**. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is 69° . It follows that the measure of angle D is also 69° . It's given that angle E is a right angle. Therefore, the measure of angle E is 90° . Let x represent the measure, in degrees, of angle F . Since the measures of the angles in a triangle sum to 180° , it follows that $69 + 90 + x = 180$, or $159 + x = 180$. Subtracting **159** from both sides of this equation yields $x = \mathbf{21}$. Therefore, the measure, in degrees, of angle F is **21**.

Question Difficulty: Medium

Question ID 0f2297cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 0f2297cc

In convex pentagon $ABCDE$, segment AB is parallel to segment DE . The measure of angle B is 139 degrees, and the measure of angle D is 174 degrees. What is the measure, in degrees, of angle C ?

ID: 0f2297cc Answer

Correct Answer: 47

Rationale

The correct answer is 47 . It's given that the measure of angle B is 139 degrees. Therefore, the exterior angle formed by extending segment AB at point B has measure $180 - 139$, or 41 , degrees. It's given that segment AB is parallel to segment DE . Extending segment BC at point C and extending segment DE at point D until the two segments intersect results in a transversal that intersects two parallel line segments. One of these intersection points is point B , and let the other intersection point be point X . Since segment AB is parallel to segment DE , alternate interior angles are congruent. Angle CXD and the exterior angle formed by extending segment AB at point B are alternate interior angles. Therefore, the measure of angle CXD is 41 degrees. It's given that the measure of angle D in pentagon $ABCDE$ is 174 degrees. Therefore, angle CDX has measure $180 - 174$, or 6 , degrees. Since angle C in pentagon $ABCDE$ is an exterior angle of triangle CDX , it follows that the measure of angle C is the sum of the measures of angles CDX and CXD . Therefore, the measure, in degrees, of angle C is $6 + 41$, or 47 .

Alternate approach: A line can be created that's perpendicular to segments AB and DE and passes through point C . Extending segments AB and DE at points B and D , respectively, until they intersect this line yields two right triangles. Let these intersection points be point X and point Y , and the two right triangles be triangle BXC and triangle DYC . It's given that the measure of angle B is 139 degrees. Therefore, angle CBX has measure $180 - 139$, or 41 , degrees. Since the measure of angle CBX is 41 degrees and the measure of angle BXC is 90 degrees, it follows that the measure of angle XCB is $180 - 90 - 41$, or 49 , degrees. It's given that the measure of angle D is 174 degrees. Therefore, angle YDC has measure $180 - 174$, or 6 , degrees. Since the measure of angle YDC is 6 degrees and the measure of angle CYD is 90 degrees, it follows that the measure of angle DCY is $180 - 90 - 6$, or 84 , degrees. Since angles XCB , DCY , and angle C in pentagon $ABCDE$ form segment XY , it follows that the sum of the measures of those angles is 180 degrees. Therefore, the measure, in degrees, of angle C is $180 - 49 - 84$, or 47 .

Question Difficulty: Hard

Question ID 1a5eb1b1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 1a5eb1b1

The side length of a square is **55 centimeters (cm)**. What is the area, **in cm²**, of the square?

- A. 110
- B. 220
- C. 3,025
- D. 12,100

ID: 1a5eb1b1 Answer

Correct Answer: C

Rationale

Choice C is correct. The area **A , in square centimeters (cm²)**, of a square with side length **s , in cm**, is given by the formula **$A = s^2$** . It's given that the square has a side length of **55 cm**. Substituting **55** for **s** in the formula **$A = s^2$** yields **$A = 55^2$** , or **$A = 3,025$** . Therefore, the area, **in cm²**, of the square is **3,025**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, **in cm**, of the square, not its area, **in cm²**.

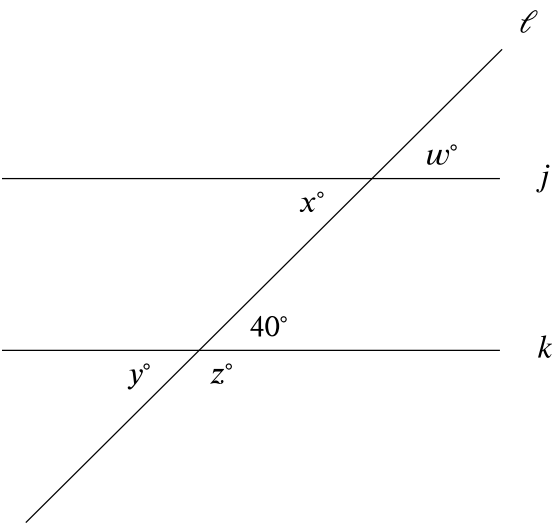
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 28788ce0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 28788ce0



Note: Figure not drawn to scale.

In the figure shown, line ℓ intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- A. $w = 40$
- B. $x = 140$
- C. $y = 40$
- D. $z = 140$

ID: 28788ce0 Answer

Correct Answer: A

Rationale

Choice A is correct. In the figure shown, lines j and k are parallel if and only if a pair of corresponding angles are congruent. It's given that one angle has a measure of 40° and that the corresponding angle has a measure of w° . Therefore, $w = 40$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring x° and the angle measuring 40° are alternate interior angles. Thus, if lines j and k are parallel, x is equal to 40 , not 140 .

Choice C is incorrect. The angle measuring y° and the angle measuring 40° are vertical angles. Thus, $y = 40$, whether lines j and k are parallel or not.

Choice D is incorrect. The angle measuring z° is supplementary to the angle measuring 40° . Thus, $z = 180 - 40$, or $z = 140$, whether lines j and k are parallel or not.

Question ID 965a3451

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 965a3451

Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is 648 square inches, and the area of rectangle $EFGH$ is 72 square inches. The length of the longest side of rectangle $ABCD$ is 36 inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

- A. 4
- B. 9
- C. 12
- D. 36

ID: 965a3451 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that rectangle $ABCD$ is similar to rectangle $EFGH$. Therefore, if the length of each side of rectangle $ABCD$ is k times the length of the corresponding side of rectangle $EFGH$, then the area of rectangle $ABCD$ is k^2 times the area of rectangle $EFGH$. It's given that the area of rectangle $ABCD$ is 648 square inches and the area of rectangle $EFGH$ is 72 square inches. It follows that $k^2 = \frac{648}{72}$, or $k^2 = 9$. Taking the square root of each side of this equation yields $k = \sqrt{9}$, or $k = 3$. It follows that the length of each side of rectangle $ABCD$ is 3 times the length of the corresponding side of rectangle $EFGH$. It's given that the length of the longest side of rectangle $ABCD$ is 36 inches. Therefore, 36 inches is 3 times the length of the longest side of rectangle $EFGH$, and the longest side of rectangle $EFGH$ is equal to $\frac{36}{3}$, or 12, inches.

Choice A is incorrect. This is the length, in inches, of the longest side of a rectangle with side lengths that are $\frac{1}{9}$ the corresponding side lengths of rectangle $ABCD$, rather than a rectangle with an area that is $\frac{1}{9}$ the area of rectangle $ABCD$.

Choice B is incorrect. This is the factor by which the area of rectangle $ABCD$ is larger than the area of rectangle $EFGH$, not the length, in inches, of the longest side of rectangle $EFGH$.

Choice D is incorrect. This is the length, in inches, of the longest side of rectangle $ABCD$, not rectangle $EFGH$.

Question Difficulty: Hard

Question ID a3c7621f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: a3c7621f

The area of a rectangle is **57** square inches. The length of the longest side of the rectangle is **19** inches. What is the length, in inches, of the shortest side of this rectangle?

ID: a3c7621f Answer

Correct Answer: 3

Rationale

The correct answer is **3**. The area of a rectangle can be calculated by multiplying the length of its longest side by the length of its shortest side. It's given that the area of the rectangle is **57** square inches and the length of the longest side of the rectangle is **19** inches. Let x represent the length, in inches, of the shortest side of this rectangle. It follows that $57 = 19x$. Dividing both sides of this equation by **19** yields $3 = x$. Therefore, the length, in inches, of the shortest side of the rectangle is **3**.

Question Difficulty: Medium

Question ID 176eea04

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 176eea04

The area of a square is **64** square inches. What is the side length, in inches, of this square?

- A. 8
- B. 16
- C. **64**
- D. 128

ID: 176eea04 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the area of a square is **64** square inches. The area **A**, in square inches, of a square is given by the formula $A = s^2$, where **s** is the side length, in inches, of the square. Substituting **64** for **A** in this formula yields $64 = s^2$. Taking the positive square root of both sides of this equation yields $8 = s$. Thus, the side length, in inches, of this square is **8**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the area, in square inches, of the square, not the side length, in inches, of the square.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 361dc27b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 361dc27b



Rectangle $ABCD$ shown has a length of **7** units and a width of **2** units. What is the area, in square units, of rectangle $ABCD$?

- A. **5**
- B. **9**
- C. **14**
- D. **28**

ID: 361dc27b Answer

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a rectangle is given by the formula $A = \ell w$, where ℓ is the length and w is the width of the rectangle. It's given that rectangle $ABCD$ shown has a length of **7** units and a width of **2** units. Substituting **7** for ℓ and **2** for w in the formula $A = \ell w$ yields $A = (7)(2)$, or $A = 14$. Therefore, the area, in square units, of rectangle $ABCD$ is **14**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

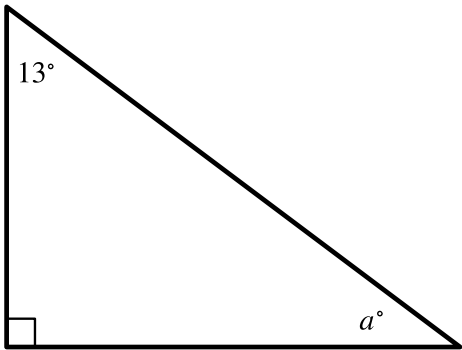
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 33cb02bf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 33cb02bf



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of a ?

- A. 13
- B. 77
- C. 90
- D. 103

ID: 33cb02bf Answer

Correct Answer: B

Rationale

Choice B is correct. The triangle shown is a right triangle, where the interior angle shown with a right angle symbol has a measure of 90° . It's shown that the other two interior angles measure 13° and a° . The sum of the measures of the interior angles of a triangle is 180° ; therefore, $90 + 13 + a = 180$. Combining like terms on the left-hand side of this equation yields $103 + a = 180$. Subtracting 103 from both sides of this equation yields $a = 77$.

Choice A is incorrect. This is the measure, in degrees, of the other acute interior angle of the right triangle, not the value of a .

Choice C is incorrect. This is the measure, in degrees, of the right angle of the right triangle, not the value of a .

Choice D is incorrect. This is the sum of the measures, in degrees, of the other two interior angles of the right triangle, not the value of a .

Question Difficulty: Easy

Question ID cdd40566

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: cdd40566

Triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z respectively. The length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC . The measure of side AB is 16 . What is the measure of side XY ?

- A. 14
- B. 16
- C. 18
- D. 32

ID: cdd40566 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z , respectively. Therefore, side AB corresponds to side XY . Since the length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC , it follows that the measure of side XY is 2 times the measure of side AB . Thus, since the measure of side AB is 16 , then the measure of side XY is $2(16)$, or 32 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure of side AB , not side XY .

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID c921127e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: c921127e

A circle has a radius of **43** meters. What is the area, in square meters, of the circle?

- A. $\frac{43\pi}{2}$
- B. 43π
- C. 86π
- D. $1,849\pi$

ID: c921127e Answer

Correct Answer: D

Rationale

Choice D is correct. The area, **A**, of a circle is given by the formula $A = \pi r^2$, where **r** is the radius of the circle. It's given that the circle has a radius of **43** meters. Substituting **43** for **r** in the formula $A = \pi r^2$ yields $A = \pi(43)^2$, or $A = 1,849\pi$. Therefore, the area, in square meters, of the circle is **1,849π**.

Choice A is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{\frac{43}{2}}$ meters.

Choice B is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{43}$ meters.

Choice C is incorrect. This is the circumference, in meters, of the circle.

Question Difficulty: Medium

Question ID cf5ba93c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: cf5ba93c

The length of one side of an equilateral triangle is **28** inches. What is the perimeter, in inches, of this triangle?

- A. **28**
- B. **31**
- C. **84**
- D. **112**

ID: cf5ba93c Answer

Correct Answer: C

Rationale

Choice C is correct. The perimeter of a triangle is the sum of the lengths of the three sides of the triangle. It's given that the length of one side of an equilateral triangle is **28** inches. Since the lengths of the three sides of an equilateral triangle are equal, it follows that the perimeter, in inches, of this equilateral triangle is **$28 + 28 + 28$** , or **84**.

Choice A is incorrect. This is the length of one side, not the perimeter, of this equilateral triangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the perimeter of a square, not an equilateral triangle, with side lengths of **28** inches.

Question Difficulty: Easy

Question ID 0ee7cc7c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 0ee7cc7c

A triangular prism has a height of **8 centimeters (cm)** and a volume of **216 cm³**. What is the area, **in cm²**, of the base of the prism? (The volume of a triangular prism is equal to ***Bh***, where ***B*** is the area of the base and ***h*** is the height of the prism.)

ID: 0ee7cc7c Answer

Correct Answer: 27

Rationale

The correct answer is **27**. It's given that a triangular prism has a volume of **216 cubic centimeters (cm³)** and the volume of a triangular prism is equal to ***Bh***, where ***B*** is the area of the base and ***h*** is the height of the prism. Therefore, **216 = *Bh***. It's also given that the triangular prism has a height of **8 cm**. Therefore, ***h* = 8**. Substituting **8** for ***h*** in the equation **216 = *Bh*** yields **216 = *B*(8)**. Dividing both sides of this equation by **8** yields **27 = *B***. Therefore, the area, **in cm²**, of the base of the prism is **27**.

Question Difficulty: Hard

Question ID b64cd60b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: b64cd60b

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A ?

- A. 43
- B. 86
- C. 129
- D. 1,849

ID: b64cd60b Answer

Correct Answer: D

Rationale

Choice D is correct. The area of a circle can be found by using the formula $A = \pi r^2$, where A is the area and r is the radius of the circle. It's given that the radius of circle A is $3n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi(3n)^2$, or $9\pi n^2$. It's also given that the radius of circle B is $129n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi(129n)^2$, or $16,641\pi n^2$. Dividing the area of circle B by the area of circle A gives $\frac{16,641\pi n^2}{9\pi n^2}$, which simplifies to 1,849. Therefore, the area of circle B is 1,849 times the area of circle A .

Choice A is incorrect. This is how many times greater the radius of circle B is than the radius of circle A .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the coefficient on the term that describes the radius of circle B .

Question Difficulty: Hard

Question ID 03dbf471

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 03dbf471

Square A has side lengths that are **246** times the side lengths of square B. The area of square A is **k** times the area of square B. What is the value of **k** ?

- A. **60,516**
- B. **492**
- C. **246**
- D. **123**

ID: 03dbf471 Answer

Correct Answer: A

Rationale

Choice A is correct. The area of a square is **s^2** , where **s** is the side length of the square. Therefore, the area of square B is **b^2** , where **b** is the side length of square B. It's given that square A has side lengths that are **246** times the side lengths of square B. Therefore, the side length of square A can be represented by the expression **$246b$** . It follows that the area of square A is **$(246b)^2$** , or **$60,516b^2$** . It's given that the area of square A is **k** times the area of square B, so **$60,516b^2 = kb^2$** . Therefore, the value of **k** is **60,516**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

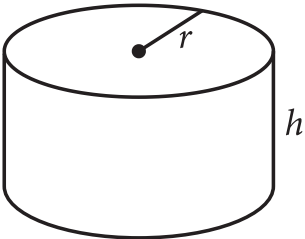
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 27630f60

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 27630f60



The figure shown is a right circular cylinder with a radius of r and height of h . A second right circular cylinder (not shown) has a volume that is **392** times as large as the volume of the cylinder shown. Which of the following could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder?

- A. $R = 8r$ and $H = 7h$
- B. $R = 8r$ and $H = 49h$
- C. $R = 7r$ and $H = 8h$
- D. $R = 49r$ and $H = 8h$

ID: 27630f60 Answer

Correct Answer: C

Rationale

Choice C is correct. The volume of a right circular cylinder is equal to $\pi a^2 b$, where a is the radius of a base of the cylinder and b is the height of the cylinder. It's given that the cylinder shown has a radius of r and a height of h . It follows that the volume of the cylinder shown is equal to $\pi r^2 h$. It's given that the second right circular cylinder has a radius of R and a height of H . It follows that the volume of the second cylinder is equal to $\pi R^2 H$. Choice C gives $R = 7r$ and $H = 8h$. Substituting $7r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi(7r)^2(8h)$, or $\pi(49r^2)(8h)$, which is equivalent to $\pi(392r^2h)$, or $392(\pi r^2 h)$. This expression is equal to **392** times the volume of the cylinder shown, $\pi r^2 h$. Therefore, $R = 7r$ and $H = 8h$ could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder.

Choice A is incorrect. Substituting $8r$ for R and $7h$ for H in the expression that represents the volume of the second cylinder yields $\pi(8r)^2(7h)$, or $\pi(64r^2)(7h)$, which is equivalent to $\pi(448r^2h)$, or $448(\pi r^2 h)$. This expression is equal to **448**, not **392**, times the volume of the cylinder shown.

Choice B is incorrect. Substituting $8r$ for R and $49h$ for H in the expression that represents the volume of the second cylinder yields $\pi(8r)^2(49h)$, or $\pi(64r^2)(49h)$, which is equivalent to $\pi(3,136r^2h)$, or $3,136(\pi r^2 h)$. This expression is equal to **3,136**, not **392**, times the volume of the cylinder shown.

Choice D is incorrect. Substituting $49r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi(49r)^2(8h)$, or $\pi(2,401r^2)(8h)$, which is equivalent to $\pi(19,208r^2h)$, or $19,208(\pi r^2 h)$. This expression is equal to **19,208**, not **392**, times the volume of the cylinder shown.

Question Difficulty: Hard

Question ID 35e9579e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 35e9579e

A rectangle has a length of **3** units and a width of **39** units. Which expression gives the area, in square units, of this rectangle?

- A. $2(3 + 39)$
- B. $2(3 \cdot 39)$
- C. $3 + 39$
- D. $3 \cdot 39$

ID: 35e9579e Answer

Correct Answer: D

Rationale

Choice D is correct. The area of a rectangle is given by lw , where l is the length of the rectangle and w is the width of the rectangle. It's given that a rectangle has a length of **3** units and a width of **39** units. It follows that the area of the rectangle is $3 \cdot 39$ square units. Therefore, the expression that gives the area, in square units, of this rectangle, is $3 \cdot 39$.

Choice A is incorrect. This expression gives the perimeter, in units, of this rectangle.

Choice B is incorrect and may result from conceptual errors.

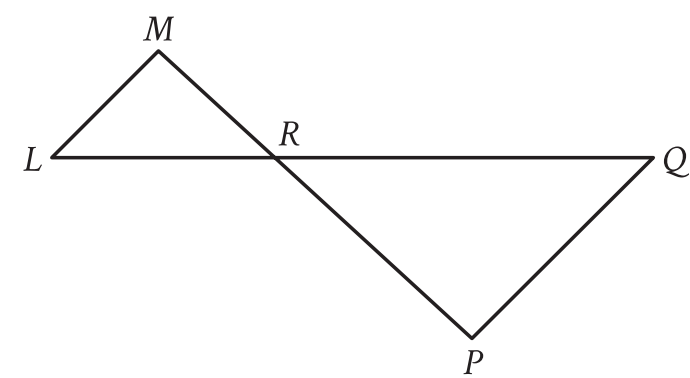
Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID cd9eb419

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: cd9eb419



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

- A. $\frac{576}{49}$
- B. $\frac{144}{7}$
- C. 63
- D. $\frac{441}{4}$

ID: cd9eb419 Answer

Correct Answer: D

Rationale

Choice D is correct. The figure shows that $\angle MRL$ and $\angle PRQ$ are vertical angles. Since vertical angles are congruent, $\angle MRL$ and $\angle PRQ$ are congruent. It's given that $\overline{LM}$ is parallel to $\overline{PQ}$. The figure also shows that $\overline{LQ}$ intersects $\overline{LM}$ and $\overline{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, $\angle MLR$ and $\angle PQR$ are congruent. Since $\triangle LMR$ and $\triangle PQR$ have two pairs of congruent angles, the triangles are similar, where $\overline{MR}$ in $\triangle LMR$ corresponds to $\overline{PR}$ in $\triangle PQR$. Since the areas of similar triangles are proportional to the squares of the lengths of corresponding sides, it follows that $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = \left(\frac{RP}{MR}\right)^2$. It's given that the lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively, and the area of $\triangle LMR$ is 36 square units. Substituting 14 for RP , 8 for MR , and 36 for **area of $\triangle LMR$** in the equation $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = \left(\frac{RP}{MR}\right)^2$ yields $\frac{\text{area of } \triangle PQR}{36} = \left(\frac{14}{8}\right)^2$, which is equivalent to $\frac{\text{area of } \triangle PQR}{36} = \left(\frac{7}{4}\right)^2$, or $\frac{\text{area of } \triangle PQR}{36} = \frac{49}{16}$. Multiplying each side of this equation by 36 yields **area of $\triangle PQR$** = $36\left(\frac{49}{16}\right)$, or **area of $\triangle PQR$** = $\frac{441}{4}$. Therefore, the area of $\triangle PQR$ is $\frac{441}{4}$ square units.

Choice A is incorrect. This is the result of multiplying 36 by $\left(\frac{8}{14}\right)^2$ instead of by $\left(\frac{14}{8}\right)^2$.

Choice B is incorrect. This is the result of multiplying 36 by $\frac{8}{14}$ instead of by $\left(\frac{14}{8}\right)^2$.

Choice C is incorrect. This is the result of multiplying 36 by $\frac{14}{8}$ instead of by $\left(\frac{14}{8}\right)^2$.

Question ID 69e2199f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 69e2199f

Rectangle $ABCD$ has an area of **280** square centimeters. The length of the rectangle is **10** centimeters. What is the width, in centimeters, of the rectangle?

- A. **10**
- B. **28**
- C. **76**
- D. **140**

ID: 69e2199f Answer

Correct Answer: B

Rationale

Choice B is correct. The area A of a rectangle is the product of its length l and its width w . It follows that $A = lw$. It's given that rectangle $ABCD$ has an area of **280** square centimeters and the length of the rectangle is **10** centimeters. Substituting **280** for A and **10** for l in the equation $A = lw$ yields $280 = 10w$. Dividing both sides of this equation by **10** yields $28 = w$. Therefore, the width, in centimeters, of the rectangle is **28**.

Choice A is incorrect. This is the length, in centimeters, not the width, in centimeters, of the rectangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is half the area, in square centimeters, not the width, in centimeters, of the rectangle.

Question Difficulty: Easy

Question ID fb30fb60

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: fb30fb60

Triangle R has an area of **80 square centimeters (cm^2)**. Square S has side lengths of **4 cm**. What is the total area of triangle R and square S, **in cm^2** ?

- A. **42**
- B. **44**
- C. **84**
- D. **96**

ID: fb30fb60 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that triangle R has an area of **80 cm^2** . The area of a square is s^2 , where s is the side length of the square. It's given that square S has side lengths of **4 cm**. It follows that the area, in **cm^2** , of square S is **4²**, or **16**. Therefore, the total area, in **cm^2** , of triangle R and square S is **80 + 16**, or **96**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 4281df6c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 4281df6c

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of $\overline{XY}$?

- A. 2
- B. 18
- C. 55
- D. 162

ID: 4281df6c Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that triangle XYZ is similar to triangle TUV . Therefore, each side of triangle XYZ is k times its corresponding side of triangle TUV , where k is a constant. It follows that the perimeter of triangle XYZ is k times the perimeter of triangle TUV . It's also given that $\overline{TU}$ corresponds to $\overline{XY}$ and the length of $\overline{TU}$ is 18. Let x represent the length of $\overline{XY}$. It follows that $x = 18k$. The table shows that the perimeters of triangles TUV and XYZ are 37 and 333, respectively. It follows that $333 = 37k$, or $9 = k$. Substituting 9 for k in the equation $x = 18k$ yields $x = (18)(9)$, or $x = 162$. Therefore, the length of $\overline{XY}$ is 162.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the length of $\overline{TU}$, not the length of $\overline{XY}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 60c17d6e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 60c17d6e

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is **10** feet tall and has a shadow that is **5** feet long. At the same time, the shadow of the other tree is **2** feet long. How tall, in feet, is the other tree?

- A. **3**
- B. **4**
- C. **8**
- D. **27**

ID: 60c17d6e Answer

Correct Answer: B

Rationale

Choice B is correct. Each tree and its shadow can be modeled using a right triangle, where the height of the tree and the length of its shadow are the legs of the triangle. At a given point in time, the right triangles formed by two nearby trees and their respective shadows will be similar. Therefore, if the height of the other tree is x , in feet, the value of x can be calculated by solving the proportional relationship $\frac{10 \text{ feet tall}}{5 \text{ feet long}} = \frac{x \text{ feet tall}}{2 \text{ feet long}}$. This equation is equivalent to $\frac{10}{5} = \frac{x}{2}$, or $2 = \frac{x}{2}$. Multiplying each side of the equation $2 = \frac{x}{2}$ by **2** yields **4** = x . Therefore, the other tree is **4 feet** tall.

Choice A is incorrect and may result from calculating the difference between the lengths of the shadows, rather than the height of the other tree.

Choice C is incorrect and may result from calculating the difference between the height of the **10**-foot-tall tree and the length of the shadow of the other tree, rather than calculating the height of the other tree.

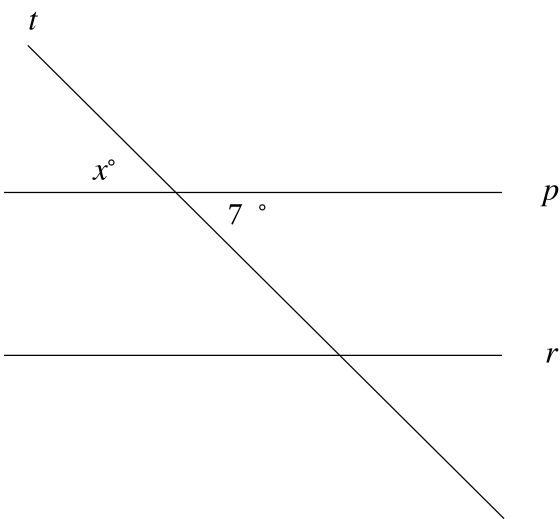
Choice D is incorrect and may result from a conceptual or calculation error.

Question Difficulty: Medium

Question ID c42eb820

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c42eb820



Note: . igure not drawn to scale.

In the figure, line *p* is parallel to line *r*, and line *t* intersects both lines. What is the value of *x*?

- A. 36
- B. 72
- C. 180
- D. 252

ID: c42eb820 Answer

Correct Answer: B

Rationale

Choice B is correct. In the figure shown, the angles with measures *x*° and 72° are vertical angles. Vertical angles have equal measure, so the two marked angles have equal measure. Therefore, the value of *x* is 72.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

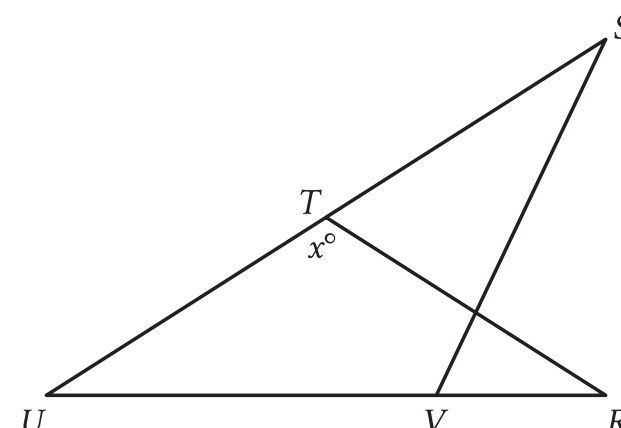
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 5a2734d6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5a2734d6



Note: Figure not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is 29° , and the measure of angle RVS is 41° . What is the value of x ?

ID: 5a2734d6 Answer

Correct Answer: 156

Rationale

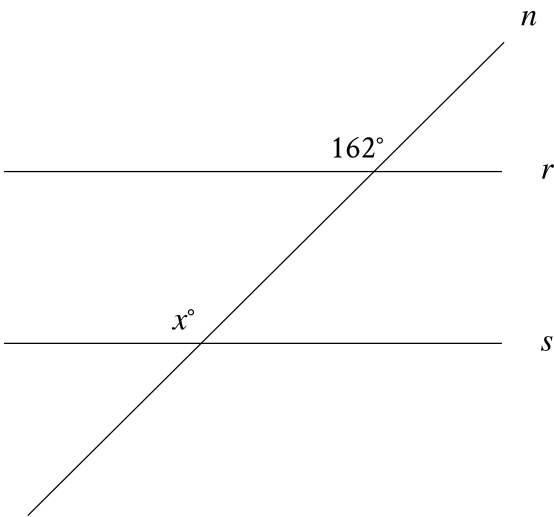
The correct answer is **156**. In the figure shown, the sum of the measures of angle UVS and angle RVS is 180° . It's given that the measure of angle RVS is 41° . Therefore, the measure of angle UVS is $(180 - 41)^\circ$, or 139° . The sum of the measures of the interior angles of a triangle is 180° . In triangle UVS , the measure of angle UVS is 139° and it's given that the measure of angle VST is 29° . Thus, the measure of angle VUS is $(180 - 139 - 29)^\circ$, or 12° . It's given that $RT = TU$. Therefore, triangle TUR is an isosceles triangle and the measure of VUS is equal to the measure of angle TRU . In triangle TUR , the measure of angle VUS is 12° and the measure of angle TRU is 12° . Thus, the measure of angle UTR is $(180 - 12 - 12)^\circ$, or 156° . The figure shows that the measure of angle UTR is x° , so the value of x is **156**.

Question Difficulty: Hard

Question ID c9eaf5d3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c9eaf5d3



Note: Figure not drawn to scale.

In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

ID: c9eaf5d3 Answer

Correct Answer: 162

Rationale

The correct answer is **162**. It's given that line r is parallel to line s . Since line n intersects both lines r and s , it's a transversal. The angles in the figure marked as 162° and x° are on the same side of the transversal, where one is an interior angle with line s as a side, and the other is an exterior angle with line r as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that the value of x is **162**.

Question Difficulty: Easy

Question ID 49570665

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 49570665

A right triangle has legs with lengths of **24** centimeters and **21** centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

ID: 49570665 Answer

Correct Answer: 113

Rationale

The correct answer is **113**. It's given that the legs of a right triangle have lengths **24** centimeters and **21** centimeters. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. It follows that if h represents the length, in centimeters, of the hypotenuse of the right triangle, $h^2 = 24^2 + 21^2$. This equation is equivalent to $h^2 = 1,017$. Taking the square root of each side of this equation yields $h = \sqrt{1,017}$. This equation can be rewritten as $h = \sqrt{9 \cdot 113}$, or $h = \sqrt{9} \cdot \sqrt{113}$. This equation is equivalent to $h = 3\sqrt{113}$. It's given that the length of the triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$. It follows that the value of d is **113**.

Question Difficulty: Hard

Question ID a3df7802

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a3df7802

In triangle ABC , $AB = 4,680$ millimeters (mm) and $BC = 4,680$ mm. Which statement is sufficient to prove that triangle ABC is equilateral?

- A. $AC = 4,680$ mm
- B. $AC = 468$ mm
- C. $AC = 46.8$ mm
- D. $AC = 4.68$ mm

ID: a3df7802 Answer

Correct Answer: A

Rationale

Choice A is correct. In an equilateral triangle, all three sides have the same length. It's given that in triangle ABC , $AB = 4,680$ mm and $BC = 4,680$ mm. Therefore, if $AC = 4,680$ mm, then all three sides of triangle ABC have the same length, so triangle ABC is equilateral. Therefore, $AC = 4,680$ mm is sufficient to prove that triangle ABC is equilateral.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 5e651b6d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 5e651b6d

The area of a rectangle is **630** square inches. The length of the rectangle is **70** inches. What is the width, in inches, of this rectangle?

- A. **9**
- B. **70**
- C. **315**
- D. **560**

ID: 5e651b6d Answer

Correct Answer: A

Rationale

Choice A is correct. The area **A** , in square inches, of a rectangle is the product of its length **ℓ** , in inches, and its width **w** , in inches; thus, **$A = \ell w$** . It's given that the area of a rectangle is **630** square inches and the length of the rectangle is **70** inches. Substituting **630** for **A** and **70** for **ℓ** in the equation **$A = \ell w$** yields **$630 = 70w$** . Dividing both sides of this equation by **70** yields **$9 = w$** . Therefore, the width, in inches, of this rectangle is **9**.

Choice B is incorrect. This is the length, not the width, in inches, of the rectangle.

Choice C is incorrect. This is half the area, in square inches, not the width, in inches, of the rectangle.

Choice D is incorrect. This is the difference between the area, in square inches, and the length, in inches, of the rectangle, not the width, in inches, of the rectangle.

Question Difficulty: Easy

Question ID 3e339853

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 3e339853

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

ID: 3e339853 Answer

Correct Answer: 1681

Rationale

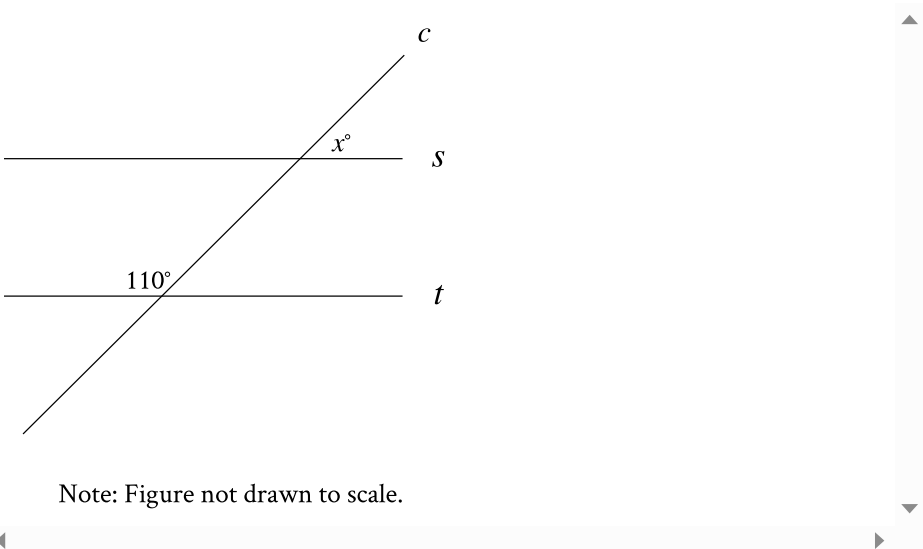
The correct answer is **1,681**. It's given that trapezoid A and trapezoid B are similar. When two trapezoids are similar, the scale factor between their corresponding side lengths is k , and the scale factor between their areas is k^2 . It's given that the length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B, so the scale factor between their corresponding side lengths is **41**. It follows that the scale factor between the areas is **41²**, or **1,681**. Therefore, the area of trapezoid A is **1,681** times as large as the area of trapezoid B.

Question Difficulty: Hard

Question ID c3cae165

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c3cae165



In the figure shown, line c intersects parallel lines s and t . What is the value of x ?

ID: c3cae165 Answer

Correct Answer: 70

Rationale

The correct answer is **70**. Based on the figure, the angle with measure 110° and the angle vertical to the angle with measure x° are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure x° also has measure x° . It's given that lines s and t are parallel. Therefore, same side interior angles between lines s and t are supplementary. It follows that $x + 110 = 180$. Subtracting 110 from both sides of this equation yields $x = 70$.

Question Difficulty: Medium

Question ID fb51f672

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: fb51f672

What is the area, in square centimeters, of a rectangle with a length of **34 centimeters (cm)** and a width of **29 cm**?

ID: fb51f672 Answer

Correct Answer: 986

Rationale

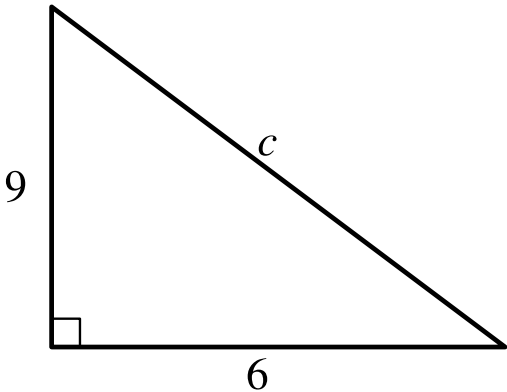
The correct answer is **986**. The area, A , of a rectangle is given by $A = \ell w$, where ℓ is the length of the rectangle and w is its width. It's given that the length of the rectangle is **34 centimeters (cm)** and the width is **29 cm**. Substituting **34** for ℓ and **29** for w in the equation $A = \ell w$ yields $A = (34)(29)$, or $A = 986$. Therefore, the area, in square centimeters, of this rectangle is **986**.

Question Difficulty: Easy

Question ID 0e997686

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 0e997686



Note: Figure not drawn to scale.

In the right triangle shown, which of the following is closest to the value of c ?

- A. 7.5
- B. 10.8
- C. 15
- D. 58.5

ID: 0e997686 Answer

Correct Answer: B

Rationale

Choice B is correct. By the Pythagorean theorem, if a right triangle has a hypotenuse with length t and legs with lengths r and s , then $r^2 + s^2 = t^2$. It's given in the right triangle shown that the legs have lengths of 9 and 6 and the hypotenuse has a length of c . Substituting 9 for r , 6 for s , and c for t in $r^2 + s^2 = t^2$ yields $9^2 + 6^2 = c^2$, or $117 = c^2$. Taking the square root of both sides of this equation yields $\pm\sqrt{117} = c$. Since the length of a side of a triangle must be positive, the value of c is $\sqrt{117}$, which is approximately equal to 10.8167. Of the choices, 10.8 is the closest to the value of c .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

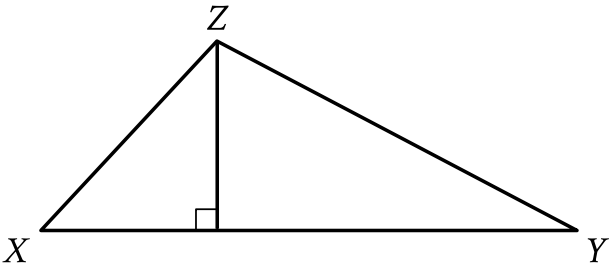
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 1cd5be79

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 1cd5be79



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 54° . The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ ?

- A. 247
- B. 494
- C. $247 \sin 54^\circ$
- D. $494 \sin 54^\circ$

ID: 1cd5be79 Answer

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of a base of the triangle and h is its corresponding height. It's given that the length of $\overline{XY}$ is 26 units. This length represents a base of the triangle, so $b = 26$. Let h represent its corresponding height. It's given that the measure of angle X is 54° and the length of $\overline{XZ}$ is 19 units. It follows that $\sin 54^\circ = \frac{h}{19}$. Multiplying both sides of this equation by 19 yields $19 \sin 54^\circ = h$. Substituting 26 for b and $19 \sin 54^\circ$ for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(26)(19 \sin 54^\circ)$, or $A = 247 \sin 54^\circ$. Therefore, the area, in square units, of triangle XYZ is $247 \sin 54^\circ$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

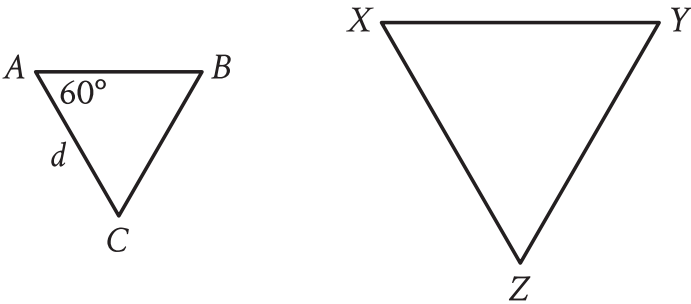
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6d21c3a6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 6d21c3a6



Note: Figures not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 3 to obtain triangle XYZ , where $d = 16$. What is the measure, in degrees, of angle X ?

- A. 20
- B. 57
- C. 60
- D. 63

ID: 6d21c3a6 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that triangle XYZ is obtained by a dilation of triangle ABC . It follows that triangle ABC is similar to triangle XYZ , where A corresponds to X . Since corresponding angles in similar triangles have the same measure and the measure of angle A is 60 degrees, it follows that the measure of angle X is also 60 degrees.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 943f6a0f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 943f6a0f

In triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . In triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

- A. The measure of angle F
- B. The measure of angle T
- C. The measure of angle F and the measure of angle T
- D. No additional information is needed.

ID: 943f6a0f Answer

Correct Answer: D

Rationale

Choice D is correct. When two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. It's given that in triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . It's also given that in triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . It follows that angle D is congruent to angle R and that angle E is congruent to angle S . Therefore, triangle DEF is similar to triangle RST and no additional information is needed.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

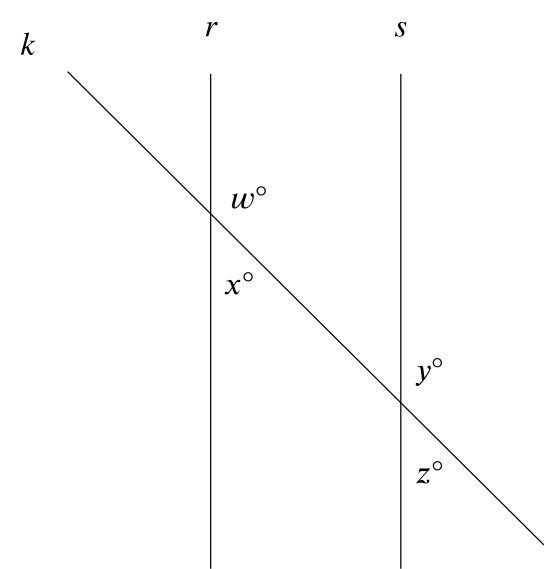
Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID fc81c29e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: fc81c29e



Note: . igure not drawn to scale.

In the figure shown, line k intersects lines r and s . If $w = 147$, which additional piece of information is sufficient to prove that lines r and s are parallel?

- A. $x = 33$
- B. $y = 147$
- C. $w + y = 180$
- D. $y + z = 180$

ID: fc81c29e Answer

Correct Answer: B

Rationale

Choice B is correct. The figure shows transversal k intersecting lines r and s , where the angles labeled w° and y° are two corresponding angles and the angles labeled x° and z° are two corresponding angles. When a transversal intersects two lines, the lines are parallel if and only if two corresponding angles are congruent. It's given that $w = 147$. If $y = 147$, then the two corresponding angles labeled w° and y° are congruent, which is sufficient to prove that lines r and s are parallel.

Choice A is incorrect. The angle labeled x° forms a linear pair with the angle labeled w° , so $x = 180 - 147$, or $x = 33$, regardless of whether lines r and s are parallel.

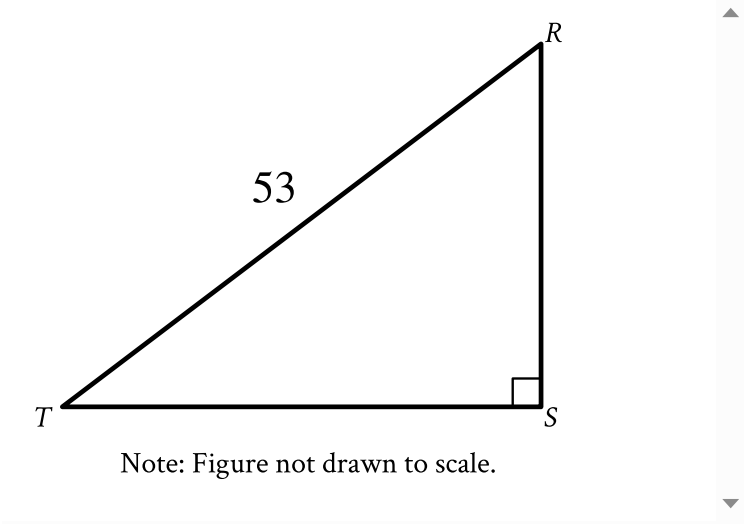
Choice C is incorrect. Substituting 147 for w in this equation yields $147 + y = 180$, or $y = 33$. Since the angle labeled w° and the angle labeled y° are corresponding angles, if w and y aren't equal, then lines r and s can't be parallel.

Choice D is incorrect. The relationship $y + z = 180$ describes a linear pair at line s , which is true regardless of whether lines r and s are parallel.

Question ID 3a0830cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 3a0830cc



In the triangle shown, $RS = \sqrt{105}$. What is the value of $\sin R$?

ID: 3a0830cc Answer

Correct Answer: .9811, 52/53

Rationale

The correct answer is $\frac{52}{53}$. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. The length of the hypotenuse of the right triangle shown is **53**. It's given that $RS = \sqrt{105}$. Therefore, the length of one of the legs of the triangle shown is $\sqrt{105}$. Let x represent TS , the length of the other leg of the triangle shown. Therefore, $53^2 = (\sqrt{105})^2 + x^2$, or $2,809 = 105 + x^2$. Subtracting **105** from both sides of this equation yields $2,704 = x^2$. Taking the positive square root of both sides of this equation yields $52 = x$. Therefore, TS , the length of the other leg of the triangle shown, is **52**. The sine of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the hypotenuse. The length of the leg opposite angle R is **52**, and the length of the hypotenuse is **53**. Therefore, the value of $\sin R$ is $\frac{52}{53}$. Note that 52/53 or .9811 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 31790611

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 31790611

The area of one of the bases of a right circular cylinder is **81** square centimeters, and the volume of the cylinder is **2,187** cubic centimeters. What is the height, in centimeters, of the cylinder?

- A. **27**
- B. **2,025**
- C. **2,106**
- D. **177,147**

ID: 31790611 Answer

Correct Answer: A

Rationale

Choice A is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where πr^2 is the area of one of the bases of the cylinder and h is the height. It's given that the area of one of the bases of the cylinder is **81** square centimeters and that the volume of the cylinder is **2,187** cubic centimeters. Substituting **2,187** for V and **81** for πr^2 in the formula $V = \pi r^2 h$ yields **2,187** = **81** h . Dividing both sides of this equation by **81** yields **27** = h . Therefore, the height, in centimeters, of the cylinder is **27**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 3109ecef

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 3109ecef

A right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. What is the height, in inches, of this pyramid?

- A. 56
- B. 90
- C. 106
- D. 180

ID: 3109ecef Answer

Correct Answer: A

Rationale

Choice A is correct. The total surface area of a right square pyramid is the sum of the combined surface area of its four lateral faces and the area of its square base. It's given that the right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. It follows that the area of the square base is the difference between the total surface area and the combined surface area of the four lateral faces, or **(70,560 – 38,160)** square inches, which is equal to **32,400** square inches. The area, ***B***, of the base of a square pyramid is given by the equation **$B = s^2$** , where ***s*** is the side length of the base. Substituting **32,400** for ***B*** in this equation yields **$32,400 = s^2$** . Taking the positive square root of both sides of this equation yields **180 = *s***. The four lateral faces of a square pyramid are congruent triangles. It follows that the area of each lateral face is $\frac{38,160}{4}$, or **9,540**, square inches. The area, ***A***, of each lateral face of a square pyramid is given by the equation **$A = \frac{1}{2}s\ell$** , where ***s*** is the side length of the pyramid's base and **ℓ** is the pyramid's slant height. Substituting **9,540** for ***A*** and **180** for ***s*** in this equation yields **$9,540 = \frac{1}{2}(180)\ell$** , or **$9,540 = 90\ell$** . Dividing both sides of this equation by **90** yields **106 = ℓ** . In the right square pyramid, a right triangle can be drawn, where the lengths of the legs correspond to the pyramid's height, ***h***, and half the side length of its base, $\frac{1}{2}s$, and the length of the hypotenuse corresponds to the pyramid's slant height, **ℓ** . It follows that the relationship between ***h***, ***s***, and **ℓ** is given by the equation **$h^2 + (\frac{1}{2}s)^2 = \ell^2$** . Substituting **180** for ***s*** and **106** for **ℓ** in this equation yields **$h^2 + ((\frac{1}{2})(180))^2 = (106)^2$** , or **$h^2 + 8,100 = 11,236$** . Subtracting **8,100** from both sides of this equation yields **$h^2 = 3,136$** . Taking the positive square root of both sides of the equation yields ***h* = 56**. Therefore, the height, in inches, of this pyramid is **56**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pyramid's slant height, not its height, in inches.

Choice D is incorrect. This is the side length of the pyramid's base, not its height, in inches.

Question Difficulty: Hard

Question ID 34ea807c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 34ea807c

Square A has side lengths that are **166** times the side lengths of square B. The area of square A is ***k*** times the area of square B. What is the value of ***k***?

ID: 34ea807c Answer

Correct Answer: 27556

Rationale

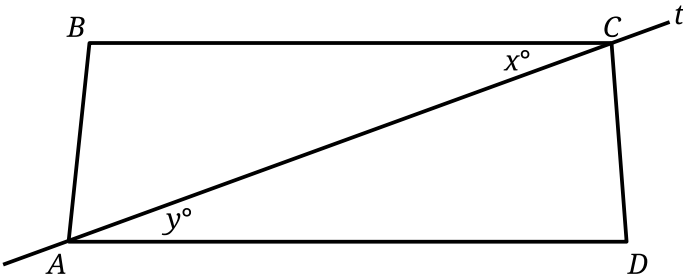
The correct answer is **27,556**. The area of a square is **s^2** , where ***s*** is the side length of the square. Let ***x*** represent the length of each side of square B. Substituting ***x*** for ***s*** in **s^2** yields **x^2** . It follows that the area of square B is **x^2** . It's given that square A has side lengths that are **166** times the side lengths of square B. Since ***x*** represents the length of each side of square B, the length of each side of square A can be represented by the expression **166*x***. It follows that the area of square A is **$(166x)^2$** , or **$27,556x^2$** . It's given that the area of square A is ***k*** times the area of square B. Since the area of square A is equal to **$27,556x^2$** , and the area of square B is equal to **x^2** , an equation representing the given statement is **$27,556x^2 = kx^2$** . Since ***x*** represents the length of each side of square B, the value of ***x*** must be positive. Therefore, the value of **x^2** is also positive, so it does not equal **0**. Dividing by **x^2** on both sides of the equation **$27,556x^2 = kx^2$** yields **$27,556 = k$** . Therefore, the value of ***k*** is **27,556**.

Question Difficulty: Hard

Question ID 52075564

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 52075564



Note: Figure not drawn to scale.

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C . What is the value of y ?

- A. 22
- B. 44
- C. 68
- D. 158

ID: 52075564 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{BC}$ is parallel to $\overline{AD}$. Since line t intersects $\overline{BC}$ and $\overline{AD}$, it's a transversal. The angles in the figure marked as x° and y° are on opposite sides of the transversal and lie between the parallel segments, so they are alternate interior angles. When two parallel segments are intersected by a transversal, alternate interior angles have equal measure. It's also given that $x = 22$. Therefore, the value of y is **22**.

Choice B is incorrect. This is the value of $2x$, not the value of y .

Choice C is incorrect. This is the complement of **22**, not the value of y .

Choice D is incorrect. This is the supplement of **22**, not the value of y .

Question Difficulty: Easy

Question ID 070511bc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 070511bc

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is **972 cubic centimeters (cm^3)**, and the volume of prism Q is **36 cm^3** . One edge of prism P has a length of **12 cm**. What is the length, in **cm**, of the corresponding edge of prism Q?

ID: 070511bc Answer

Correct Answer: 4

Rationale

The correct answer is **4**. It's given that right rectangular prism P is similar to right rectangular prism Q. Therefore, there is a positive constant **k** such that the length of each edge of prism P is **k** times the length of the corresponding edge of prism Q and the volume of prism P is **k^3** times the volume of prism Q. It's given that the volume of prism P is **972 cm^3** and the volume of prism Q is **36 cm^3** . It follows that **$972 = 36k^3$** . Dividing both sides of this equation by **36** yields **$\frac{972}{36} = k^3$** , or **$k^3 = 27$** . Taking the cube root of both sides of this equation yields **$k = 3$** . It follows that the length of each edge of prism P is **3** times the length of the corresponding edge of prism Q. It's given that one edge of prism P has a length of **12 cm**. It follows that **12 cm** is **3** times the length of the corresponding edge of prism Q. Thus, the length, in **cm**, of the edge of prism Q corresponding to the edge of prism P with a length of **12 cm** is **$\frac{12}{3}$** , or **4**.

Question Difficulty: Hard

Question ID 72ea5f55

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 72ea5f55

Each base of a right rectangular prism has a length of **19** inches and a width of **8** inches. The prism has a volume of **2,736** cubic inches. What is the height, in inches, of the prism?

- A. **18**
- B. **27**
- C. **144**
- D. **152**

ID: 72ea5f55 Answer

Correct Answer: A

Rationale

Choice A is correct. The volume, V , of a rectangular prism is given by the formula $V = \ell wh$, where ℓ is the length of the base, w is the width of the base, and h is the height of the prism. It's given that each base of a right rectangular prism has a length of **19** inches and a width of **8** inches, and that the prism has a volume of **2,736** cubic inches. Substituting **19** for ℓ , **8** for w , and **2,736** for V in the formula $V = \ell wh$ gives $2,736 = (19)(8)(h)$, or $2,736 = 152h$. Dividing each side of this equation by **152** yields $18 = h$. Therefore, the height, in inches, of the prism is **18**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

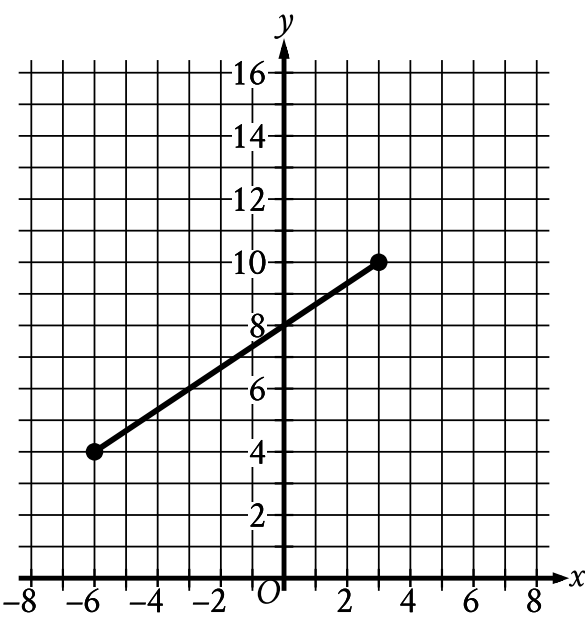
Choice D is incorrect. This is the area, in square inches, of the base of the prism, not the height, in inches, of the prism.

Question Difficulty: Medium

Question ID 716b70c1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 716b70c1



The line segment shown in the xy -plane represents one of the legs of a right triangle. The area of this triangle is $36\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

- A. 12
- B. 24
- C. $3\sqrt{13}$
- D. $18\sqrt{13}$

ID: 716b70c1 Answer

Correct Answer: B

Rationale

Choice B is correct. The length of a segment in the xy -plane can be found using the distance formula, $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, where (x_1, y_1) and (x_2, y_2) are the endpoints of the segment. The segment shown has endpoints at $(-6, 4)$ and $(3, 10)$. Substituting $(-6, 4)$ and $(3, 10)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the distance formula yields $\sqrt{(3 - (-6))^2 + (10 - 4)^2}$, or $\sqrt{9^2 + 6^2}$, which is equivalent to $\sqrt{81 + 36}$, or $\sqrt{117}$. Let x represent the length, in units, of the other leg of this triangle. The area, A , of a right triangle can be calculated using the formula $A = \frac{1}{2}bh$, where b and h are the lengths of the legs of the triangle. It's given that the area of the triangle is $36\sqrt{13}$ square units. Substituting $36\sqrt{13}$ for A , $\sqrt{117}$ for b , and x for h in the formula $A = \frac{1}{2}bh$ yields $36\sqrt{13} = \frac{1}{2}(\sqrt{117})(x)$. Multiplying both sides of this equation by 2 yields $72\sqrt{13} = x\sqrt{117}$. Dividing both sides of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{13}}{\sqrt{117}} = x$. Multiplying the numerator and denominator

on the left-hand side of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{1,521}}{117} = x$, or $\frac{72(39)}{117} = x$, which is equivalent to $\frac{2,808}{117} = x$, or $24 = x$. Therefore, the length, in units, of the other leg of this triangle is **24**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $3\sqrt{13}$ is equivalent to $\sqrt{117}$, which is the length, in units, of the line segment shown in the xy-plane, not the length, in units, of the other leg of the triangle.

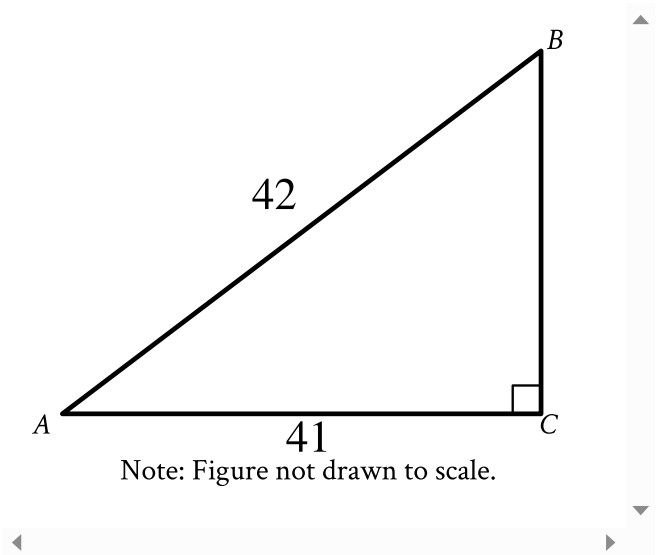
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a97f2bbe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: a97f2bbe



What is the value of $\cos A$ in the triangle shown?

- A. $\frac{42}{41}$
- B. $\frac{41}{42}$
- C. $\frac{1}{42}$
- D. $\frac{1}{41}$

ID: a97f2bbe Answer

Correct Answer: B

Rationale

Choice B is correct. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to angle A is 41 , and the length of the hypotenuse is 42 . Therefore, $\cos A = \frac{41}{42}$.

Choice A is incorrect. This is the value of $\frac{1}{\cos A}$.

Choice C is incorrect and may result from conceptual or calculation errors.

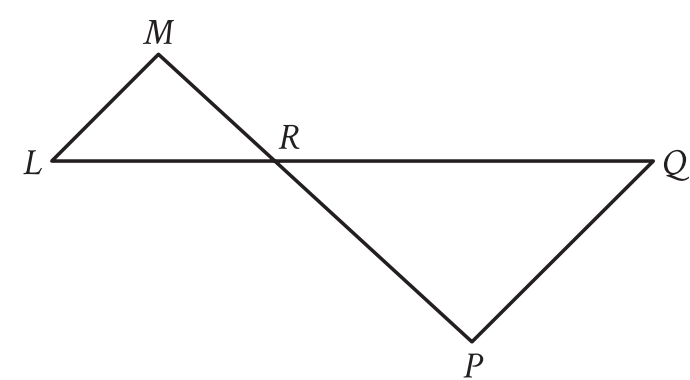
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8211c0f2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 8211c0f2



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 6, 7, and 11, respectively. What is the length of $\overline{LQ}$?

- A. $\frac{119}{11}$
- B. $\frac{77}{6}$
- C. $\frac{113}{6}$
- D. $\frac{119}{6}$

ID: 8211c0f2 Answer

Correct Answer: D

Rationale

Choice D is correct. The figure shows that angle MRL and angle PRQ are vertical angles. Since vertical angles are congruent, angle MRL and angle PRQ are congruent. It's given that $\overline{LM}$ is parallel to $\overline{PQ}$. The figure also shows that $\overline{LQ}$ intersects $\overline{LM}$ and $\overline{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, alternate interior angles MLR and PQR are congruent. Since triangles LMR and PQR have two pairs of congruent angles, the triangles are similar. Sides LR and MR in triangle LMR correspond to sides RQ and RP , respectively, in triangle PQR . Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{RQ}{LR} = \frac{RP}{MR}$. It's given that the lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 6, 7, and 11, respectively. Substituting 6 for MR , 7 for LR , and 11 for RP in the equation $\frac{RQ}{LR} = \frac{RP}{MR}$ yields $\frac{RQ}{7} = \frac{11}{6}$. Multiplying each side of this equation by 7 yields $RQ = (\frac{11}{6})(7)$, or $RQ = \frac{77}{6}$. It's given that $\overline{LQ}$ intersects $\overline{MP}$ at point R , so $LQ = LR + RQ$. Substituting 7 for LR and $\frac{77}{6}$ for RQ in this equation yields $LQ = 7 + \frac{77}{6}$, or $LQ = \frac{119}{6}$. Therefore, the length of $\overline{LQ}$ is $\frac{119}{6}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the length of $\overline{RQ}$, not $\overline{LQ}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID 9a7ee377

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 9a7ee377

A rectangle has an area of **63** square meters and a length of **9** meters. What is the width, in meters, of the rectangle?

- A. **7**
- B. **54**
- C. **81**
- D. **567**

ID: 9a7ee377 Answer

Correct Answer: A

Rationale

Choice A is correct. The area **A** , in square meters, of a rectangle is the product of its length ℓ , in meters, and its width **w** , in meters; thus, **$A = \ell w$** . It's given that a rectangle has an area of **63** square meters and a length of **9** meters. Substituting **63** for **A** and **9** for ℓ in the equation **$A = \ell w$** yields **$63 = 9w$** . Dividing both sides of this equation by **9** yields **$7 = w$** . Therefore, the width, in meters, of the rectangle is **7**.

Choice B is incorrect. This is the difference between the area, in square meters, and the length, in meters, of the rectangle, not the width, in meters, of the rectangle.

Choice C is incorrect. This is the square of the length, in meters, not the width, in meters, of the rectangle.

Choice D is incorrect. This is the product of the area, in square meters, and the length, in meters, of the rectangle, not the width, in meters, of the rectangle.

Question Difficulty: Medium

Question ID 28cf42b0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 28cf42b0

The length of one side of square M is 6 times the length of one side of square N. The area of square N is 49 square feet. What is the area, in square feet, of square M?

- A. 1,764
- B. 504
- C. 294
- D. 252

ID: 28cf42b0 Answer

Correct Answer: A

Rationale

Choice A is correct. The area, A , of a square is given by the equation $A = s^2$, where s is the length of one side of the square. Let x represent the length, in feet, of one side of square N. It's given that the area of square N is 49 square feet; therefore, it follows that $49 = x^2$. It's also given that the length of one side of square M is 6 times the length of one side of square N. Therefore, the length of one side of square M can be represented by the expression $6x$. Substituting $6x$ for s in the equation $A = s^2$ yields $A = (6x)^2$, or $A = 36x^2$. Since $49 = x^2$, substituting 49 for x^2 in the equation $A = 36x^2$ yields $A = 36(49)$, or $A = 1,764$. Therefore, the area, in square feet, of square M is 1,764.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 0e6301d0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 0e6301d0

Quadrilaterals $PQRS$ and $WXYZ$ are similar, where P , Q , and R correspond to W , X , and Y , respectively. The measure of $\angle S$ is 135° , $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

- A. 5°
- B. 27°
- C. 45°
- D. 135°

ID: 0e6301d0 Answer

Correct Answer: D

Rationale

Choice D is correct. Corresponding angles in similar figures have equal measure. It's given that quadrilaterals $PQRS$ and $WXYZ$ are similar and that P , Q , and R correspond to W , X , and Y . It follows that $\angle S$ corresponds to $\angle Z$. It's also given that the measure of $\angle S$ is 135° . Therefore, the measure of $\angle Z$ is 135° .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

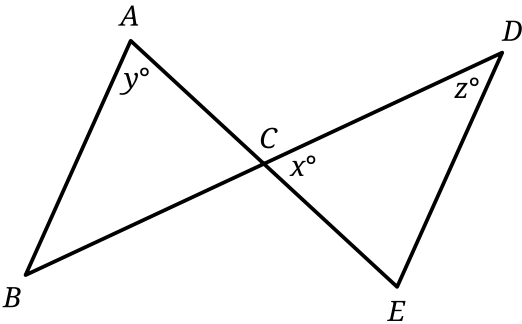
Choice C is incorrect. This is the supplement of the measure of $\angle Z$, not the measure of $\angle Z$.

Question Difficulty: Hard

Question ID 65975012

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 65975012



Note: Figure not drawn to scale.

In the figure, $\overline{AE}$ intersects $\overline{DB}$ at point C and $\overline{AB}$ is parallel to $\overline{DE}$, creating triangles ABC and CDE . If $x = 63$ and $y = 2x - 47$, what is the value of z ?

- A. 38
- B. 47
- C. 63
- D. 79

ID: 65975012 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{AE}$ intersects $\overline{DB}$ at point C and that $\overline{AB}$ is parallel to $\overline{DE}$, creating triangles ABC and CDE . Since $\overline{AB}$ is parallel to $\overline{DE}$, angles BAC and DEC are opposite interior angles and are congruent. It follows that the measure of angle DEC is y° . This means the three interior angles of triangle CDE have measures of x° , z° , and y° , respectively. It's given that $x = 63$ and $y = 2x - 47$. Substituting 63 for x in the equation $y = 2x - 47$ yields $y = 2(63) - 47$, or $y = 79$. Therefore, the three interior angles of triangle CDE have measures 63° , z° , and 79° , respectively. The sum of the interior angles of a triangle is 180° . Therefore, $63 + z + 79 = 180$, or $z + 142 = 180$. Subtracting 142 from both sides of this equation yields $z = 38$. Therefore, the value of z is 38 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of x , not z .

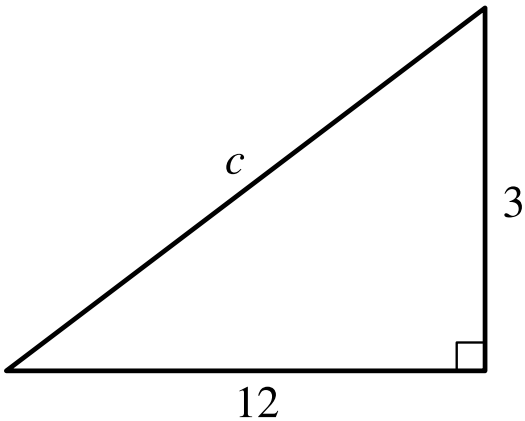
Choice D is incorrect. This is the value of y , not z .

Question Difficulty: Medium

Question ID b180d087

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: b180d087



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the triangle shown?

- A. $12 \cdot 3 = c$
- B. $12 + 3 = c$
- C. $12^2 + 3^2 = c^2$
- D. $12^2 - 3^2 = c^2$

ID: b180d087 Answer

Correct Answer: C

Rationale

Choice C is correct. The Pythagorean theorem states that for a right triangle, $a^2 + b^2 = c^2$, where a and b represent the lengths of the legs and c represents the length of the hypotenuse. It's given that the right triangle shown has legs with lengths **12** and **3** and a hypotenuse with length c . Substituting **12** for a and **3** for b in the formula $a^2 + b^2 = c^2$ yields $12^2 + 3^2 = c^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 00f5029f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 00f5029f

A rectangle has a length of **56** inches and a width of **28** inches. What is the area, in square inches, of the rectangle?

- A. **28**
- B. **84**
- C. **168**
- D. **1,568**

ID: 00f5029f Answer

Correct Answer: D

Rationale

Choice D is correct. The area, ***A***, of a rectangle can be found using the formula **$A = \ell w$** , where **ℓ** represents the length and **w** represents the width of the rectangle. It's given that a rectangle has a length of **56** inches and a width of **28** inches. Substituting **56** for **ℓ** and **28** for **w** in the formula **$A = \ell w$** yields **$A = (56)(28)$** , or **$A = 1,568$** . Therefore, the area, in square inches, of the rectangle is **1,568**.

Choice A is incorrect. This is the width, in inches, not the area, in square inches, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 18e8b39b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 18e8b39b

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF ?

ID: 18e8b39b Answer

Correct Answer: 135/8, 16.87, 16.88

Rationale

The correct answer is $\frac{135}{8}$. It's given that triangles ABC and DEF are similar and each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . For two similar triangles, if each side length of the first triangle is k times the corresponding side length of the second triangle, then the area of the first triangle is k^2 times the area of the second triangle. Therefore, the area of triangle ABC is 4^2 , or 16 , times the area of triangle DEF . It's given that the area of triangle ABC is 270 square inches. Let a represent the area, in square inches, of triangle DEF . It follows that 270 is 16 times a , or $270 = 16a$. Dividing both sides of this equation by 16 yields $\frac{270}{16} = a$, which is equivalent to $\frac{135}{8} = a$. Thus, the area, in square inches, of triangle DEF is $\frac{135}{8}$. Note that 135/8, 16.87, and 16.88 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 8eb0dbdd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 8eb0dbdd

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(9x - 560)^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-18x + w)^\circ$. What is the value of w ?

ID: 8eb0dbdd Answer

Correct Answer: 1660

Rationale

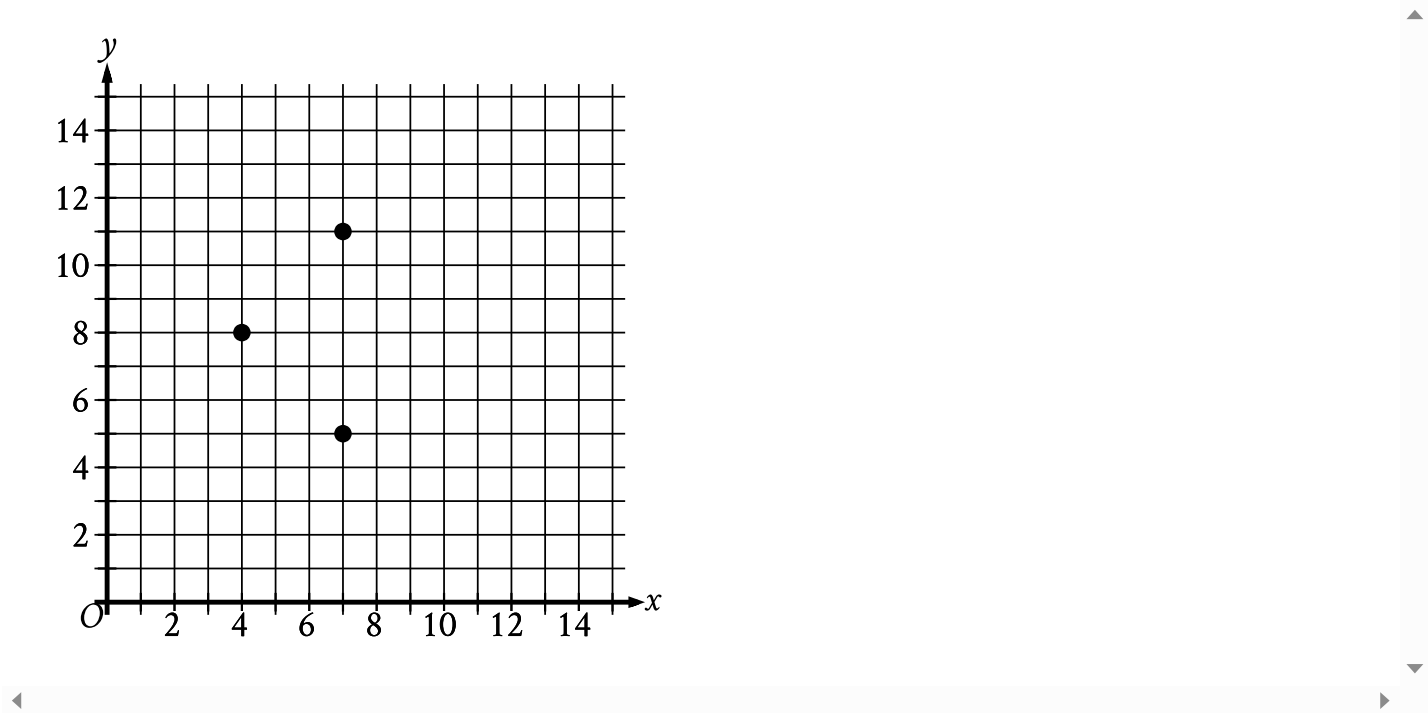
The correct answer is **1,660**. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. When two parallel lines are intersected by a transversal line, the angles formed have the following properties: two adjacent angles are supplementary, and alternate interior angles are congruent. Therefore, each of the four acute angles have the same measure, and each of the four obtuse angles have the same measure. It's also given that the measure of one of the acute angles is $(9x - 560)^\circ$. If two angles are supplementary, then the sum of their measures is 180° . Therefore, the measure of the obtuse angle adjacent to any of the acute angles is $(180 - (9x - 560))^\circ$, or $(180 - 9x + 560)^\circ$, which is equivalent to $(-9x + 740)^\circ$. It's given that the sum of the measures of one of the acute angles and three of the obtuse angles is $(-18x + w)^\circ$. It follows that $(9x - 560) + 3(-9x + 740) = (-18x + w)$, which is equivalent to $9x - 560 - 27x + 2,220 = -18x + w$, or $-18x + 1,660 = -18x + w$. Adding $18x$ to both sides of this equation yields $1,660 = w$.

Question Difficulty: Hard

Question ID 3cdfddf0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: 3cdfddf0



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

- A. 3
- B. 6
- C. 7
- D. 9

ID: 3cdfddf0 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point (7, 8) is halfway between the points (7, 5) and (7, 11) and is a distance of 3 units from each of those two points. The point (7, 8) is also a distance of 3 units from (4, 8). Because the point (7, 8) is the same distance from all three given points, it must be the center of the circle. The radius of a circle is the distance from the center to any point on the circle. Since that distance is 3, it follows that the radius of the circle is 3. The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the circle is $2(\pi)(3)$, or 6π . It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is 6.

Choice A is incorrect. This is the radius of the circle, not the value of k in the expression $k\pi$.

Choice C is incorrect. This is the x-coordinate of the center of the circle, not the value of k in the expression $k\pi$.

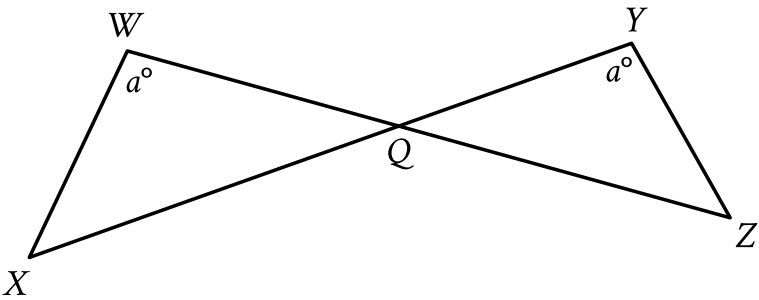
Choice D is incorrect. This is the value of k for which $k\pi$ represents the area of the circle, in square units, not the circumference of the circle, in units.

Question Difficulty: Hard

Question ID 57754a45

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 57754a45



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . $YQ = 63$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

ID: 57754a45 Answer

Correct Answer: 54

Rationale

The correct answer is **54**. The figure shown includes two triangles, triangle WQX and triangle YQZ , such that angle WQX and angle YQZ are vertical angles. It follows that angle WQX is congruent to angle YQZ . It's also given in the figure that the measures of angle W and angle Y are a° . Therefore angle W is congruent to angle Y . Since triangle WQX and triangle YQZ have two pairs of congruent angles, triangle WQX is similar to triangle YQZ by the angle-angle similarity postulate, where $\overline{YZ}$ corresponds to $\overline{WX}$, and $\overline{YQ}$ corresponds to $\overline{WQ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{WX} = \frac{YQ}{WQ}$. It's given that $YQ = 63$, $WQ = 70$, and $WX = 60$. Substituting **63** for YQ , **70** for WQ , and **60** for WX in the equation $\frac{YZ}{WX} = \frac{YQ}{WQ}$ yields $\frac{YZ}{60} = \frac{63}{70}$. Multiplying each side of this equation by **60** yields $YZ = (\frac{63}{70})(60)$, or $YZ = 54$. Therefore, the length of $\overline{YZ}$ is **54**.

Question Difficulty: Hard

Question ID 1fafa02d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 1fafa02d

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

ID: 1fafa02d Answer

Correct Answer: 14.66, 14.67, 44/3

Rationale

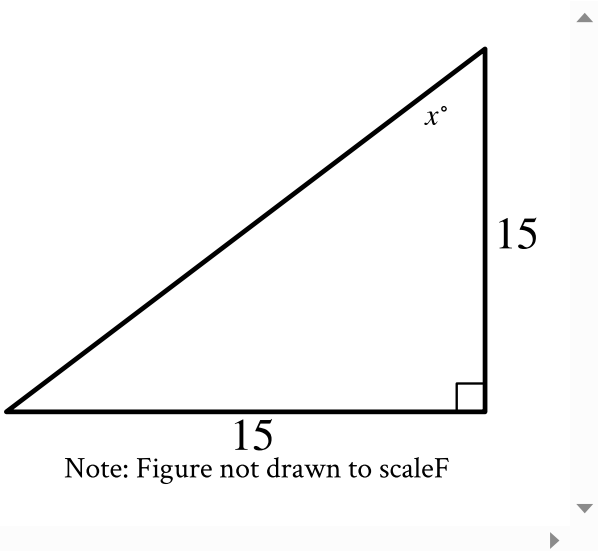
The correct answer is $\frac{44}{3}$. It's given that in triangle RST , angle T is a right angle. The area of a right triangle can be found using the formula $A = \frac{1}{2} \times \text{leg}_1 \times \text{leg}_2$, where A represents the area of the right triangle, leg_1 represents the length of one leg of the triangle, and leg_2 represents the length of the other leg of the triangle. In triangle RST , the two legs are $\overline{RT}$ and $\overline{ST}$. Therefore, if the length of $\overline{RT}$ is 72 and the area of triangle RST is 792, then $792 = \frac{1}{2}(72)(ST)$, or $792 = (36)(ST)$. Dividing both sides of this equation by 36 yields $22 = ST$. Therefore, the length of $\overline{ST}$ is 22. It's also given that point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. It follows that angle LKS is a right angle. Since triangles RST and LSK share angle S and have right angles T and K , respectively, triangles RST and LSK are similar triangles. Therefore, the ratio of the length of $\overline{RT}$ to the length of $\overline{LK}$ is equal to the ratio of the length of $\overline{ST}$ to the length of $\overline{SK}$. If the length of $\overline{RT}$ is 72 and the length of $\overline{LK}$ is 24, it follows that the ratio of the length of $\overline{RT}$ to the length of $\overline{LK}$ is $\frac{72}{24}$, or 3, so the ratio of the length of $\overline{ST}$ to the length of $\overline{SK}$ is 3. Therefore, $\frac{22}{SK} = 3$. Multiplying both sides of this equation by SK yields $22 = (3)(SK)$. Dividing both sides of this equation by 3 yields $\frac{22}{3} = SK$. Since the length of $\overline{ST}$, 22, is the sum of the length of $\overline{SK}$, $\frac{22}{3}$, and the length of $\overline{KT}$, it follows that the length of $\overline{KT}$ is $22 - \frac{22}{3}$, or $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID e2ae617a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: e2ae617a



In the triangle shown, what is the value of x ?

ID: e2ae617a Answer

Correct Answer: 45

Rationale

The correct answer is **45**. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of **15**. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are **45°** . It follows that the value of x is **45**.

Question Difficulty: Medium

Question ID 67b1d8bb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 67b1d8bb

In $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . What is the measure of $\angle Z$?

- A. 43°
- B. 89°
- C. 91°
- D. 179°

ID: 67b1d8bb Answer

Correct Answer: C

Rationale

Choice C is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180° . It's given that in $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . It follows that the measure of $\angle Z$ is $(180 - 23 - 66)^\circ$, or 91° .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID ad993e24

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: ad993e24

In right triangle QRS , the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4 . What is the length of side RS ?

ID: ad993e24 Answer

Correct Answer: 36

Rationale

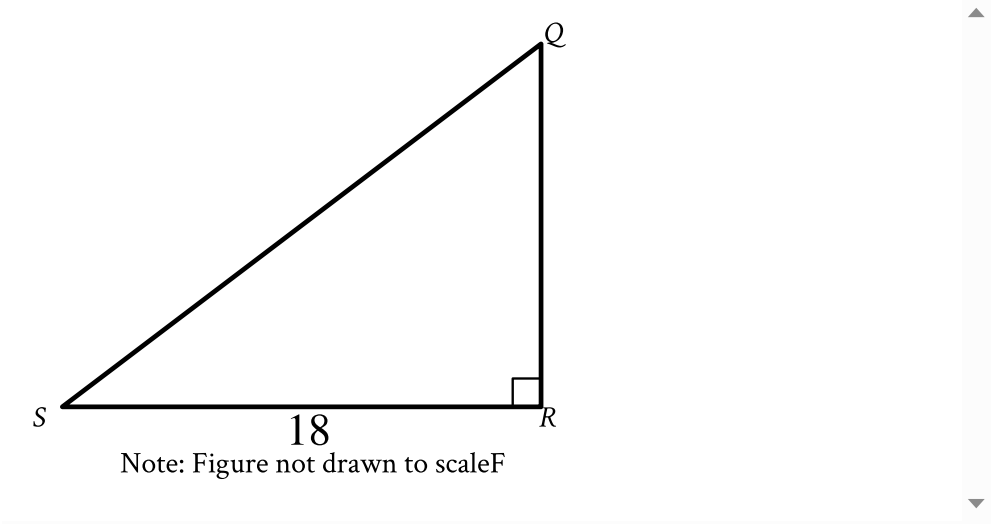
The correct answer is **36**. It's given that side SQ is the hypotenuse of right triangle QRS . It's also given that the length of side SQ is $4\sqrt{82}$ and the length of side QR is 4 . It follows that sides QR and RS are the legs of triangle QRS . The Pythagorean theorem states that for a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Let b represent the length of side RS . By the Pythagorean theorem, $4^2 + b^2 = \left(4\sqrt{82}\right)^2$, or $16 + b^2 = 1,312$. Subtracting **16** from both sides of this equation yields $b^2 = 1,296$. Taking the positive square root of both sides of this equation yields $b = 36$. Therefore, the length of side RS is **36**.

Question Difficulty: Hard

Question ID 1c902848

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 1c902848



In triangle QRS shown, $QR < RS$. Which expression represents the length of QS ?

- A. $18 \cos Q$
- B. $18 \sin Q$
- C. $\frac{18}{\cos Q}$
- D. $\frac{18}{\sin Q}$

ID: 1c902848 Answer

Correct Answer: D

Rationale

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 18. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS , an expression representing QS can be found by substituting 18 for RS in this equation, which yields $\sin Q = \frac{18}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 18$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{18}{\sin Q}$. Therefore, the expression $\frac{18}{\sin Q}$ represents the length of QS .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 77361a06

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 77361a06



In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

- A. $x < 115$
- B. $x > 115$
- C. $x + y < 180$
- D. $x + y > 180$

ID: 77361a06 Answer

Correct Answer: B

Rationale

Choice B is correct. In the figure shown, the angle measuring y° is congruent to its vertical angle formed by lines s and m , so the measure of the vertical angle is also y° . The vertical angle forms a same-side interior angle pair with the angle measuring x° . It's given that lines r and s are parallel. Therefore, same-side interior angles in the figure are supplementary, which means the sum of the measure of the vertical angle and the measure of the angle measuring x° is 180° , or $x + y = 180$. Subtracting x from both sides of this equation yields $y = 180 - x$. Substituting $180 - x$ for y in the inequality $y < 65$ yields $180 - x < 65$. Adding x to both sides of this inequality yields $180 < 65 + x$. Subtracting 65 from both sides of this inequality yields $115 < x$, or $x > 115$. Thus, if $y < 65$, it must be true that $x > 115$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $x + y$ must be equal to, not less than, 180 .

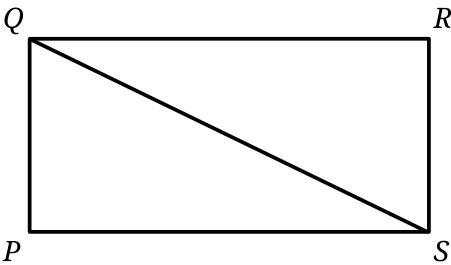
Choice D is incorrect. $x + y$ must be equal to, not greater than, 180 .

Question Difficulty: Hard

Question ID aeb838ed

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: aeb838ed



Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS . What is the length of side QR ?

ID: aeb838ed Answer

Correct Answer: 38

Rationale

The correct answer is **38**. It's given that in rectangle $PQRS$, the length of side PS is 8 more than the length of side RS . In a rectangle, opposite sides are congruent. Since side QR is opposite side PS , it follows that side QR is congruent to side PS . Thus, the length of side QR is 8 more than the length of side RS . Let x represent the length of side RS . It follows that the length of side QR is represented by the expression $x + 8$. Since all angles in a rectangle are right angles, angle R is a right angle. It follows that side QR , side RS , and line segment QS form a right triangle, where line segment QS is the hypotenuse and sides QR and RS are the legs. It's given that the length of line segment QS is $\sqrt{2,344}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting x for a , $x + 8$ for b , and $\sqrt{2,344}$ for c in this equation yields $x^2 + (x + 8)^2 = (\sqrt{2,344})^2$, or $x^2 + (x + 8)^2 = 2,344$. Expanding the left-hand side of this equation yields $x^2 + x^2 + 16x + 64 = 2,344$, or $2x^2 + 16x + 64 = 2,344$. Dividing both sides of this equation by 2 yields $x^2 + 8x + 32 = 1,172$. To complete the square, subtracting 16 from both sides of this equation yields $x^2 + 8x + 16 = 1,156$. Since the left-hand side of this equation is a perfect square trinomial, it follows that $x^2 + 8x + 16 = 1,156$ is equivalent to $(x + 4)^2 = 1,156$. Taking the positive square root of both sides of this equation yields $x + 4 = 34$. Subtracting 4 from both sides of this equation yields $x = 30$. Since the length of side QR is represented by the expression $x + 8$, it follows that the length of side QR is $30 + 8$, or **38**.

Question Difficulty: Hard

Question ID bdf1d4ad

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Geometry and Trigonometry	Area and volume	

ID: bdf1d4ad

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is **58 square centimeters (cm^2)**, and the surface area of right rectangular prism Y is **1,450 cm^2** . The volume of right rectangular prism Y is **1,250 cubic centimeters (cm^3)**. What is the sum of the volumes, **in cm^3** , of right rectangular prism X and right rectangular prism Y?

ID: bdf1d4ad Answer

Correct Answer: 1260

Rationale

The correct answer is **1,260**. Since it's given that prisms X and Y are similar, all the linear measurements of prism Y are **k** times the respective linear measurements of prism X, where **k** is a positive constant. Therefore, the surface area of prism Y is **k^2** times the surface area of prism X and the volume of prism Y is **k^3** times the volume of prism X. It's given that the surface area of prism Y is **1,450 cm^2** , and the surface area of prism X is **58 cm^2** , which implies that **$1,450 = 58k^2$** . Dividing both sides of this equation by **58** yields **$\frac{1,450}{58} = k^2$** , or **$k^2 = 25$** . Since **k** is a positive constant, **$k = 5$** . It's given that the volume of prism Y is **1,250 cm^3** . Therefore, the volume of prism X is equal to **$\frac{1,250}{k^3} \text{ cm}^3$** , which is equivalent to **$\frac{1,250}{5^3} \text{ cm}^3$** , or **10 cm^3** . Thus, the sum of the volumes, in **cm^3** , of the two prisms is **1,250 + 10**, or **1,260**.

Question Difficulty: Hard